Syllabus

Introduction to Control Systems . Laplace transforms. Modeling of mechanical, liquid-level and electrical control systems. Block diagram Reduction and obtaining transfer function. Calculation of time response of first-order and second-order .Applying different types of controllers (P, I, D, PI.PD and PID).Analyzing system stability using Routh's and Nyquist criteria. Drawing Bode Plot and

Text book: K. Ogata Modern control Engineering. 5th edition

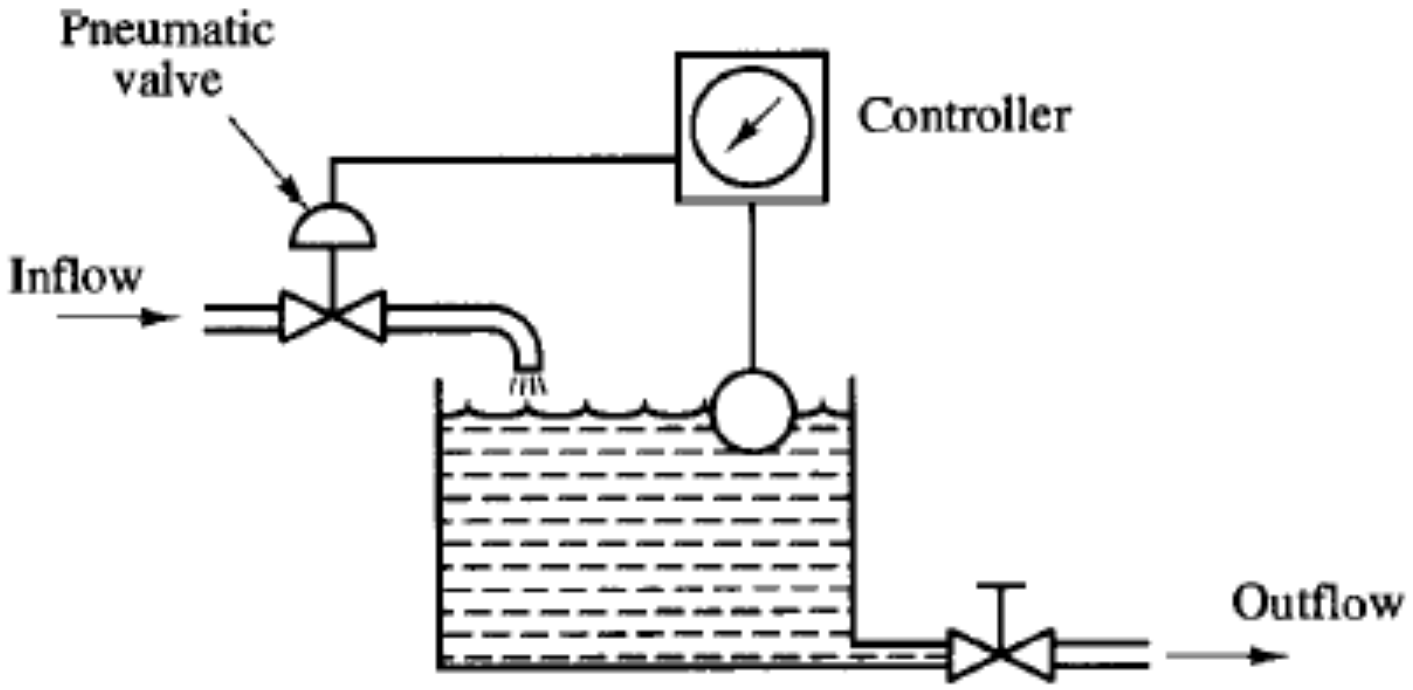
Introduction to Control Systems

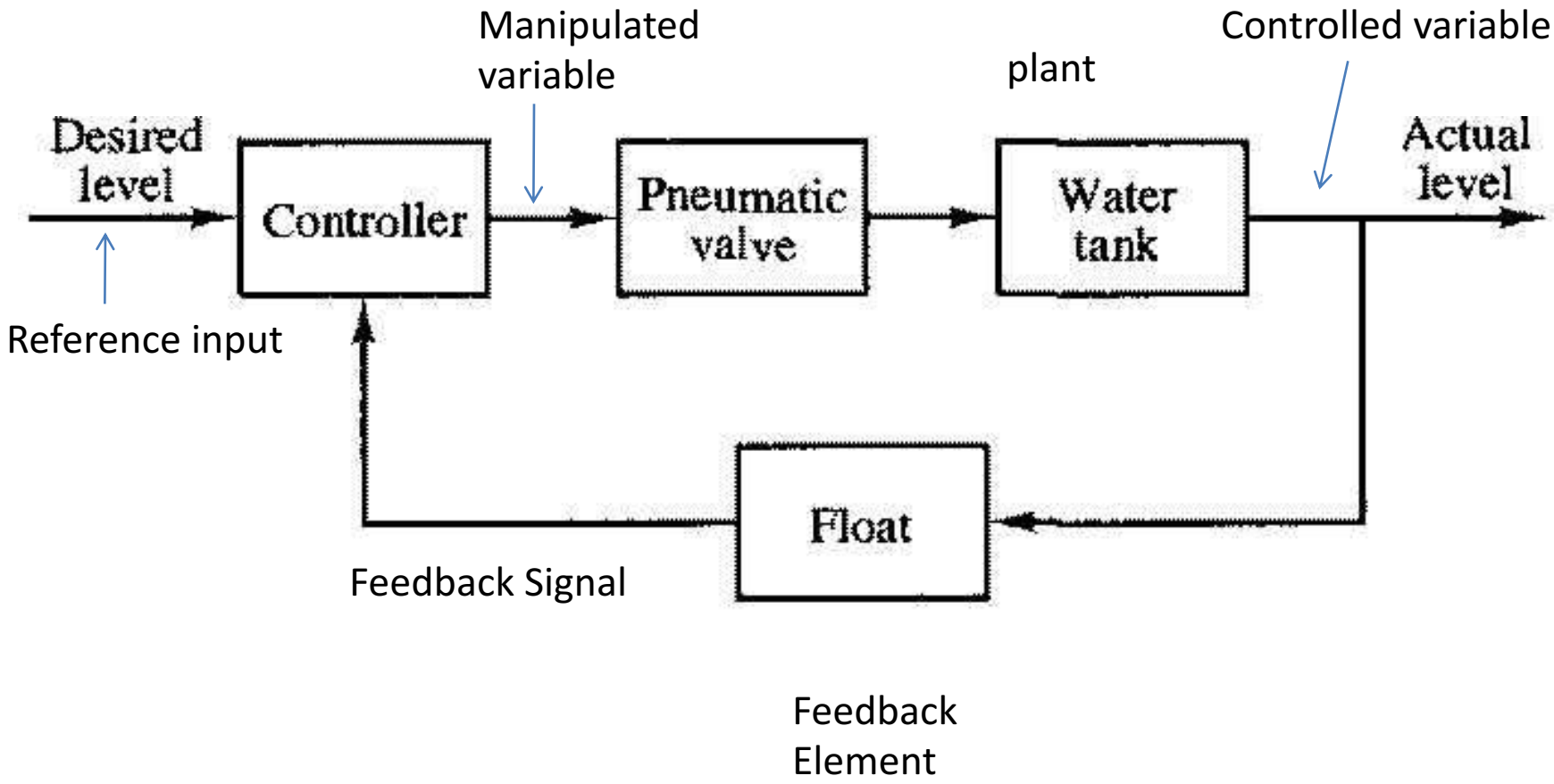
applications

- Robotic systems
- Missile guidance systems
- Industrial processes
- Transportations

Liquid-level Control system

level control is very common in water treatment, chemical, petrochemical, biomedical, food & bevarage, Pharmaceutical etc. industries .




Manipulated variable
plant
Controlled variable
Desired level
Controller
Pneumatic valve
Water tank
Actual level
Reference input
Float
Feedback Signal
Feedback Element

Definitions.

- *controlled* variable is the quantity or condition that is measured and controlled. the controlled variable is the output of the system.
- **The *manipulated* variable** is the quantity or condition that is varied by the controller so as to affect the value of the controlled variable.

- ***Control*** means measuring the value of the controlled variable of the system and applying the manipulated variable to the system to correct or limit deviation of the measured value from a desired value.
- ***Plant*** physical object to be controlled (such as a mechanical device, a heating furnace, a chemical reactor, or a spacecraft)

- ***Processes.*** defines a process to be a natural, progressively continuing operation or development marked by a series of gradual changes that succeed one another in a relatively fixed way and lead toward a particular result or end;
- ***Systems.*** A system is a combination of components that act together and perform a certain objective.

- ***Disturbances.*** A disturbance is a signal that tends to adversely affect the value of the output of a system. If a disturbance is generated within the system, it is called ***internal,*** while an ***external*** disturbance is generated outside the system and is an input.

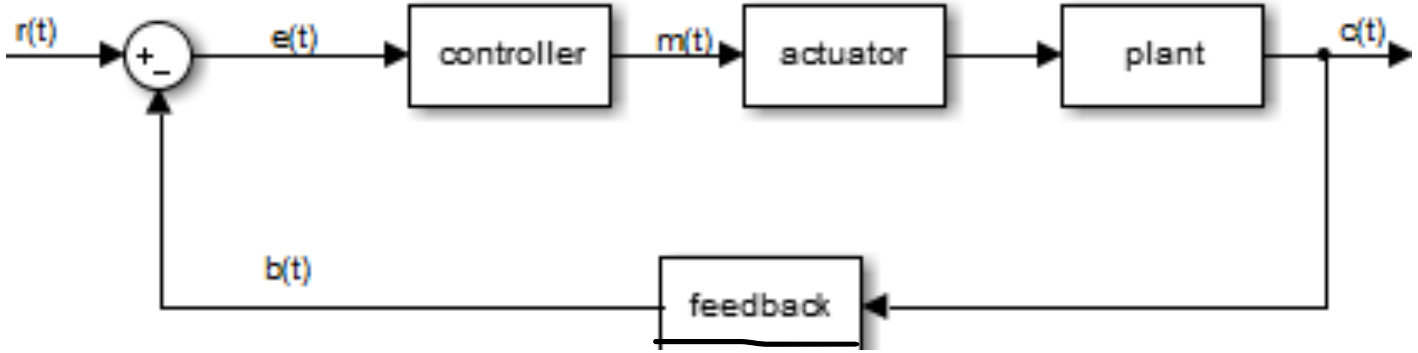
- ***Feedback Control*** Feedback control refers to an operation that, in the presence of disturbances, tends to reduce the difference between the output of a system and some reference input and that does so on the basis of this difference.

Closed-Loop control Versus Open-Loop control

- **Feedback control** systems. A system that maintains a prescribed relationship between the output and the reference input by comparing them and using the difference as a means of control is called a *feedback control system*

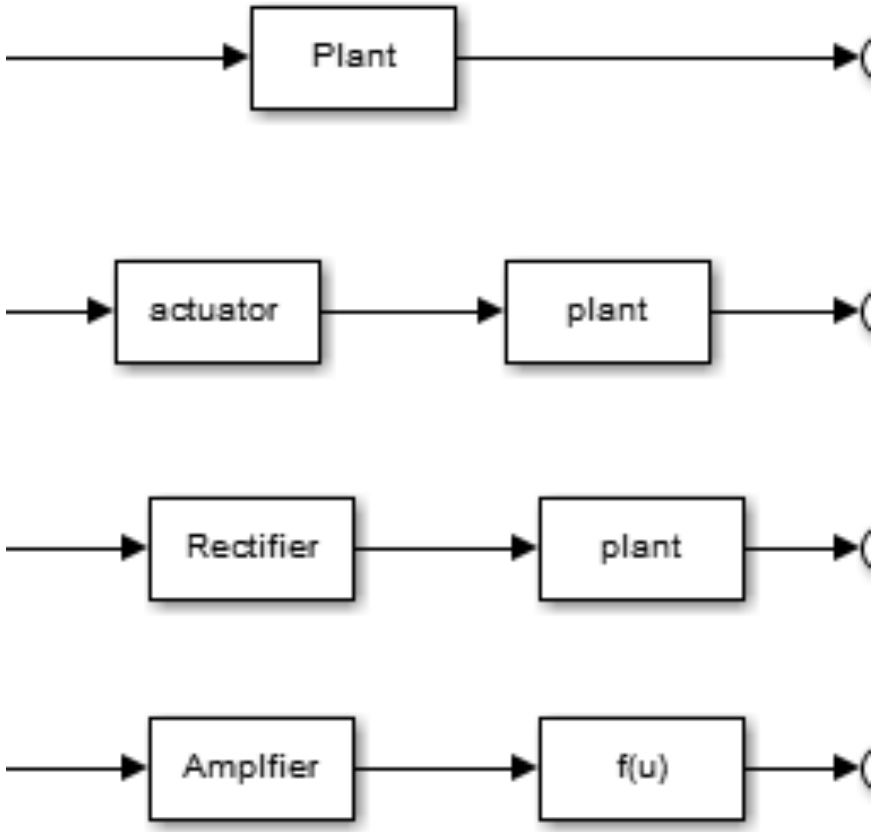
Closed-loop control systems

- **Closed-loop control systems.** Feedback control systems are often referred to as *closed-loop control systems.*
- In practice, the terms feedback control and closed-loop control are used interchangeably.
- In a closed-loop control system the actuating error signal, which is the difference between the input signal and the feedback signal , is fed to the controller so as to reduce the error and bring the output of the system to a desired value.



open-loop control systems

- open-loop control systems. Those systems in which the output has no effect on the control action are called *open-loop control systems.* In other words, in an open-loop control system the output is neither measured nor fed back for comparison with the input.



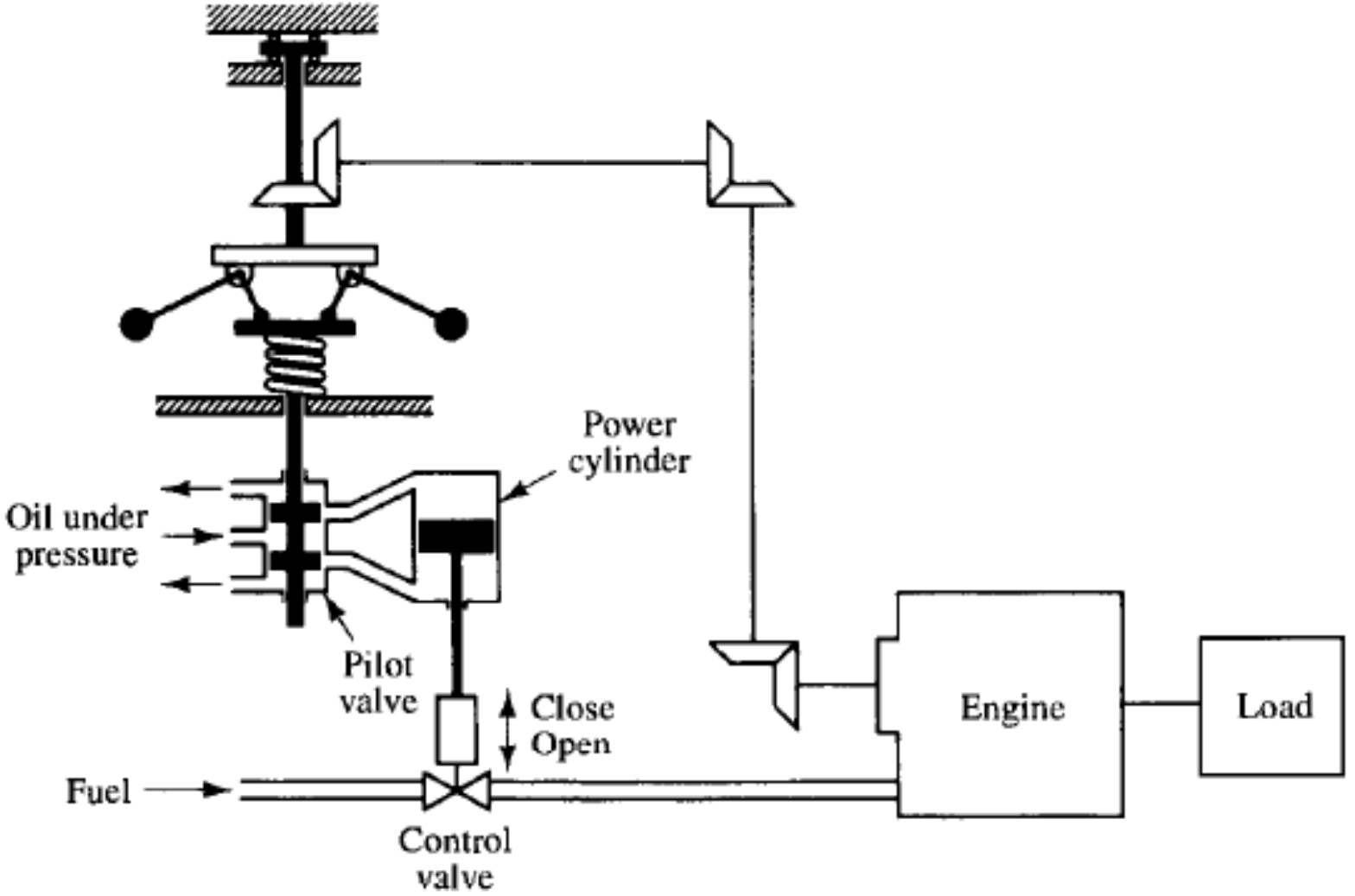
- In the presence of disturbances, an open-loop control system will not perform the desired task.
- Open-loop control can be used, in practice, only if the relationship between the input and output is known and if there are neither internal nor external disturbances.

- Closed-loop versus open-loop control systems.

  possible to use relatively inaccurate and inexpensive components to obtain the accurate control of a given plant, whereas doing so is impossible in the open-loop case.

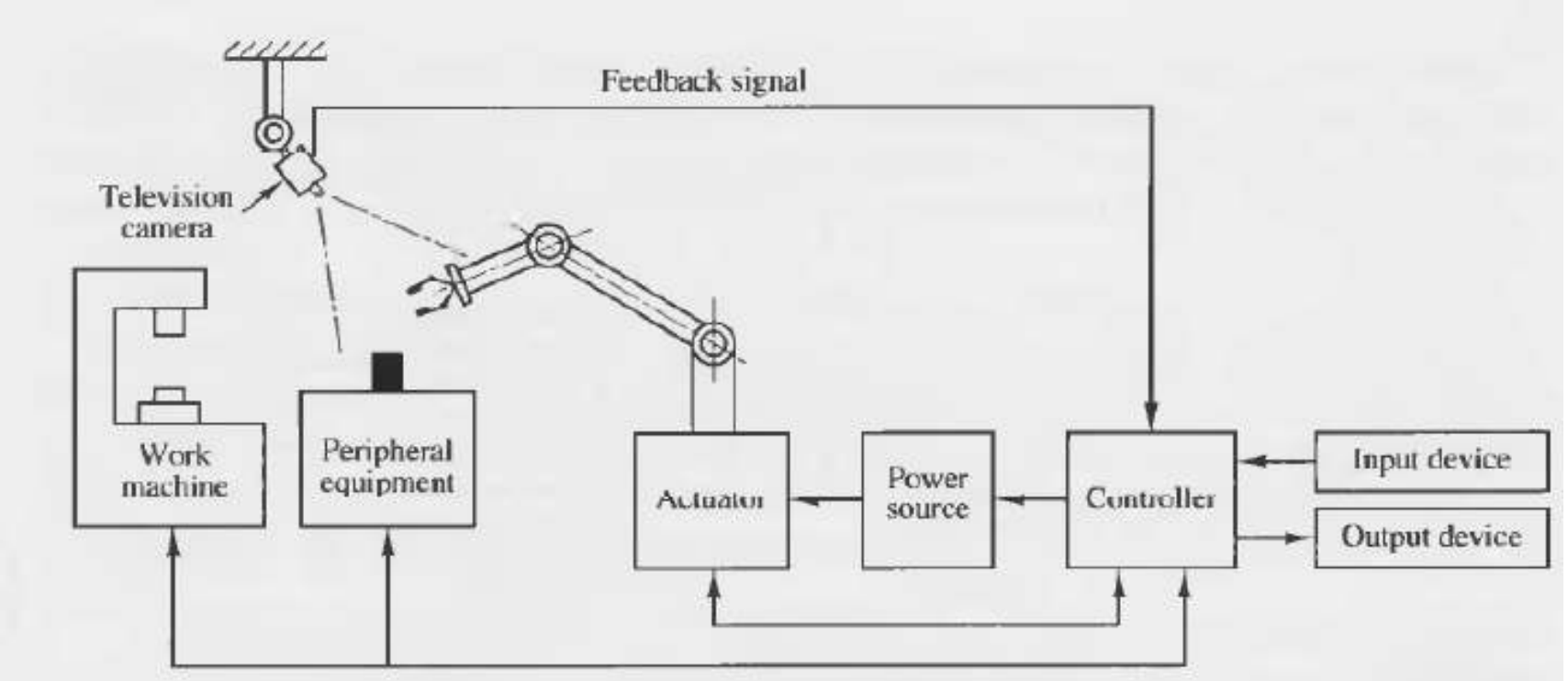
  From the point of view of stability, the open-loop control system is easier to build because system stability is not a major problem. On the other hand, stability *is* a major problem in the closed-loop control system,

Speed control system.



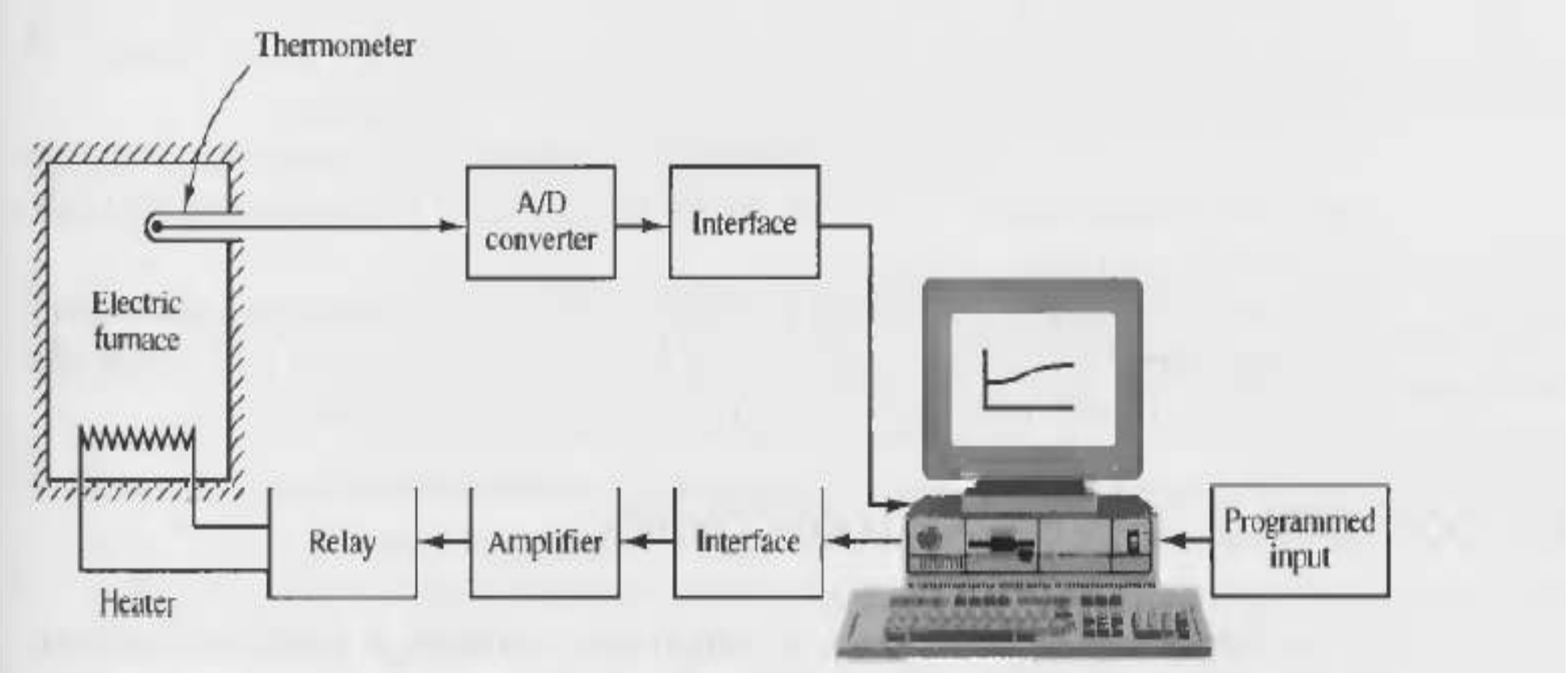
- The amount or fuel admitted to the engine is adjusted according to the difference between the desired and the actual engine speeds. The sequence of actions may be stated as follows:
- The speed governor is adjusted such that, at the desired speed, no pressured oil will flow into either side of the power cylinder.
- If the actual speed drops below the desired value due to disturbance, then the decrease in the centrifugal force of the speed governor causes the control valve to move downward, supplying more fuel, and the speed of the engine increases until the desired value is reached,
- On the other hand, if the speed of the engine increases above the desired value, then the increase in the centrifugal force of the governor causes the control valve to move upward. This decreases the supply of fuel, and the speed of the engine decreases until the desired value is reached.

Robot control system



- the computerized robot recognizes the presence and orientation of each mechanical part by a pattern-recognition process that consists of reading the code numbers attached to it. Then the robot picks up the part and moves it to an appropriate place for assembling, and there it assembles several parts into a component. A well-programmed digital computer acts as a controller.

Temperature control system



- The temperature in the electric furnace is measured by a thermometer, which is an analog device.
- The analog temperature is converted to a digital temperature by an *AID* converter.
- The digital temperature is fed to a controller through an interface.
- This digital temperature is compared with the programmed input temperature, and if there is any error, the controller sends out a signal to
  the heater

UNIT 2

Laplace Transform

- Why Laplace Transform
- Definition of Laplace Transform
- Existence of Laplace transform
- Laplace transform Table
- Laplace Transform properties
- Inverse Laplace Transform

Why Laplace Transform

- **Solving linear differential equations**
- **Obtaining transfer functions**
- **Converts many functions into algebraic functions of s**
- **Converts LDEs into algebraic equations of s**
- **Use graphical techniques for predicting a system performance with out solving DE**

Definition of Laplace Transform

Let us define

$f(t)$ = a function of time t such that $f(t) = 0$ for $t < 0$

s = a complex variable

$\mathcal{L}$ = an operational symbol indicating that the quantity that it prefixe is to be transformed by the Laplace integral $\int_0^\infty e^{-st}\,dt$

$F(s)$ = Laplace transform of $f(t)$

Then the Laplace transform of $f(t)$ is given by

$$\mathcal{L}[f(t)] = F(s) = \int_0^\infty e^{-st}\,dt[f(t)] = \int_0^\infty f(t)e^{-st}\,dt$$

EXAMPLES

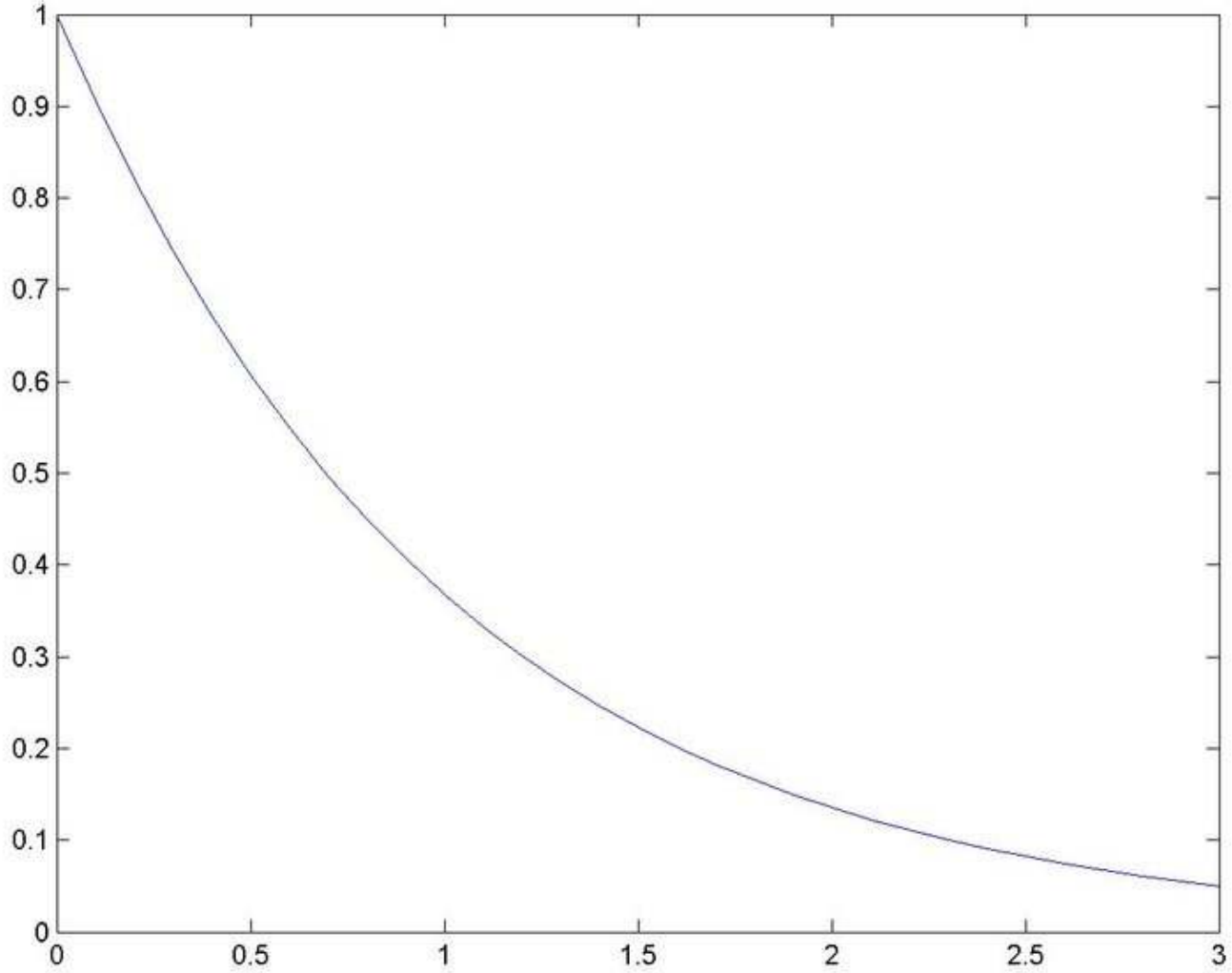
Exponential function. Consider the exponential function

$$f(t) = 0, \qquad \text{for } t < 0$$

$$= Ae^{-\alpha t}, \qquad \text{for } t \geq 0$$

where A and α are constants. The Laplace transform of this exponential function can be obtained as follows:

$$\mathcal{L}[Ae^{-\alpha t}] = \int_0^\infty Ae^{-\alpha t}e^{-st}\,dt = A\int_0^\infty e^{-(\alpha+s)t}\,dt = \frac{A}{s+\alpha}$$



Damped exponential function

Step function. Consider the step function

$$f(t) = 0, \qquad \text{for } t < 0$$

$$= A, \qquad \text{for } t > 0$$



where A is a constant. Note that it is a special case of the exponential function $Ae^{-\alpha t}$, where $\alpha = 0$. The step function is undefined at $t = 0$. Its Laplace transform is given by

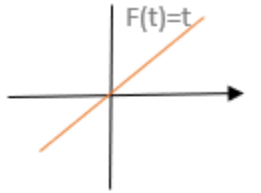
$$\mathcal{L}[A] = \int_0^\infty Ae^{-st}\,dt = \frac{A}{s}$$

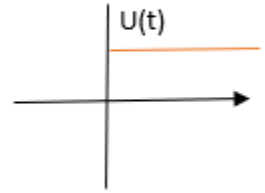
Ramp function. Consider the ramp function

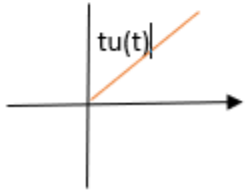
$$f(t) = 0, \qquad \text{for } t < 0$$
$$= At, \qquad \text{for } t \geq 0$$



OR







where A is a constant. The Laplace transform of this ramp function is obtained as

$$\mathcal{L}[At] = \int_0^\infty Ate^{-st}\,dt = At\frac{e^{-st}}{-s}\bigg|_0^\infty - \int_0^\infty \frac{Ae^{-st}}{-s}\,dt$$

$$= \frac{A}{s}\int_0^\infty e^{-st}\,dt = \frac{A}{s^2}$$

Sinusoidal function. The Laplace transform of the sinusoidal function

$$f(t) = 0, \qquad \text{for } t < 0$$

$$= A \sin \omega t, \qquad \text{for } t \geq 0$$

where A and ω are constants, is obtained as follows. Referring to Equation (2–3), $\sin \omega t$ can be written

$$\sin \omega t = \frac{1}{2j}(e^{j\omega t} - e^{-j\omega t})$$

Hence

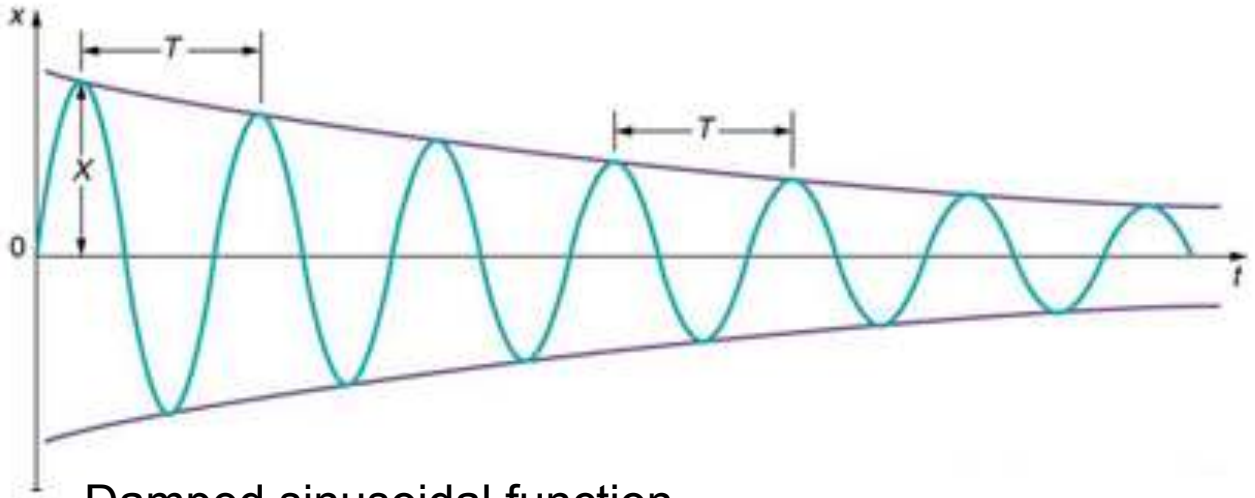
$$\mathcal{L}[A \sin \omega t] = \frac{A}{2j}\int_0^{\infty} (e^{j\omega t} - e^{-j\omega t})e^{-st}\, dt$$

$$= \frac{A}{2j}\frac{1}{s - j\omega} - \frac{A}{2j}\frac{1}{s + j\omega} = \frac{A\omega}{s^2 + \omega^2}$$

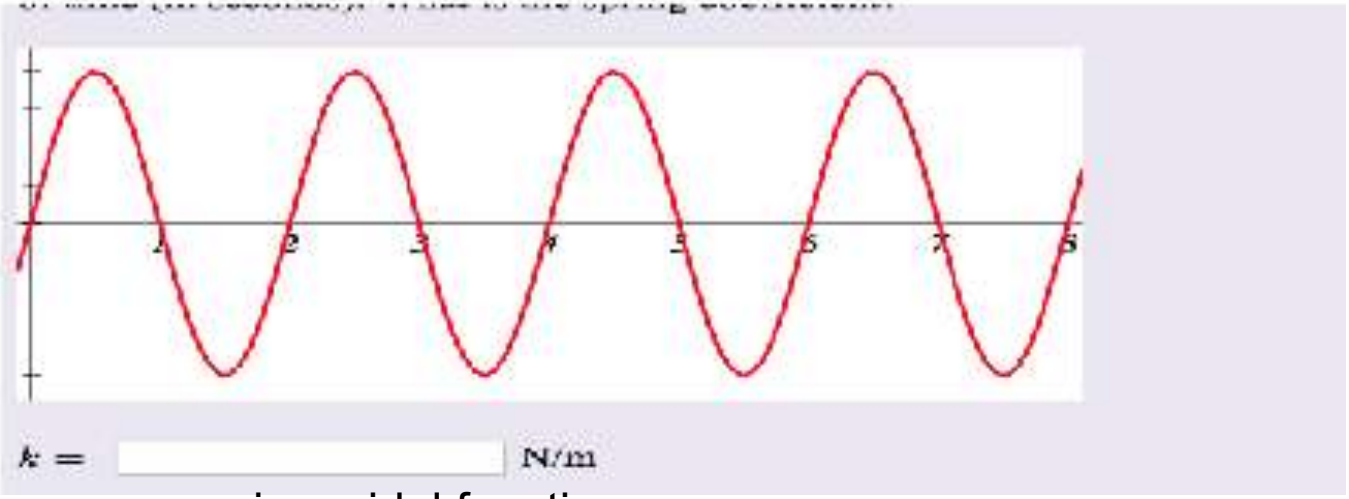
Laplace Transform Pairs(Laplace Transform Table)

function	Waveform $f(t)$	Transform F(s)
Impulse	$\delta(t)$	1
unit step	$u(t)$	$\frac{1}{s}$
unit ramp	$tu(t)$	$\frac{1}{s^2}$
Exponential	$[e^{-\alpha t}]u(t)$	$\frac{1}{s+\alpha}$
Damped ramp	$[te^{-\alpha t}]u(t)$	$\frac{1}{(s+\alpha)^2}$
Sine	$[\sin \beta t]u(t)$	$\frac{\beta}{s^2+\beta^2}$
Cosine	$[\cos \beta t]u(t)$	$\frac{s}{s^2+\beta^2}$
Damped sine	$[e^{-\alpha t} \sin \beta t]u(t)$	$\frac{\beta}{(s+\alpha)^2+\beta^2}$
Damped cosine	$[e^{-\alpha t} \cos \beta t]u(t)$	$\frac{(s+\alpha)}{(s+\alpha)^2+\beta^2}$

Inputs only Inputs and output Output only



Damped sinusoidal function

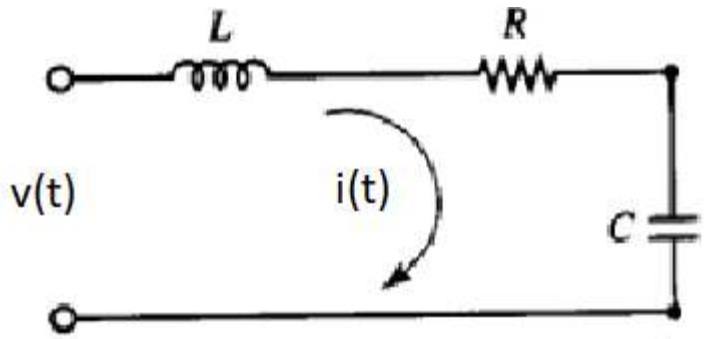


sinusoidal function

Laplace Transform Properties

Addition/Scaling	$L[af_1(t) \pm bf_2(t)] = aF_1(s) \pm bF_2(s)$
Differentiation	$L\left[\frac{d}{dt} f(t)\right] = sF(s) - f(0\pm)$
Integration	$L\left[\int f(t)dt\right] = \frac{F(s)}{s} + \frac{1}{s}\left[\int f(t)dt\right]_{t=0\pm}$
Convolution	$\int_0^t f_1(t-t)f_2(t)dt = F_1(s)F_2(s)$
Initial-value theorem	$f(0+) = \lim_{s\to\infty} sF(s)$
Final-value theorem	$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$

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Circuit equation in time domain

$$v(t) = Ri(t) + L\frac{di(t)}{dt} + \frac{1}{C}\int i(t)dt$$

Applying Laplace transform to the forgoing equation

$$\mathcal{L}[\boldsymbol{v(t)}] = \mathcal{L}[\boldsymbol{Ri(t)}] + \mathcal{L}[\boldsymbol{L\frac{di(t)}{dt}}] + \mathcal{L}[\boldsymbol{\frac{1}{C}\int i(t)dt}]$$

We obtain

$$V(s) = RI(s) + L[sI(s) - i_0] + \frac{1}{C}[\frac{1}{s}I(s) + \frac{1}{s}\int i(t)dt_{t=0}]$$

For zero initial condition

$$V(s) = RI(s) + LsI(s) + \frac{1}{C}\frac{1}{s}I(s)$$

The transfer function

$$\frac{\boldsymbol{I(s)}}{\boldsymbol{V(s)}} = \frac{\boldsymbol{Cs}}{\boldsymbol{LCs^2+RCs+1}}$$ **=G(s)**

$$I(s) = G(s)V(s)$$

INVERSE LAPLACE TRANSFORMATION

- **the inverse Laplace transform can be obtained using the inversion integral**

$$\mathcal{L}^{-1}[F(s)] = f(t) = \frac{1}{2\pi j}\int_{c-j\infty}^{c+j\infty} F(s)e^{st}\,ds, \qquad \text{for } t > 0$$

- ***However, the inversion integral is complicated***

The inverse Laplace Transform can be calculated in a few ways. If the function whose inverse Laplace Transform you are trying to calculate is in the table, you are done. Otherwise, we will use partial fraction expansion (PFE); it is also called partial fraction decomposition. If you have never used partial fraction expansions you may wish to read a background article, but you can probably continue without it.

Inverse Laplace Transform by Partial Fraction Expansion

This technique uses Partial Fraction Expansion to split up a complicated fraction into forms that are in the Laplace Transform table. As you read through this section, you may find it helpful to refer to the review section on partial fraction expansion techniques. The text below assumes you are familiar with that material.

- **For problems in control systems analysis, *F(s),* the Laplace transform of f(t), frequently occurs in the form**

$$F(s) = \frac{B(s)}{A(s)}$$

- **where *A(s)* and *B(s)* are polynomials in s. In the expansion of *F(s)* into a partial-fraction form, it is important that the highest power of *s* in *A(s)* be greater than the highest power of *s* in *B(s),***
- **If such is not the case, the numerator *B(s)* must be divided by the denominator *A(s)* in order to produce a polynomial ins plus a remainder**

If F(s) is broken up into components,

$$F(s) = F_1(s) + F_2(s) + \cdots + F_n(s)$$

and if the inverse Laplace transforms of F1(s),F2(s), . . . , Fn(s) are read available, the

$$\begin{aligned}\mathcal{L}^{-1}[F(s)] &= \mathcal{L}^{-1}[F_1(s)] + \mathcal{L}^{-1}[F_2(s)] + \cdots + \mathcal{L}^{-1}[F_n(s)] \\ &= f_1(t) + f_2(t) + \cdots + f_n(t)\end{aligned}$$

$$F(s) = \frac{B(s)}{A(s)} = \frac{K(s + z_1)(s + z_2)\cdots(s + z_m)}{(s + p_1)(s + p_2)\cdots(s + p_n)},$$

$-z_1, -z_2, \ldots -z_m$ are the zeros of F(s)

$-p_1, -p_2, \ldots -p_n$ are the poles of F(s)

Important Formulas for Solving Quadratic Equations

The following list of important formulas is helpful to solve quadratic equations.

- The standard form of a quadratic equation is $ax^2 + bx + c = 0$
- The formula to find the roots of the quadratic equation is

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- The discriminant of the quadratic equation is $D = b^2 - 4ac$
- For $D > 0$ the roots are real and different (distinct).
- For $D = 0$ the roots are real and equal.(multiple or repeated)
- For $D < 0$ the roots are complex-conjugate

- Consider

There are three cases to consider in doing the partial fraction expansion of F(s).

Case 1: **F(s) has all non repeated simple roots.(distinct poles)**

$$F(s)=\frac{s+2}{s(s+1)}$$

Case 2: **F(s) has complex poles:**

$$F(s)=\frac{s+2}{(s+1+j2)(s+1-j2)}$$

Case 3: **F(s) has repeated poles. (multiple poles)**

$$F(s)=\frac{s+2}{(s+1)(s+1)}$$

Case #1 Partial-fraction expansion when *F(s)* involves distinct poles only

Partial Fraction Expansion

$$F(s) = \frac{a_1}{s + p_1} + \frac{a_2}{s + p_2} + \cdots + \frac{a_n}{s + p_n}$$

Where a_1 , a_2 and a_3 are the residues and calculated Using residue method or common denominator method

Residue method: Multiply both sides by (s+a_1)

$$(s+p_1)F(s) = a_1 + \frac{a_2(s+p_1)}{s+p_2} + \ldots + \frac{a_n(s+p_1)}{s+p_n}$$

Cancelling common terms and set s= -P_1

$$(s+p_1)F(s)\big|s = -p_1 = a_1$$

We see that all the expanded terms drop out with the exception of a_k. Thus the residue a_k is found from

$$a_k = \left[(s + p_k) \frac{B(s)}{A(s)} \right]_{s=-p_k}$$

Since

$$\mathcal{L}^{-1}\left[\frac{a_k}{s + p_k} \right] = a_k e^{-p_k t}$$

$f(t)$ is obtained as

$$f(t) = \mathcal{L}^{-1}[F(s)] = a_1 e^{-p_1 t} + a_2 e^{-p_2 t} + \cdots + a_n e^{-p_n t}, \qquad \text{for } t \geq 0$$

$Given\ \ G(s)\frac{1}{(s+1)(s+2)}\ and R(S)=\frac{2}{s}$, Then the output is

$$F(s)=\frac{2}{s(s+1)(s+2)}=\frac{A1}{s}+\frac{A2}{s+1}+\frac{A3}{s+2}$$

Multiplying both sides by s

$$\frac{2s}{s(s+1)(s+2)}=\frac{sA1}{s}+\frac{sA2}{s+1}+\frac{sA3}{s+2}$$

Canceling the common terms and replacing s by zero

$$\frac{2s}{s(s+1)(s+2)}=\frac{sA1}{s}+\frac{sA2}{s+1}+\frac{sA3}{s+2}$$

S=0

We have $A1=\frac{2}{(s+1)(s+2)}$

S=0

And A1=1

$$A2 = \frac{2}{s(s+2)}$$

S=-1

A2=-2

$$A3 = \frac{2}{s(s+1)}$$

S=-2

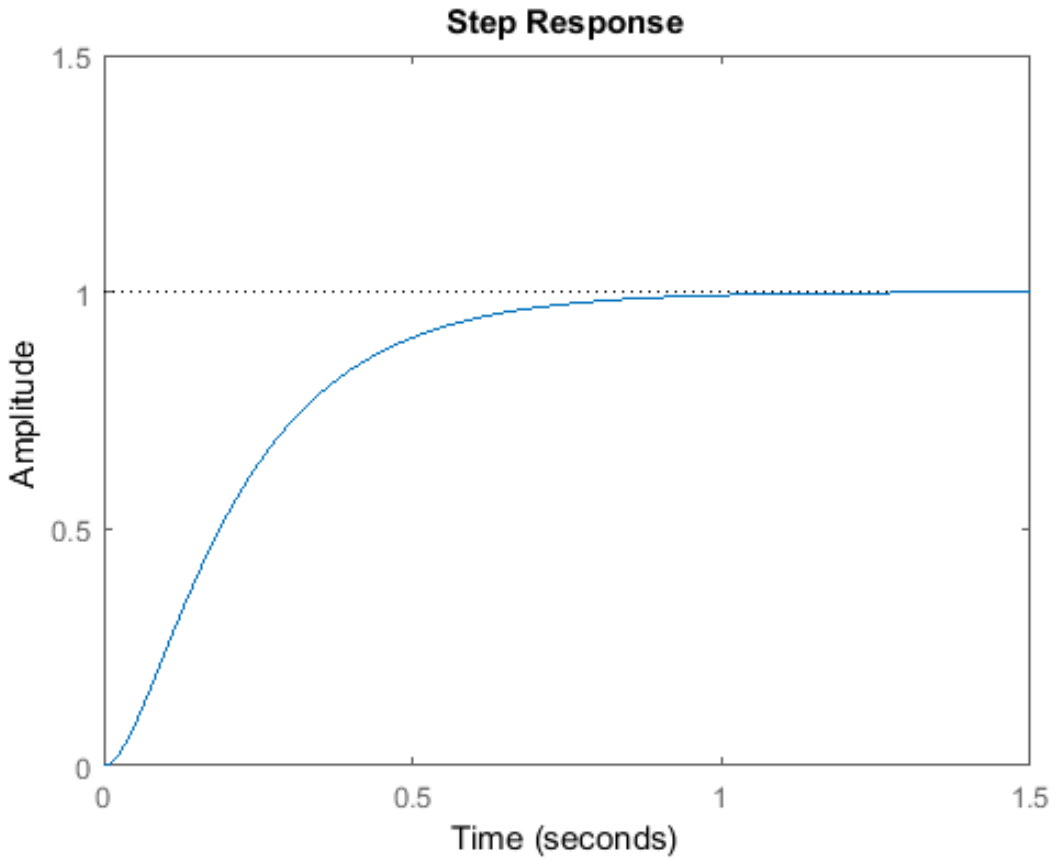
A3=+1

$$F(s) = \frac{1}{s} - \frac{2}{s+1} + \frac{1}{s+2}$$

$f(t) = 1u(t) - 2e^{-t} + e^{-2t}$

$f(0) = \lim_{s \to \infty} sF(s) = 0$

$f(\infty) = \lim_{s \to 0} sF(s) = 1$



If all poles of the function F(s) are distinct except zero pole. The time function f(t) will be an exponential

Partial-Fraction Expansion

$$H(s) = \frac{10s}{(s+4)(s+9)} = \frac{K_1}{s+4} + \frac{K_2}{s+9}$$

$$K_1 = (s+4)\left.\frac{10s}{(s+4)(s+9)}\right|_{s=-4} = \left.\frac{10s}{s+9}\right|_{s=-4} = \frac{-40}{5} = -8$$

$$K_2 = (s+9)\left.\frac{10s}{(s+4)(s+9)}\right|_{s=-9} = \left.\frac{10s}{s+4}\right|_{s=-9} = \frac{-90}{-5} = 18$$

$$H(s) = \frac{-8}{s+4} + \frac{18}{s+9} = \frac{-8s-72+18s+72}{(s+4)(s+9)} = \frac{10s}{(s+4)(s+9)}$$ Check.

$$h(t) = -8e^{-4t} + 18e^{-9t}\,u(t)$$

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Case #2 Partial-fraction expansion when *F(s)* involves complex poles

- Since the initial equations are real, complex poles will always appear as complex conjugate pole pairs
- Complex poles pairs in the partial fraction expansion of the Laplace transform

$$F(s) = \frac{A_1 s + A_2}{s^2 + a s + b} + F_1(s)$$

Method of completing the square

- Break the denominator into two parts

$$s^2 + as + b = s^2 + 2\alpha s + \alpha^2 + \beta^2 = (s+\alpha)^2 + \beta^2$$

The numerator then becomes

$$A_1 s + A_2 = A_1(s+\alpha) + B_1\beta$$

The denominator is okay, but break the numerator into two parts

$$F(s)=\frac{A_1(s+\alpha)}{(s+\alpha)^2+\beta^2}+\frac{B_1\beta}{(s+\alpha)^2+\beta^2}+F_1(s)$$

The form is now a damped cosine and sine!

$$f(t)=e^{-\alpha t}\left[A_1\cos(\beta t)+B_1\sin(\beta t)\right]+f_1(t)$$

: Complex Conjugate Poles (Method 2)

Given: $G(S) = \dfrac{13}{(s^2 + 4s + 13)}$; and $\mathrm{R(s)} = \dfrac{2}{s}$. Then the output is

$$F(s) = \frac{26}{s\,(s^2+4s+13)} = \frac{26}{s(s+2+j3)(s+2-j3)} = \frac{26}{s\{(s+2)^2+3^2\}}$$

Partial Fraction $F(s) = \dfrac{26}{s\,(s^2+4s+13)} = \dfrac{A}{s} + \dfrac{Bs+C}{s^2+4s+13} = \dfrac{As^2+4As+13A+Bs^2+Cs}{s\,(s^2+4s+13)}$

Taking common denominators $F(s) = \dfrac{26}{s(s^2+4s+13)} = \dfrac{As^2+4As+13A+Bs^2+Cs}{s(s^2+4s+13)}$

Equating both numerators $26 = As^2 + 4As + 13A + Bs^2 + Cs$

Equating the coefficients of same power

S^2 : 0=A+B

S^1 : 0=4A+C → Solving we obtain A=2 ; B=-2; C=-8

S^0 : 26=13A

And

$$F(s) = = \frac{2}{s} + \frac{-2s-8}{s^2+4s+13} = \; = = \frac{2}{s} - 2\frac{s+4}{(s+2)^2+3^2} =$$

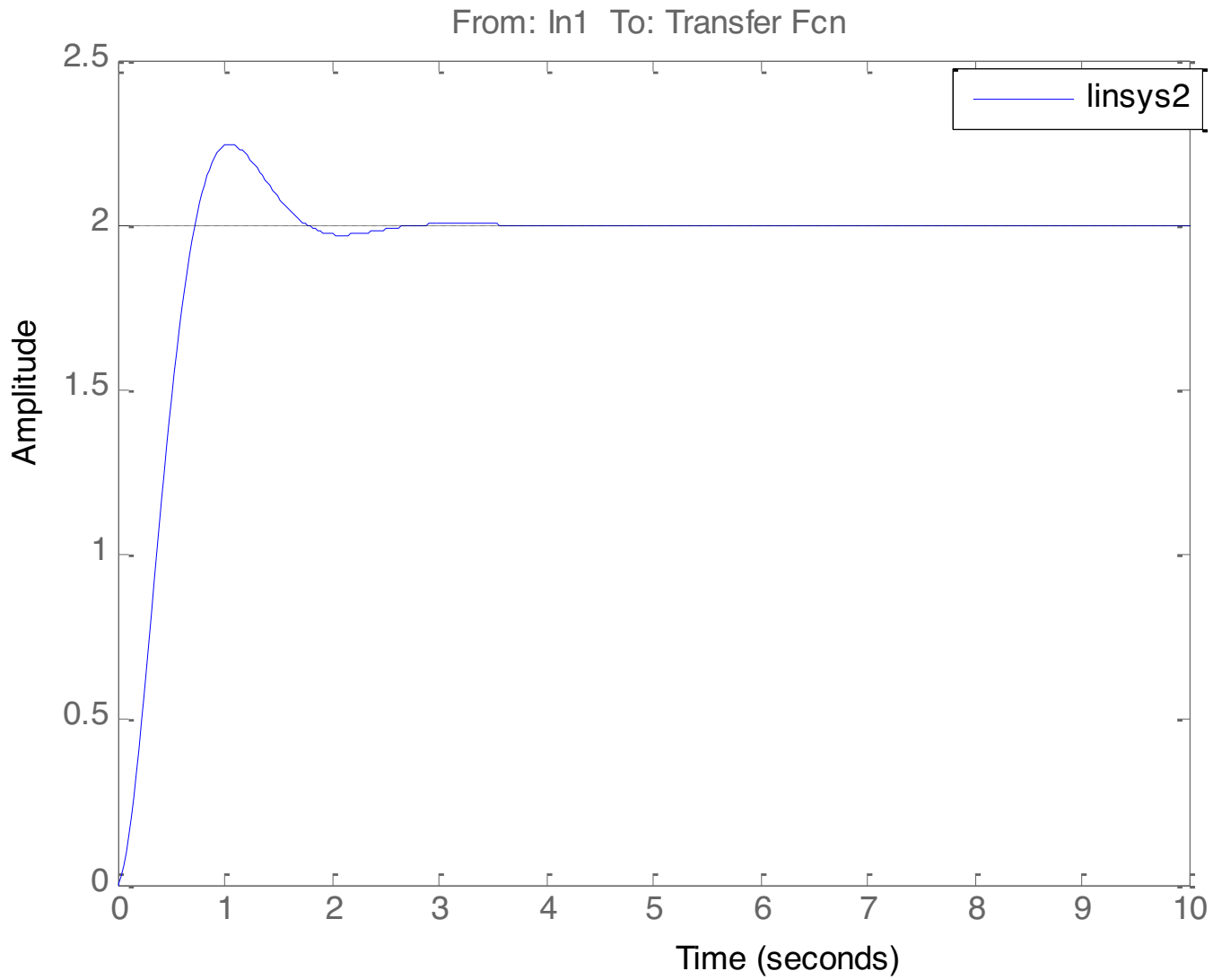
$$= \frac{2}{s} - 2\frac{s+2}{(s+2)^2+3^2} - 2\frac{2\times {}^3/_3}{(s+2)^2+3^2\}}$$

$$= \frac{2}{s} - 2\frac{s+2}{(s+2)^2+3^2} - \frac{4}{3}\frac{3}{(s+2)^2+3^2\}}$$

Finally the inverse Laplace transform is

$$\text{f(t)} = \; 2\text{u(t)} - 2e^{-2t}\cos(3t) - \frac{4}{3}e^{-2t}\sin(3t) = 2\text{u(t)} - e^{-2t}\{2(\cos(3t) + \frac{4}{3}\sin(3t)\}$$

f(0)=0 and f(∞) =2



If all poles of the function F(s) are complex conjugate, except zero pole. The time function f(t) will be an damped sinusoi

Special Case :Imaginary Conjugate Poles (Method 2)

Given G(s)=$\frac{1}{(s^2+9)}$; and R(S) = $\frac{1}{s}$ Then the output is

$$F(s) = \frac{1}{s(s^2+9)}$$

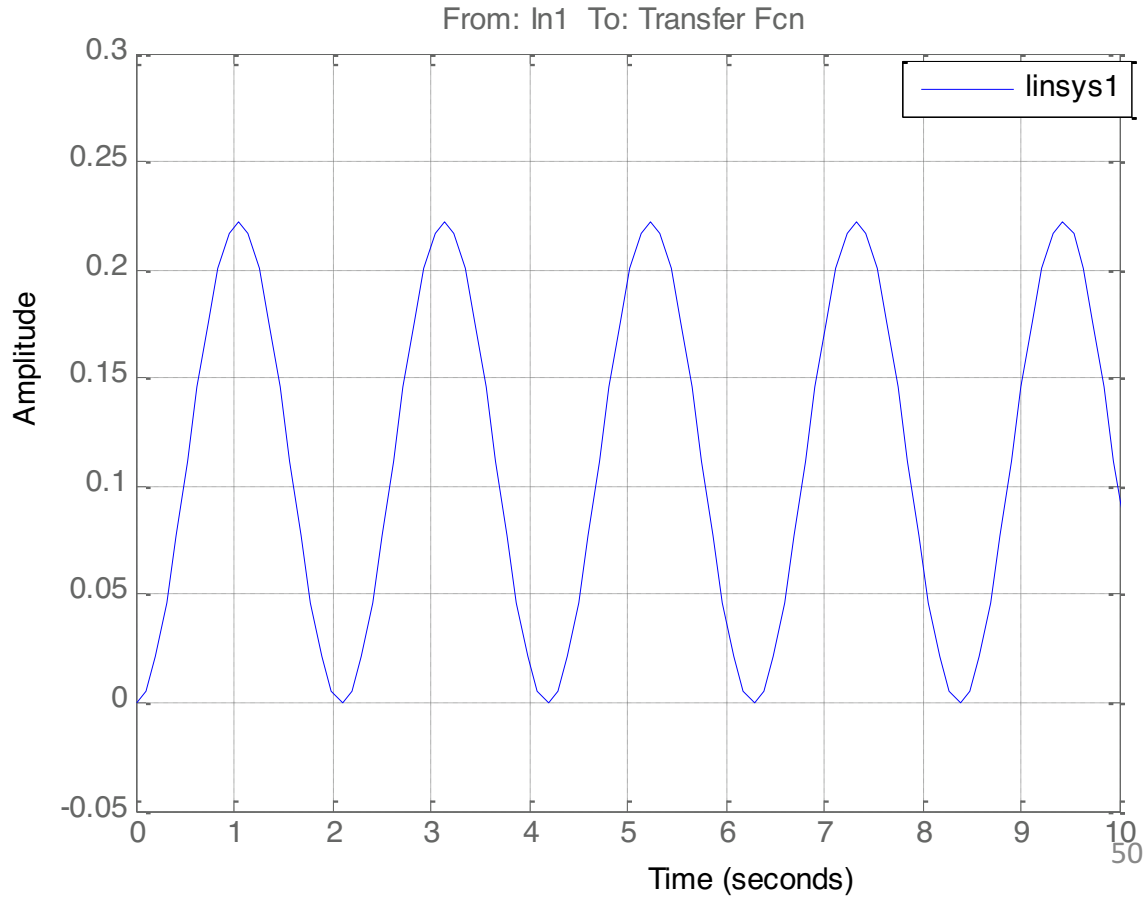
$The\ poles\ are\ 0, \pm j3$

$$F(s)=\frac{1}{s(s^2+9)}=\frac{A}{s}+\frac{Bs+C}{s^2+9}=\frac{As^2+9A+Bs^2+Cs}{s(s^2+9)}$$

$We\ obtain\ A=\frac{1}{9}, B=-\frac{1}{9}, C=0$

$And \qquad f(t)=\frac{1}{9}$u(t) - $\frac{1}{9}$cos3t

If all poles of the function F(s) are imaginary conjugate except zero pole. The time function f(t) will be sinusoidal



Case #3 Partial-fraction expansion when *F(s)* involves multiple poles

- Instead of discussing the general case, we shall use an example to show how to obtain the partial- fraction expansion of *F(s).*

Consider the following $G(s) = \frac{1}{(s+1)^2}; and\ R(s) = \frac{1}{s}. Then\ the\ output\ is$

$$F(s) = \frac{1}{s(s+1)^2} = \frac{A}{s} + \frac{B}{(s+1)^2} + \frac{C}{s+1}$$

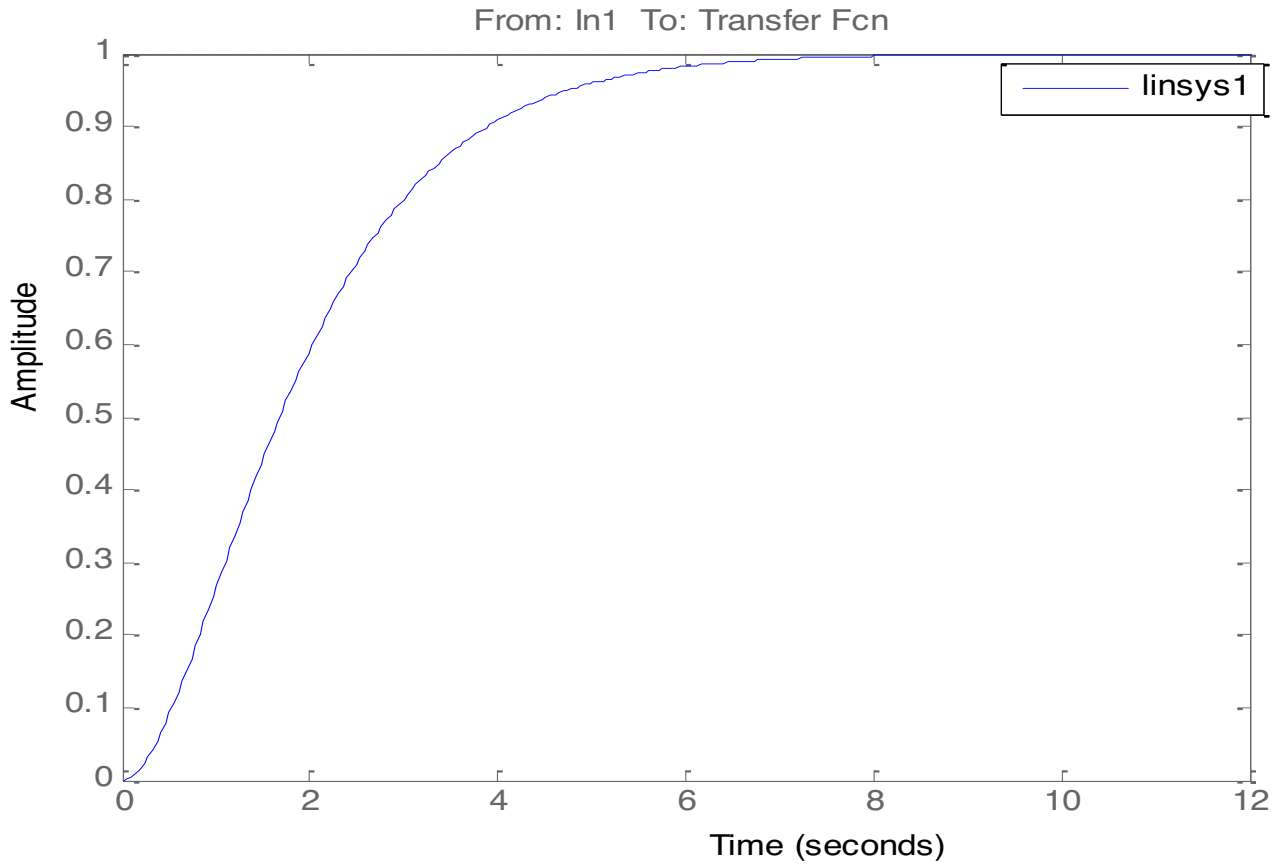
Taking common denominators and equating numerators

$$1 = As^2 + 2As + A + Bs + cs^2 + Cs$$

Equating like powers of "s" gives us:

A=1 ; B=-1; C=-1

The inverse Laplace $f(t)=u(t)-e^{-t}-te^{-t}=u(t)-e^{-t}(t+1)$



If all poles of the function F(s) are multiple except zero pole. The time function f(t) will be damped ramp

SOLVING LINEAR DIFFERENTIAL EQUATIONS

- **In solving linear, time-invariant, differential equations by the Laplace transform method, two steps are involved.**

  1. By taking the Laplace transform of each term in the given differential equation, convert the differential equation into an algebraic equation in s and obtain the expression for the Laplace transform of the dependent variable by rearranging the algebraic equation.

- 2. The time solution of the differential equation is obtained by finding the inverse Laplace transform of the dependent variable.

- Example

Find the solution of the differential equation

$$\ddot{x} + 2\dot{x} + 5x = 3, \qquad x(0) = 0, \qquad \dot{x}(0) = 0$$

$$s^2X(s) + 2sX(s) + 5X(s) = \frac{3}{s}$$

Solving for $X(s)$, we find

$$X(s) = \frac{3}{s(s^2 + 2s + 5)} = \frac{3}{5}\frac{1}{s} - \frac{3}{5}\frac{s + 2}{s^2 + 2s + 5}$$

$$= \frac{3}{5}\frac{1}{s} - \frac{3}{10}\frac{2}{(s + 1)^2 + 2^2} - \frac{3}{5}\frac{s + 1}{(s + 1)^2 + 2^2}$$

$$x(t) = \mathcal{L}^{-1}[X(s)]$$

$$= \frac{3}{5}\mathcal{L}^{-1}\left[\frac{1}{s}\right] - \frac{3}{10}\mathcal{L}^{-1}\left[\frac{2}{(s+1)^2+2^2}\right] - \frac{3}{5}\mathcal{L}^{-1}\left[\frac{s+1}{(s+1)^2+2^2}\right]$$

$$= \frac{3}{5} - \frac{3}{10}e^{-t}\sin 2t - \frac{3}{5}e^{-t}\cos 2t, \qquad \text{for } t \geq 0$$

UNIE 3

Mathematical Modeling of Dynamic Systems

- Mathematical Models
- linear and nonlinear systems.
- transfer function
- block diagrams
- Examples

Modeling of physical systems

The control systems can be represented with a set of mathematical equations known as **mathematical model**.

These models are useful for analysis and design of control systems. Analysis of control system means finding the output when we know the input and mathematical model.

Design of control system means finding the mathematical model when we know the input and the output.

The following mathematical models are mostly used.

- Differential equation model
- Transfer function model
- State space model

Steps in Modeling

- Understand the physical systems and its components
- Make appropriate simplifying assumptions
- Use basic principles to formulate the mathematical model
- Write differential and algebraic equations describing the model
- Check the model for validity

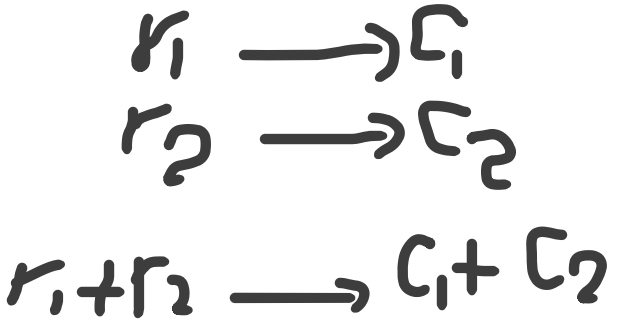
What will the model be used for

- Solution of the differential and algebraic equations allows system response and performance to analyzed and designed
- The Laplace transform will be applied to the model to allow convenient manipulations and dynamic analysis
- Input-output relationships for system and components will be obtained
- Controller models will be designed that can be implemented in hardware

Simplicity versus accuracy

- . It is possible to improve the accuracy of a mathematical model by increasing its complexity.
- we must make a compromise between the simplicity of the model and the accuracy of the results of the analysis.
- In deriving a reasonably simplified mathematical model, we frequently find it necessary to ignore certain inherent physical properties of the system.
- In particular, it is always necessary to ignore certain nonlinearities and distributed parameters that may be present in the physical system.

Linear systems



- A system is linear if the principle of superposition applies.
- The principle of superposition states that the response produced by the concurrent application of two different input functions is the sum of the two individual responses
- The system is linear if it is represented by linear differential equation

(dy(t)/dt)^2+4dy(t)/dt+6y(t)=r(t)

Linear time-invariant systems

- A differential equation is linear if the coefficients are constants or not functions only of the independent variable.
- Dynamic systems that are composed of linear time-invariant lumped-parameter components may be described by linear time-invariant (constant- coefficient) differential equations, Such systems are called *linear time-invariant* (or *linear constant-coefficient)* systems.

(dy(t)/dt)^2+m(t)dy(t)/dt+6y(t)=r(t)

linear time-varying systems

- Systems that are represented by differential equations whose coefficients are functions of time are called *linear time-varying* systems. An example of a time-varying control system is a spacecraft control system. (The mass of a spacecraft changes due to fuel consumption.)

$$(dy(t)/dt)^2+m(t)dy(t)/dt+6y(t)=r(t)$$

Nonlinear systems

- A system is nonlinear if the principle of superposition does not apply.
- Thus, for a nonlinear system the response to two inputs cannot be calculated by treating one input at a time and adding the results.
- **If a nonlinearity does not suffer any changes with time**, the system is nonlinear time-invariant
- **If a nonlinearity changes with time**, the system is nonlinear time-varying
-

examples

$$y^{(5)}(t)-2y''(t)+5y'(t)-y(t)=f'''(t)+f''(t)-2f(t$$

Linear time-invariant

$$y'''(t)+y''(t)+ty(t)=f(t)$$

Linear time-varying

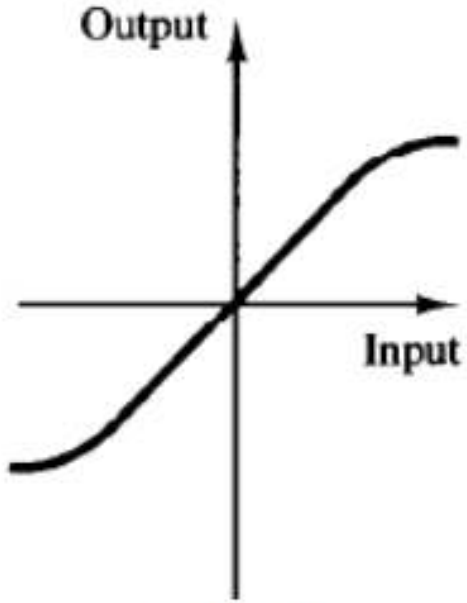
$$y''(t)+2y(t)y'(t)+y(t)=f'(t)+3f(t)$$

Nonlinear time-invariant

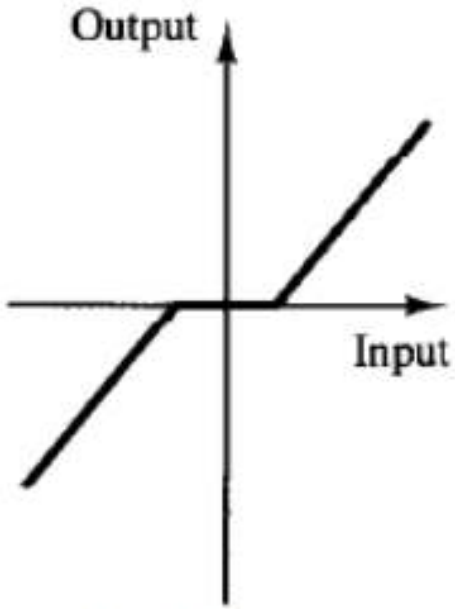
$$t^2y''(t)+y^2(t)=f(t)$$

Nonlinear time-varying

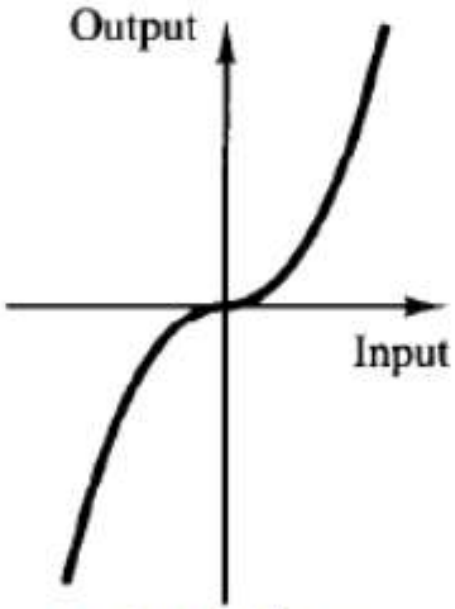
Examples of nonlinearities



saturation



Dead-zone



square law

TRANSFER FUNCTION

- **transfer functions are commonly used to characterize the input—output relationships of components or systems that can be described by linear, time-invariant, differential equations.**

- **The *transfer function* of a linear, time-invariant, differential equation system is defined as the ratio of the Laplace transform of the output (response function) to the Laplace transform of the input (driving function) under the assumption that all initial conditions are zero.**

Consider a system defined :

$$a_0 y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1}\dot{y} + a_n y$$

$$= b_0 x^{(m)} + b_1 x^{(m-1)} + \cdots + b_{m-1}\dot{x} + b_m x \qquad (n \geq m)$$

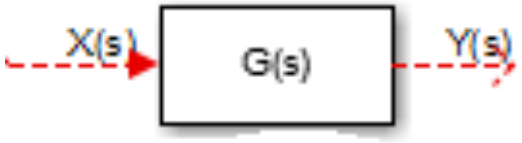
where y is the output of the system and x is the input

The transfer function of this system is obtained by taking the Laplace transforms of both sides of Equation under the assumption that all initial conditions are zero, or

$$\text{Transfer function} = G(s) = \left.\frac{\mathcal{L}[\text{output}]}{\mathcal{L}[\text{input}]}\right|_{\text{zero initial conditions}}$$

$$= \frac{Y(s)}{X(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \cdots + b_{m-1} s + b_m}{a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n}$$

$$G(s) = \frac{Y(s)}{X(s)}; .or \ldots . Y(s) = G(s)X(s)$$



Comments on transfer function

- **The transfer function of a system is a mathematical model in that it is an operational method of expressing the differential equation that relates the output variable to the input variable.**
- **The transfer function is a property of a system itself, independent of the magnitude and nature of the input or driving function.**
- **The transfer function includes the units necessary to relate the input to the output; however, it does not provide any information concerning the physical structure of the system. (The transfer functions of many physically different systems can be identical.)**

- **If the transfer function of a system is known, the output or response can be studied for various forms of inputs with a view toward understanding the nature of the system.**

- **If the transfer function of a system is unknown, it may be established experimentally by introducing known inputs and studying the output of the system.**

- **a transfer function gives a full description of the dynamic characteristics of the system, as distinct from its physical description.**

IMPULSE-
RESPONSE FUNCTION

- Impulse-response function. Consider the output (response) of a system to a unit impulse input when the initial conditions are zero, Since the Laplace transform of the unit-impulse function is unity, the Laplace transform of the output of the system is
-

$$Y(s) = G(s)$$

- The inverse Laplace transform of the output gives the impulse response of the system. The inverse Laplace transform of G*(s),* or

- is called the impulse-response function. This function g(t) is also called the weighting function of the system.

$$\mathscr{L}^{-1}[G(s)] = g(t)$$

To derive the transfer function

1. Write the differential equation for the system.

2. Take the Laplace transform of the differential equation, assuming all initial conditions are zero.

3. Take the ratio of the output C*(s)* to the input R*(s).* This ratio is the transfer function.

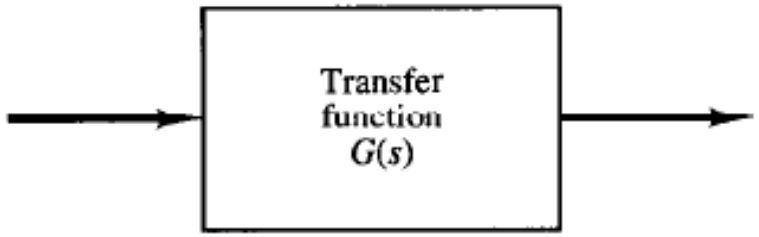
BLOCK DIAGRAMS

- A *block diagram* of a system is a pictorial representation of the functions performed by each component and of the flow of signals.
- Such diagram depicts the interrelationships that exist among the various components. Differing from a purely abstract mathematical representation,
- a block diagram has the advantage of indicating more realistically the signal flows of the actual system.

- In a block diagram all system variables are linked to each other through functional blocks.

- The *functional block* or simply *block* is a symbol for the mathematical operation on the input signal to the block that produces the output.
- The transfer functions of the components are usually entered in the corresponding blocks, which are connected by arrows to indicate the direction of the flow of signals.

Element of a block diagram.



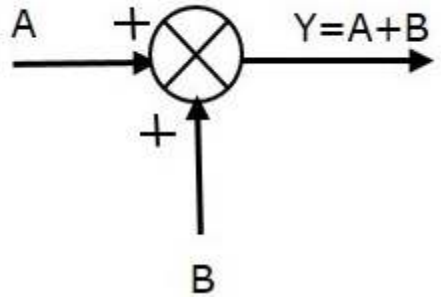
The arrowhead pointing toward the block indicates the input.

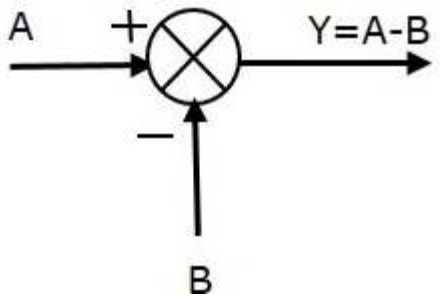
the arrowhead leading away from the block represents the output.

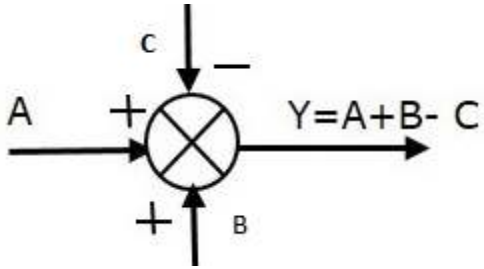
Such arrows are referred to as *signals.*

Summing Point

The summing point is represented with a circle having cross (X) inside it. It has two or more inputs and single output. It produces the algebraic sum of the inputs. It also performs the summation or subtraction or combination of summation and subtraction of the inputs based on the polarity of the inputs.

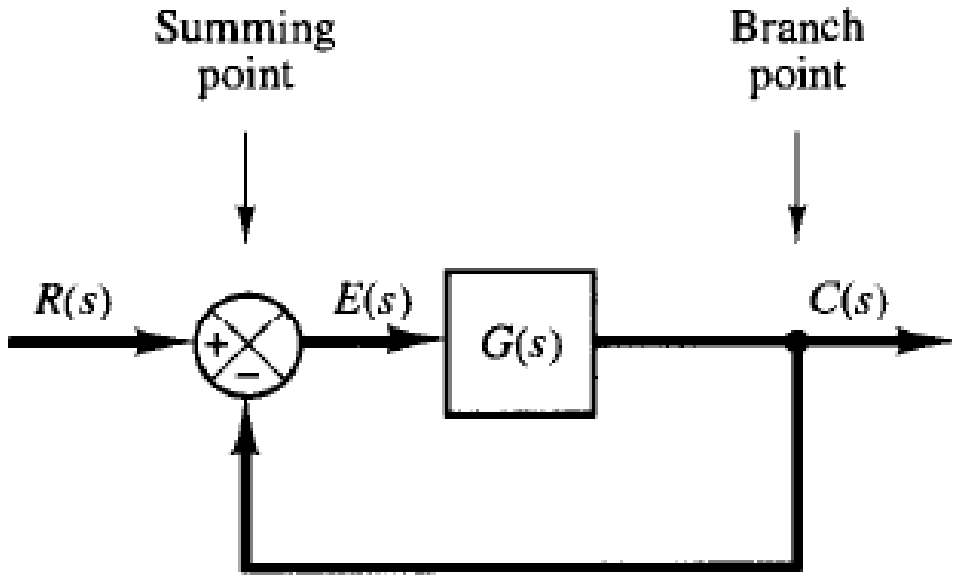




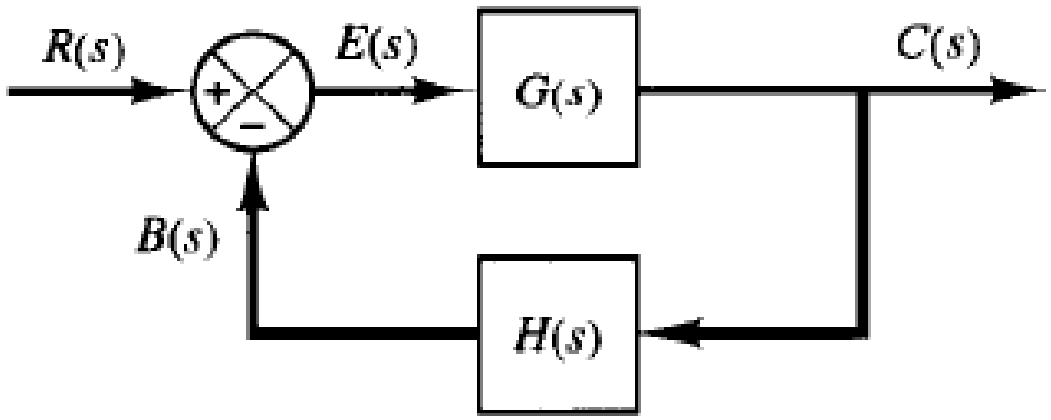


Branch Point

- ***branch point (take off point)*** **is a point from which the signal from a block goes concurrently to other blocks or summing points.**



Block diagram of a closed-loop system.



B(s) the feedback signal
E*(s)* actuating error signal
G(S) Feed forward transfer function
G(s)H(s) Open-loop transfer function
C(S)/R(s) Closed-loop transfer function

Transfer function only used for linear time invariant systems mainly single input single output(SISO)

- *Modeling in State Space*

Modern control theory versus conventional control theory.

- **Modern control theory is applicable to multiple-input—multiple-output systems, which may be linear or nonlinear, time invariant or time varying,**

- **conventional control theory is applicable only to linear time- invariant single-input—single-output systems.**

- **modern control theory is essentially a time-domain approach, while conventional control theory is a complex frequency- domain approach.**

definitions

- State. The state of a dynamic system is the smallest set of variables (called *state variables)* such that the knowledge of these variables at t = to, together with the knowledge of the input for $t \geq t_0$, completely determines the behavior of the system for any time t ≥to.

- **State variables.** The state variables of a dynamic system are the variables making up the smallest set of variables that determine the state of the dynamic system. If at least *n* variables x1, x2,. . . , xn, are needed to completely describe the behavior of a dynamic system (so that once the input is given for t ≥ to and the initial state at t = t0 is specified, the future state of the system is completely determined), then such n variables are a set of state variables.

- **State vector.** If *n* state variables are needed to completely describe the behavior of a given system, then these *n* state variables can be considered the *n* components of a vector **x**. Such a vector is called a *state vector.*
- State space. The n-dimensional space whose coordinate axes consist of the x1 axis, x2 axis,.. . , xn, axis is called a *state space.*

- State-space equations. In state-space analysis we are concerned with three types of variables that are involved in the modeling of dynamic systems:
- input variables,
- output variables,
- and state variables

STATE-SPACE REPRESENTATION OF DYNAMIC SYSTEMS

Case #1 The input does not involve derivative terms

- By use of vector-matrix notation, an nth-order differential equation may be expressed by a first- order vector-matrix differential equation. If *n* elements of the vector are a set of state variables, then the vector-matrix differential equation is a *state* equation. In this section we shall present methods for obtaining state-space representations of continuous- time systems.
  - Assume that a multiple-input—multiple-output system involves
  - *n state variables* $x_1(t), x_2(t),..., x_n(t)$.
  - Assume also that there are *r* inputs $u_1(t), u_2(t), \ldots, u_r(t)$
  - and *m* outputs $y_1(t), y_2(t),..., y_m(t)$.
  - Then the system may be described by

$$\dot{x}_1(t) = f_1(x_1, x_2, \ldots, x_n; u_1, u_2, \ldots, u_r; t)$$
$$\dot{x}_2(t) = f_2(x_1, x_2, \ldots, x_n; u_1, u_2, \ldots, u_r; t)$$
$$\vdots$$
$$\dot{x}_n(t) = f_n(x_1, x_2, \ldots, x_n; u_1, u_2, \ldots, u_r; t)$$

The outputs $y_1(t), y_2(t), \ldots, y_m(t)$ of the system may be given by

$$y_1(t) = g_1(x_1, x_2, \ldots, x_n; u_1, u_2, \ldots, u_r; t)$$
$$y_2(t) = g_2(x_1, x_2, \ldots, x_n; u_1, u_2, \ldots, u_r; t)$$
$$\vdots$$
$$y_m(t) = g_m(x_1, x_2, \ldots, x_n; u_1, u_2, \ldots, u_r; t)$$

If we define

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \cdot \\ \cdot \\ \cdot \\ x_n(t) \end{bmatrix}, \qquad \mathbf{f}(\mathbf{x}, \mathbf{u}, t) = \begin{bmatrix} f_1(x_1, x_2, \ldots, x_n; u_1, u_2, \ldots, u_r; t) \\ f_2(x_1, x_2, \ldots, x_n; u_1, u_2, \ldots, u_r; t) \\ \cdot \\ \cdot \\ \cdot \\ f_n(x_1, x_2, \ldots, x_n; u_1, u_2, \ldots, u_r; t) \end{bmatrix}$$

$$\mathbf{y}(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ \cdot \\ \cdot \\ \cdot \\ y_m(t) \end{bmatrix}, \qquad \mathbf{g}(\mathbf{x}, \mathbf{u}, t) = \begin{bmatrix} g_1(x_1, x_2, \ldots, x_n; u_1, u_2, \ldots, u_r; t) \\ g_2(x_1, x_2, \ldots, x_n; u_1, u_2, \ldots, u_r; t) \\ \cdot \\ \cdot \\ \cdot \\ g_m(x_1, x_2, \ldots, x_n; u_1, u_2, \ldots, u_r; t) \end{bmatrix},$$

$$\mathbf{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ \cdot \\ \cdot \\ \cdot \\ u_r(t) \end{bmatrix}$$

- then the state equations and the output equations become

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}, \mathbf{u}, t)$$

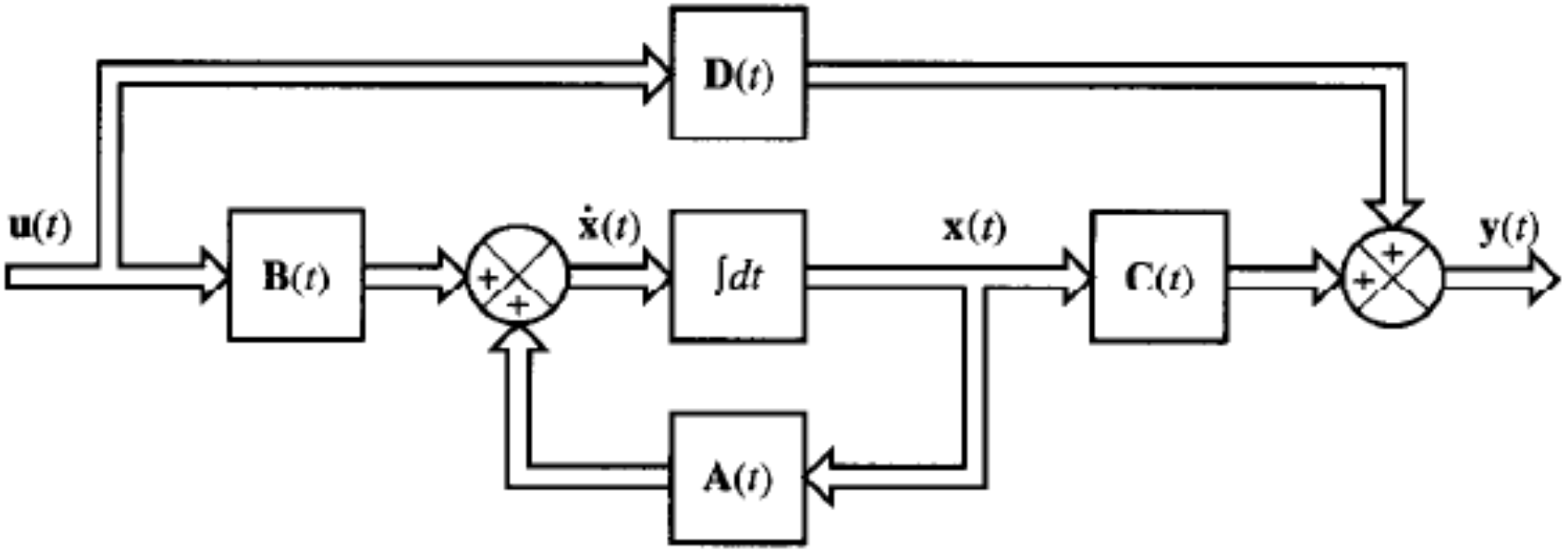
$$\mathbf{y}(t) = \mathbf{g}(\mathbf{x}, \mathbf{u}, t)$$

And the linearized equations

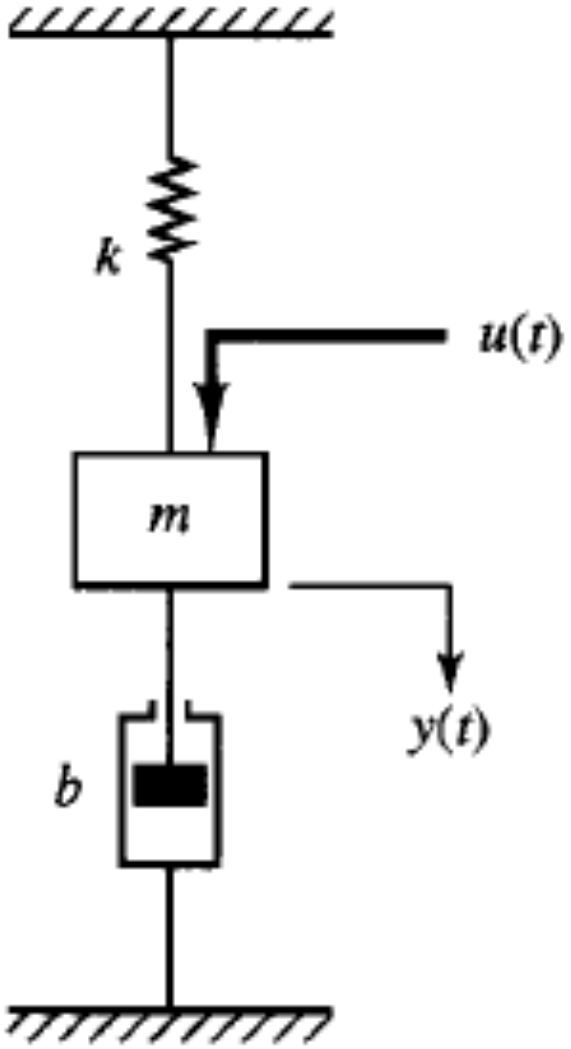
$$\dot{\mathbf{x}}(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{B}(t)\mathbf{u}(t)$$

$$\mathbf{y}(t) = \mathbf{C}(t)\mathbf{x}(t) + \mathbf{D}(t)\mathbf{u}(t)$$

- Where:
- A(t) the state matrix, (n*n)
- B(t) the input matrix, (n*r)
- C(t) the output matrix,(m*n)
- D(t) the direct transmission matrix.(m*r)
- A block diagram representation of These Equations


D(t)
u(t)
B(t)
ẋ(t)
∫dt
x(t)
C(t)
y(t)
A(t)

EXAMPLE Mechanical system



u(t) **is the input to the system,**

the displacement y(t) of the mass is the output.

. This system is a single-input—single-output system.

From the diagram, the system equation is

From the diagram, the system equation is

$$m\ddot{y} + b\dot{y} + ky = u$$

Let us define state variables x1 (t) and *x2(t)* as $x_1(t)=y(t)$

$$x_2(t) = \dot{y}(t)$$

Then we obtain

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \frac{1}{m}(-ky - b\dot{y}) + \frac{1}{m}u$$

- Or

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\frac{k}{m}x_1 - \frac{b}{m}x_2 + \frac{1}{m}u$$

The output equation is $y = x_1$

In a vector-matrix form

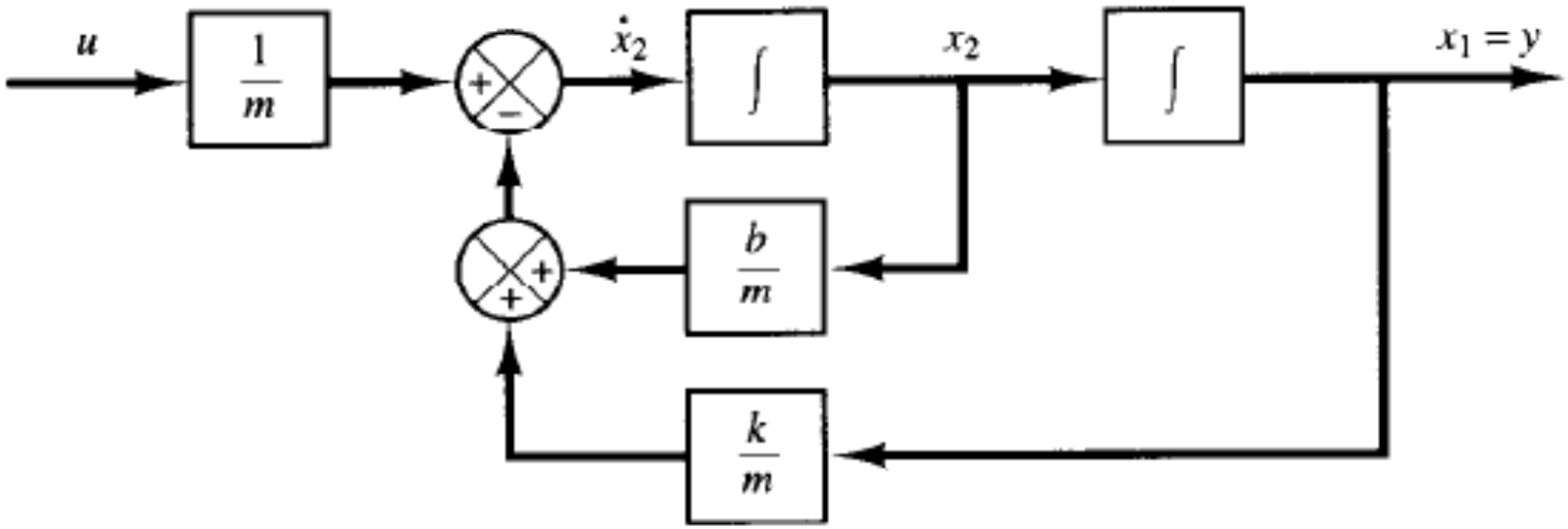
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

block diagram for the system

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\frac{k}{m}x_1 - \frac{b}{m}x_2 + \frac{1}{m}u$$



From the given state space model , obtain the matrices A,B,C and D

$$x_1^{/}. = 2x1 + 5x2 + 3x3 + 2r1$$

$$x_2^{/} = -6x1 - 2x2 + 5r2$$

$$x_3^{/} = x1 - 3x2 - 4x3 \text{ -}$$

$$y1 = 4r2 + 6x1$$

$$y2 = 9x3 + 2r2$$

$$A = \begin{matrix} 2 & 5 & 3 \\ -6 & -2 & 0 \\ 1 & -3 & -4 \end{matrix} \qquad B = \begin{matrix} 2 & 0 \\ 0 & 5 \\ 0 & 0 \end{matrix}$$

$$C = \begin{matrix} 6 & 0 & 0 \\ 0 & 0 & 9 \end{matrix} \qquad D = \begin{matrix} 0 & 4 \\ 0 & 2 \end{matrix}$$

Solving Differential equation by Simulation

Consider the following linear differential equation

$$\dddot{y}(t) + a\ddot{y}(t) + b\dot{y}(t) + cy(t) = x(t)$$

Where y(t) is the output, x(t) is the input

To solve this differential equation the next steps should be done

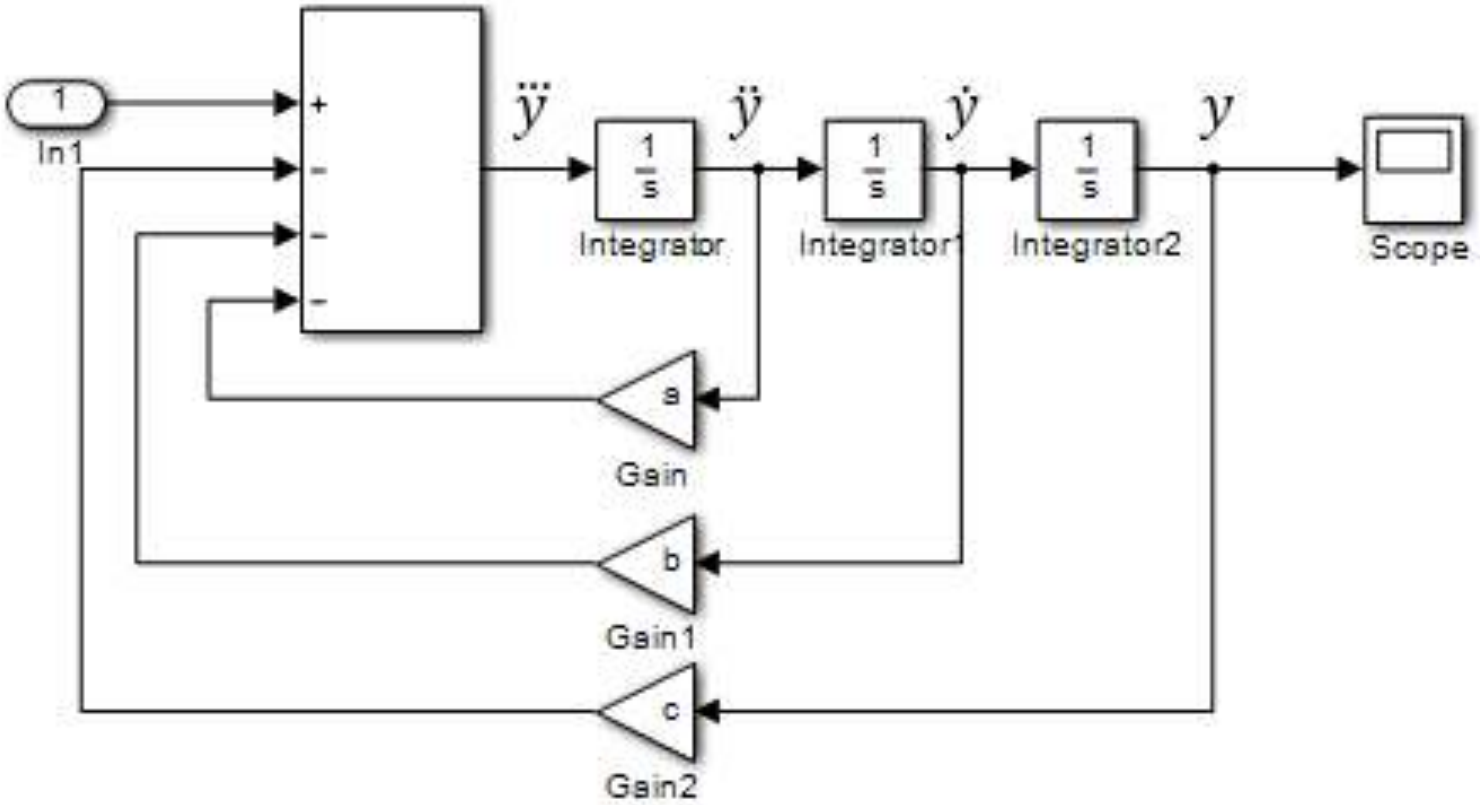
1) Rewrite this equation as

$$\dddot{y}(t) = -a\ddot{y}(t) - b\dot{y}(t) - cy(t) + x(t) \qquad (2.1)$$

2) Use a summing element to add all terms in the right-half side

3) After the summing element use a number of integrators equals the order of the equation

4) Feed back the output of each integrator to the summing element



$$\dddot{y}(t) + a\ddot{y}(t) + b\dot{y}(t) + cy(t) = x(t)$$

$$\dddot{y}(t) = -a\ddot{y}(t) - b\dot{y}(t) - cy(t) + x(t)$$

Correlation between transfer functions and state-space equations

Let us consider the system whose transfer function is given by

$$\frac{Y(s)}{U(s)} = G(s)$$

This system may be represented in state space by the following equations:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$y = \mathbf{C}\mathbf{x} + Du$$

- The Laplace transform

$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}U(s)$$
$$Y(s) = \mathbf{C}\mathbf{X}(s) + DU(s)$$

assume that x(0) is zero. Then we have

$$s\mathbf{X}(s) - \mathbf{A}\mathbf{X}(s) = \mathbf{B}U(s)$$

Or

$$(s\mathbf{I} - \mathbf{A})\mathbf{X}(s) = \mathbf{B}U(s)$$

By premultiplying $(s\mathbf{I} - \mathbf{A})^{-1}$ to both sides

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}U(s)$$

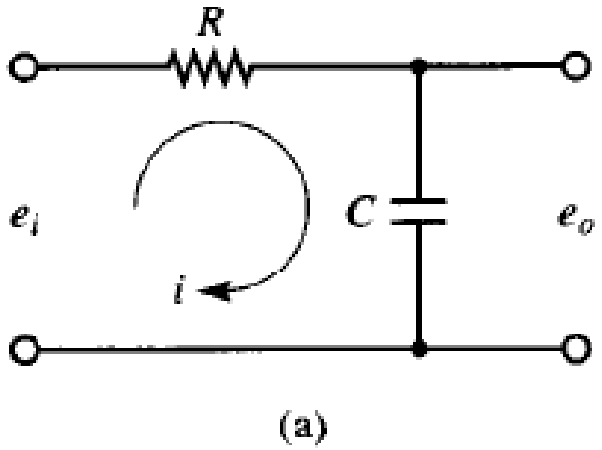
By substituting this Equation into the output equation, we get

$$Y(s) = [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + D]U(s)$$

$$G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + D$$

EXAMPLES
ELECTRICAL SYSTEMS

RC Circuit



(a)

$$i = \frac{e_i - e_o}{R}$$

$$e_o = \frac{\int i\,dt}{C}$$

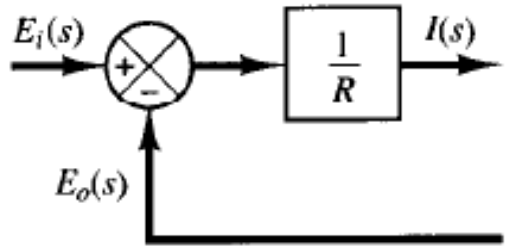
Parameters: R,C

Independent variables (inputs): ei

dependent variables (outputs): e_o, i

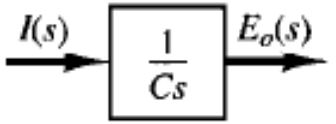
Laplace Transformed equations

$$I(s) = \frac{E_i(s) - E_o(s)}{R}$$

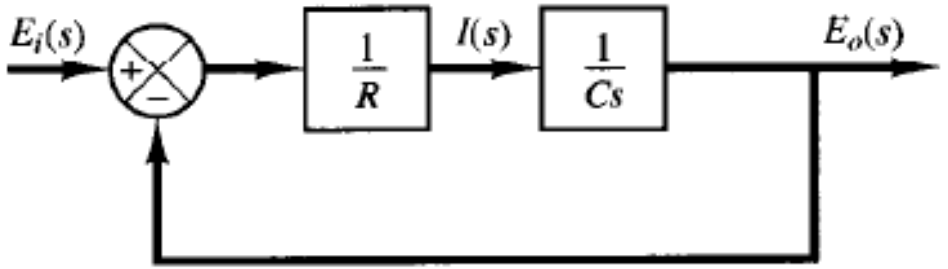


(b)

$$E_o(s) = \frac{I(s)}{Cs}$$



(c)



(d)

$$e_i = Ri + \frac{1}{C}\int i\,dt \longrightarrow E_i(s) = RI(s) + \frac{1}{Cs}I(s) \longrightarrow \frac{I(s)}{E_i(s)} = \frac{Cs}{RCs+1}$$

$E_i(s)$ → $\frac{Cs}{RCs+1}$ → $I(s)$

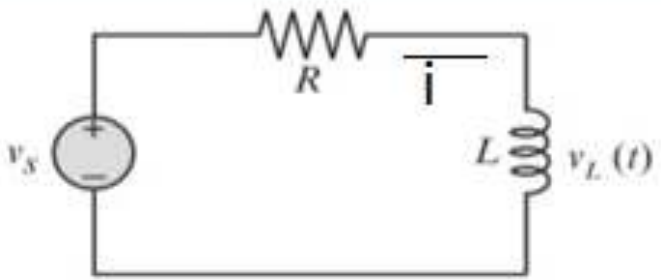
$$e_o = \frac{1}{C}\int i\,dt \longrightarrow E_o(s) = \frac{1}{Cs}I(s) \longrightarrow \frac{E_o(s)}{I(s)} = \frac{1}{cs}$$

$I(s)$ → $\frac{1}{cs}$ → $E_o(s)$

$E_i(s)$ → $\frac{Cs}{RCs+1}$ → $I(s)$ → $I(s)$ → $\frac{1}{cs}$ → $E_o(s)$

$$\longrightarrow \frac{E_o(s)}{E_i(s)} = \frac{1}{RCs+1}$$

RL Series Circuit



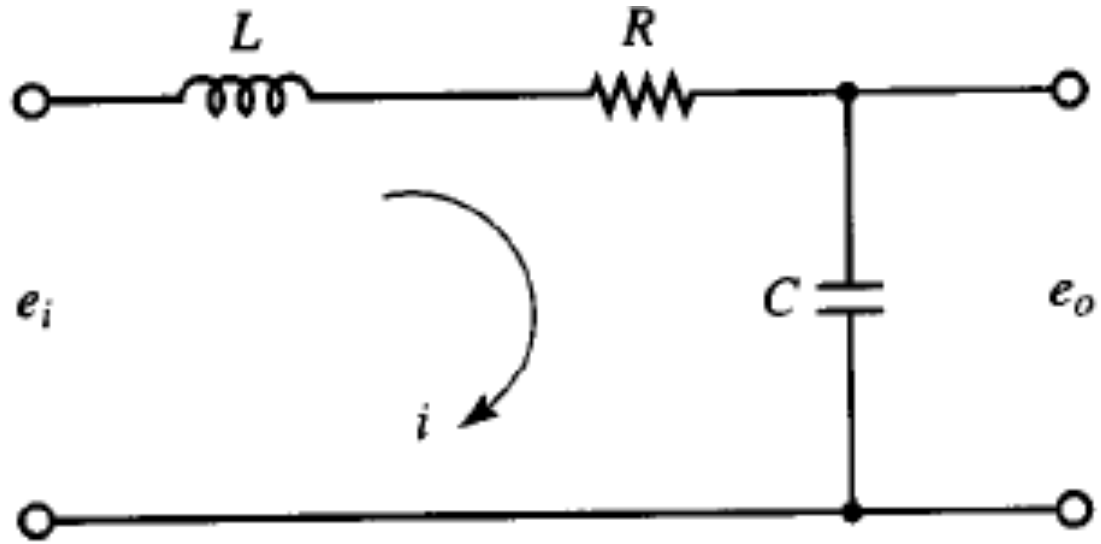
$$v_s = Ri + L\frac{di}{dt} \longrightarrow V_s(s) = RI(s) + LsI(s) \longrightarrow \frac{I(s)}{V_s(s)} = \frac{1}{Ls+R}$$

$$v_L(t) = L\frac{di}{dt} \longrightarrow V_L(s) = LsI(s) \longrightarrow \frac{V_L(s)}{I(s)} = Ls$$

$V_s(s)$ → $\frac{1}{Ls+R}$ → $I(s)$ → LS → $V_L(s)$

LRC circuit

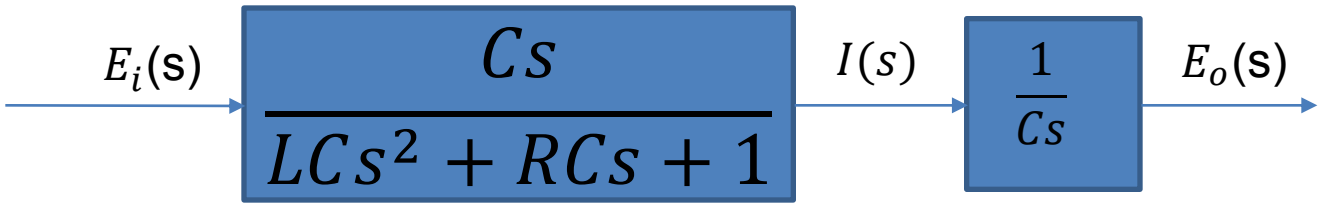
- Consider the electrical circuit shown . The circuit consists of an inductance *L* (henry), a resistance *R* (ohm), and a capacitance *C* (farad). Applying Kirchhoff's voltage law to the system, we obtain the following equations:



$$L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt = e_i \longrightarrow LsI(s) + RI(s) + \frac{1}{C}\frac{1}{s}I(s) = E_i(s)$$

$$\frac{1}{C}\int i\,dt = e_o \qquad \frac{1}{C}\frac{1}{s}I(s) = E_o(s)$$

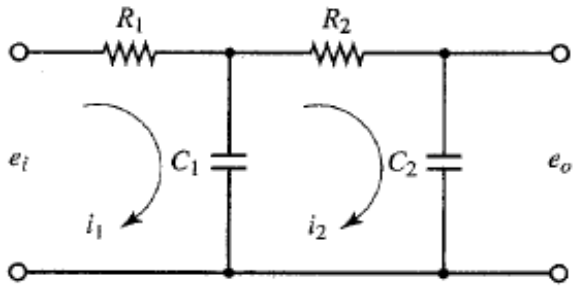
$$\frac{I(s)}{Ei(s)} = \frac{Cs}{LCs^2+RCs+1} \qquad \frac{Eo(s)}{I(s)} = \frac{1}{Cs}$$



If e_i is assumed to be the input and e_o the output, then the transfer function of this system is found to

$$\frac{E_o(s)}{E_i(s)} = \frac{1}{LCs^2 + RCs + 1}$$

Transfer functions of cascaded elements



First loop equation

$$R_1 i_1 + \frac{1}{C_1}\int i_1\,dt - \frac{1}{C_1}\int i_2\,dt = e_i$$

Taking forward Laplace transform

$$\frac{1}{C_1 s} I_1(s) + R_1 I_1(s) - \frac{1}{C_1 s} I_2(s) = E_i(s)$$

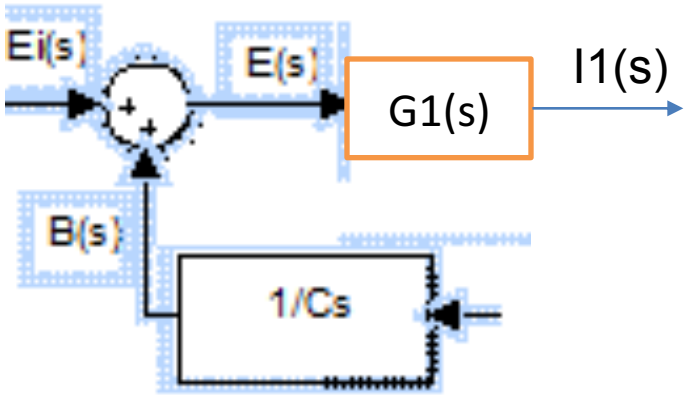
Rearrange results in

$$I_1(s)\frac{C_1 R_1 s + 1}{C_1 s} = \frac{1}{C_1 s} I_2(s) + E_i(s) = E(s)$$

Transfer function

$$I_1(s)/E(s) = \frac{C_1 s}{R_1 C_1 s + 1} = G_1(s)$$

In block diagram form

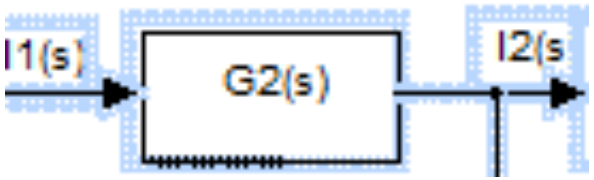


Second loop equation $0= R_2 i_2 + \frac{1}{C_2}\int i_2\,dt + \frac{1}{C_1}\int i_2\,dt - \frac{1}{C_1}\int i_1\,dt$

Taking forward Laplace transform and rearrange we obtain the transfer function of the second loop

$$\frac{I_2(s)}{I_1(s)} = \frac{C_2}{R_2C_1C_2s+C_1+C_2} = G_2(s)$$

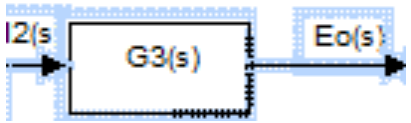
And the block diagram



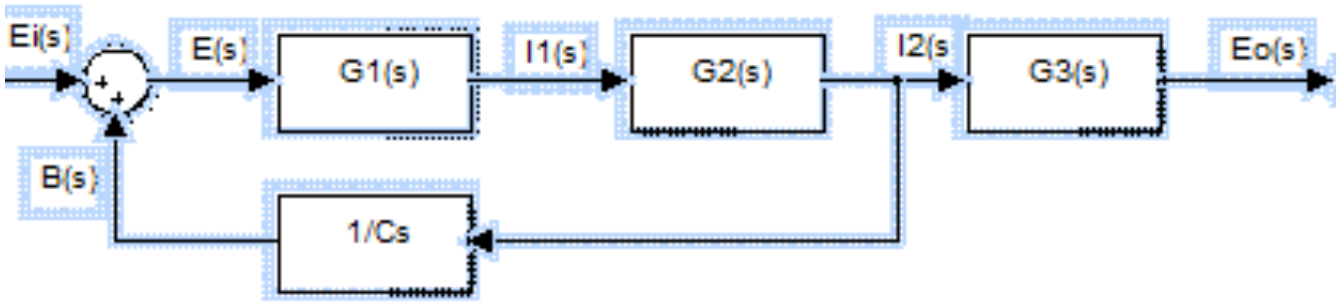
Voltage equation a cross the capacitor C_2 $\quad e_o = \frac{1}{C_2}\int i_2 dt$

Taking forward Laplace transform and rearrange we obtain the transfer function $\quad \frac{E_o}{I_2} = \frac{1}{C_2 s} = G_3(s)$

The block diagram



Finally ,connection all blocks together results in block diagram of the system



The transfer function of the system

$$\frac{E_o(s)}{E_i(s)} = \frac{1}{(R_1C_1s+1)(R_2C_2s+1)+R_1C_2s}$$

$$= \frac{1}{R_1C_1R_2C_2s^2+(R_1C_1+R_2C_2+R_1C_2)s+1}$$

Liquid-level system

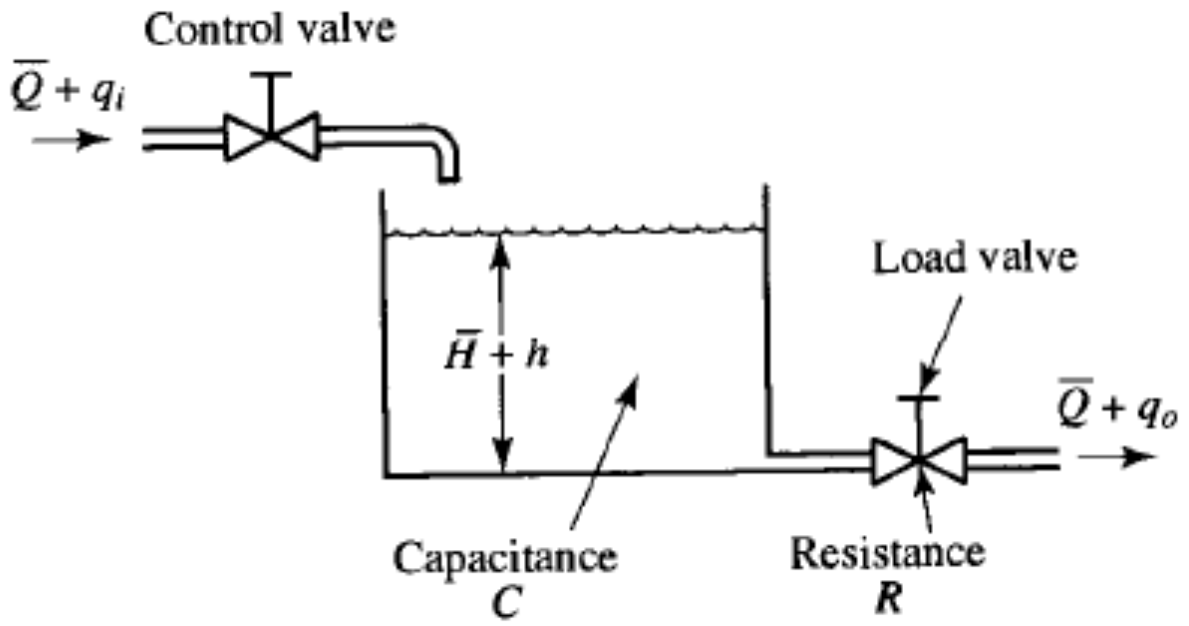
$\bar{Q}$ = steady-state flow rate (before any change has occurred), m^3/sec

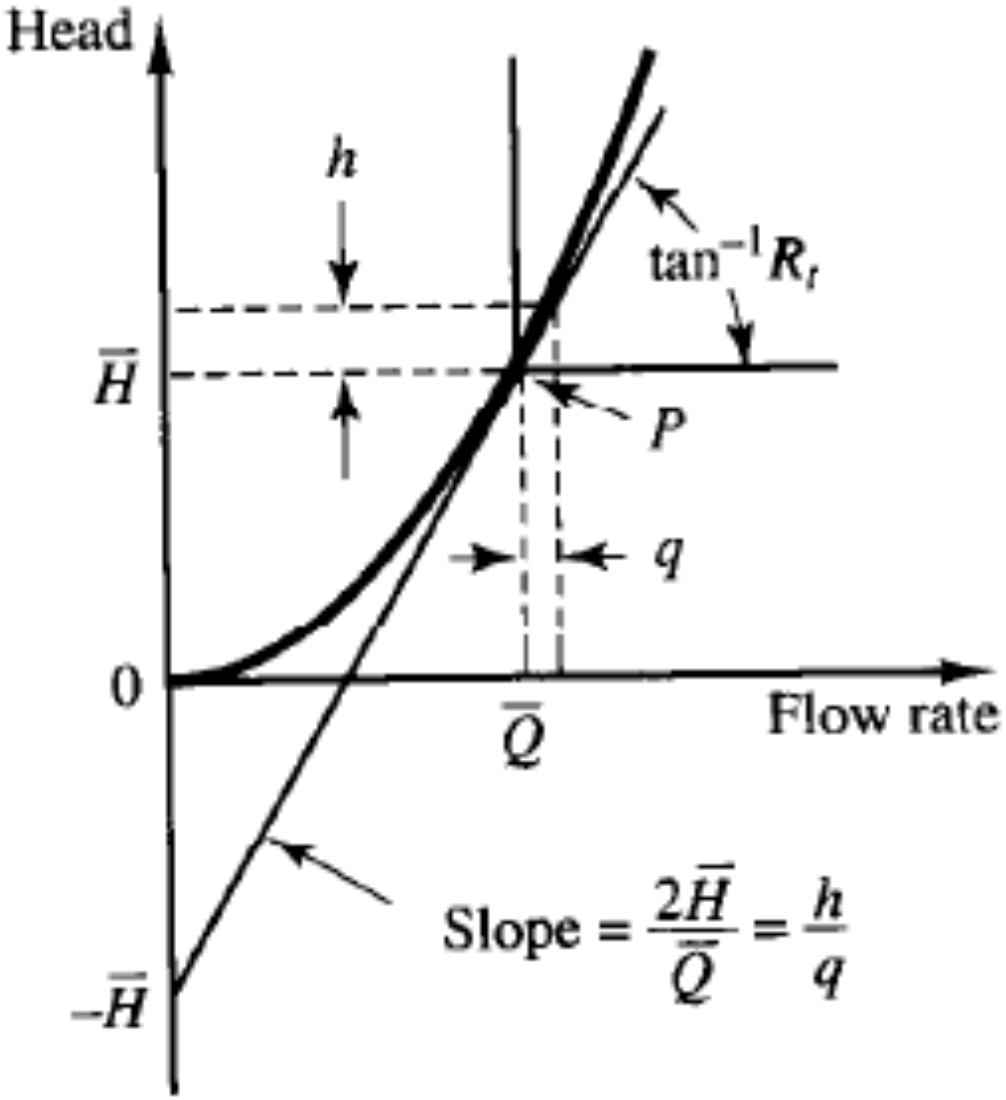
q_i = small deviation of inflow rate from its steady-state value, m^3/sec

q_o = small deviation of outflow rate from its steady-state value, m^3/sec

$\bar{H}$ = steady-state head (before any change has occurred), m

h = small deviation of head from its steady-state value, m


Control valve
$\bar{Q} + q_i$
$\bar{H} + h$
Load valve
$\bar{Q} + q_o$
Capacitance
C
Resistance
R


Head
h
$\tan^{-1} R_f$
$\bar{H}$
P
q
0
$\bar{Q}$
Flow rate
$-\bar{H}$
Slope = $\frac{2\bar{H}}{\bar{Q}} = \frac{h}{q}$

- For linear or linearized system

$$C\,dh = (q_i - q_o)\,dt$$

From the definition of resistance, the relationship between q_o and h is given by

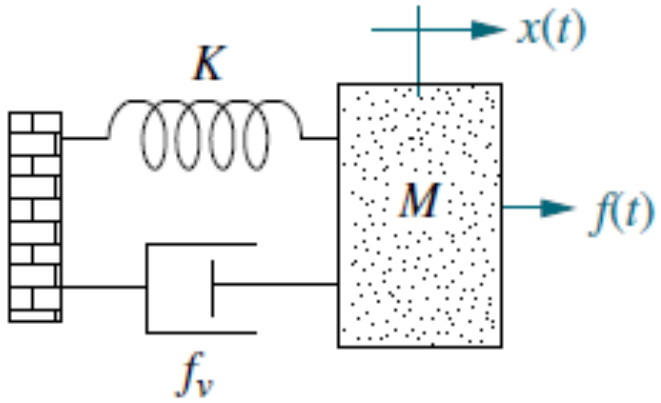
$$q_o = \frac{h}{R}$$

The differential equation for this system for a constant value of R becomes

$$RC\frac{dh}{dt} + h = Rq_i$$

$$(RCs + 1)H(s) = RQ_i(s) \qquad \frac{H(s)}{Q_i(s)} = \frac{R}{RCs + 1}$$

Modeling Mechanical Systems
Translational System



$$M\frac{d^2x(t)}{dt^2}+f_v\frac{dx(t)}{dt}+Kx(t)=f(t)$$

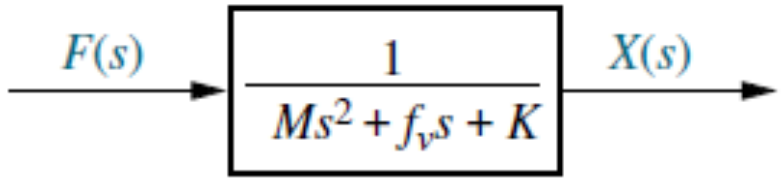
- take the Laplace transform, assuming zero initial conditions,

$$Ms^2X(s)+f_vsX(s)+KX(s)=F(s) \quad \rightarrow (Ms^2+f_vs+K)X(s)=F(s)$$

- solving for the transfer
- function yields

$$G(s)=\frac{X(s)}{F(s)}=\frac{1}{Ms^2+f_vs+K}$$

Block Diagram



Rotational Mechanical System

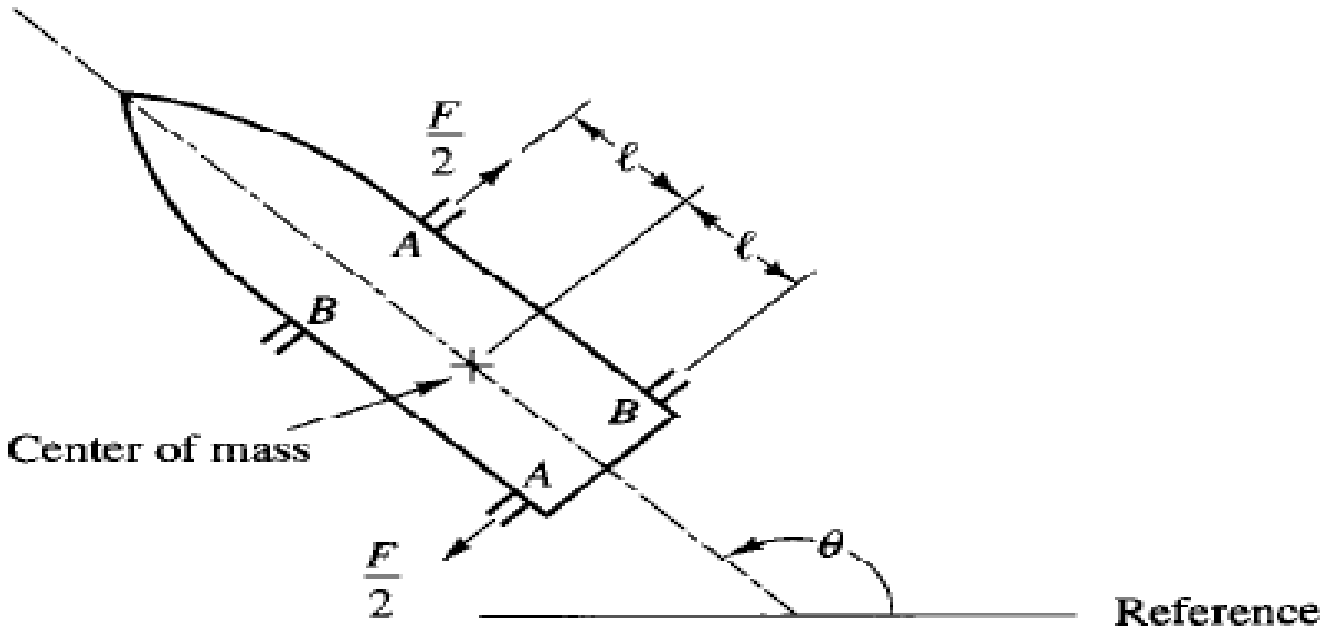
- **Consider the satellite attitude control system . The diagram shows the control of only the yaw angle θ.**

- **Small jets apply reaction forces to rotate the satellite body into the desired position. The two skew symmetrically placed jets denoted by *A* or *B* operate in pairs.**

- **Assume that each jet thrust is *F/2* and a torque *T = Fl* is applied to the system. The jets are applied for a certain time duration and thus the torque can be written as *T(t).***

- **The moment of inertia about the axis of rotation at the center of mass is *J.***
  Let us obtain the transfer function of this system by assuming that torque *T(t)* is the input, and the angular displacement θ*(t)* of the satellite is the output.

- **MECHANICAL SYSTEM**



Schematic diagram of a satellite attitude control system.

- Applying Newton's second law to the present system and noting that there is no friction in the environment of the satellite, we have

$$J\frac{d^2\theta}{dt^2} = T$$

Taking the Laplace transform of both sides of this last equation and assuming all initial conditions to be zero yields

$$Js^2\Theta(s) = T(s) \longrightarrow \frac{\Theta(s)}{T(s)} = \frac{1}{Js^2}$$

Electrical and Mechanical Systems Analogs

- An electric circuit that is analogous to a system from another discipline is called an **electric circuit analog**.
- The mechanical systems with which we worked can be represented by equivalent electric circuits.
- Analogs can be obtained by comparing the equations of motion of a mechanical system, with either electrical mesh or nodal equations.
- When compared with mesh equations, the resulting electrical circuit is called a **series analog**.

RLC series circuit equation

$$L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt = e_i$$

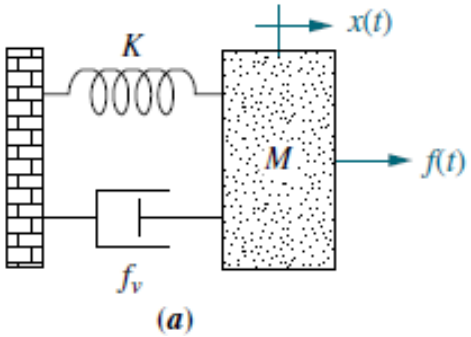
Translational mechanical system equation

$$M\frac{d^2x(t)}{dt^2} + f_v\frac{dx(t)}{dt} + Kx(t) = f(t)$$

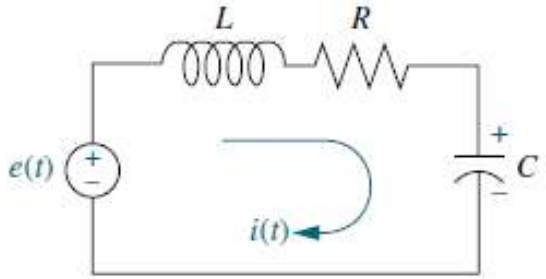
Replacing displace met x(t) by velocity v(t) = dx(t)/dt the equation of mechanical system is

$$M\dot{v}\,(t) + f_c\,v(t) + k\int v(t)dt = f(t)$$

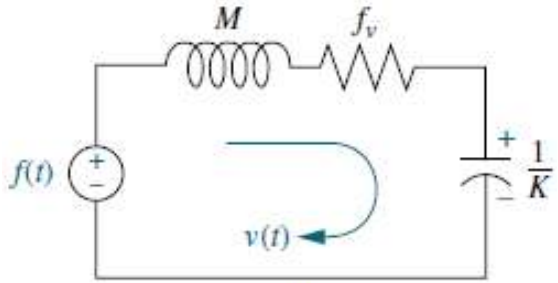
Series Analog(Force -Voltage Analog)



(a)



(b)



(c)

Equation of motion of the above translational mechanical system is;

Kirchhoff's mesh equation for the above simple series RLC network is;

a direct analogy

$$M\dot{v}\,(t)+f_c\,v(t)+k\int v(t)dt = f(t)$$

$$L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt = e_i$$

mass = M ⟶ inductor = M henries

viscous damper = f_v ⟶ resistor = f_v ohms

spring = K ⟶ capacitor = $\frac{1}{K}$ farads

applied force = $f(t)$ ⟶ voltage source = $f(t)$

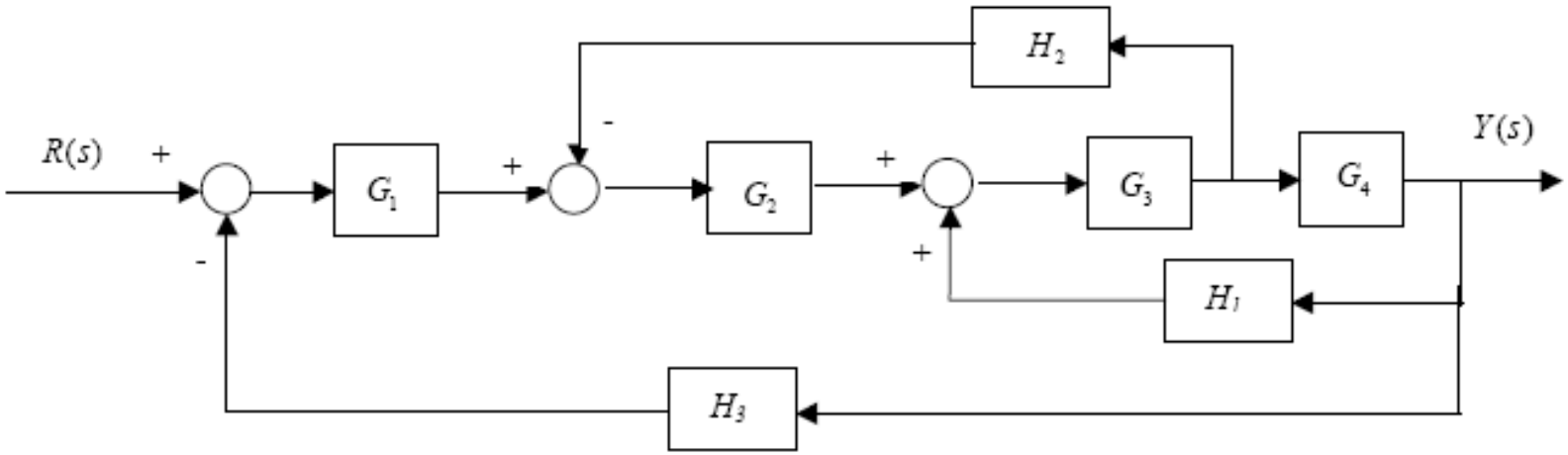
velocity = $v(t)$ ⟶ mesh current = $v(t)$

(d)

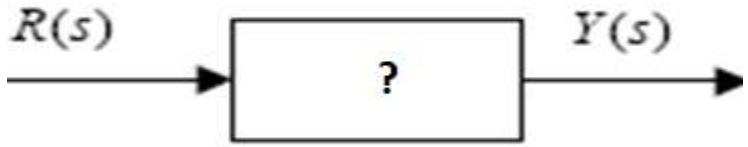
Block Diagram reduction (Manipulation)

- We often represent control systems using block diagrams. A block diagram consists of blocks that represent transfer functions of the different variables of interest.
- If a block diagram has many blocks, not all of which are in cascade, then it is useful to have rules for rearranging the diagram such that you end up with only one block.
- Reduction of Block Diagram means replacing all blocks and summing points by a single block relating the output and the input of the system

For example, we would want to transform the following diagram

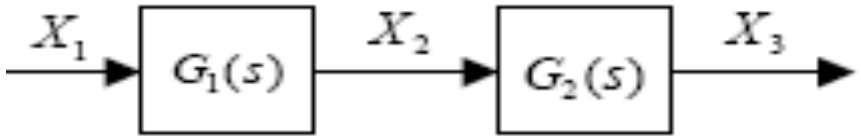


To the next form



Block Diagram Transformations Rules

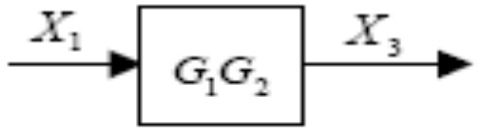
1. Combining blocks in cascade(series connections)



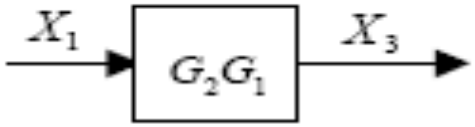
X_2=$G_1(s)X_1$

X_3=$G_2(s)X_2$

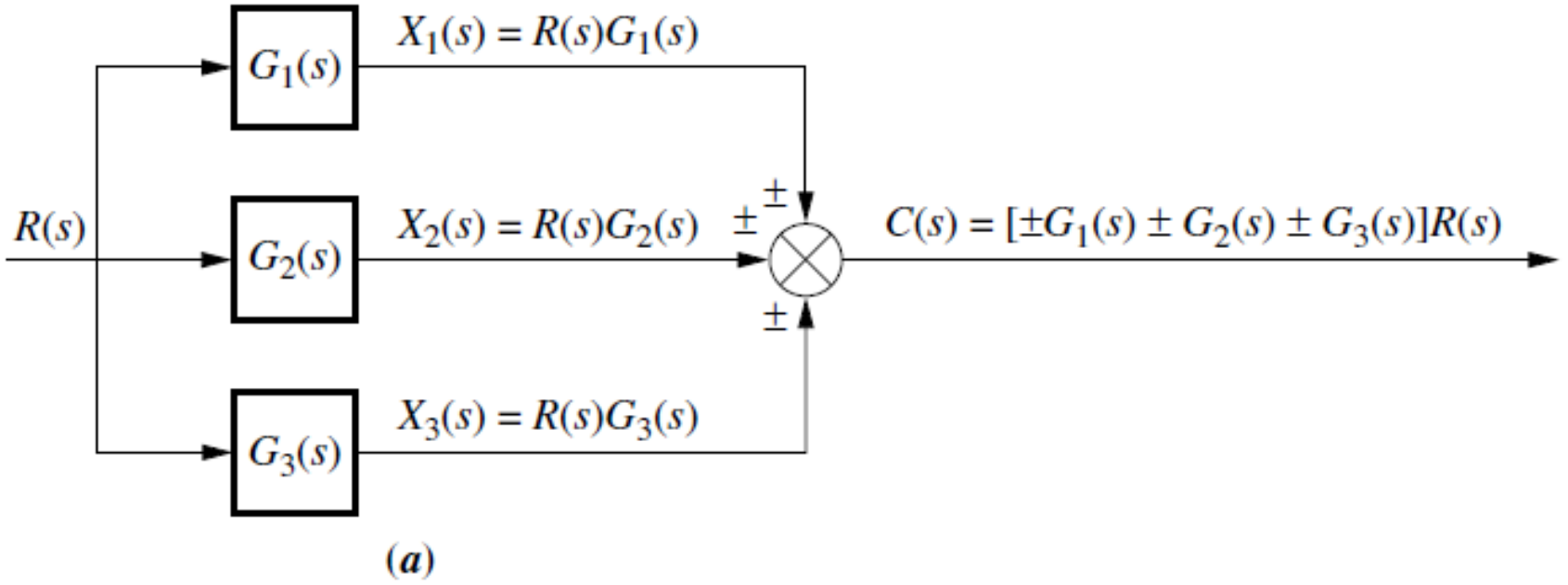
X_3=$G_2(s)G_{1(s)}X_1$



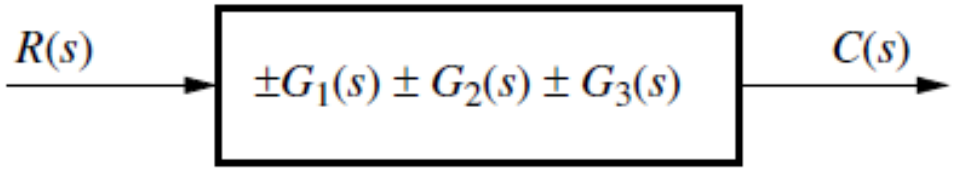
or



2.Parallel Connection:

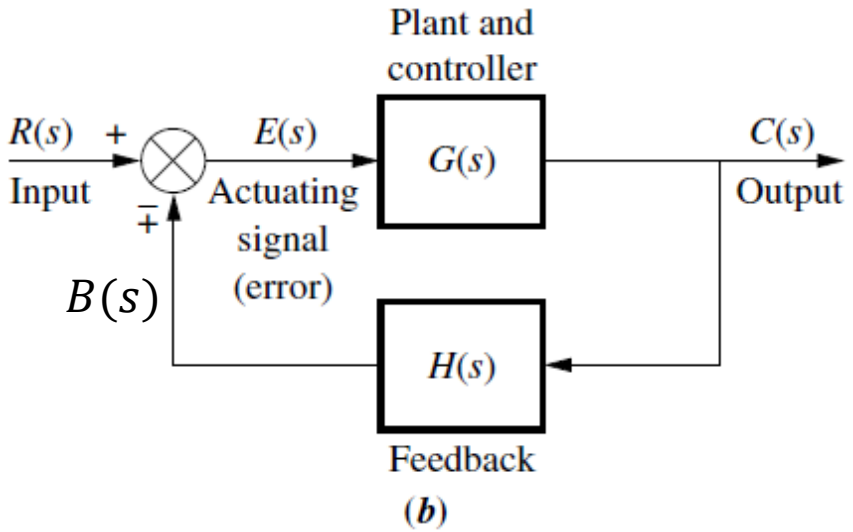


(a)



(b)

3.Feedback Connection :



Negative feedback

$$C(s) = G(s)E(s)$$

$$B(s)=H(s)C(s)$$

$$E(s)=R(s) - B(s)$$

Solving We obtain ,the equivalent or closed-loop transfer function is

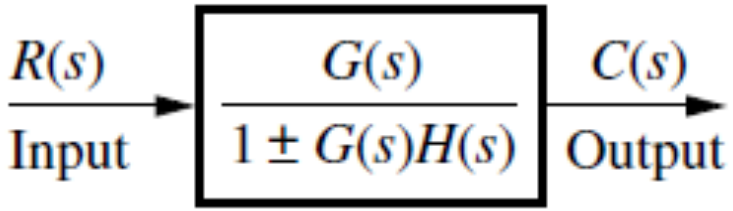
$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

Positive feedback

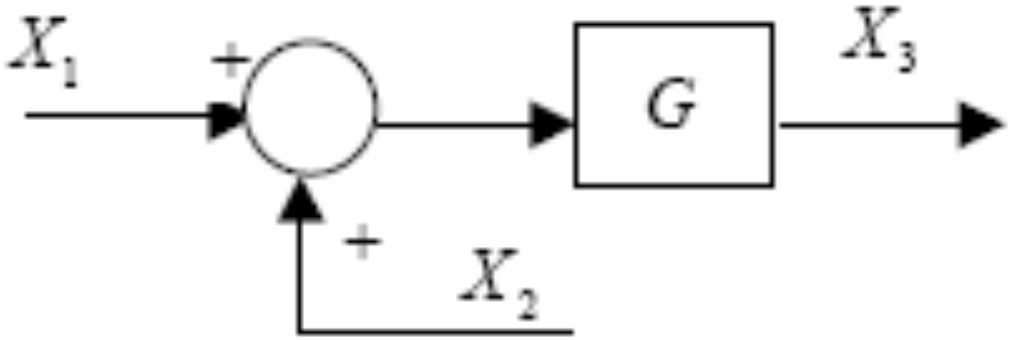
C(s)=G(s)E(s)
B(s)=H(s)C(s)
E(s)=R(s)+B(s)

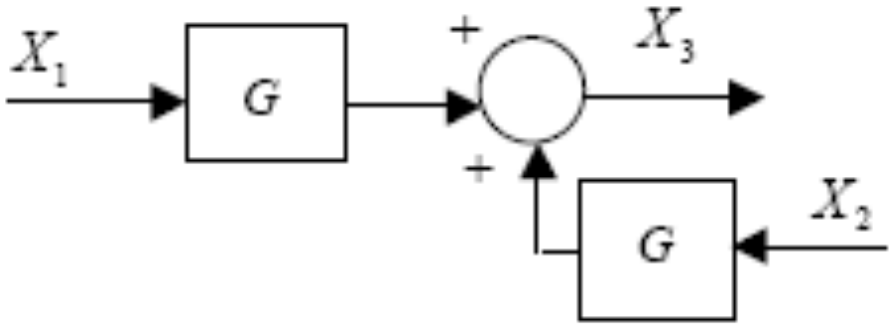
Solving We obtain ,the equivalent or closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 - G(s)H(s)}$$

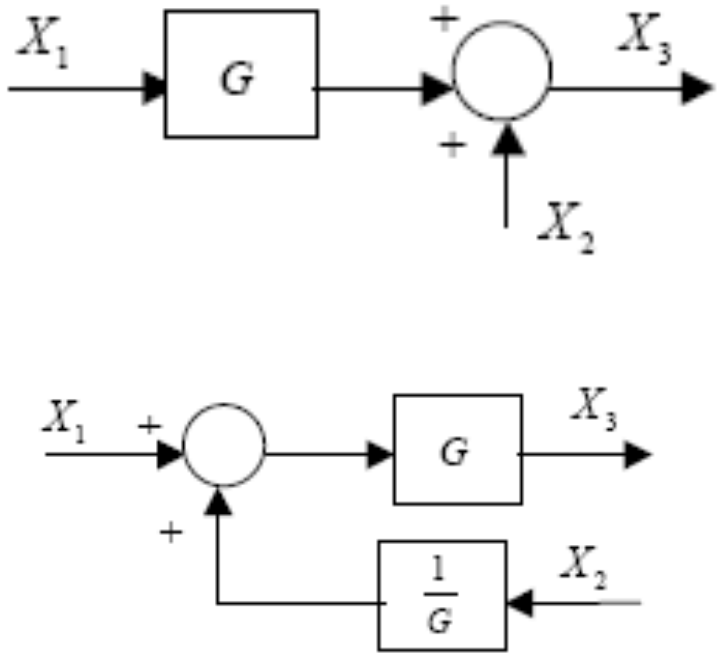


4. Moving summing Point
A. behind a block

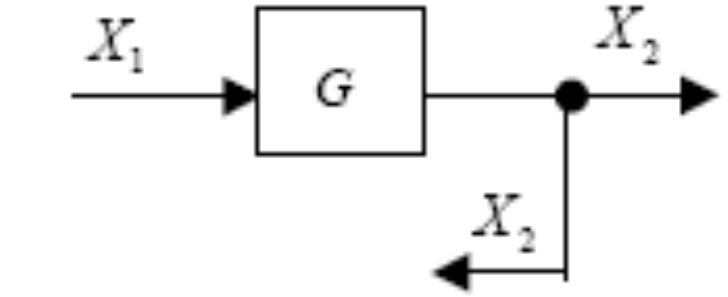


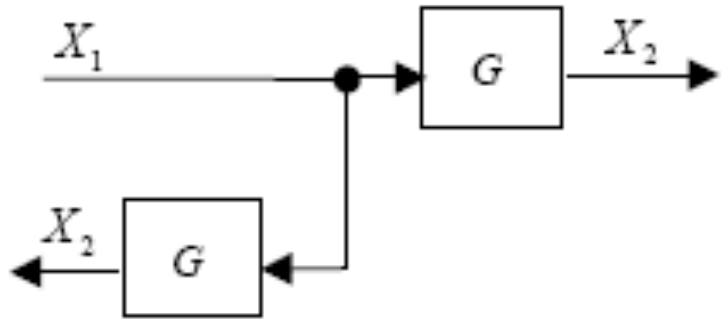


4. Moving a summing point
B. ahead of a block

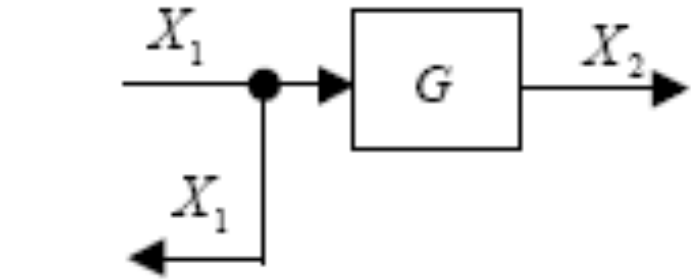


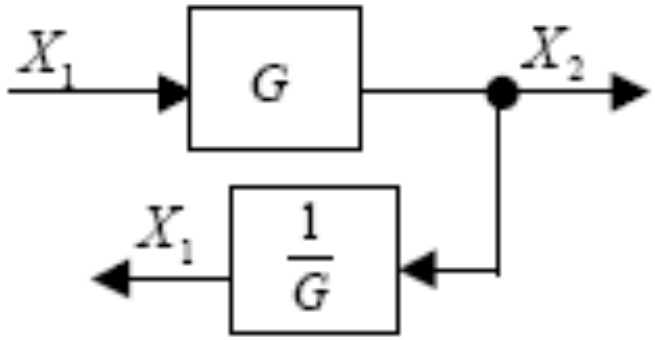
5. Moving a pickoff point
A. ahead of a block



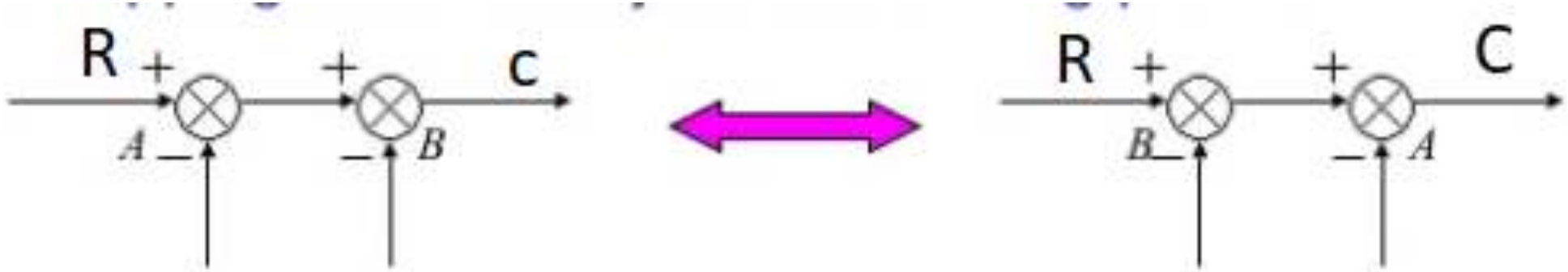


5. Moving a pickoff point
B. behind a block





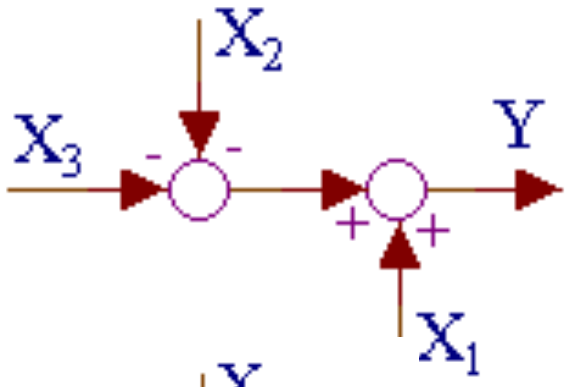
6. replacing summing points



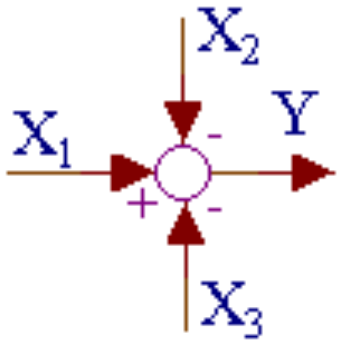
C=R-A-B

C=R-A-B

7. Combining summing points

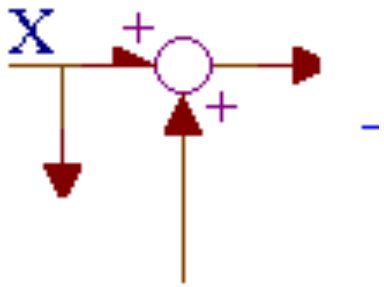


$$Y = X_1 - X_2 - X_3$$

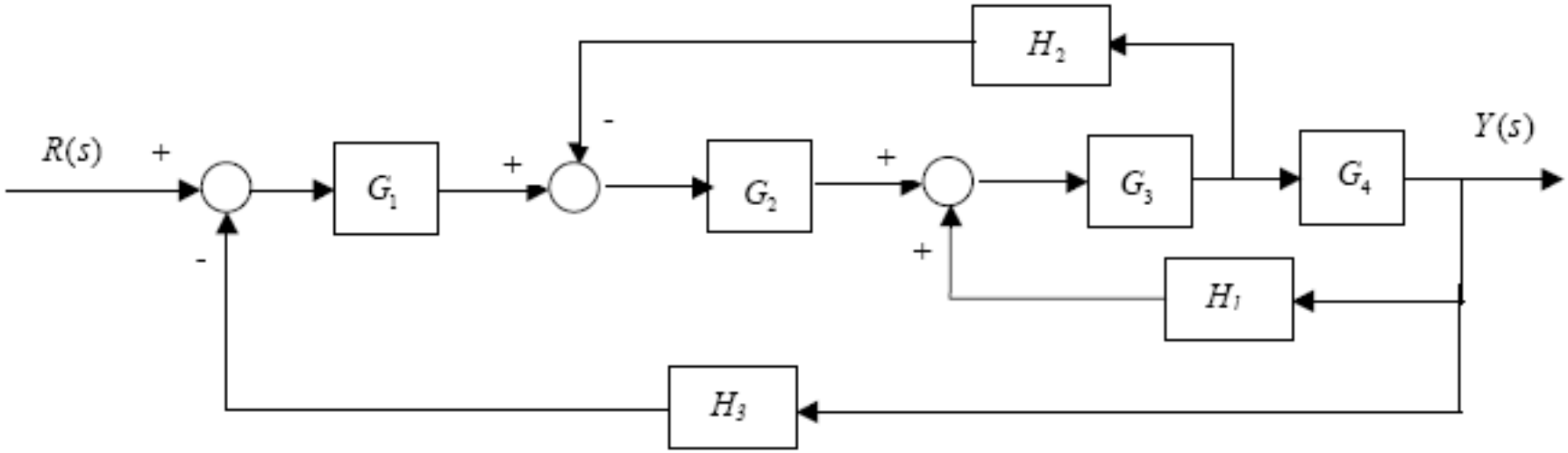


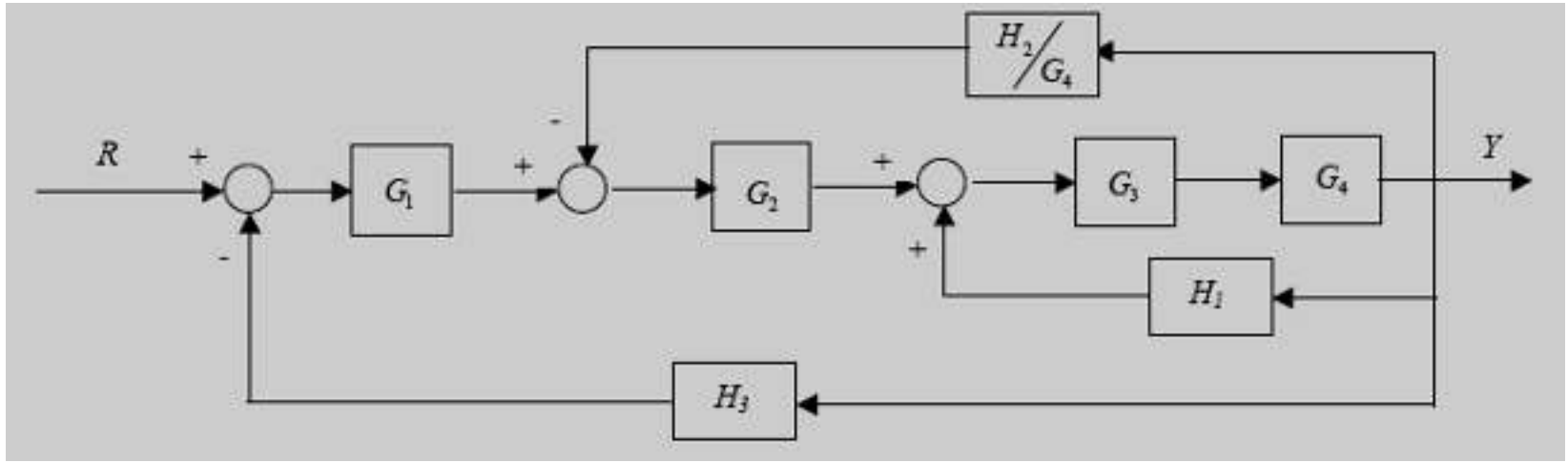
$$Y = X_1 - X_2 - X_3$$

Attention Don't use
This Moving branch point through summing point



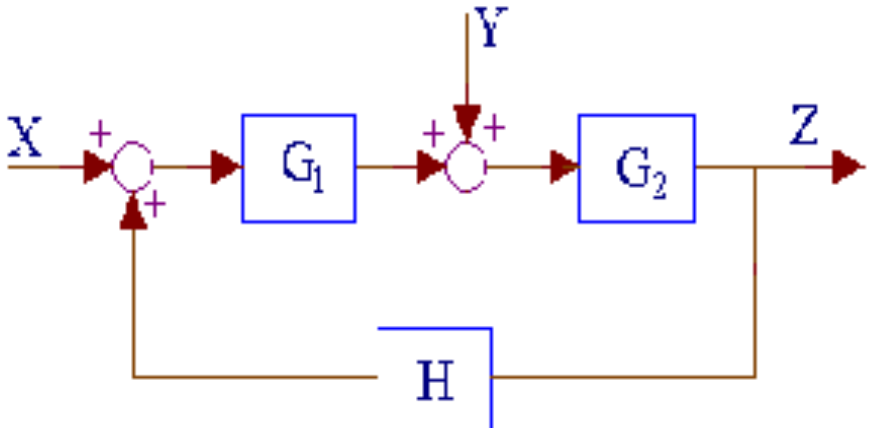
Example
reduce the following block diagram

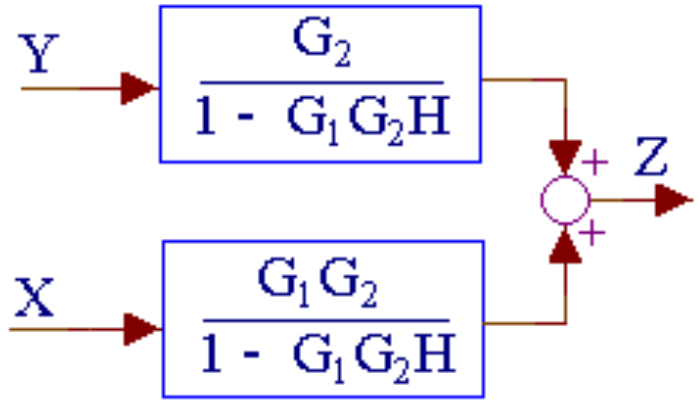




$$R \longrightarrow \boxed{\frac{G_1\ G_2\ G_3\ G_4}{1 - G_3\ G_4\ H_1 + G_2\ G_3\ H_2 + G_1\ G_2\ G_3\ G_4\ H_3}} \longrightarrow Y$$

Example



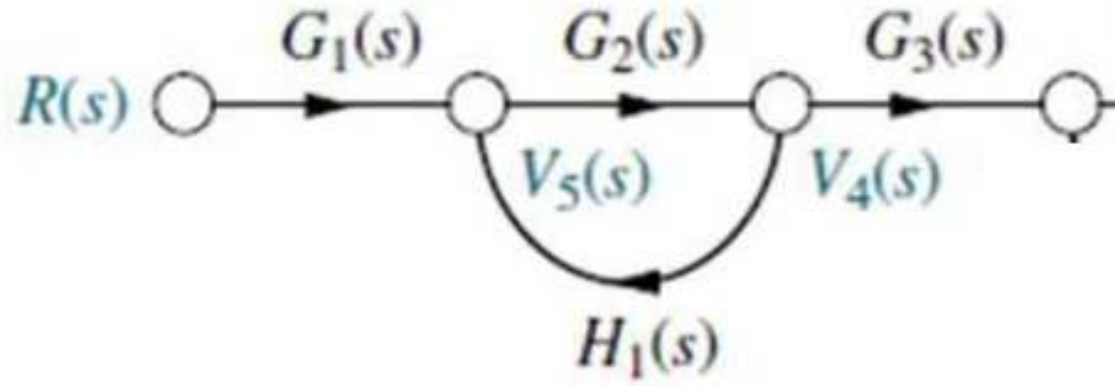


Signal-flow graph

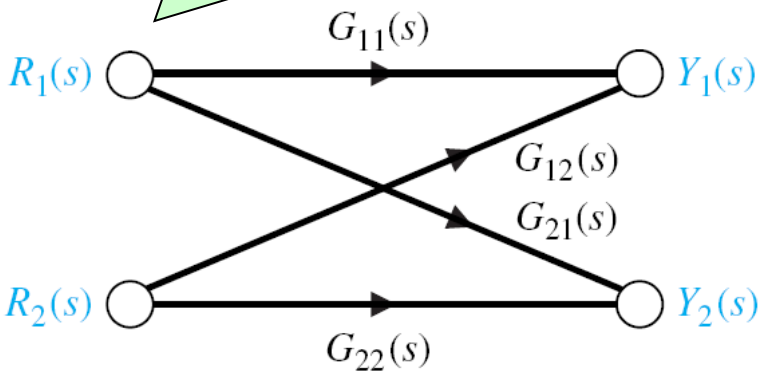
- Alternative method to block diagram representation, developed by S.J.Mason.
- Advantage: the availability of a flow graph gain formula, also called Mason's gain formula.
- A signal-flow graph consists of a network in which nodes are connected by directed branches.
- It depicts the flow of signals from one point of a system to another and gives the relationships among the signals. Note that the signal flows in only one direction.

Definitions

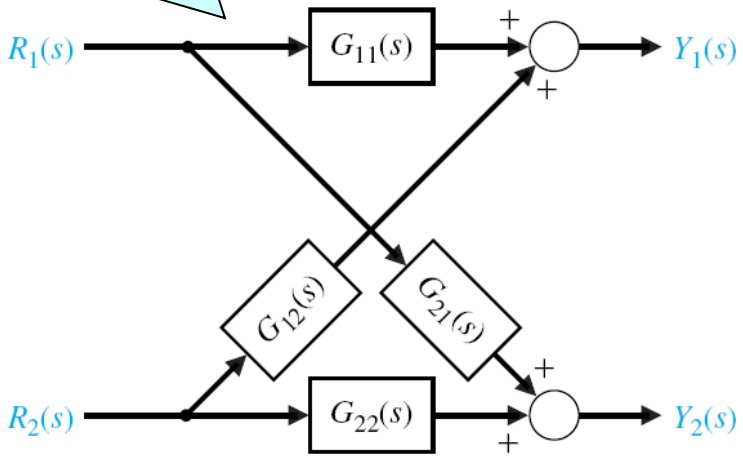
- **Node** - **a point representing a signal or variable.**
- **Branch** – **unidirectional line segment joining two nodes.**
- **Path** – **a branch or a continuous sequence of branches that can be traversed from one node to another node.**
- **Loop** – **a closed path that originates and terminates on the same node and along the path no node is met twice.**
- **Nontouching loops** – **two loops are said to be nontouching if they do not have a common node.**



Signal-flow graph of interconnected system

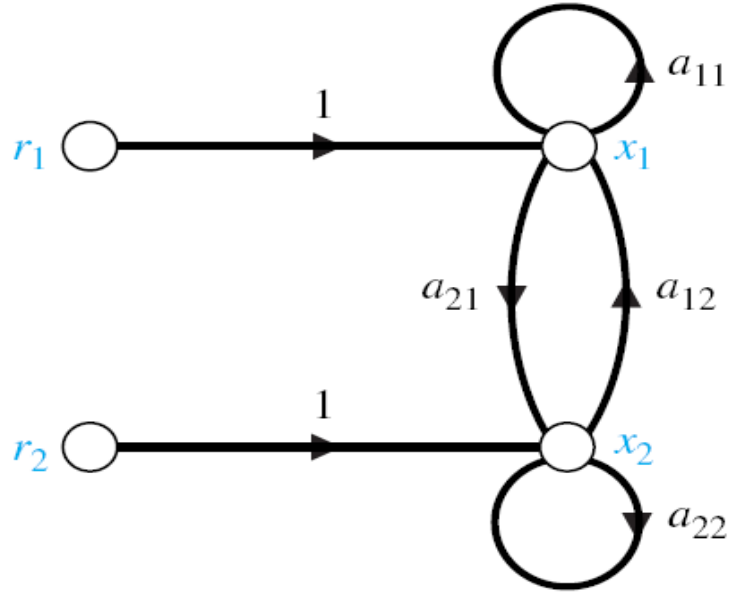


Corresponding block diagram



$$a_{11}x_1 + a_{12}x_2 + r_1 = x_1$$
$$a_{21}x_1 + a_{22}x_2 + r_2 = x_2$$

Signal-flow graph of two algebraic equations



Mason's gain formula

The linear dependence T_{ij} between the independent variable x_i (also called the input variable) and a dependent variable x_j is

$$T_{ij} = \frac{\Sigma_k P_{ijk} \Delta_{ijk}}{\Delta},$$

P_{ijk} = kth path from variable x_i to variable x_j,

Δ = determinant of the graph,

Δ_{ijk} = cofactor of the path P_{ijk},

and the summation is taken over all possible k paths from x_i to x_j. The cofactor Δ_{ijk} is the determinant with the loops touching the kth path removed. The determinant Δ is

$$\Delta = 1 - \sum_{n=1}^{N} L_n + \sum_{m=1,q=1}^{M,Q} L_m L_q - \sum L_r L_s L_t + \cdots,$$

where L_q equals the value of the qth loop transmittance. Therefore the rule for evaluating Δ in terms of loops $L_1, L_2, L_3, \ldots, L_N$ is

$\Delta = 1$ − (sum of all different loop gains)

+ (sum of the gain products of all combinations of two nontouching loops)

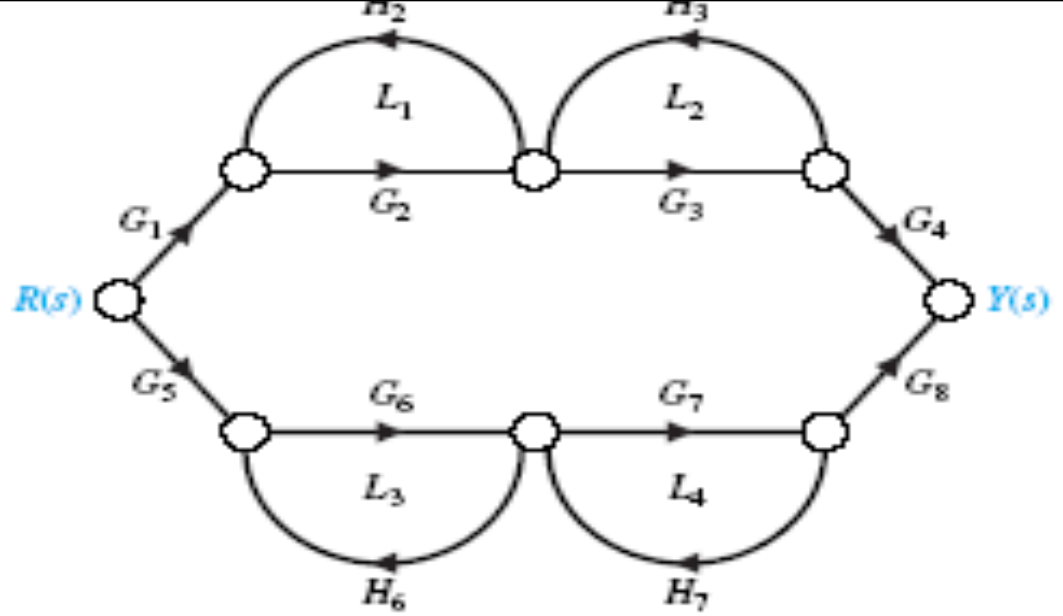
− (sum of the gain products of all combinations of three nontouching loops)

+ $\cdots$.

Write the gain formula in a simplified form: $T = \frac{\sum_k P_k \Delta_k}{\Delta}$

Systematic approach:

1. Calculate forward path transfer function P_k for each forward path *k*
2. Calculate all loop transfer functions
3. Consider nontouching loops 2 at a time
4. Consider nontouching loops 3 at a time
5. etc
6. Calculate Δ from steps 2,3,4 and 5
7. Calculate Δ_k as portion of Δ not touching forward path *k*



1.Calculate forward path transfer function *Pk* for each forward path *k*.

$P_1 = G_1G_2G_3G_4$ (path 1) and $P_2 = G_5G_6G_7G_8$ (path 2)

2.Calculate all loop TF's.

$L_1 = G_2H_2, \quad L_2 = H_3G_3, \quad L_3 = G_6H_6, \quad L_4 = G_7H_7$

3.Consider nontouching loops 2 at a time.

Loops L1 and L2 do not touch Loops L3 and L4

4. Consider nontouching loops 3 at a time.

   None.

5. Calculate Δ from steps 2,3,4.

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4) + (L_1L_3 + L_1L_4 + L_2L_3 + L_2L_4)$$

6. Calculate Δ_k as portion of Δ not touching forward path *k*.

$$\Delta_1 = 1 - (L_3 + L_4) \quad \text{and} \quad \Delta_2 = 1 - (L_1 + L_2)$$

The TF of the system is $\dfrac{Y(s)}{R(s)} = T(s) = \dfrac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$

الى هنا مادة امتحان منتصف الفصل

UNIT FOUR
Time response of Control Systems

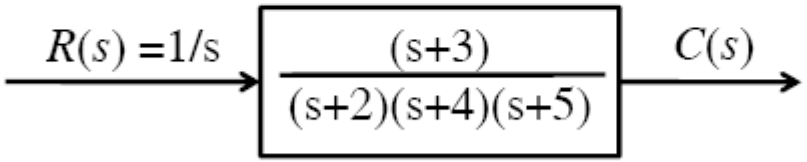
Transient response and steady-state response

- If the output of **control system** for an input varies with respect to time, then it is called the **time response** of the **control system**.
- The time response of a control system consists of two parts:
- transient response means that which goes from the initial state to the final state.(free or natural response)
- steady-state response, we mean the manner in which the system output behaves as t approaches infinity.(forced response)

C(s)=G(s)R(s)

Response vs. Poles

- *Given the system, write the output and specify the forced and free (natural) parts of the solution.*



1. *By inspection, each system pole generates an exponential as part of the free (natural) response. The input's pole generates the forced response. Thus,*

$$C(s) \equiv \underbrace{\frac{K_1}{s}}_{\text{Forced response}} + \underbrace{\frac{K_2}{(s+2)} + \frac{K_3}{(s+4)} + \frac{K_4}{(s+5)}}_{\text{Free (Natural) response}}$$

2. *Taking the inverse Laplace transform, we get*

$$c(t) \equiv \underbrace{K_1}_{\text{Forced response}} + \underbrace{K_2 e^{-2t} + K_3 e^{-4t} + K_4 e^{-5t}}_{\text{Free (Natural) response}}$$

typical input signals(control signals or test signals)

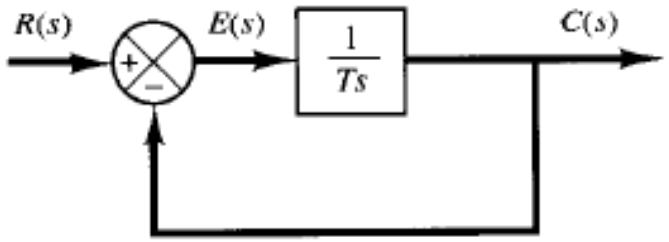
- typical input signals to use for analyzing system characteristics may he determined by the form of the input that the system will be subjected to most frequently under normal operation.
- If the inputs to a control system are gradually changing functions of time, then a ramp function of time may be a good test signal.
- if a system is subjected to sudden disturbances, a step function of time may be a good test signal;
- and for a system subjected to shock inputs, an impulse function may be best.

- Once a control system is designed on the basis of test signals, the performance of the system in response to actual inputs is generally satisfactory. The use of such test signals enables one to compare the performance of all systems on the same basis.

FIRST-ORDER SYSTEMS

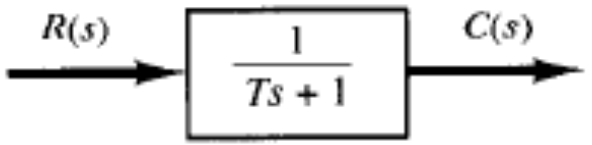
- Consider the first-order system shown . Physically, this system may represent an *RC* circuit, thermal system, or the like.

$$\frac{C(s)}{R(s)} = \frac{1}{Ts + 1}$$



(a)

- A simplified block diagram is shown in Figure (b). The input—output relationship is given by



(b)

$$\frac{C(s)}{R(s)} = \frac{1}{Ts + 1}$$

In the following, we shall analyze the system responses to such inputs as the unit-step, unit-ramp, and unit-impulse functions. The initial conditions are assumed to be zero.

Unit-step time response of first-order systems

- Since the Laplace transform of the unit-step function is 1/s, substituting R(s) , we obtain

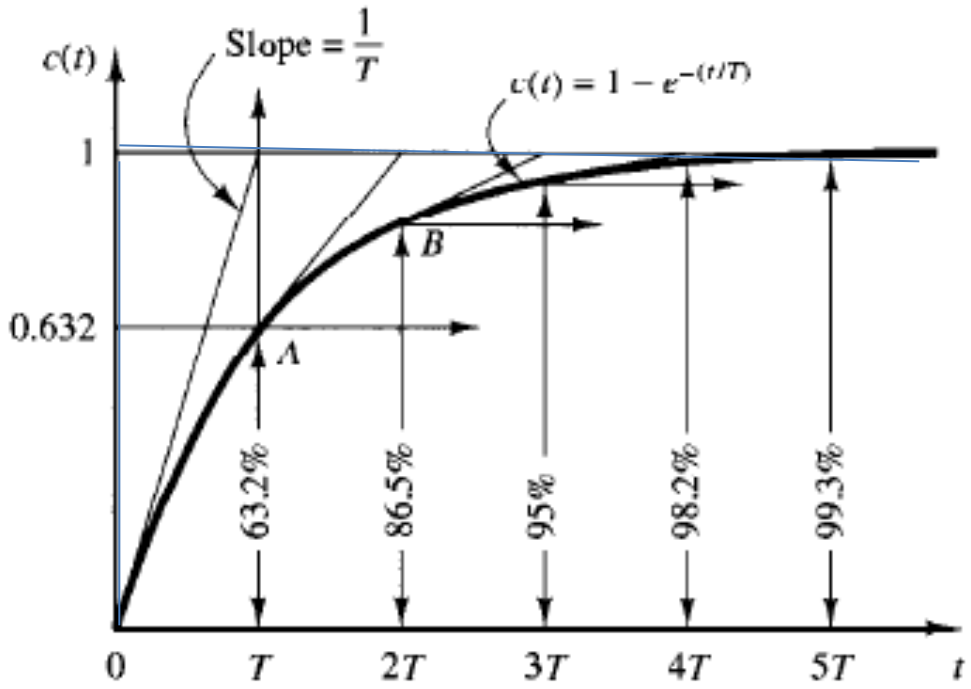
$$C(s) = \frac{1}{Ts + 1}\frac{1}{s}$$

- Expanding *C(s)* into partial fractions gives

$$C(s) = \frac{1}{s} - \frac{T}{Ts + 1} = \frac{1}{s} - \frac{1}{s + (1/T)}$$

$$c(t) = 1 - e^{-t/T}, \qquad \text{for } t \geq 0$$

t(SEC)	0	T	2T	3T	4T	inf
c(T)	0	63.2%	86.5%	95%	98.2%	1



$$e(t) = r(t) - c(t) = e^{-t/T}$$

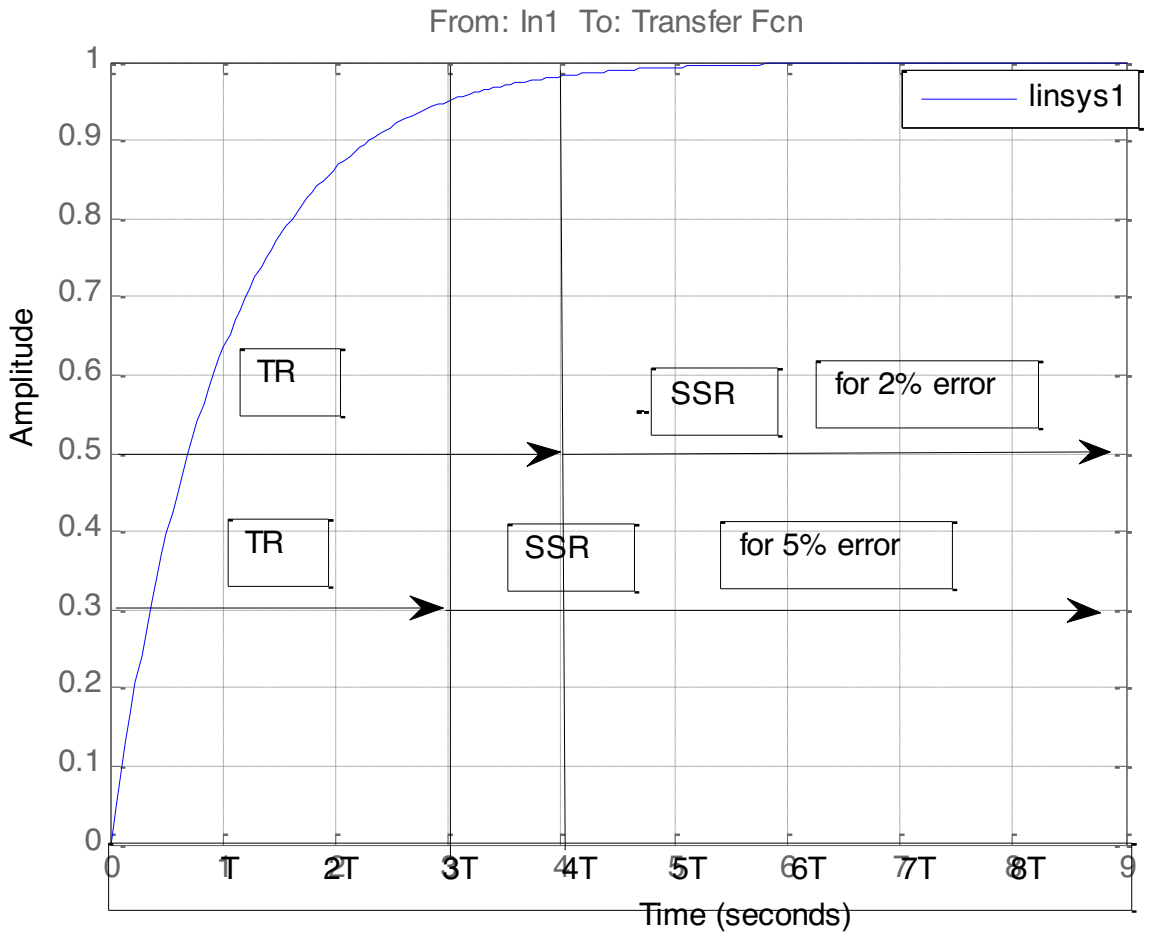
COMMENTS

- smaller the time constant *T,* faster the system response.
- Another important characteristic of the exponential response curve is that the slope of the tangent line at **t** = 0 is 1/T. since

$$\left.\frac{dc}{dt} = \frac{1}{T} e^{-t/T} \right|_{t=0} = \frac{1}{T}$$

- The output **would** reach the final value at $t = T$ if **it** maintained its initial speed of response

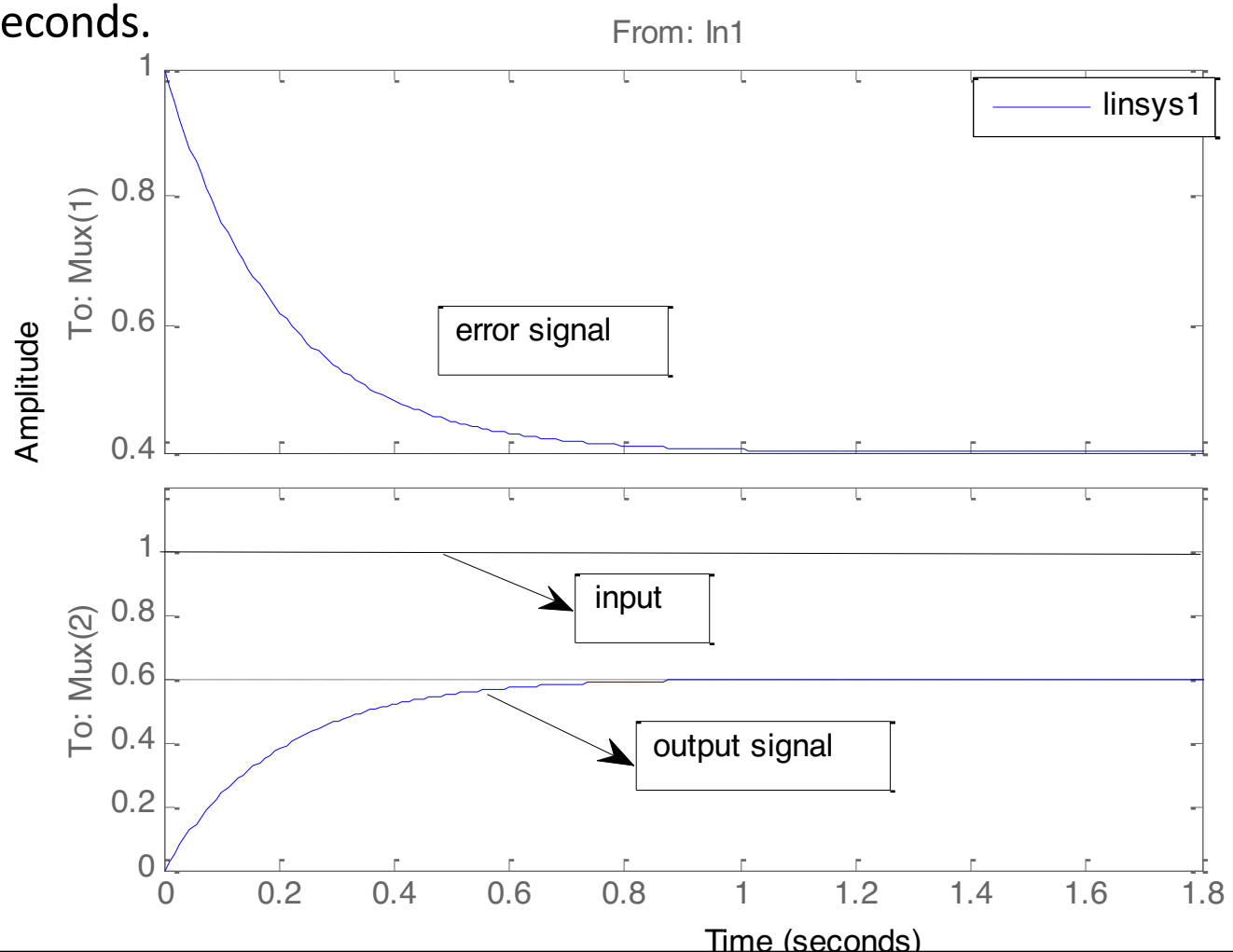
- the steady state is reached mathematically only after an infinite time.
- In practice, however, a reasonable estimate of the response time is the length of time the response curve needs to reach the 2% line of the final value, or four time constants.



- The first order system given below.

$$G(s) \quad \frac{3}{s \quad 5} \quad \frac{3/5}{1/5s \quad 1}$$

- D.C Gain of the system is 3/5 and time constant is 1/5 seconds.



Unit-ramp time response of first-order systems

- Since the Laplace transform of the unit-ramp function is 1/s^2, we obtain the output of the system as

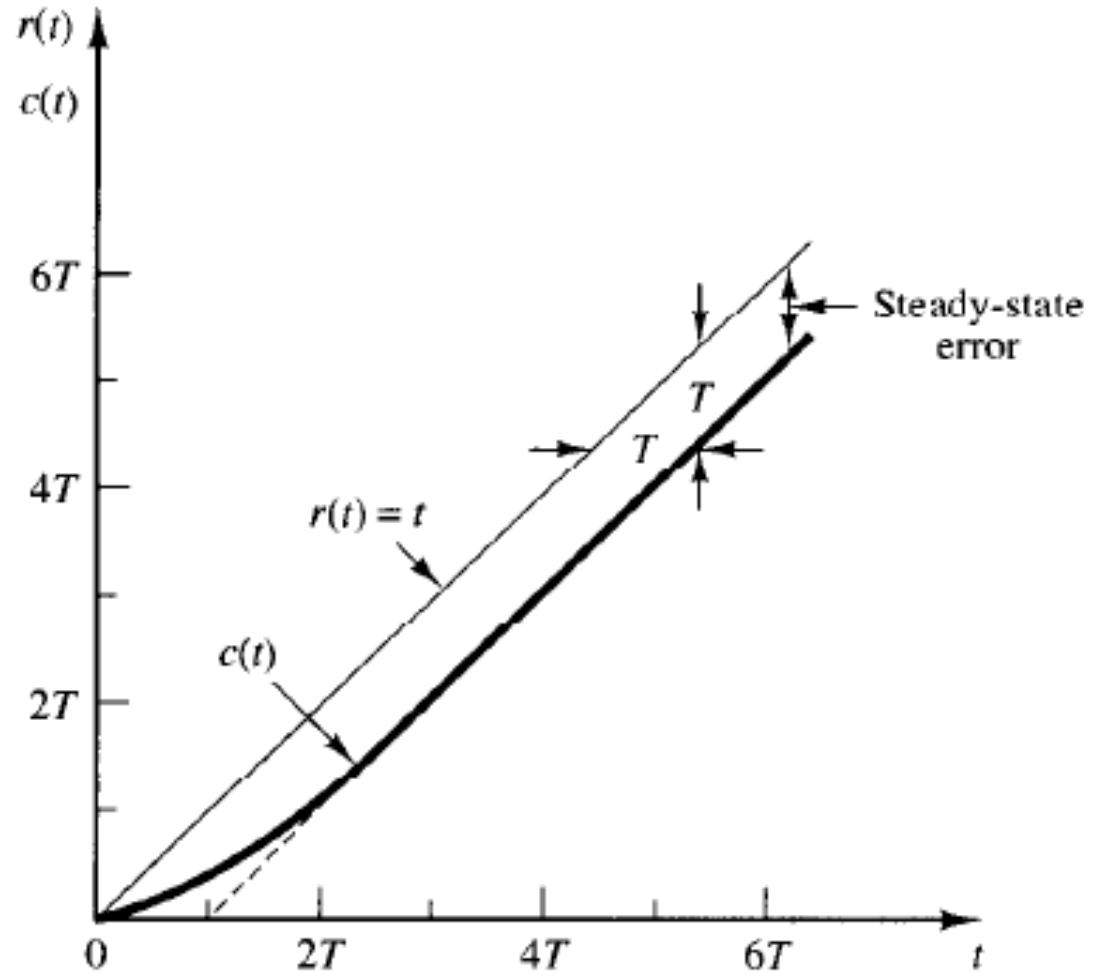
$$C(s) = \frac{1}{Ts+1}\frac{1}{s^2} = \frac{1}{s^2} - \frac{T}{s} + \frac{T^2}{Ts+1}$$

$$c(t) = t - T + Te^{-t/T}, \qquad \text{for } t \geq 0$$

The error signal $e(t)$ is then

$$e(t) = r(t) - c(t)$$
$$= T(1 - e^{-t/T})$$

$$e(\infty) = T$$

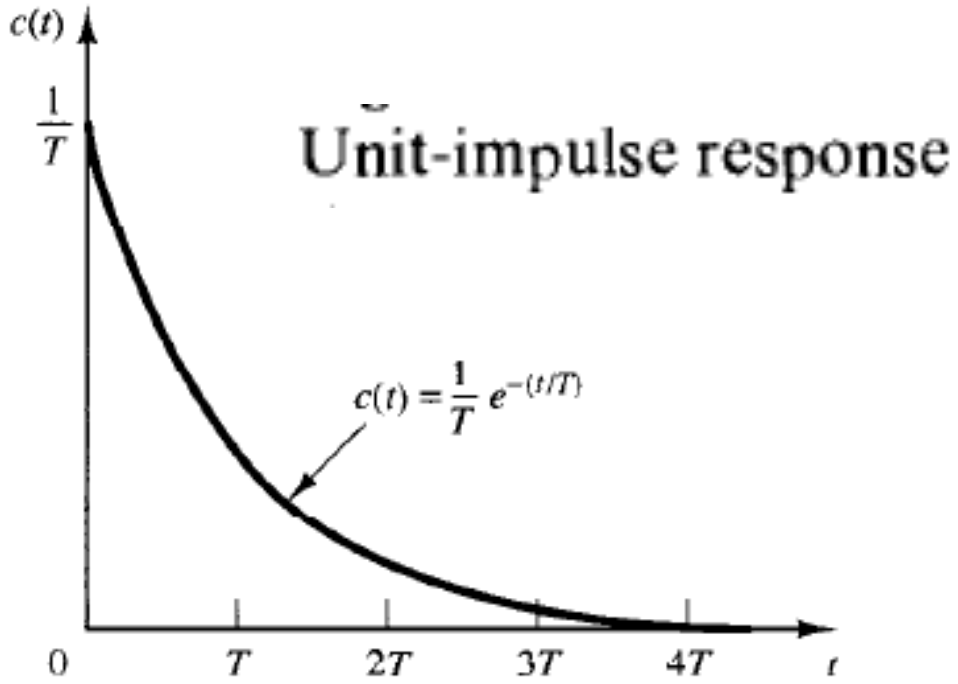


Unit-ramp response of the system

Unit-impulse response of first-order systems

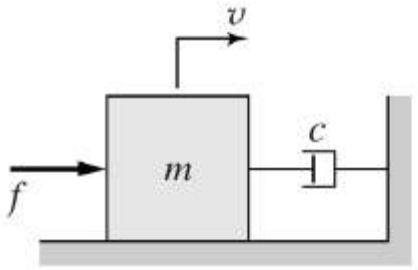
- Since the Laplace transform of the unit-impulse function is R(s)=1, we obtain the output of the system

$$c(t) = \frac{1}{T} e^{-t/T}, \qquad \text{for } t \geq 0$$



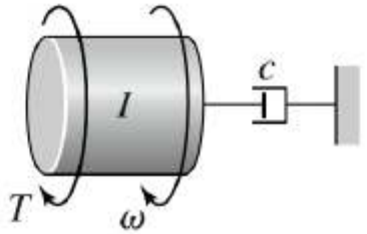
- **Comparison of the system responses to these three inputs clearly indicates that :**
- **the response to the derivative of an input signal can be obtained by differentiating the response of the system to the original signal.**
- **the response to the integral of the original signal can be obtained by integrating the response of the system to the original signal**
- **This is a property of linear time-invariant systems.**

Examples of First-Order Systems



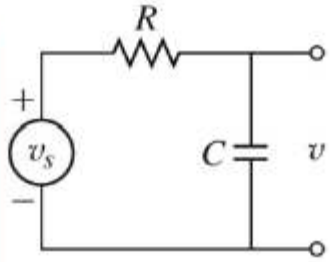
$$m\frac{dv}{dt}+cv=f$$

$$\tau=\frac{m}{c}$$



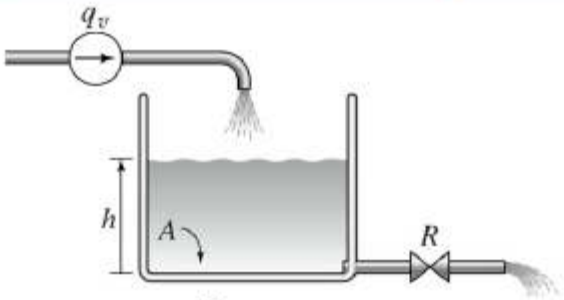
$$I\frac{d\omega}{dt}+c\omega=T$$

$$\tau=\frac{I}{c}$$



$$RC\frac{dv}{dt}+v=v_s$$

$$\tau=RC$$



$$AR\frac{dh}{dt}+gh=Rq_v$$

$$\tau=\frac{AR}{g}$$

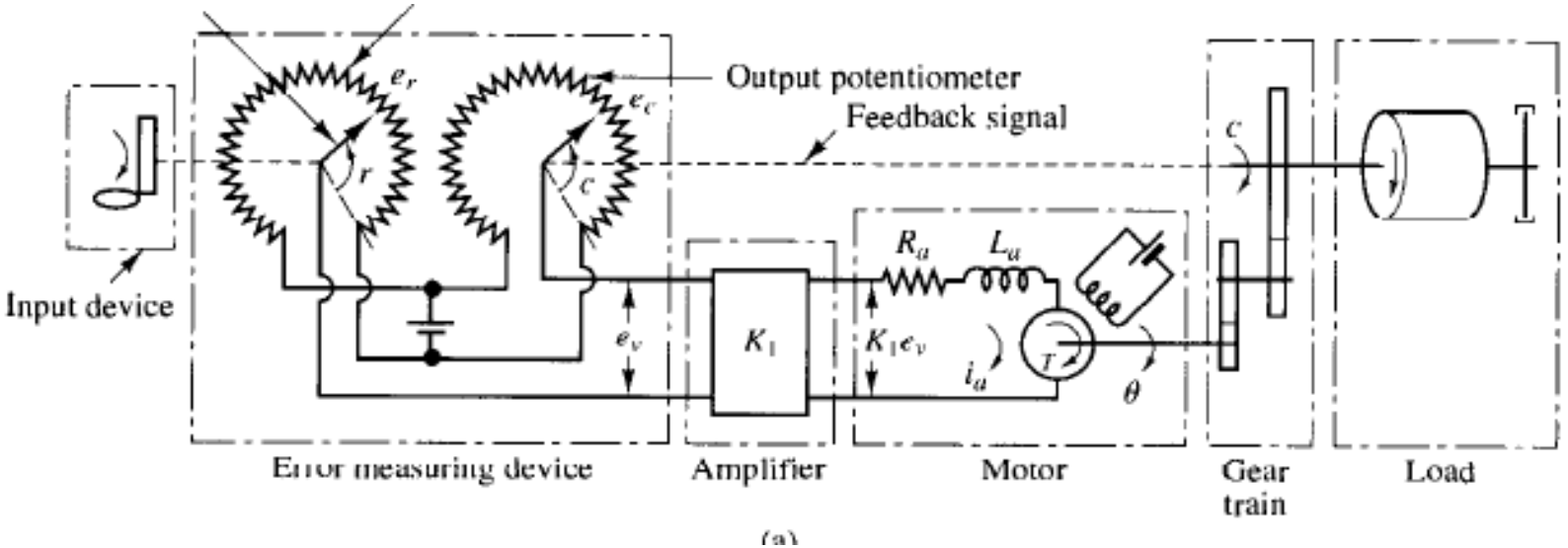


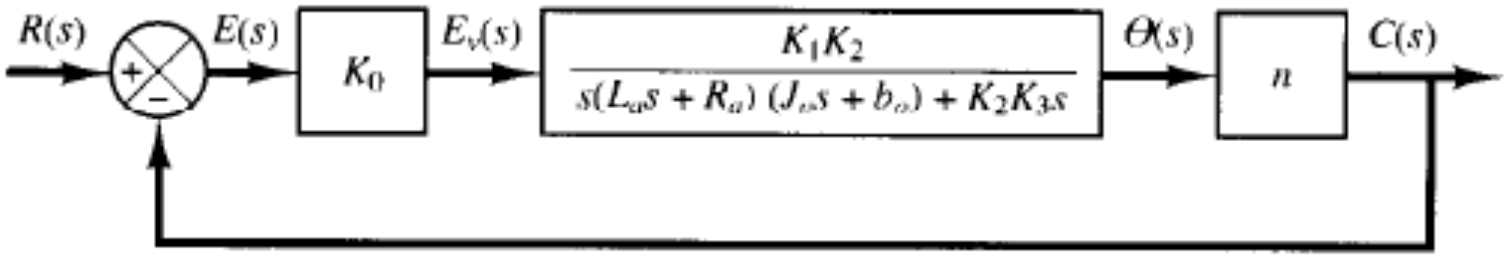
$$mc_pR\frac{dT}{dt}+T=T_b$$

$$\tau=mc_pR$$

SECOND-ORDER SYSTEMS

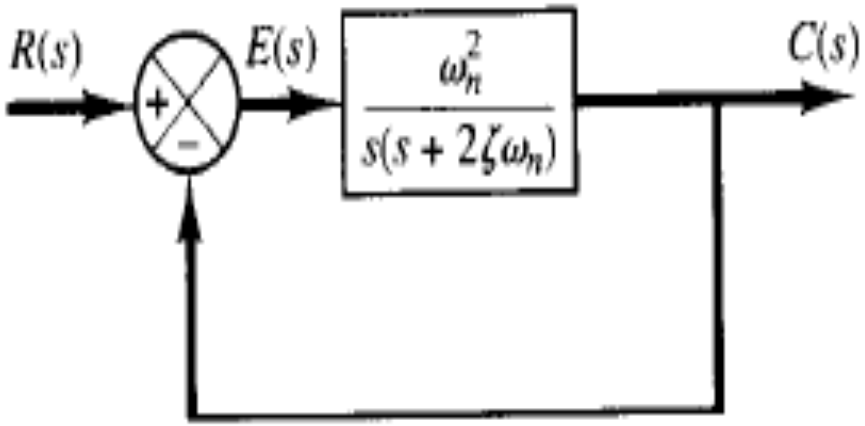
Example: servo system





The closed-loop transfer function of a second order system and the block diagram are

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$



- The time response of the second-order system can then be described in terms of two parameters
- ω_n −undamped natural frequency of the second order system, which is the frequency of oscillation of the system without damping.
- ξ(zeta)- damping ratio of the second order system, which is a measure of the degree of resistance to change in the system output.
- We shall now solve for the response of the system unit-step input. We shall consider four different cases:
- The poles of the system are

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

- Complex conjugate poles the response is underdamped (0 **< ζ <** 1),
- Multiple poles the response is critically damped (ζ = 1),
- Distinct poles the response is overdamped (ζ> 1)
- Imaginary poles the response is ζ = 0, the transient response does not die out.

Underdamped case ($0 < \zeta < 1$):

$$C(s) = \frac{1}{s} \cdot \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

Applying partial fraction on Equation 3 gives

$$C(s) = \frac{1}{s} - \frac{s + 2\xi\omega_n}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

The denominator term $s^2 + 2\xi\omega_n s + \omega_n^2$ of Equation can be written as

$$s^2 + 2\xi\omega_n s + \omega_n^2 = (s + \xi\omega_n)^2 + \omega_n^2(1-\xi^2)$$

Also,

$$\omega_d^2 = \omega_n^2(1-\xi^2)$$

Then

$$C(s) = \frac{1}{s} - \frac{s + \xi\omega_n}{(s + \xi\omega_n)^2 + \omega_d^2} - \frac{\xi\omega_n}{\omega_d} \cdot \frac{\omega_d}{(s + \xi\omega_n)^2 + \omega_d^2}$$

$$\mathscr{L}^{-1}C(s) = \mathscr{L}^{-1}\left[\frac{1}{s} - \frac{s+\xi\omega_n}{(s+\xi\omega_n)^2+\omega_d^2} - \frac{\xi\omega_n}{\omega_d}\cdot\frac{\omega_d}{(s+\xi\omega_n)^2+\omega_d^2}\right]$$

$$c(t) = 1 - e^{-\xi\omega_n t}.\cos\omega_d t - \frac{\xi\omega_n}{\omega_d}.e^{-\xi\omega_n t}.\sin\omega_d t$$

$$c(t) = 1 - e^{-\xi\omega_n t}\left[\cos\omega_d t + \frac{\xi}{\sqrt{1-\xi^2}}.\sin\omega_d t\right]$$

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\left[\sqrt{1-\xi^2}.\cos\omega_d t + \xi\sin\omega_d t\right]$$

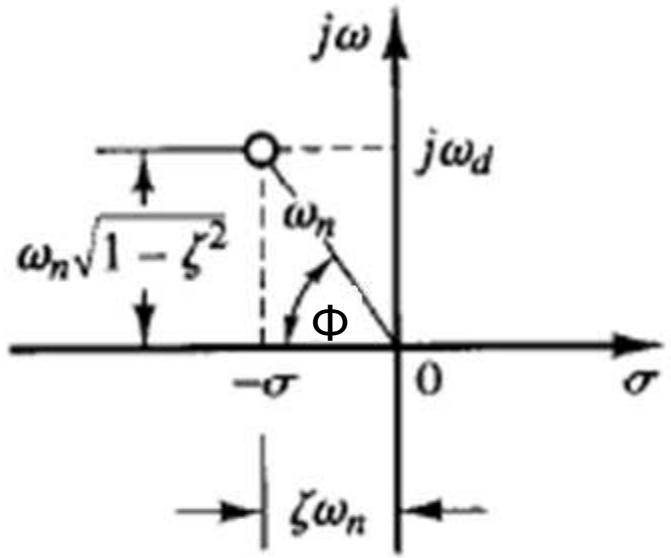
Put $\sqrt{1-\xi^2} = \sin\phi$

$\therefore \cos\phi = \xi$

$$\tan\phi = \frac{\sqrt{1-\xi^2}}{\xi}$$

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}[sin\theta\cos(\omega_d t) + cos\theta\sin(\omega_d t)]$$

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\sin(\omega_d t + \phi)$$



If (0 < ζ < I) the poles of the system are complex conjugate and the time response is underdamped

Using the identity,

$$A\sin\omega t + B\cos\omega t = \sqrt{A^2 + B^2}\sin\left[\omega t + \tan^{-1}\frac{B}{A}\right]$$

Now, comparing term $\xi\sin\omega_d t + \sqrt{1-\xi^2}\cos\omega_d t$ and term $A\sin\omega t + B\cos\omega t$ of Equation 8 gives,

$$A = \xi\,;\, B = \sqrt{1-\xi^2}\,;\, \omega = \omega_d$$

2.undamped case ζ=0

And the output is $\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + \omega_n^2}$

$$C(s) = \frac{1}{s}\frac{\omega_n^2}{(s^2 + \omega_n^2)}$$

Taking partial fraction

$$C(s) = \frac{A}{s} + \frac{Bs + C}{(s^2 + \omega_n^2)}$$

$$\frac{\omega_n^2}{s(s^2 + \omega_n^2)} = \frac{A}{s} + \frac{Bs + C}{(s^2 + \omega_n^2)}$$

$$\frac{\omega_n^2}{s(s^2 + \omega_n^2)} = \frac{A(s^2 + \omega_n^2) + (Bs + C)s}{s(s^2 + \omega_n^2)}$$

On simplifying

$$\omega_n^2 = A(s^2 + \omega_n^2) + (Bs + C)s$$

$$\omega_n^2 = s^2(A + B) + A\omega_n^2 + Cs$$

Comparing the constant

$$\omega_n^2 = A\omega_n^2$$

$$A = 1$$

Comparing the coefficient of s^2

$$A + B = 0$$

$$1 + B = 0$$

$$B = -1$$

Comparing the coefficient of s

$$C = 0$$

Substituting the values in the partial fraction, we will get

$$C(s) = \frac{1}{s} - \frac{s}{(s^2 + \omega_n^2)}$$

Taking Inverse Laplace transform of the over equation

$$c\,(t) = L^{-1}\left[\frac{1}{s} - \frac{s}{(s^2 + \omega_n^2)}\right]$$

Thus

$$c\,(t) = (1 - \cos \omega_n t)\, u(t)$$

3.Critically damped case ($\zeta = 1$)

So, the transfer function of the critically damped system:

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\omega_n s + \omega_n^2}$$

In the case of unit step signal,

$$C(s) = R(s)\frac{\omega_n^2}{s^2 + 2\omega_n s + \omega_n^2}$$

On substituting

$$C(s) = \frac{1}{s}\frac{\omega_n^2}{(s + \omega_n)^2}$$

Taking partial fraction

$$C(s) = \frac{A}{s} + \frac{B}{s+\omega_n} + \frac{C}{(s+\omega_n)^2}$$

$$\frac{\omega_n^2}{s(s+\omega_n)^2} = \frac{A(s+\omega_n)^2 + Bs(s+\omega_n) + Cs}{s(s+\omega_n)^2}$$

$$\omega_n^2 = As^2 + 2A\omega_n s + A\omega_n^2 + Bs^2 + Bs\omega_n + Cs$$

$$\omega_n^2 = s^2(A+B) + s(2A\omega_n + B\omega_n + C) + A\omega_n^2$$

Comparing the constant

$$\omega_n^2 = A\omega_n^2$$

$$\mathrm{A} = 1$$

Further comparing the coefficient of s

$$2A\omega_n + B\omega_n + C = 0$$
$$2\omega_n - \omega_n + C = 0$$
$$C = -\omega_n$$

Now on substituting the values in partial fraction

$$C(s) = \frac{1}{s} - \frac{1}{s+\omega_n} - \frac{\omega_n}{(s+\omega_n)^2}$$

Taking inverse Laplace transform

$$c(t) = L^{-1}[\frac{1}{s} - \frac{1}{s+\omega_n} - \frac{\omega_n}{(s+\omega_n)^2}]$$

Thus, we will get

$$\mathrm{c(t)} = (1 - e^{-\omega_n t} - \omega_n t e^{-\omega_n t})u(t)$$

4.overdamped case (ζ> 1):

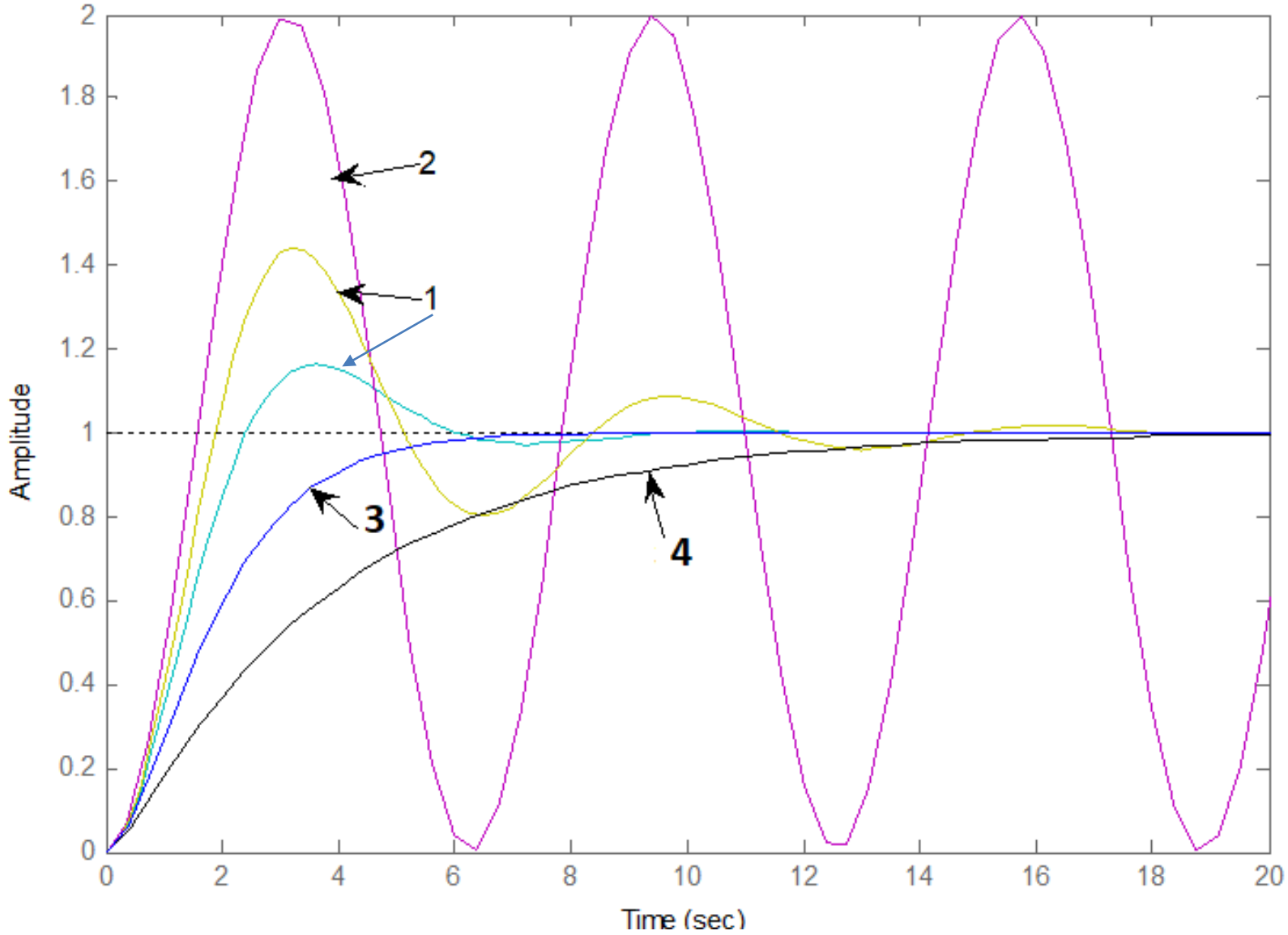
- In this case, the two poles are negative real and udequal. For a unit-step input

$$C(s) = \frac{\omega_n^2}{(s + \zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1})(s + \zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1})s}$$

$$c(t) = 1 + \frac{1}{2\sqrt{\zeta^2 - 1}(\zeta + \sqrt{\zeta^2 - 1})} e^{-(\zeta + \sqrt{\zeta^2 - 1})\omega_n t}$$

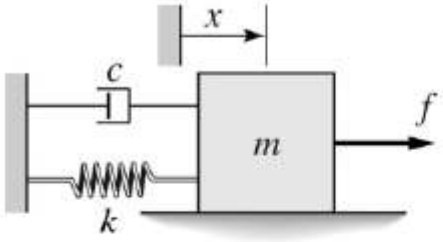
$$- \frac{1}{2\sqrt{\zeta^2 - 1}(\zeta - \sqrt{\zeta^2 - 1})} e^{-(\zeta - \sqrt{\zeta^2 - 1})\omega_n t}$$

If (ζ > I) the poles of the system are distinct and the time response is overdamped

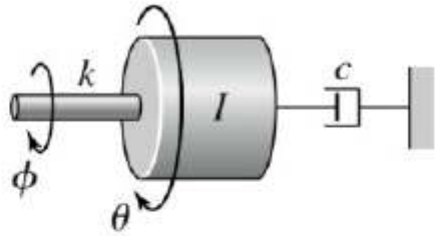


unit-step time response

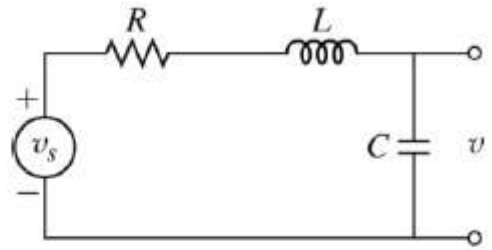
Examples of Second-Order Systems



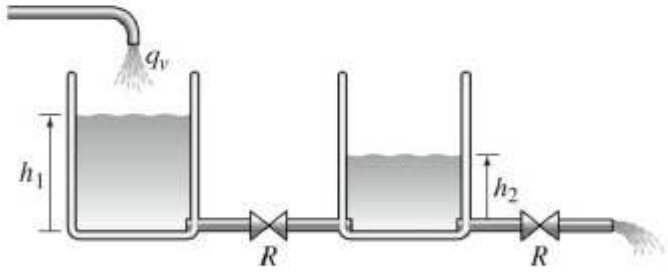
$$m\frac{d^2x}{dt^2}+c\frac{dx}{dt}+kx=f$$



$$I\frac{d^2\theta}{dt^2}+c\frac{d\theta}{dt}+k\theta=k\phi$$

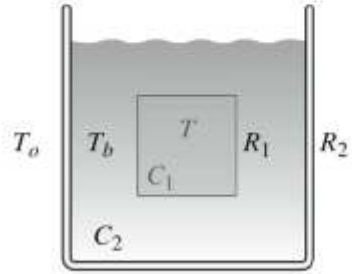


$$LC\frac{d^2v}{dt^2}+RC\frac{dv}{dt}+v=v_s$$



$$RA\frac{dh_1}{dt}+g(h_1-h_2)=Rq_v$$

$$RA\frac{dh_2}{dt}+g(h_2-h_1)+gh_2=0$$



$$R_1C_1\frac{dT}{dt}+T=T_b$$

$$R_1R_2C_2\frac{dT_b}{dt}+(R_1+R_2)T_b=R_2T+R_1T_o$$

Given the transfer function below, find natural frequency ω_n and damping ratio ζ.

$$G(s) = \frac{36}{s^2 + 4.2s + 36}$$

Solution

The general form of the second-order transfer function is

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Comparing the equations we will get :$\boldsymbol{\omega_n^2 = 36,}$ or natural frequency $\boldsymbol{\omega_n = 6}$

$\boldsymbol{2\zeta\omega_n = 4.2,}$ substituting the value of ω_n we will get, damping ratio $\boldsymbol{\zeta = 0.35.}$

Time domain specifications of control systems

1. DELAY TIME (t_d): The delay time is the time required for the response to reach 50% of the final value in first time.
2. RISE TIME (t_r): It is time required for the response to rise from 10% to 90% of its final value for over-damped systems and 0 to 100% for under-damped systems.
3. PEAK TIME (t_p): The peak time is the time required for the response to reach the first peak of the time response or first peak overshoot.

4. MAXIMUM OVERSHOOT (M_p):

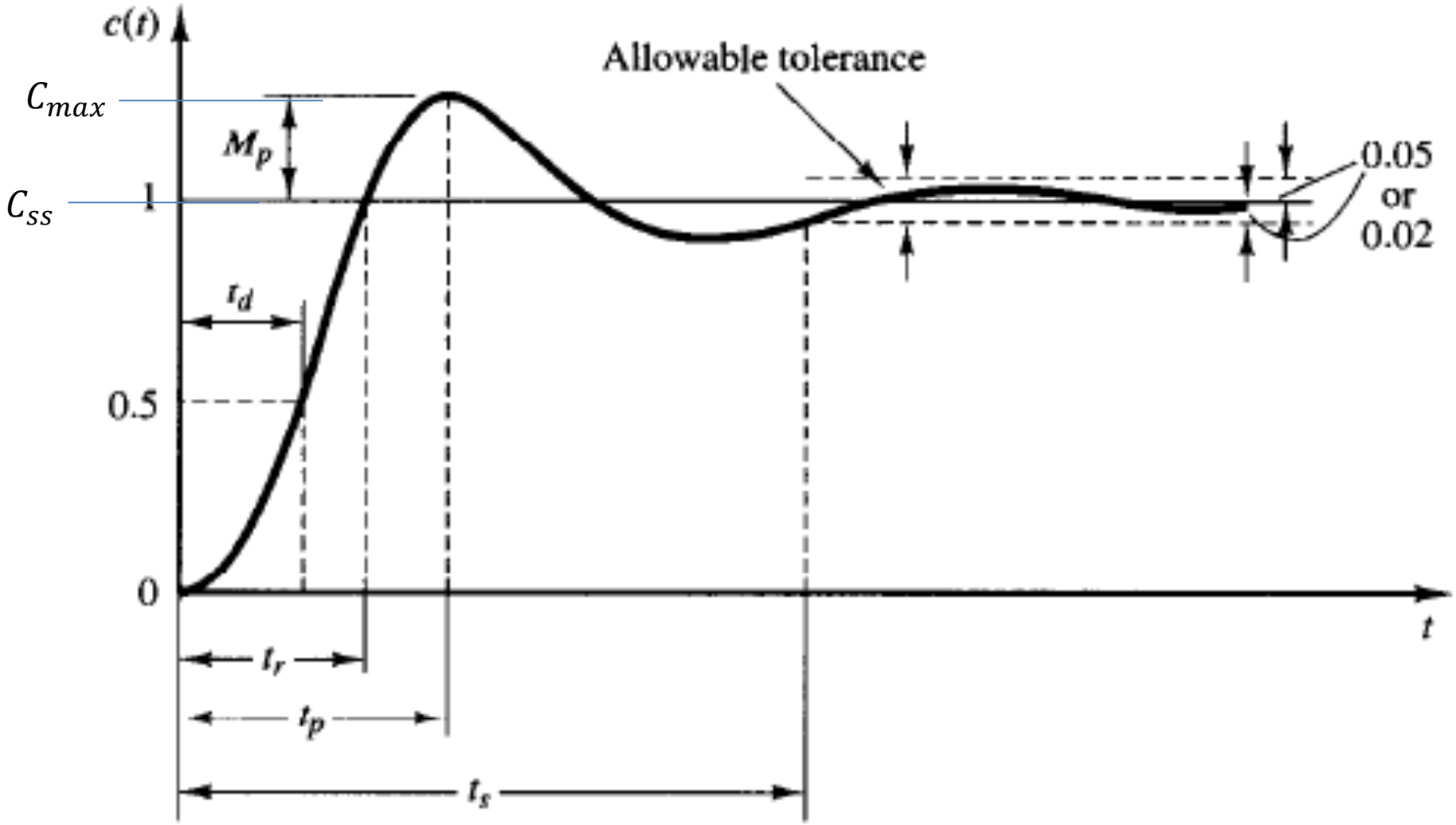
$$M_p = \frac{C_{max} - C_{ss}}{C_{ss}} \times 100\%$$

5. SETTLING TIME (t_s):

The settling time is defined as the time required for the transient response to reach and stay within the prescribed percentage error.

6. STEADY STATE ERROR (e_{ss}): It is difference between actual output and desired output as time 't' tends to infinity.

$$e_{ss} = \underset{t \to \infty}{Limit}\left[r(t) - c(t)\right]$$



Time domain specifications

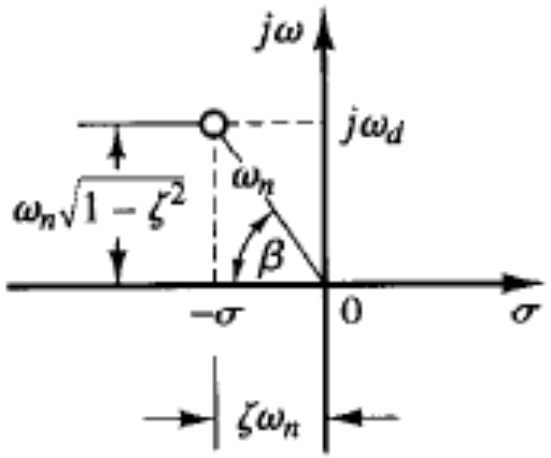
1) Rise time, T_r. Time the response takes to rise from 0 to 100%

$$c(t)\Big|_{t\ T_r} \quad 1 \quad \underbrace{\frac{1}{-}e^{\ {}_n t}}_{0} \ \underbrace{\sin\ {}_d t}_{0} \quad 1$$

$$\sin(\ {}_d T_r\ ) \quad 0$$

$${}_d T_r \qquad \sin^{\ 1}(0)$$

$$T_r \qquad \frac{}{{}_n\sqrt{1\qquad {}^2}}$$



- Peak time Referring to the same output, we may obtain the peak time by differentiating *c(t)* with respect to time and letting this derivative equal zero. Since

$$\frac{dc}{dt} = \zeta\omega_n e^{-\zeta\omega_n t}\left(\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right) + e^{-\zeta\omega_n t}\left(\omega_d \sin\omega_d t - \frac{\zeta\omega_d}{\sqrt{1-\zeta^2}}\cos\omega_d t\right)$$

and the cosine terms in this last equation cancel each other, *dc/dt,* evaluated at **t = tp,** can be simplified to

$$\left.\frac{dc}{dt}\right|_{t=t_p} = (\sin\omega_d t_p)\frac{\omega_n}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t_p} = 0$$

This last equation yields the following equation:

$$\sin \omega_d t_p = 0 \longrightarrow \omega_d t_p = 0, \pi, 2\pi, 3\pi, \ldots$$

Since the peak time corresponds to the first peak overshoot,

$$t_p = \frac{\pi}{\omega_d}$$

- **Maximum overshoot**

The maximum overshoot occurs at the peak time or at $t =$ tp . Thus, Mp is obtained as

$$M_p = c(t_p) - 1$$

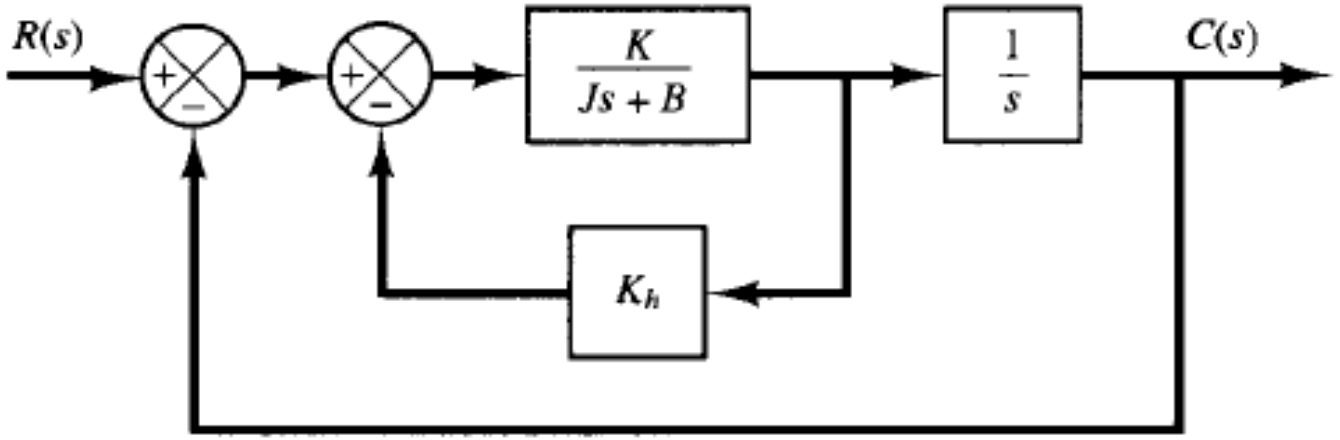
$$= -e^{-\zeta\omega_n(\pi/\omega_d)}\left(\cos\pi + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\pi\right)$$

$$= e^{-(\sigma/\omega_d)\pi} = e^{-(\zeta/\sqrt{1-\zeta^2})\pi}$$

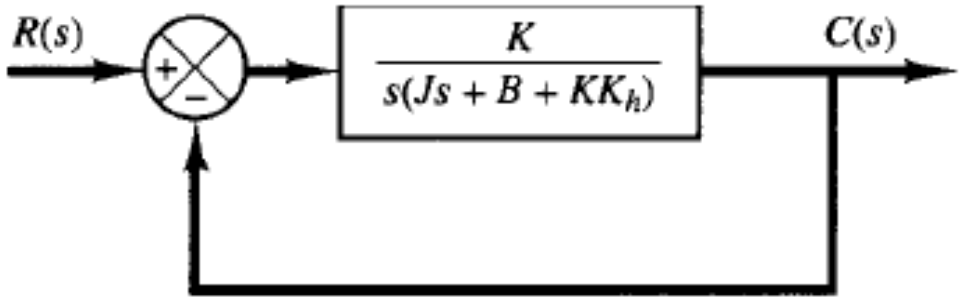
- **Settling time**
- The speed of decay of the transient response depends on the value of the time constant obtained from the following
- $e^{-t/T} = e^{-\zeta\omega_n t}$
- Equating the orders: T= $1/\varsigma\omega_n$
- And the setting time
- $$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n} \qquad \text{(2\% criterion)}$$

  $$\text{or} \qquad t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n} \qquad \text{(5\% criterion)}$$

Example: Servo system with velocity feedback



(a)



(b)

$$\frac{C(s)}{R(s)} = \frac{K}{Js^2 + (B + KK_h)s + K}$$

$$= \frac{K/J}{s^2 + \left[\frac{B+KK_h}{J}\right]s + K/J}$$

For the system, determine the values of gain *K* and velocity feedback constant *Kh* so that the maximum overshoot in the unit-step response is 0.2 and the peak time is 1 sec. With these values of *K* and *Kh*, obtain the rise time and settling time. Assume that *J* =1 kg-m2 and *B* = 1 N-m/rad/sec.

$$M_p = e^{-(\zeta/\sqrt{1-\zeta^2})\pi}$$

This value must be 0.2. Thus,

$$e^{-(\zeta/\sqrt{1-\zeta^2})\pi} = 0.2$$

or

$$\frac{\zeta\pi}{\sqrt{1-\zeta^2}} = 1.61$$

which yields

$$\zeta = 0.456$$

The peak time t_p is specified as 1 sec; therefore, from Equation (4–29),

$$t_p = \frac{\pi}{\omega_d} = 1$$

or

$$\omega_d = 3.14$$

Since ζ is 0.456, ω_n is

$$\omega_n = \frac{\omega_d}{\sqrt{1-\zeta^2}} = 3.53$$

Since the natural frequency ω_n is equal to $\sqrt{K/J}$,

$$K = J\omega_n^2 = \omega_n^2 = 12.5 \text{ N-m}$$

Then, K_h is, from Equation (4–34),

$$K_h = \frac{2\sqrt{KJ}\zeta - B}{K} = \frac{2\sqrt{K}\zeta - 1}{K} = 0.178 \text{ sec}$$

Rise time t_r: From Equation (4–28), the rise time t_r is

$$t_r = \frac{\pi - \beta}{\omega_d}$$

where

$$\beta = \tan^{-1} \frac{\omega_d}{\sigma} = \tan^{-1} 1.95 = 1.10$$

Thus, t_r is

$$t_r = 0.65 \text{ sec}$$

Settling time t_s: For the 2% criterion,

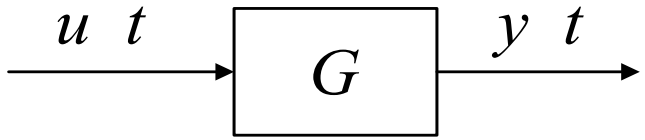
$$t_s = \frac{4}{\sigma} = 2.48 \text{ sec}$$

For the 5% criterion,

$$t_s = \frac{3}{\sigma} = 1.86 \text{ sec}$$

Higher order systems & Stability

- Higher order systems



$$Y(s) = G(s)U(s)$$

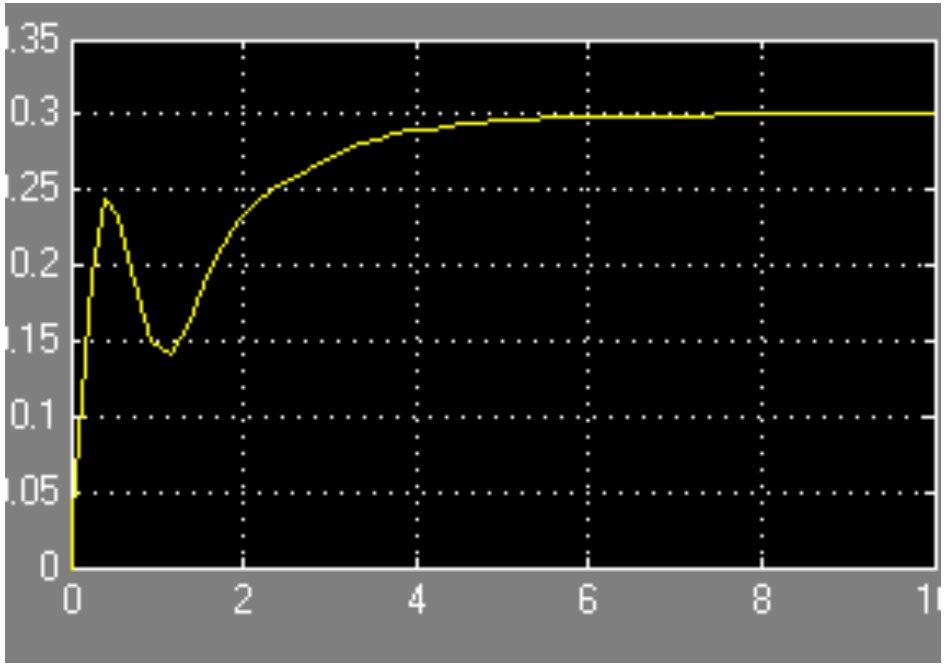
$$G(s) = \frac{K\prod_{i=1}^{m}(s - z_i)}{\prod_{j=1}^{q}(s - p_i)\prod_{k=1}^{r}(s^2 + 2\zeta_{nk}\omega_{nk}s + \omega_{nk}^2)}$$

$$Y(s) = \frac{a}{s}\prod_{k=1}^{q}\frac{a_k}{(s - p_k)}\prod_{k=1}^{r}\frac{b_k(s + 2\zeta_{nk}\omega_{nk}) + c_k\omega_{dk}^2}{s^2 + 2\zeta_{nk}\omega_{nk}s + \omega_{nk}^2}$$

Where $d_k = \omega_{nk}\sqrt{1-\zeta_k^2}$

$$y(t) = a_0 + \sum_{k=1}^{q} a_k e^{p_k t} + \sum_{k=1}^{r} e^{\zeta_k \omega_{nk} t}\left(b_k \cos \omega_{dk} t + c_k \sin \omega_{dk} t\right)$$

Unit step response of higher order system →



Reduction The Order Of Higher Order System

The order of higher system can be reduced to a second order system if it has dominant closed-loop poles

$$G(s)=\frac{1}{(s+a)(s^2+2\sigma s+\omega_n^2)}$$

If

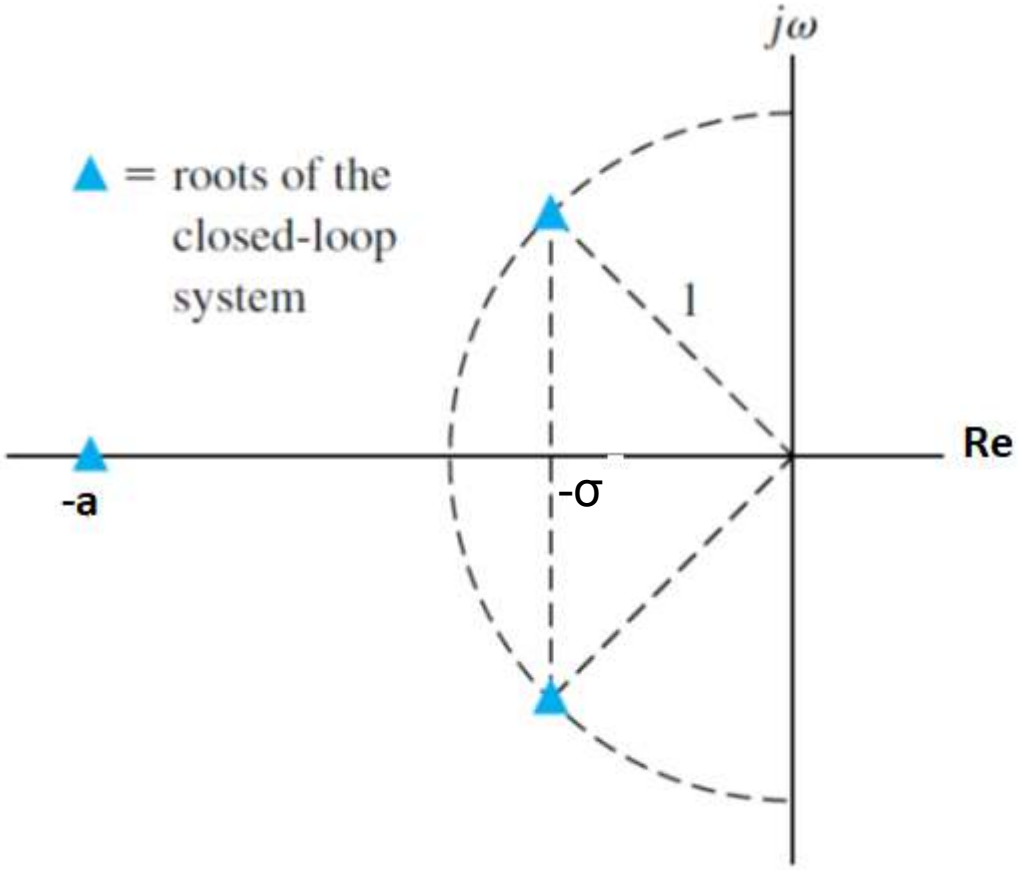
1, There are two complex conjugate poles

2. $\frac{a}{\sigma} \geq 5$

3. There are no zeros nearby the complex conjugate poles

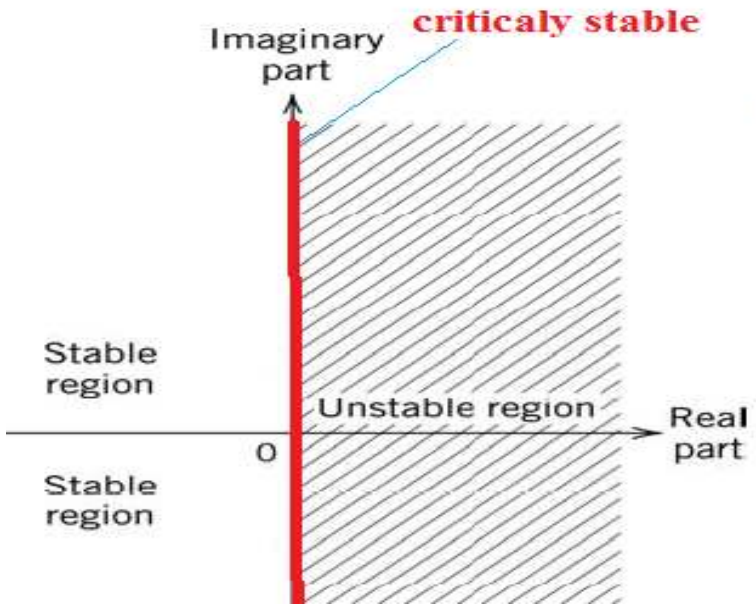
The transfer function is

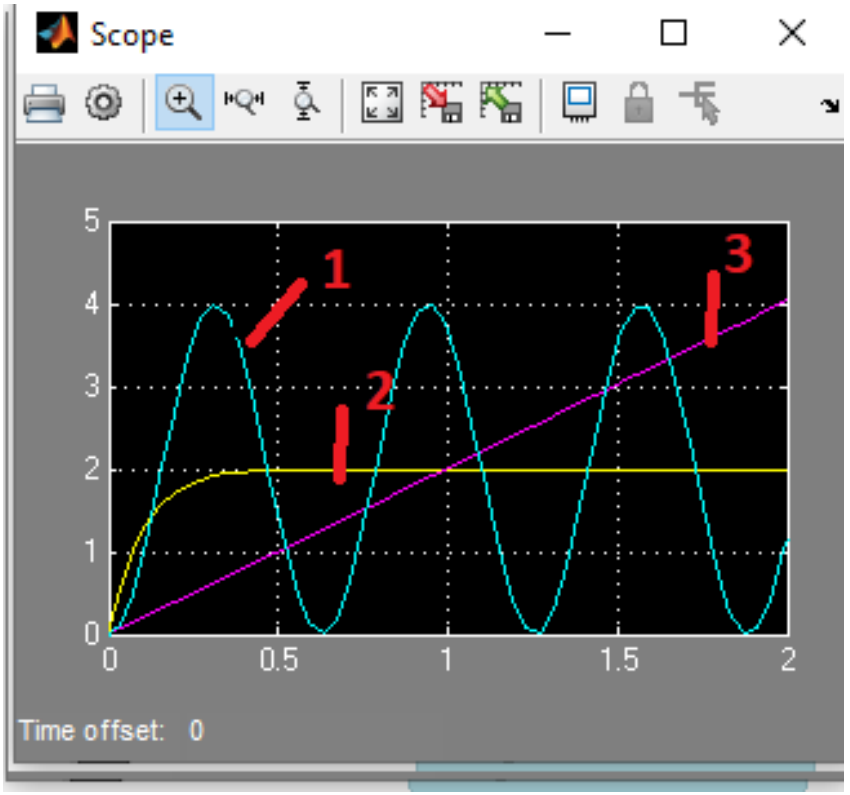
$$G(s)\approx \frac{1}{(a)(s^2+2\sigma s+\omega_n^2)}$$



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- **Definition of stability. A** linear system is said to be stable if the output response is bounded for all bounded inputs. Otherwise, it is said to be unstable.
- For negative pole F(s)= 1/(s+a) f(t)= e^-at stable term
- For positive pole F(s)= 1/(s-a) f(t)= e^+at unstable term
- Instead of poles we use roots of characteristic equation





1 critically stable
2.Stable
3.unstable

Routh's Stability Criterion

- A *necessary condition for stability* requires that ***all the coefficients be positive.***

A necessary (but not sufficient) condition for stability is that ***all*** **the coefficients of the polynomial characteristic equation be positive.**

A ***necessary and sufficient condition*** for Stability

- Routh's formulation requires the computation of a triangular array that is a function of the coefficients of the polynomial characteristic equation.

A **system is stable** if and only if ***all*** **the elements of the first column** of the Routh array are **positive**

Characteristic Equation

- Consider an nth-order system whose the characteristic equation (*which is also the denominator of the transfer function*) is:

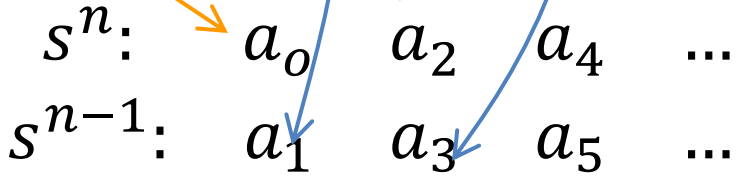
$$a(s) = a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + \cdots + a_{n-1} s^1 + a_n s^0 = 0$$

Method for determining the Routh array(table)

- Consider the characteristic equation:

$$a(s) = a_0 \times s^n + a_1 s^{n-1} + a_2 s^{n-2} + a_3 s^{n-3} + \cdots + a_{n-1} s^1 + a_n \quad s^0$$

- First arrange the coefficients of the characteristic equation in two rows, beginning with the first and second coefficients and followed by the even-numbered and odd-numbered coefficients:



Routh array: method (cont'd)

- Then add subsequent rows to complete the **Routh array:**

s^n:	a_o	a_2	a_4	…
s^{n-1}:	a_1	a_3	a_5	…
s^{n-2}:	b_1	b_2	b_3	…
s^{n-3}:	c_1	c_2	c_3	…
⋮	⋮	⋮	⋮	
s^2:	$*$	$*$		
s^1:	$*$			
s^0:	$*$			

?!

Routh array: method (cont'd)

- Compute elements for the 3rd row:

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1},$$

$$b_2 = \frac{a_1 a_4 - a_0 \times a_5}{a_1},$$

$$b_3 = \frac{a_1 a_6 - a_0 a_7}{a_1}$$

s^n:	a_o	a_2	a_4	…
s^{n-1}:	a_1	a_3	a_5	…
s^{n-2}:	b_1	b_2	b_3	…
s^{n-3}:	c_1	c_2	c_3	…
⋮	⋮	⋮	⋮	
s^2:	*	*		
s^1:	*			
s^0:	*			

Routh array: method (cont'd)

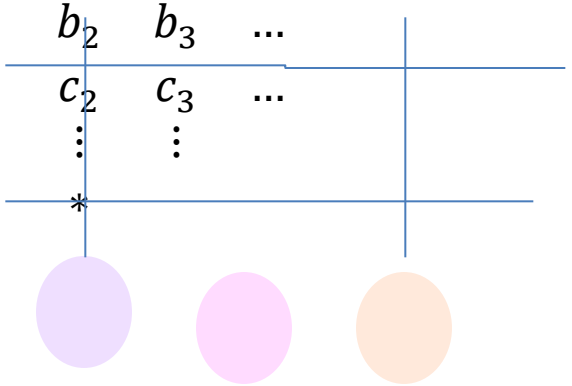
- Compute elements for the 4th row:

$$c_1 = \frac{b_1 a_3 - a_1 b_2}{b_1}$$

$$c_2 = \frac{b_1 a_5 - a_1 b_3}{b_1}$$

$$c_3 = \frac{b_1 a_7 - a_1 b_4}{b_1}$$

s^n:	$a0$	a_2	a_4	…
s^{n-1}:	a_1	a_3	a_5	…
s^{n-2}:	b_1	b_2	b_3	…
s^{n-3}:	c_1	c_2	c_3	…
⋮	⋮	⋮	⋮	
s^2:	$*$	$*$		
s^1:	$*$			
s^0:	$*$			



Example 1:

Given the characteristic equation,

$$a(s) = s^6 + 4s^5 + 3s^4 - 2s^3 + s^2 + 4s + 4$$

is the system described by this characteristic equation stable?

Answer:

- One coefficient (-2) is negative.

- Therefore, the system **does not** satisfy the necessary condition for stability.

Example 2:

Given the characteristic equation,

$$a(s) = s^6 + 4s^5 + 3s^4 + 2s^3 + s^2 + 4s + 4$$

is the system described by this characteristic equation stable?

Answer:

- All the coefficients are positive and nonzero.
- Therefore, the system **satisfies** the necessary condition for stability.
- We should determine whether any of the coefficients of the first column of the Routh array are negative.

Example 2 (cont'd): Routh array

$$a(s) = s^6 + 4s^5 + 3s^4 + 2s^3 + s^2 + 4s + 4$$

s^6:	1	3	1	4
s^5:	4	2	4	0
s^4:	?	?	?	
s^3:	?	?	?	
s^2:	?	?		
s^1:	?	?		
s^0:	?			

Example 2 (cont'd): Routh array

$$a(s) = s^6 + 4s^5 + 3s^4 + 2s^3 + s^2 + 4s + 4$$

s^6:	1	3	1	4
s^5:	4	2	4	0
s^4:	5/2	0	4	
s^3:	?	?	?	
s^2:	?	?		
s^1:	?	?		
s^0:	?			

$$b_1 = \frac{4 * 3 - 1 * 2}{4} = \frac{5}{2}$$

$$b_2 = \frac{4 * 1 - 1 * 4}{4} = 0$$

$$b_3 = \frac{4 * 4 - 1 * 0}{4} = 4$$

Example 2 (cont'd): Routh array

$$a(s) = s^6 + 4s^5 + 3s^4 + 2s^3 + s^2 + 4s + 4$$

s^6:	1	3	1	4
s^5:	4	2	4	0
s^4:	5/2	0	4	
s^3:	2	$-12/5$	0	
s^2:	?	?		
s^1:	?	?		
s^0:	?			

$$c_1 = \frac{2.5 * 2 - 4 * 0}{2.5} = 2$$

$$c_2 = \frac{2.5 * 4 - 4 * 4}{2.5} = -12/5$$

$$c_3 = \frac{2.5 * 0 - 4 * 0}{2.5} = 0$$

Example 2 (cont'd): Routh array

$$a(s) = s^6 + 4s^5 + 3s^4 + 2s^3 + s^2 + 4s + 4$$

s^6:	1	3	1	4
s^5:	4	2	4	0
s^4:	5/2	0	4	
s^3:	2	−12/5	0	
s^2:	3	4		
s^1:	−76/15	0		
s^0:	4			

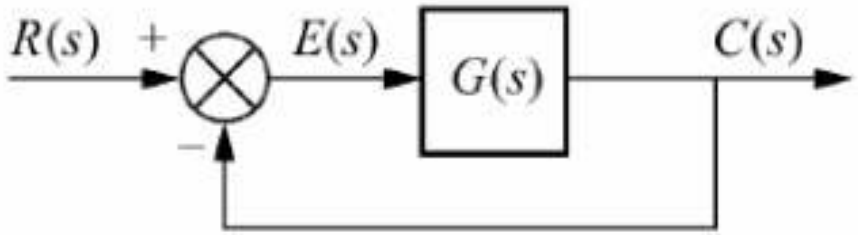
The elements of the 1st column **are not all** positive:

the system is unstable

Stability versus Parameter Range

- Given that G(s) is the open loop transfer function for a unity feedback system, find the range of K to yield a stable system

$$G(s) = \frac{K(s+20)}{s(s+2)(s+3)}$$



- Solution

$$T(s)=\frac{G(s)}{1+G(s)}=\frac{K(s+20)}{s^3+5s^2+(6+K)s+20K}$$

s^3	1	$(6+K)$
s^2	5	$20K$
s^1	$\frac{30-15K}{5}$	
s^0	$20K$	

From the s^1 row, $K<2$. From the s^0 row, $K>0$. Thus, for stability, $0<K<2$.

Example 3 (cont'd)

$$1 + K\frac{s+1}{s(s-1)(s+6)} = 0$$

- Expressing the characteristic equation in polynomial form, we obtain:

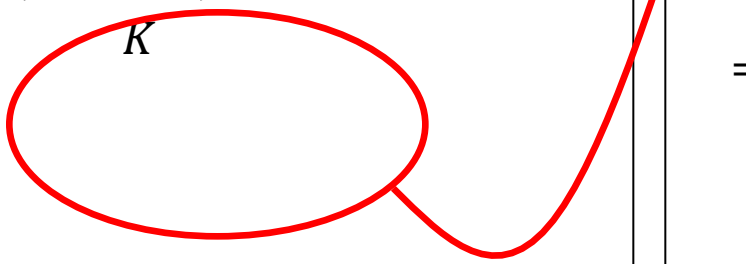
$$s^3 + 5s^2 + (K-6)s + K = 0$$

Example 3 (cont'd)

$$a(s) = s^3 + 5s^2 + (K-6)s + K$$

- The corresponding Routh array is:

s^3:	1	$K-6$
s^2:	5	K
s^1:	$(4K-30)/5$	
s^0:	K	



- Therefore, the **system is stable if and only if**

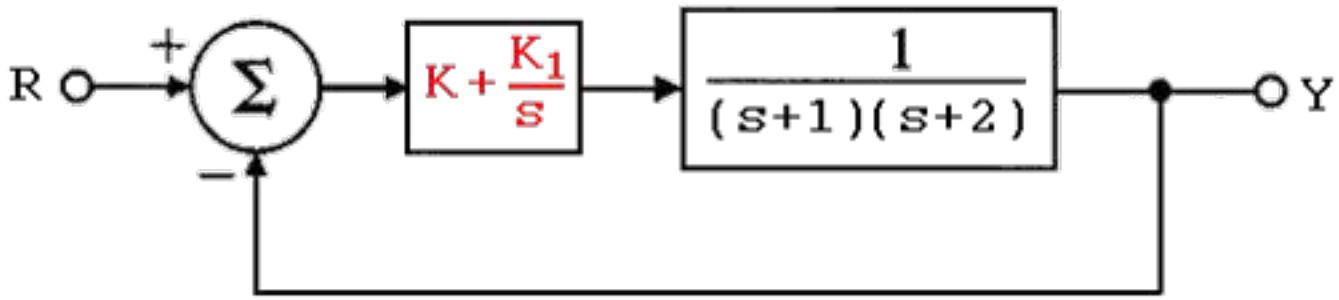
$$\frac{4K-30}{5} > 0 \ and \ K > 0$$

$$\Rightarrow \quad K > 7.5 \ and \ K > 0$$

$$\Rightarrow \quad K > 7.5$$

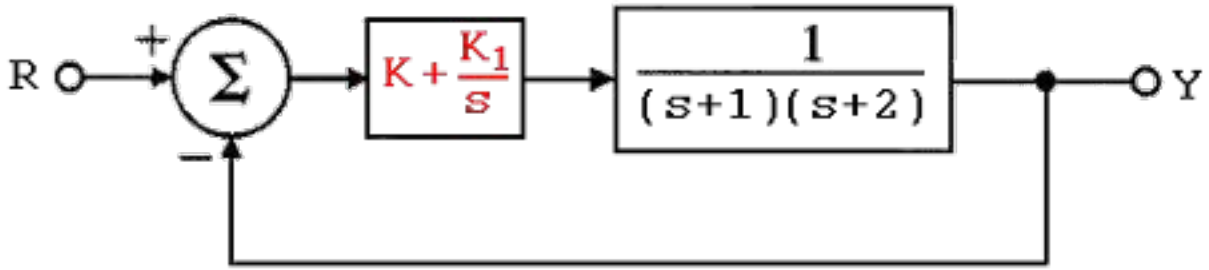
Example 4: Stability versus Two Parameter Range

Consider a Proportional-Integral (PI) control such as:



Find the range of the controller gains (K, K_1) so that the PI feedback system is stable.

Example 4 (cont'd)



- The characteristic equation for the system is given by:

$$1 + \left(K + \frac{K_1}{s}\right)\frac{1}{(s+1)(s+2)} = 0$$

Example 4 (cont'd)

$$1 + \left(K + \frac{K_1}{s}\right)\frac{1}{(s+1)(s+2)} = 0$$

- Expressing the characteristic equation in polynomial form, we obtain:

$$s^3 + 3s^2 + (2+K)s + K_1 = 0$$

Example 4 (cont'd)

$$s^3 + 3s^2 + (2 + K)s + K_1 = 0$$

- The corresponding Routh array is:

s^3:	1	$2 + K$
s^2:	3	K_1
s^1:	$(6 + 3K - K_1)/3$	
s^0:	K_1	

- For **stability**, we must have:

$$K_1 > 0 \; and \; K > \frac{K_1}{3} - 2$$

Special case 1: If only the first element if one of the rows is zero

- Replace zero with small positive constant ε
- Apply stability criterion by taking the limit as ε → 0

$$a(s) = s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$$

s^5:	1	2	6
s^4:	~~3~~ 1	~~6~~ 2	~~9~~ 3
s^3:	0	3	0

New s^3

s^3:	ε	3	0	← Replace 0 by ε
s^2:	$\frac{2\varepsilon - 3}{\varepsilon}$	3	0	
s :	$3 - \frac{3\varepsilon^2}{2\varepsilon - 3}$	0	0	
s^0:	3	0		

2 sign changes: +++-++

2 poles **not** in the LHP

- Zero 1st column element
  - Replace element by $\varepsilon > 0$
  - Proceed as usual
  - Example

$$d \quad s \quad s^3 \quad 3s \quad 2$$

s^3	1	3
s^2	0	2
s^1	$\frac{3 \quad 2}{}$	0
s^0	2	

2 sign changes $\Rightarrow$ 2 roots in RHP

Roots at 1,1, 2

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Special case 2: When an entire row of the Routh array is zero

- Form auxiliary equation from previous non-zero row

$$a_1(s) = \beta_1 s^{i+1} + \beta_2 s^{i-1} + \beta_3 s^{i-3} + \ldots$$

where $\{\beta_i\}$ of the $(i+1)$th row in the array

- Replace ith row by the coefficients of the derivative of the auxiliary polynomial

$$a(s) = s^5 + 5s^4 + 11s^3 + 23s^2 + 28s + 12$$

s^5 :	1	11	28	
s^4 :	5	23	12	
s^3 :	6.4	25.6	0	
s^2 :	3	12		
s :	0	0		$\leftarrow a_1(s) = 3s^2 + 12$

Calculate new s

No changes in 1st column

s :	6	0	$\leftarrow \frac{da_1(s)}{ds} = 6s$
s^0 :	12		

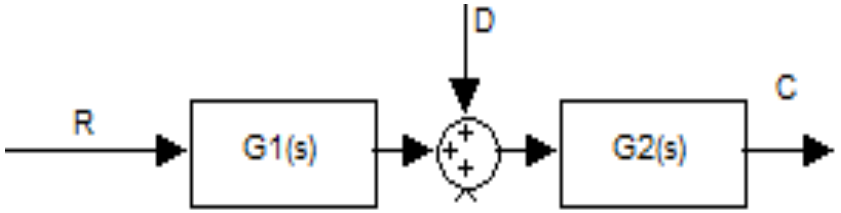
Basic control actions

- A controller compares the actual value of

output with the reference input, determines the deviation, and produces a control signal that will reduce the deviation to zero or to a small value.

- The manner in which the controller produces the control signal is called the *control action.*

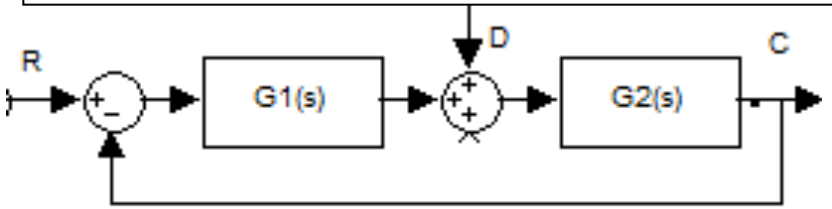
Open loop system



$$C_R = G_1(s)G_2(s)R$$

$$C_D = G_2(s)D$$

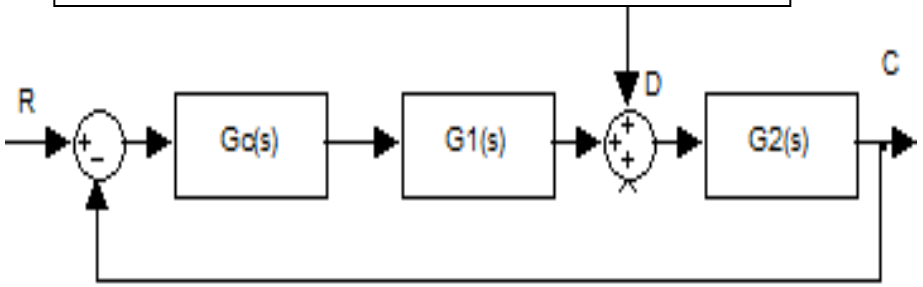
Closed loop uncontrolled system



$$C_R = \frac{G_1(s)G_2(s)}{1 + G_1(s)G_2(s)}R$$

$$C_D = \frac{G_2(s)}{1 + G_1(s)G_2(s)}D$$

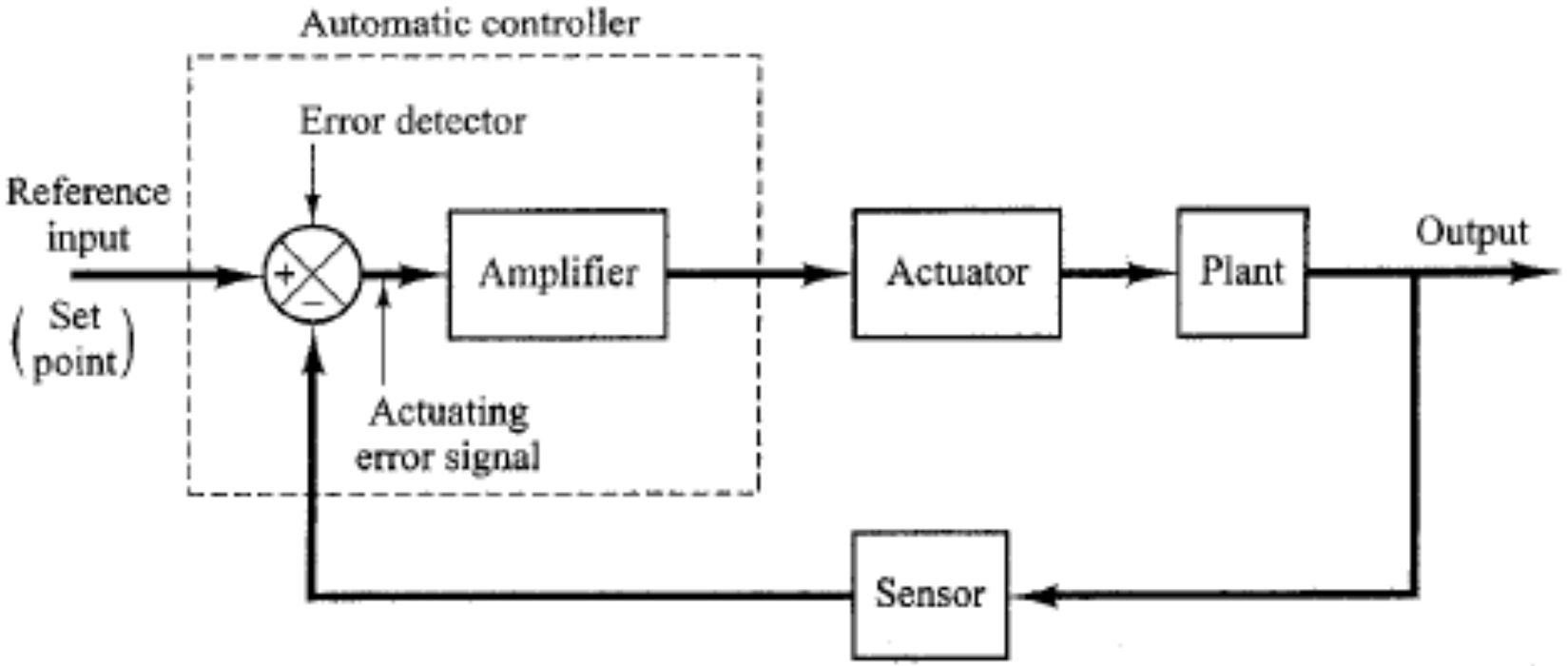
Closed loop controlled system



$$C_R = \frac{G_C(s)G_1(s)G_2(s)}{1 + G_C(s)G_1(s)G_2(s)}R$$

$$C_D = \frac{G_2(s)}{1 + G_C(s)G_1(s)G_2(s)}D$$

block diagram of an industrial control system

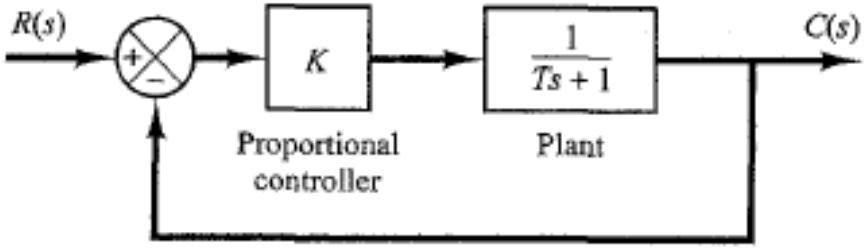


Classifications of Industrial Controllers.

- **1. Two-position or on-off controllers**
- **2. Proportional controllers**
- **3. Integral controllers**
- 4. **Proportional-plus-integral controllers**
- **5. Proportional-plus-derivative controllers**
- ***6. Proportional-plus-integral-plus-derivative controllers***

First order systems: Proportional control

The proportional controller is defined as : m(t)=K e(t)
Or M(s)/E(s)=K



For open loop system $\frac{C(s)}{R(s)} = \frac{1}{Ts+1}, \ldots . C_{ss} = 1, \ldots . E_{ss} = 0$

For uncontrolled closed loop system $\frac{C(s)}{R(s)} = \frac{1}{Ts+2} = \frac{0.5}{0.5Ts+1}, \ldots . C_{ss} = 0.5, \ldots . E_{ss} = 0.5$

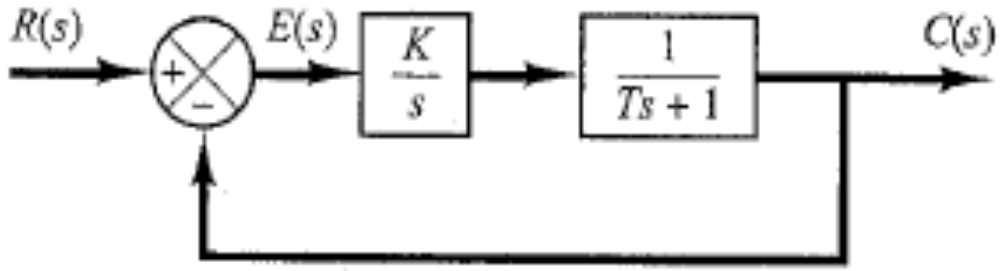
For closed loop system

$$\frac{C(s)}{R(s)} = \frac{K_{eq}}{T_{eq}s+1}, \ldots . C_{ss} = \frac{K}{1+K}, \ldots . E_{ss} = \frac{1}{1+K}$$

$$K_{eq} = \frac{K}{1+K}, \ldots . T_{eq} = \frac{T}{1+K}$$

Proportional control of first order system reduces the steady state error and the time constant both advantages

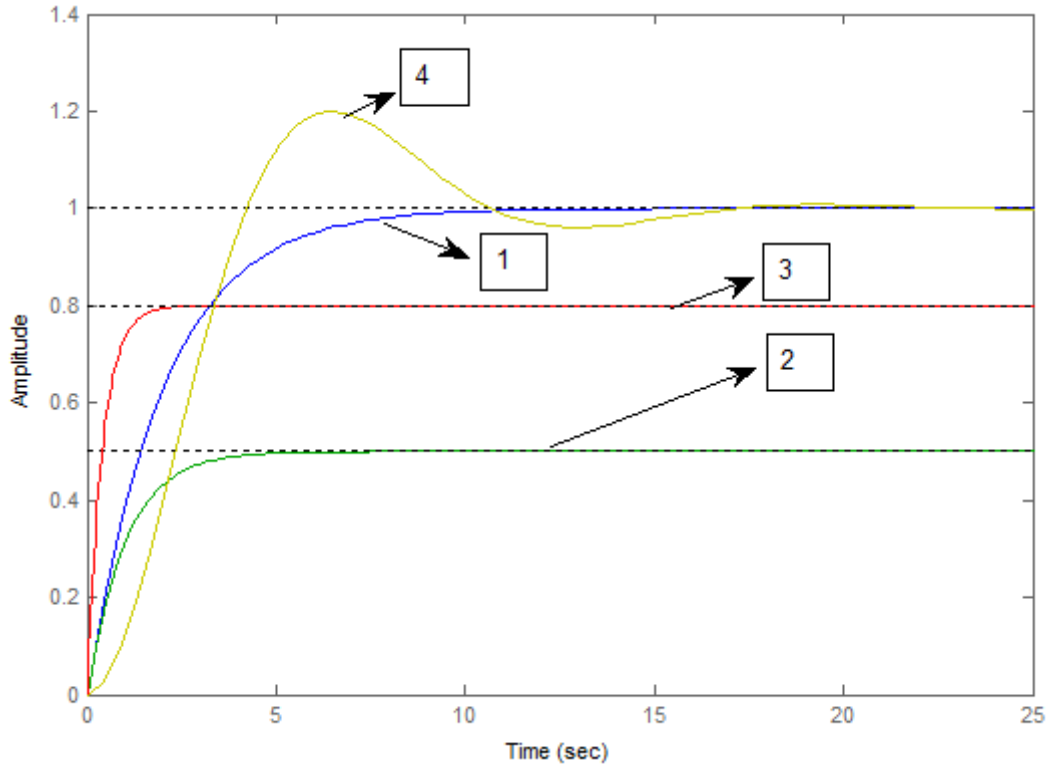
first Order System Integral Control



$$\frac{C(s)}{R(s)} = \frac{K}{s(Ts+1)+K}$$

$C_{ss} = 1, \ldots E_{ss} = 0$

Integral control of the system eliminates the steady-state error in the response to the step input and rise the order of the system by one . This Oscillates the output

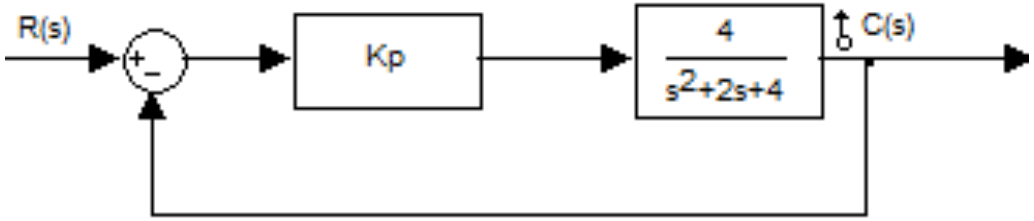


Time response of first order system:1. open loop system, 2.uncontrolled Closed loop system, 3. proportional control, 4.integral control

*Proportional controller reduce the steady state error and the time constant

*Integral controller eliminates the steady state error , rise the order of The system by one and increase the oscillations

Second Order System Proportional Control



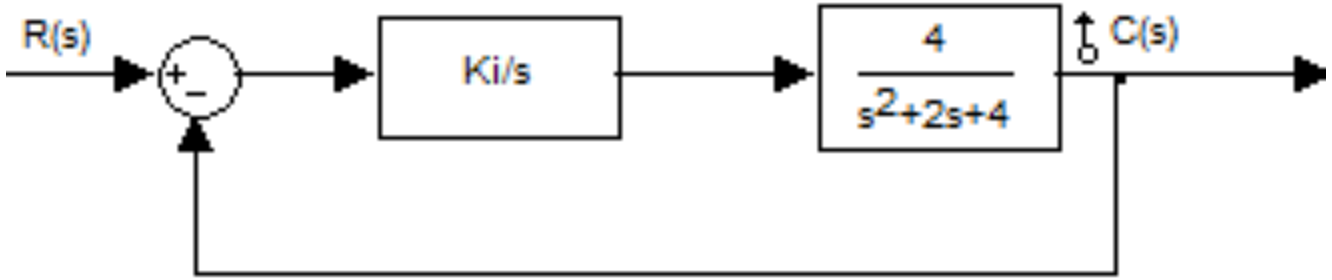
For open loop $\frac{C(s)}{R(s)} = \frac{4}{s^2+2s+4}, \ldots C_{ss} = 1, \ldots E_{ss} = 0, \ldots \varsigma = 0.5$

For uncontrolled $\frac{C(s)}{R(s)} = \frac{4}{s^2+2s+8}, \ldots C_{ss} = 0.5, \ldots E_{ss} = 0.5, \ldots \varsigma = \frac{1}{2\sqrt{2}}$

For controlled $\frac{C(s)}{R(s)} = \frac{4K_p}{s^2+2s+4+4K_p}, \ldots C_{ss} = \frac{K_p}{1+K_p}$

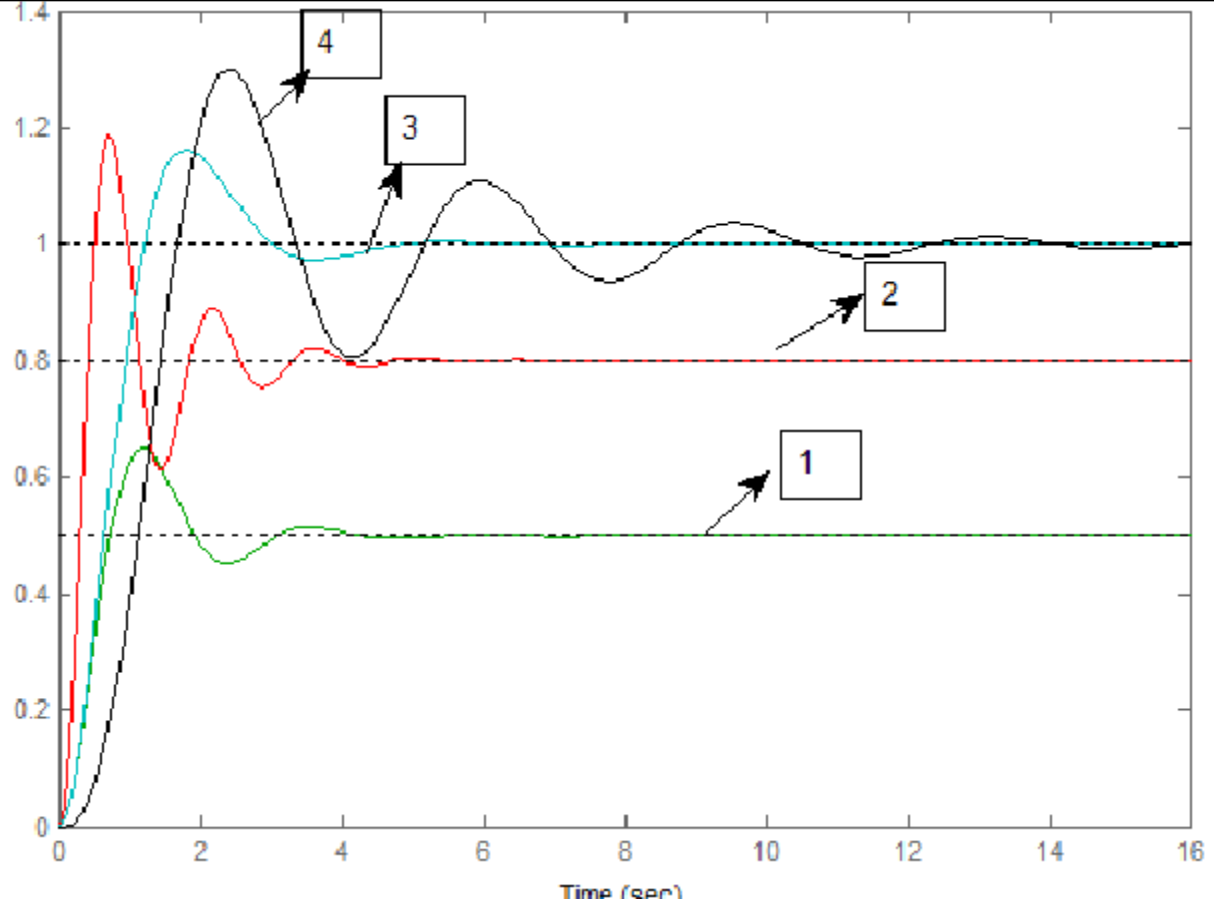
$E_{ss} = \frac{1}{1+K_p}, \ldots \varsigma = \frac{1}{\sqrt{1+k_p}}$

Second Order System
Integral Control



For controlled system $\frac{C(s)}{R(s)} = \frac{4K_i}{s^3 + 2s^2 + 4s + 4K_i}, \ldots C_{ss} = 1, \ldots E_{ss} = 0$

The controlled system is stable if $0 \prec K_i \prec 2$



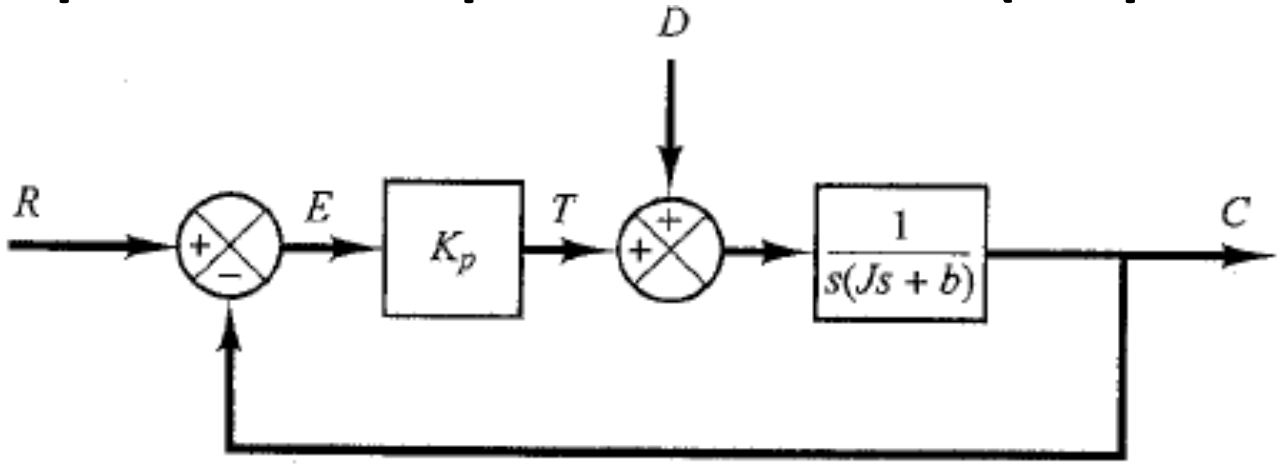
Time response of second order system

1. Uncontrolled system 2. proportional control 3. open loop system
2. 4. Integral control

*Proportional controller reduces the steady state error and the damping ratio

*Integral controller eliminates the steady state error , rise the order of The system by one and the system may be unstable

Response to Torque Disturbances (Proportional Control)



Assume that R=0, then $\frac{C(s)}{D(s)} = \frac{1}{Js^2 + bs + K_p}$ And $C_{ss} = -E_{ss} = 1/K_p$

The proportional controller reduces the steady state error caused By disturbance

If the P controller is replaced by I controller the characteristic equation of the controlled system is $Js^3 + bs^2 + K_i = 0$ and the system is unstable for all K_i , *To stabilise the system P term is added*

$$T(t) = K_p e(t) + K_i \int e(t)dt$$

$$\begin{matrix} J & 0 \\ b & K_i \\ -JK_i/b & \end{matrix}$$

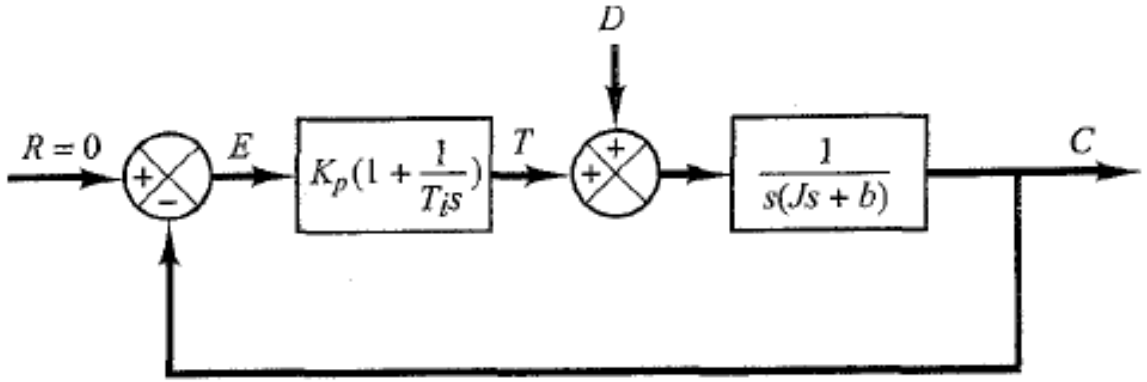
T(s)/E(s)= Kp+Ki/s=Kp(1+1/Tis), Ti =Ki/Kp

Proportional plus Integral control

$$\frac{C(s)}{D(s)} = \frac{s}{Js^3 + bs^2 + K_p s + \dfrac{K_p}{T_i}}$$

In the absence of the reference input, or $r(t) = 0$, the error signal is obtained from

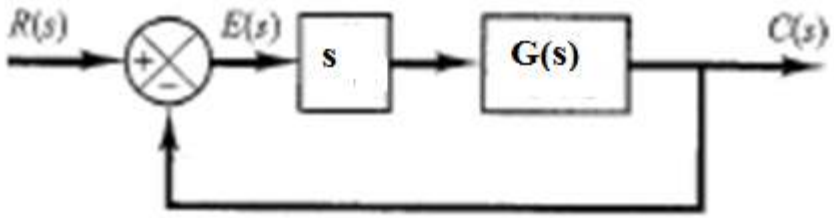
$$E(s) = -\frac{s}{Js^3 + bs^2 + K_p s + \dfrac{K_p}{T_i}} D(s)$$



Differential Control Action

$$m(t) = K_d \frac{de(t)}{d(t)}$$

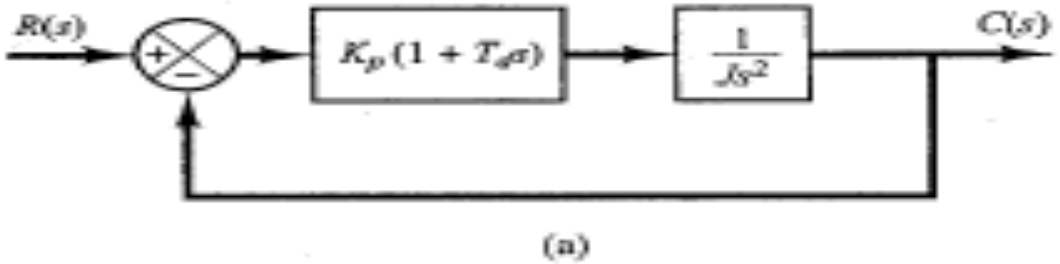
$$\frac{M(s)}{E(s)} = K_d s$$



$PD.controller$

$$m(t) = K_p e(t) + k_d \frac{de(t)}{dt} = K_p (e(t) + T_d \frac{de(t)}{dt})$$

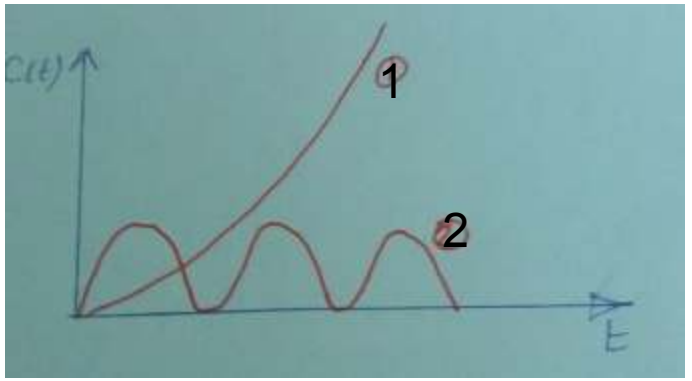
Proportional-Plus-Derivative Control



(a)

For open loop system $\frac{C(s)}{R(s)} = \frac{1}{Js^2}$ The unit step response is shown in plot (1)

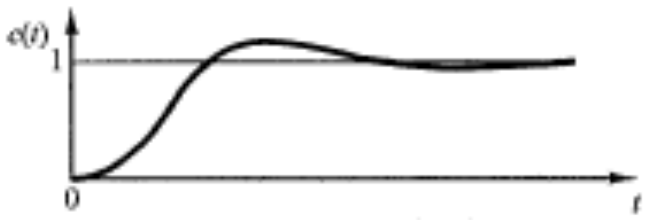
For closed loop uncontrolled system $\frac{C(s)}{R(s)} = \frac{1}{Js^2 + 1}$



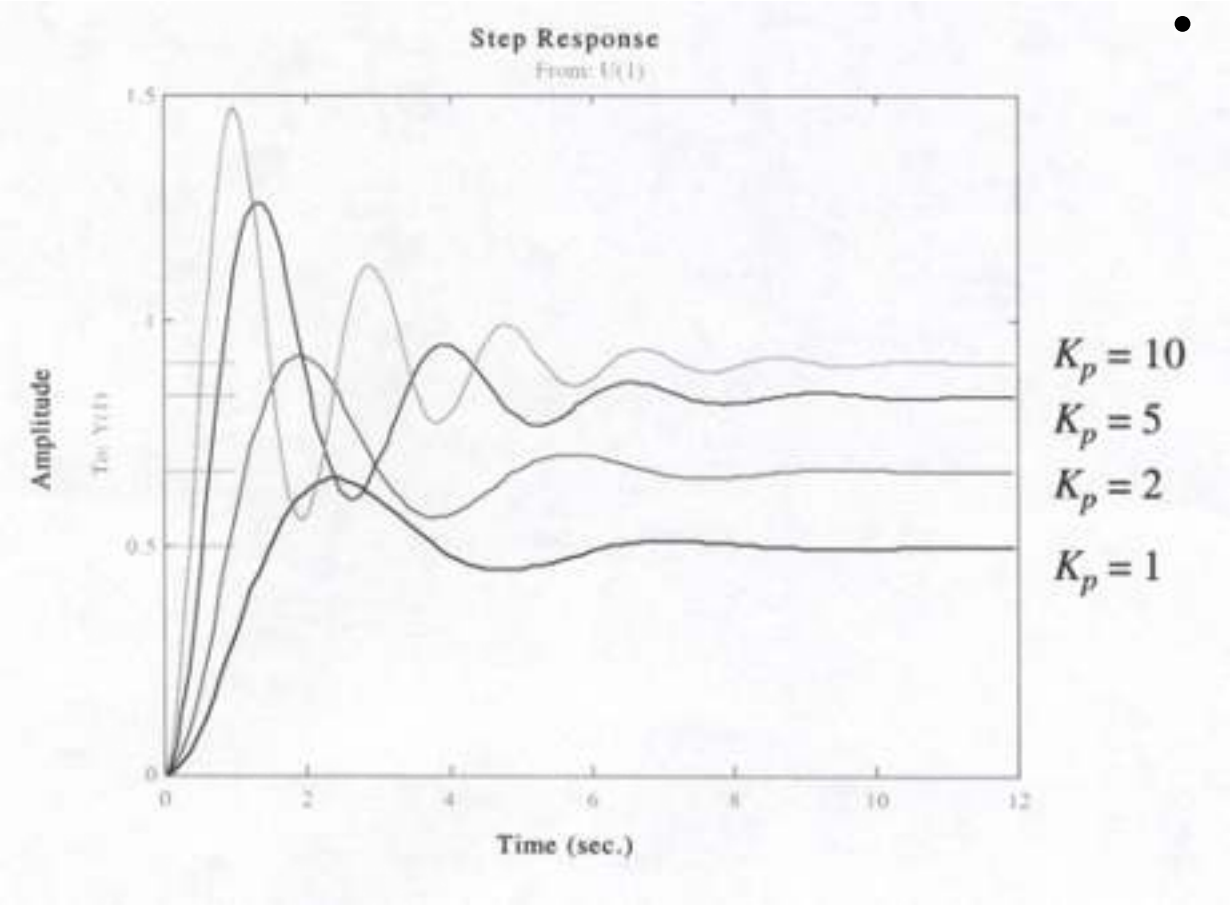
The unit step response is shown in plot (2)

$$\frac{C(s)}{R(s)} = \frac{K_p(1 + T_d s)}{Js^2 + K_p T_d s + K_p}$$

. Thus derivative control introduces a damping effect. A typical response curve ***c (t) to a unit step***



Change in gain in P controller



- **Increase in gain**:

→ Upgrade both steady-state and transient responses

→ Reduce steady-state error

→ ***Reduce stability! Of second orser system and the higer order system may be unstable***

Integral Controller

- Integral of error with a constant gain

$\rightarrow$ increase the system type by 1

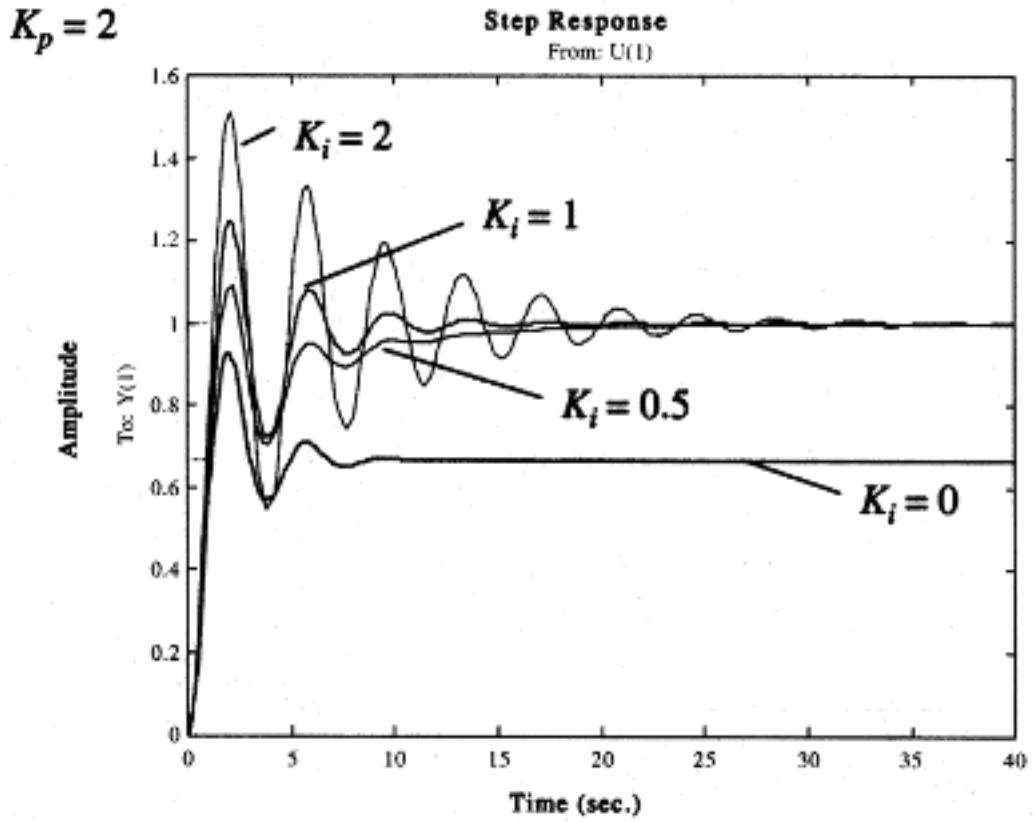
$\rightarrow$*eliminate steady-state error for*

a unit step input

$\rightarrow$ amplify overshoot and oscillations

$\rightarrow$Second and higher order systems may be unstable

Change in gain for PI controller

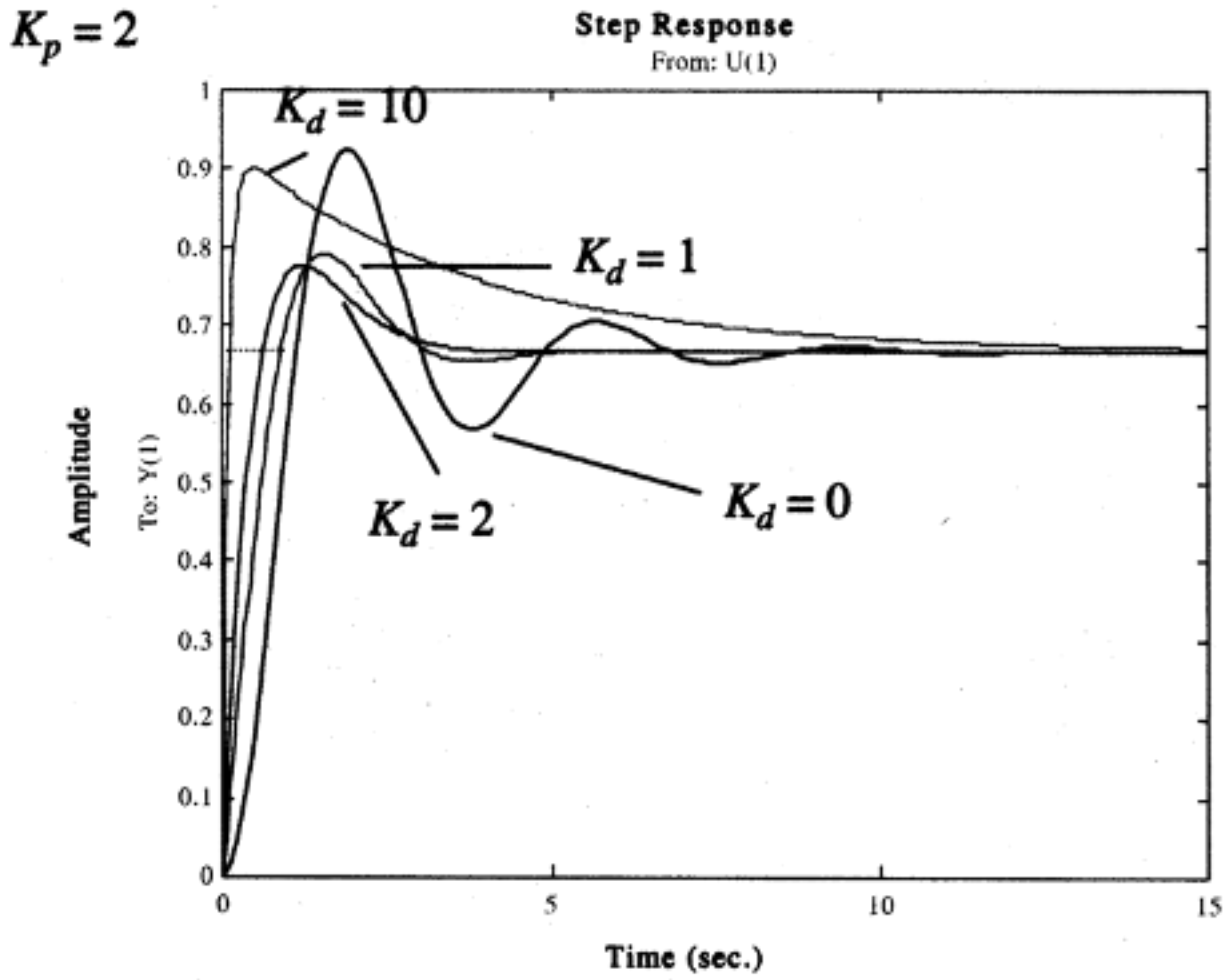


- **Increase in gain**:

  → Do not upgrade steady-state responses

  → ***Increase oscillations and overshoot! But less than I controller***

Effect of change for gain PD controller



The PD controller is mainly used to **improve the transient response of a control system**. Reduce the max. peak and settling time

Proportional plus Integral plus Derivative(PID) controller

Proportional plus Integral plus Derivative(PID) controller is given by the following expression

$$m(t) = K_p\, e(t) + K_i \int e(t)dt + K_d \frac{de(t)}{dt}$$

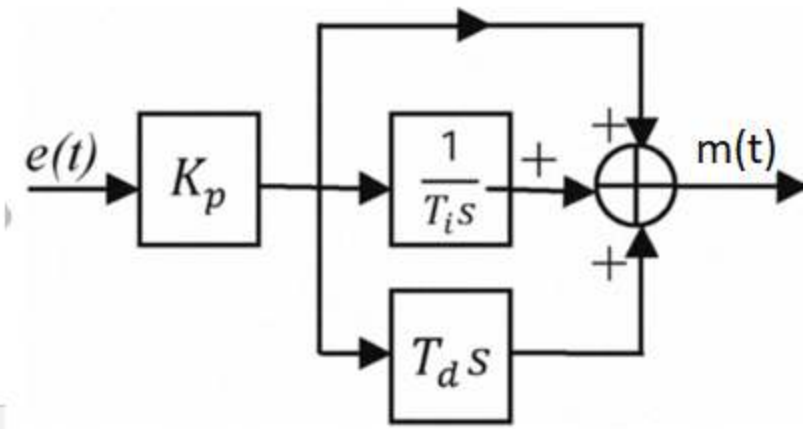
The transfer function of PID controller is given as

$$\frac{M(s)}{E(s)} = K_p + \frac{K_i}{s} + K_d s = \frac{K_d s^2 + K_p s + K_i}{s}$$

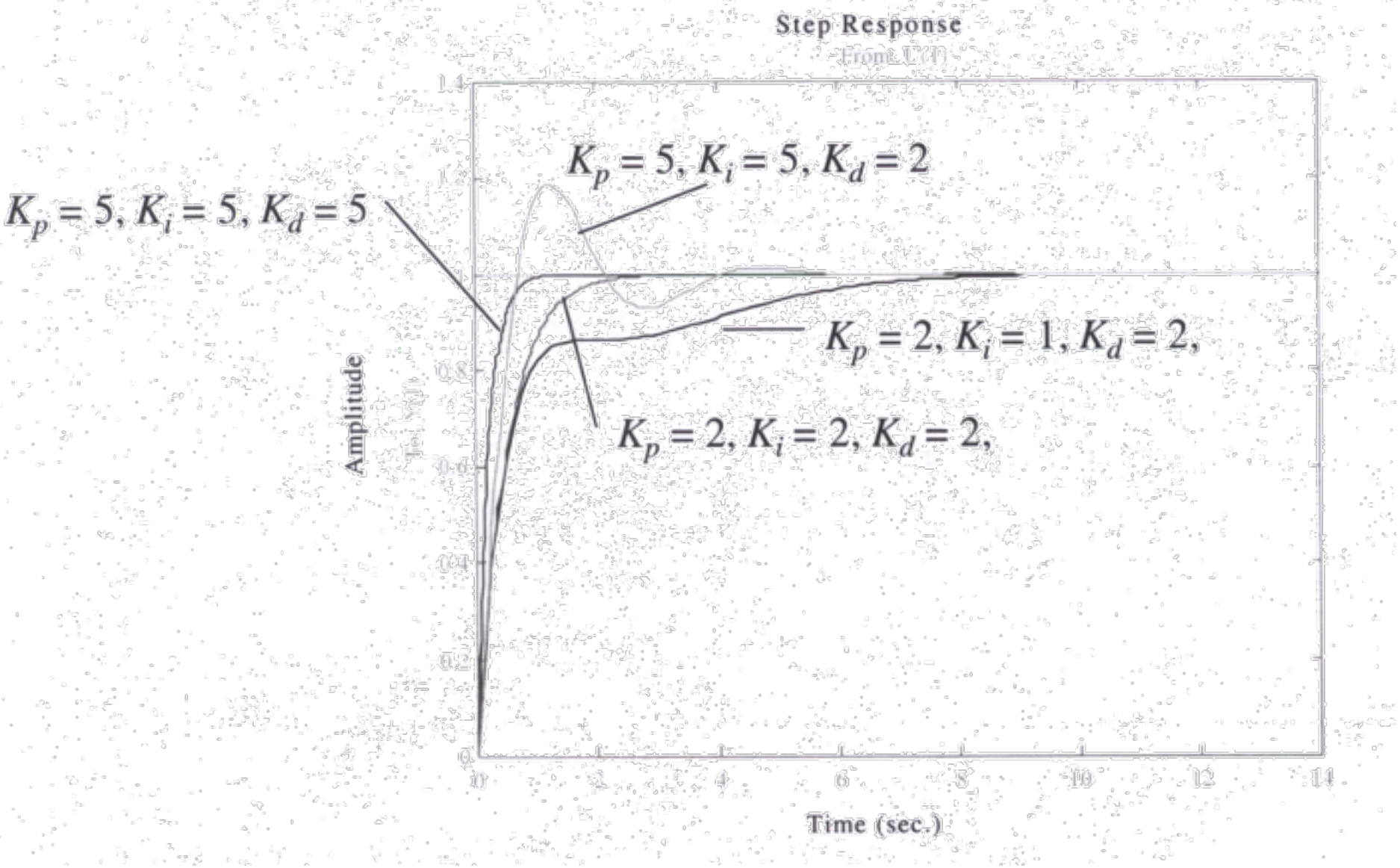
The transfer function of industrial PID controller is given by the following relationship

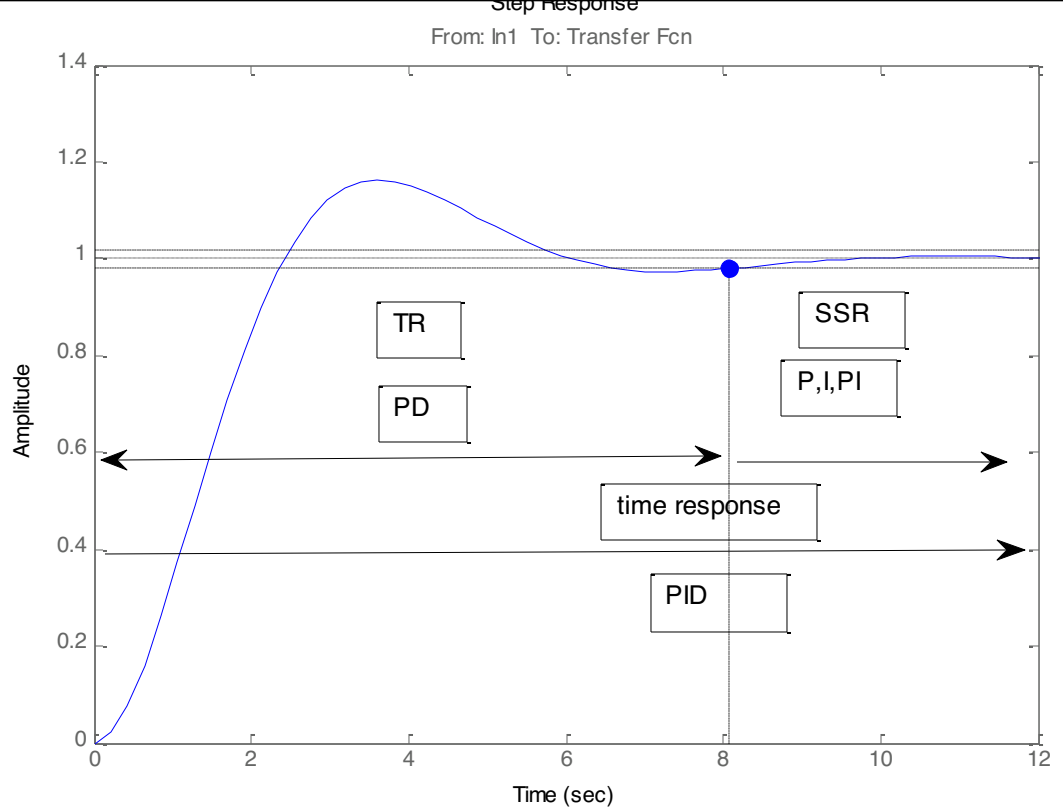
$$\frac{M(S)}{E(s)} = K_p \left(1 + \frac{1}{T_i} s + T_d s\right)$$

And the block diagram of this controller is



Changes in gains for PID Controller



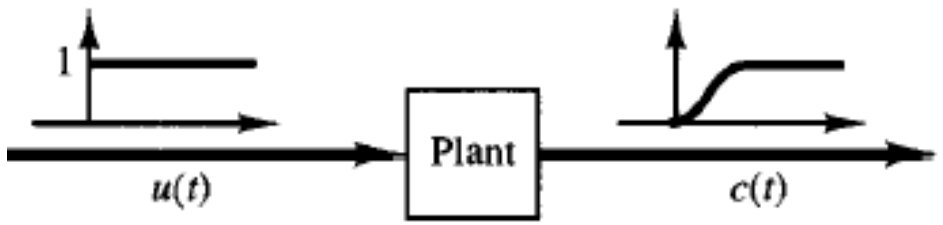


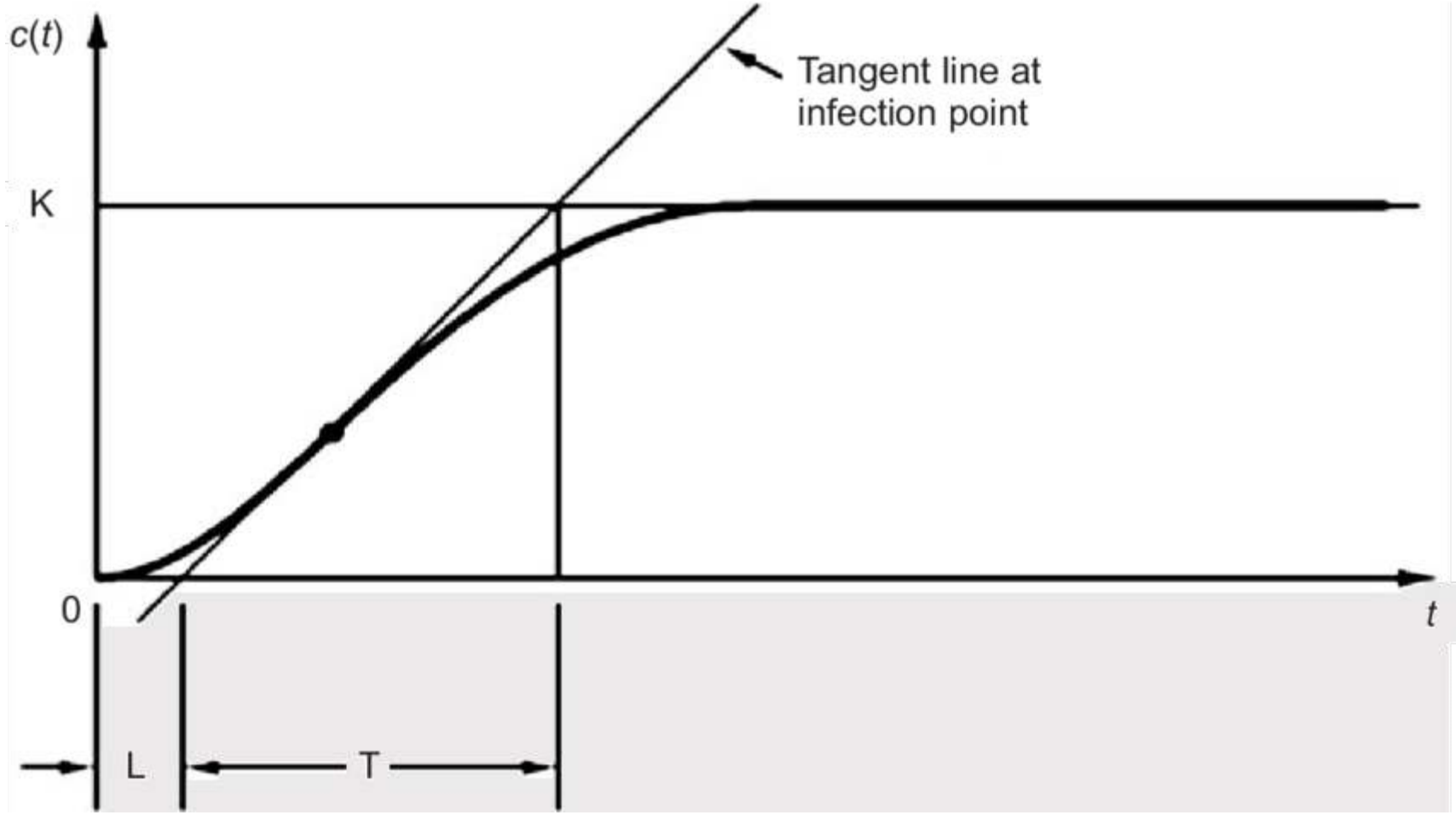
Tuning PID controller Using trial error method

1.Set Ti to infinity and Td to zero
2.Increase Kp to reduce initial steady state error to half it's value
3. Add small Ti and tune for min. overshoot
4.Add a Td so that max. peak less than 20%

Ziegler–Nichols rules for tuning PID controllers.

- These rules are used to determine Kp, Ti and Td for PID controllers
- Firs Method: The response is obtained experimentally to a unit step input. The plant involves neither integrators nor differentiators




c(t)
K
Tangent line at
infection point
0
t
L
T

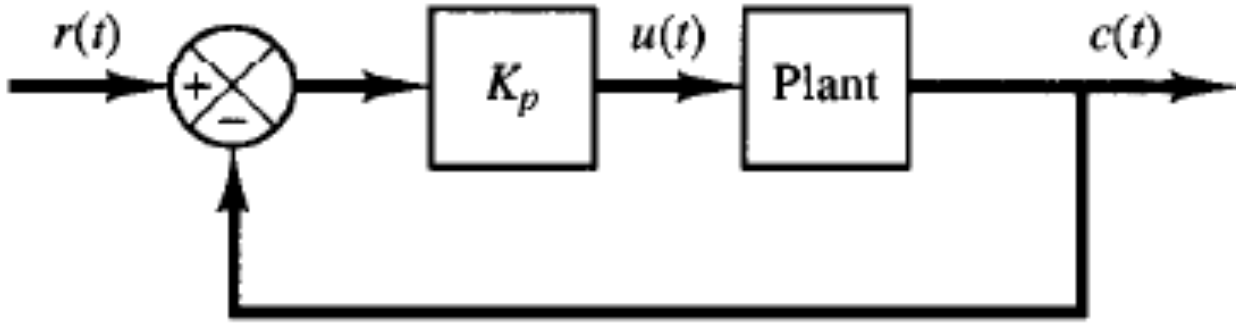
Ziegler and Nichols suggested to set the values of K_p, T_i, and T_d according to Table 10–1.

Table 10–1 Ziegler–Nichols Tuning Rule Based on Step Response of Plant (First Method)

Type of Controller	K_p	T_i	T_d
P	$\frac{T}{L}$	∞	0
PI	$0.9\frac{T}{L}$	$\frac{L}{0.3}$	0
PID	$1.2\frac{T}{L}$	$2L$	$0.5L$

Second Method

- Set Ti= inf and Td=0, increase Kp from 0 t a critical value Kcr where the output exhibits sustained oscillations.
- Use Kcr , Pcr and Table 10-2 to determine the parameters of the controller



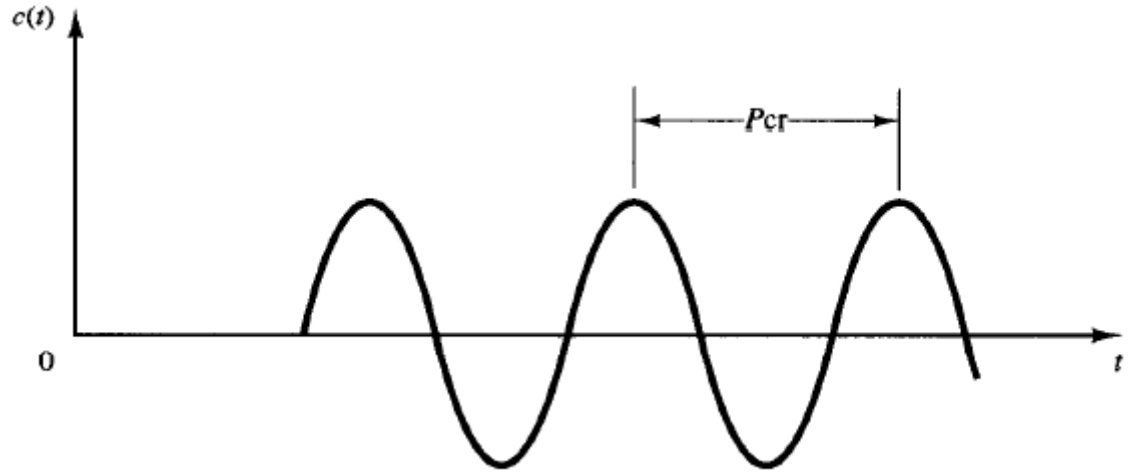


Table 10–2 Ziegler–Nichols Tuning Rule Based on Critical Gain K_{cr} and Critical Period P_{cr} (Second Method)

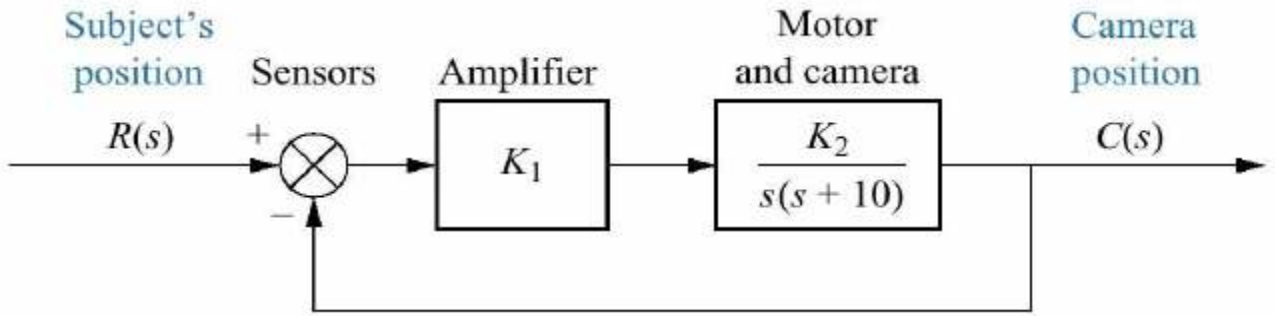
Type of Controller	K_p	T_i	T_d
P	$0.5K_{cr}$	∞	0
PI	$0.45K_{cr}$	$\frac{1}{1.2}P_{cr}$	0
PID	$0.6K_{cr}$	$0.5P_{cr}$	$0.125P_{cr}$

Root Locus Design Method

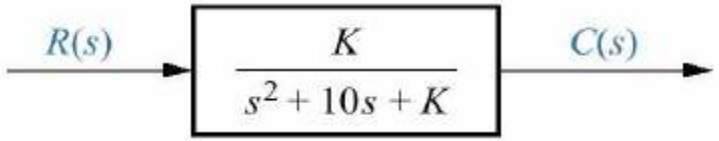
Goal

Learn a *specific technique* which shows ho***changes in one ofa system's parameter*** (usually the controller gain, *K*)

will modify the location of the closed-loop poles in the *s*-domain.



(b)

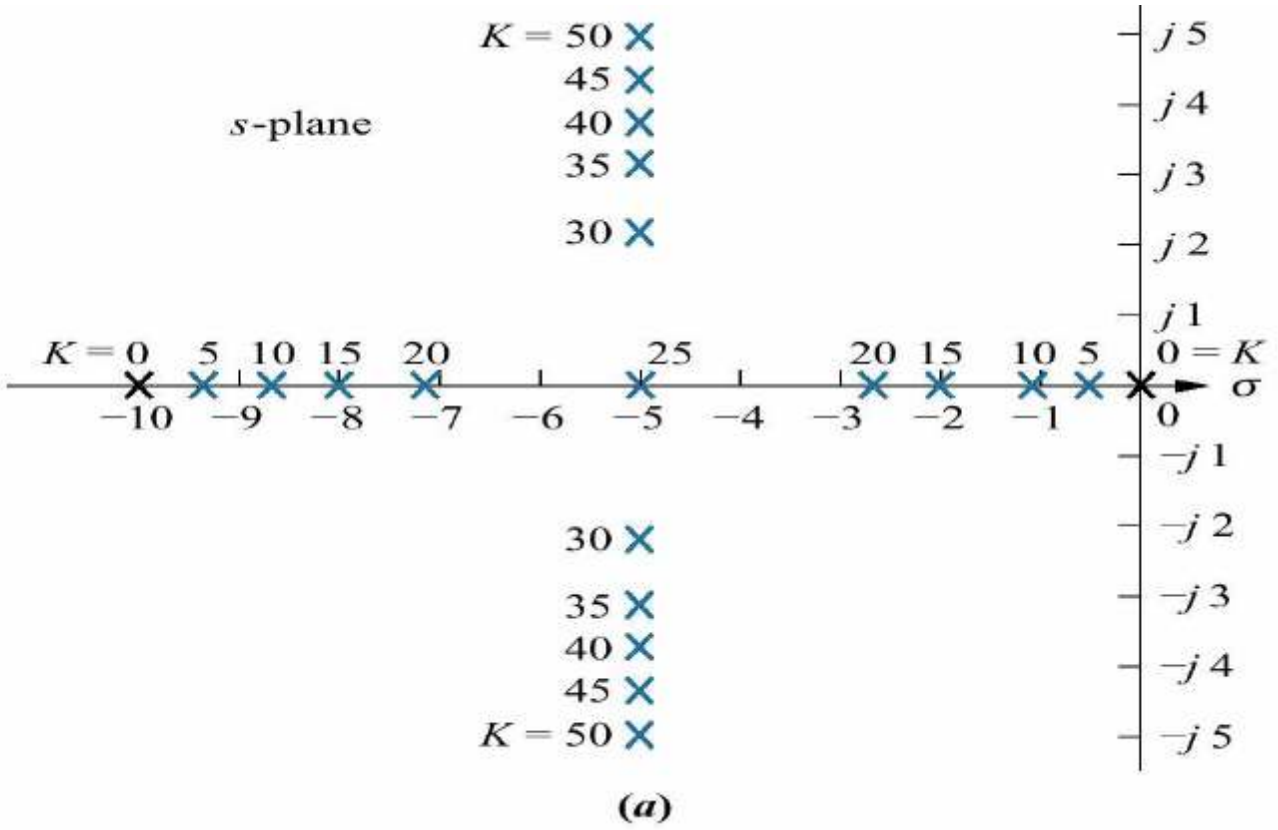


where $K = K_1K_2$

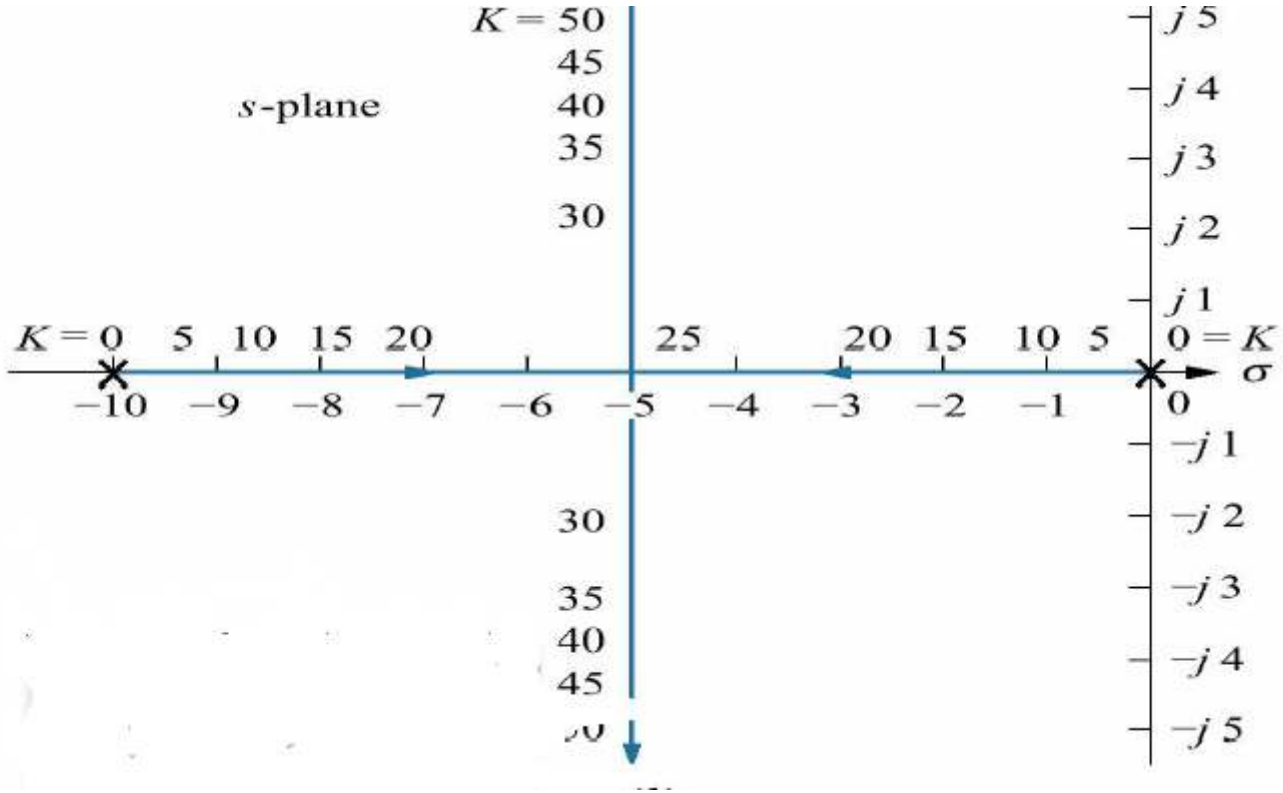
The roots of the characteristic are

$$s_{1,2} = -5 \pm \sqrt{25 - K}$$

K	Pole 1	Pole 2
0	-10	0
5	-9.47	-0.53
10	-8.87	-1.13
15	-8.16	-1.84
20	-7.24	-2.76
25	-5	-5
30	$-5 + j2.24$	$-5 - j2.24$
35	$-5 + j3.16$	$-5 - j3.16$
40	$-5 + j3.87$	$-5 - j3.87$
45	$-5 + j4.47$	$-5 - j4.47$
50	$-5 + j5$	$-5 - j5$

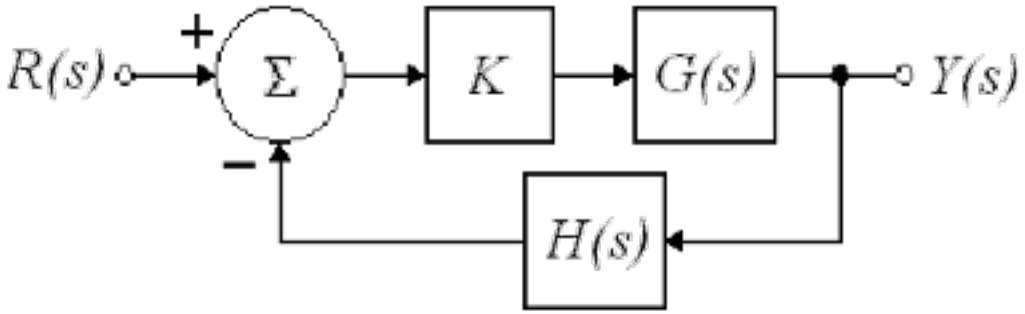

s-plane
K = 50
45
40
35
30
30
35
40
45
K = 50
j 5
j 4
j 3
j 2
j 1
−j 1
−j 2
−j 3
−j 4
−j 5
K = 0 5 10 15 20 25 20 15 10 5 0 = K
−10 −9 −8 −7 −6 −5 −4 −3 −2 −1 0
σ

(a)


s-plane
K = 50
45
40
35
30
K = 0
5
10
15
20
25
20
15
10
5
0 = K
σ
−10
−9
−8
−7
−6
−5
−4
−3
−2
−1
0
j 5
j 4
j 3
j 2
j 1
−j 1
−j 2
−j 3
−j 4
−j 5
30
35
40
45

Definition

- The closed-loop poles of the negative feedback control system: are the roots of the characteristic equation:



$$1 + KG(s)H(s) = 0$$

The root locus is the locus of the closed-loop poles when a specific parameter (usually gain, K) is varied from 0 to infinity.

Root Locus Method Foundations

- The value of *s* in the *s*-plane that make the loop gain *KG*(s)*H*(s) equal to -1 are the closed-loop poles

  (*i.e.* $1 + KG(s)H(s) = 0 \Leftrightarrow KG(s)H(s) = -1$)

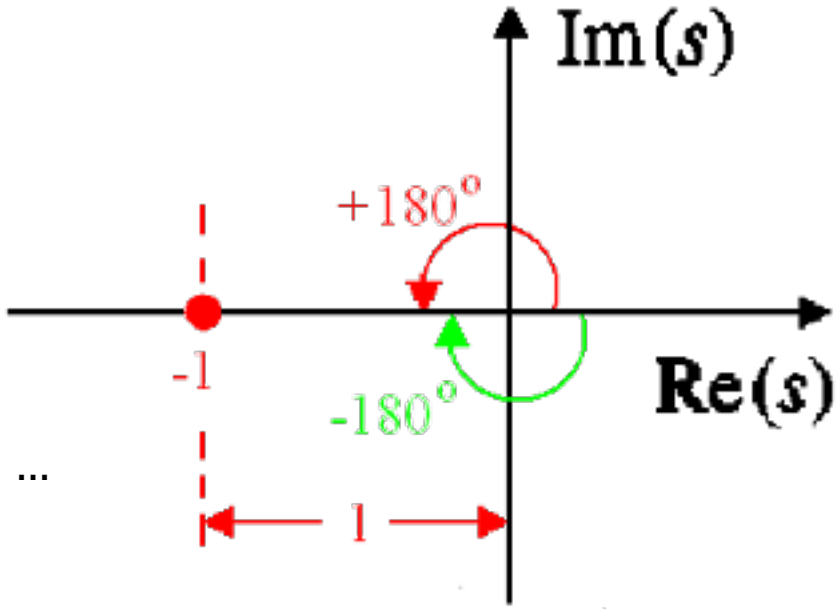
- *KG*(s)*H*(s) = -1 **can be split into two equations by equating the magnitudes and angles** of both sides of the equation.

Angle and Magnitude Conditions

$$KG(s)H(s) = -1$$

$$\Leftrightarrow \begin{cases} |KG(s)H(s)| = 1 \\ KG(s)H(s) = \pm 180^0(2l+1) \end{cases}$$

$$\Leftrightarrow \begin{cases} \angle\, |G(s)H(s)| = 1/K \\ G(s)H(s) = \pm 180^o(2l+1) \end{cases} \quad l = 0,\ 1,\ 2,\ \ldots$$



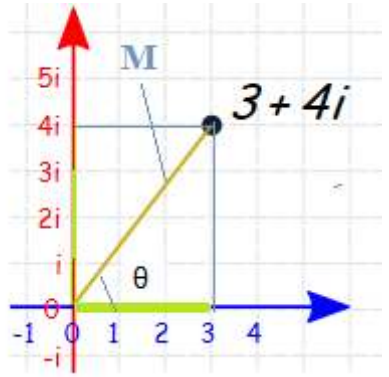
$$\angle\, \qquad\qquad\qquad\qquad l = 0,\ 1,\ 2,\ \ldots$$

Independent of K

Magnitude and phase of distinct zero assume
G(s) =s+3 and s=i4 then
G(i4)=3+i4

$$M = \sqrt{\mathrm{Re}^2 + \mathrm{Im}^2}$$

$$\theta = \tan^{-1}\frac{\mathrm{Im}}{\mathrm{Re}}$$



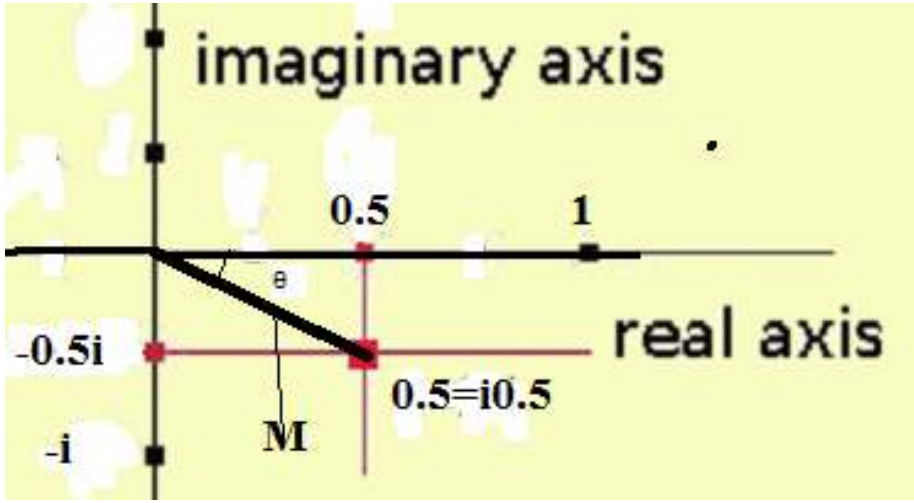
$M = 5;\ \Theta = 53.1$

Magnitude and phase of distinct pole assume

G(s)=1/(s+1) and s=i1
Then G(i1)=1/(1+i1) =0.5 –i0.5

$$M = \sqrt{\mathrm{Re}^2 + \mathrm{Im}^2}$$

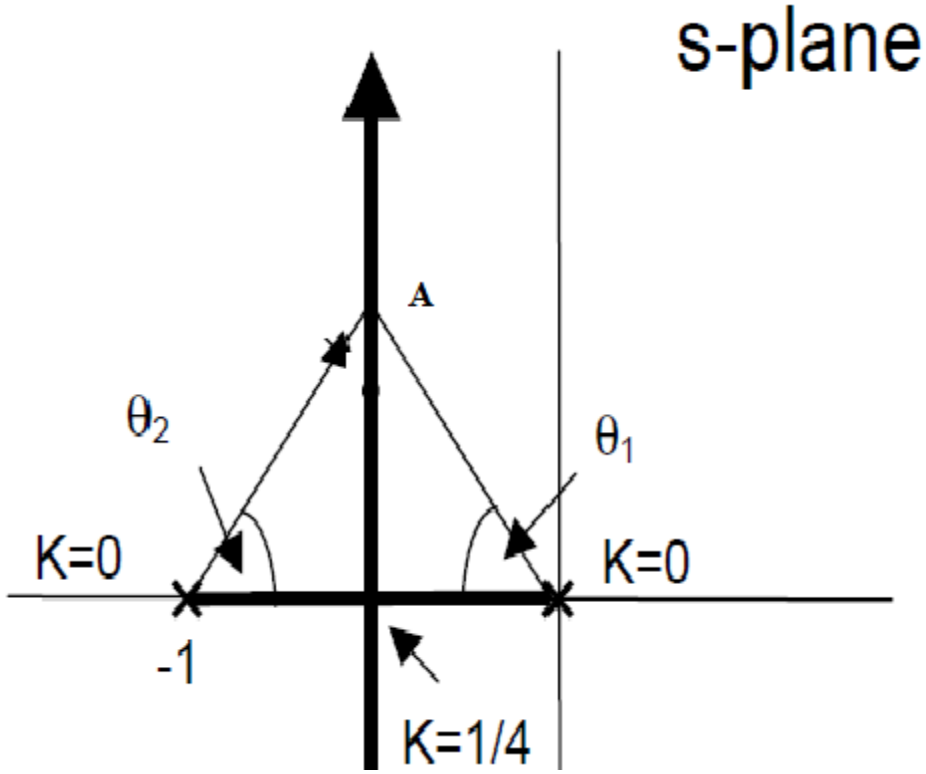
$$\theta = -\tan^{-1}\frac{\mathrm{Im}}{\mathrm{Re}}$$



M=0.7 ; $\Theta = -45$

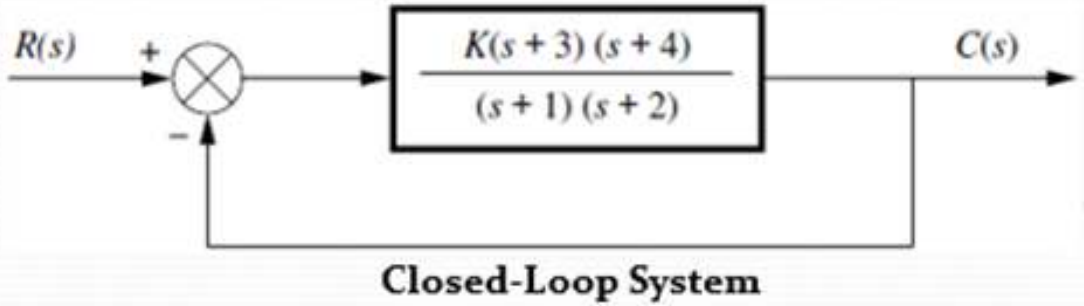
Magnitude and Phase

$$G(s)\cdot H(s) = \frac{1}{s\cdot(s+1)}$$

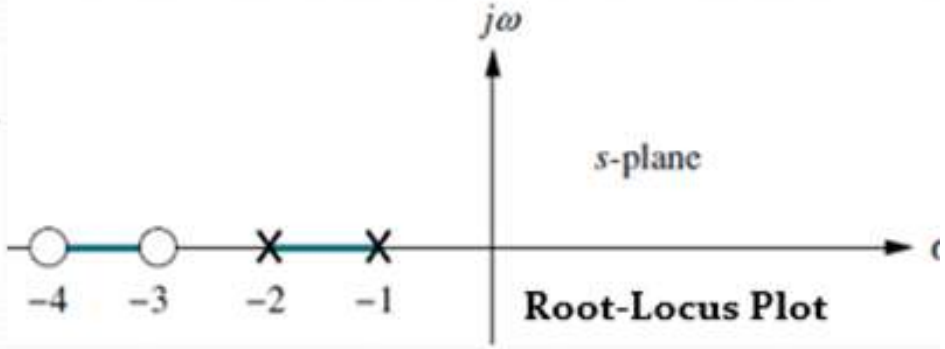


$$\angle\left(\frac{K}{s\cdot(s+1)}\right) = -\angle s - \angle s+1 = -(180-\theta_1)-\theta_2 = \pm 180$$

Example-3: Check whether point s = - 2 + j3 is a closed-loop system pole for some value of gain, *K*, or not.



Closed-Loop System



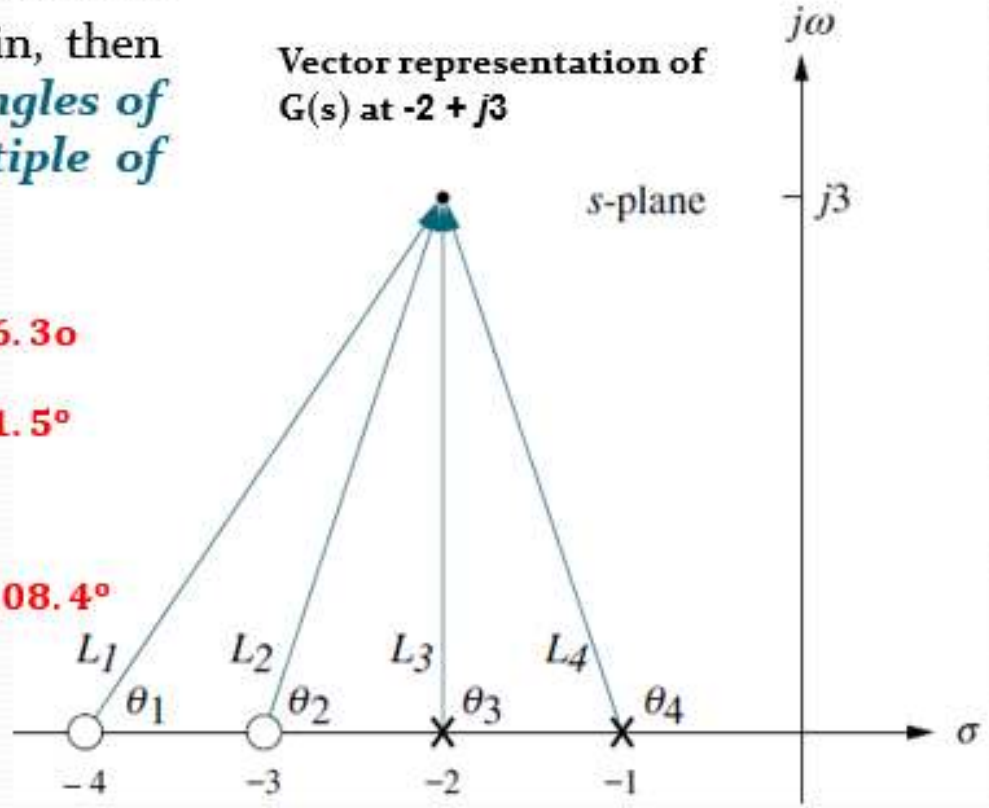
Root-Locus Plot

- For this system, the open-loop transfer function is $KG(s)H(s) = \dfrac{K(s+3)(s+4)}{(s+1)(s+2)}$
- The closed-loop transfer function is $T(s) = \dfrac{K(s+3)(s+4)}{(1+K)s^2 + (3+7K)s + (2+12K)}$
- If point s is a closed-loop system pole for some value of gain, **K**, then s must satisfy

Example-3: Continue.

- Consider the point **s = - 2 +j3**. If this point is a closed-loop pole for some value of gain, then ***the angles of the zeros minus the angles of the poles must equal an odd multiple of 180°.***

- **L1 = (-2 + j3) – (-4 + j0) = 2 + j3 = $\sqrt{13} < 56.3°$**
- **L2 = (-2 + j3) – (-3 + j0) = 1 + j3 = $\sqrt{10} < 71.5°$**
- **L3 = (-2 + j3) – (-2 + j0) = 0 + j3 = 3 < 90°**
- **L4 = (-2 + j3) – (-1 + j0) = -1 + j3 = $\sqrt{10} < 108.4°$**



Hence, $\theta_1 + \theta_2 - \theta_3 - \theta_4 = 56.31° + 71.57° - 90° - 108.43° = -70.55°$

- Therefore, **s = -2 + j3** is **NOT** a point on the root locus, or alternatively, **s = -2 + j3** is **NOT** a closed-loop pole for any value of gain ***K***.

Example-3: Continue.

- **If these calculations are repeated for another point $s = -2 + j(\sqrt{2}/2)$.** If this point is a closed-loop pole for some value of gain K, then ***the angles of the zeros minus the angles of the poles must equal an odd multiple of*** **180°.**

- L1 = $(-2 + j(\sqrt{2}/2)) - (-4 + j0) = 2 + j(\sqrt{2}/2) = 2.12 < 19.4°$
- L2 = $(-2 + j(\sqrt{2}/2)) - (-3 + j0) = 1 + j(\sqrt{2}/2) = 1.22 < 35.3°$
- L3 = $(-2 + j(\sqrt{2}/2)) - (-2 + j0) = 0 + j(\sqrt{2}/2) = \frac{\sqrt{2}}{2} < 90°$
- L4 = $(-2 + j(\sqrt{2}/2)) - (-1 + j0) = -1 + j(\sqrt{2}/2) = 1.22 < 144.7°$

Hence, $\theta_1 + \theta_2 - \theta_3 - \theta_4 = 19.4° + 35.3° - 90° - 144.7°^{+} = -180°$

So, angles do add up to 180°, therefore $s = -2 + j(\sqrt{2}/2)$ is a point on the root locus, or alternatively, $s = -2 + j(\sqrt{2}/2)$ is a closed-loop pole for some value of gain K.

- The value of gain ***K*** is calculated using Eq. (iv) as; $K = \frac{L_3 L_4}{L_1 L_2} = \frac{\frac{\sqrt{2}}{2}(1.22)}{(2.12)(1.22)} = 0.33$

Thus, the point $s = -2 + j(\sqrt{2}/2)$ is a point on the root locus for a gain K=0.33.

Rules for Construction of Root Locus

Follow these rules for constructing a root locus.

Rule 1 –

Starting points G(s)H(s)=1/K $\frac{s+4}{s(s+10)} = \frac{1}{K}$

0,-10

Terminating points -4, or infinity

Rule 2 – Locate the open loop poles and zeros in the 's' plane.

Find the number of root locus branches.

We know that the root locus branches start at the open loop poles and end at open loop zeros. So, the number of root locus branches N is equal to the number of finite open loop poles P or the number of finite open loop zeros Z, whichever is greater.

Mathematically, we can write the number of root locus branches N as

$N = P$ if $P \geq Z$

$N = Z$ if $P < Z$

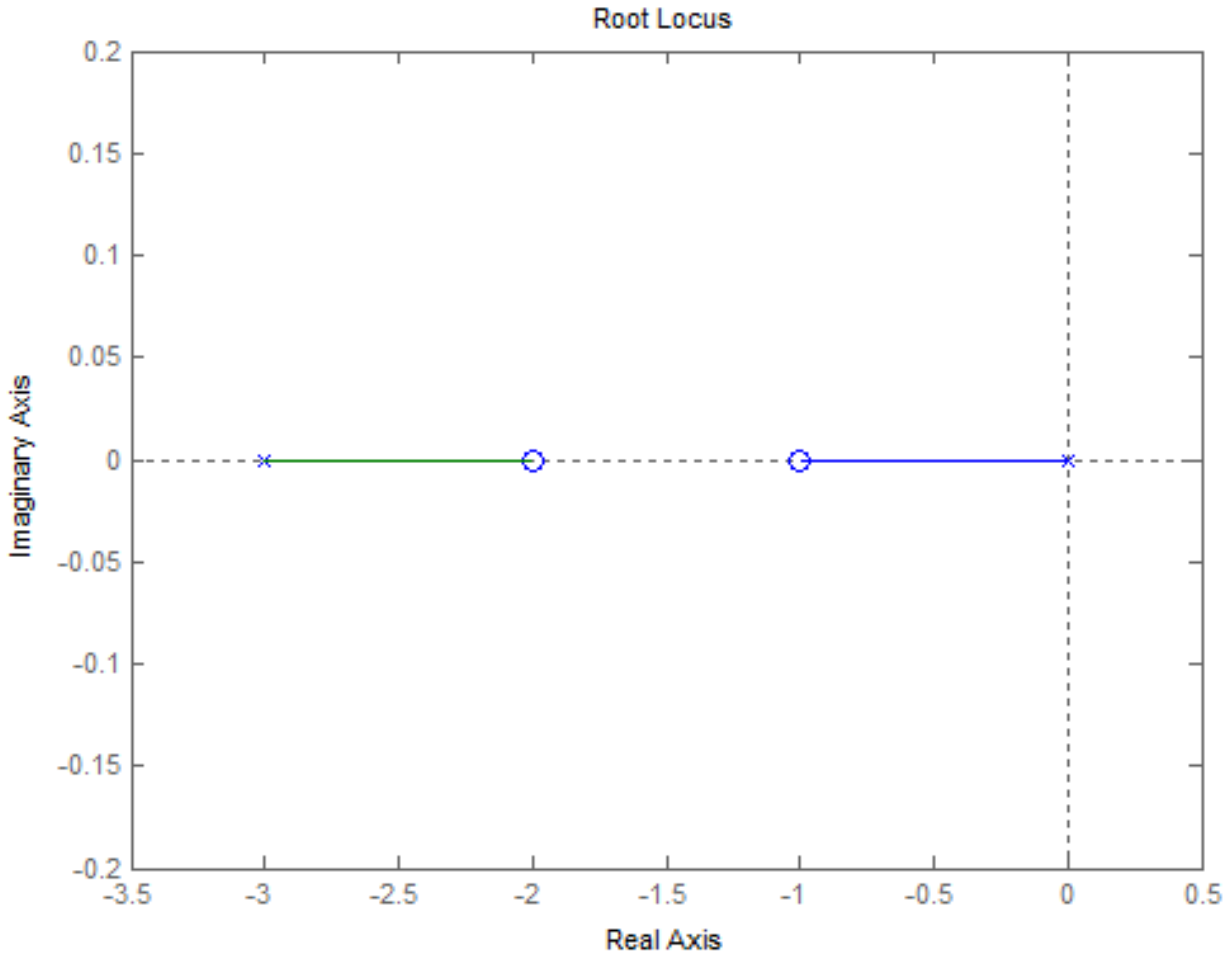
Rule 3 – Identify and draw the **real axis root locus branches**.

If the angle of the open loop transfer function at a point is an odd multiple of $\pm 180^0$, then that point is on the root locus. If odd number of the open loop poles and zeros exist to the right of a point on the real axis, then that point is on the root locus branch. Therefore, the branch of points which satisfies this condition is the real axis of the root locus branch.

Remember that root locus is symmetrical around real axis due to comlpex conjugate poles

Example: Plot a root locus of a system given as

$$G(s)H(s) = \frac{(s+1)(s+2)}{s(s+3)}$$



Rule 4 – Find the centroid and the angle of asymptotes.

If $P=Z$, then all the root locus branches start at finite open loop poles and end at finite open loop zeros.

If $P>Z$, then Z number of root locus branches start at finite open loop poles and end at finite open loop zeros and $P-Z$ number of root locus branches start at finite open loop poles and end at infinite open loop zeros.

If $P<Z$, then P number of root locus branches start at finite open loop poles and end at finite open loop zeros and $Z-P$ number of root locus branches start at infinite open loop poles and end at finite open loop zeros.

So, some of the root locus branches approach infinity, when $P\neq Z$.

Asymptotes give the direction of these root locus branches. The intersection point of asymptotes on the real axis is known as **centroid**.

We can calculate the **centroid α** by using this formula,

$$\alpha = \frac{\sum Real\ part\ of\ finite\ open\ loop\ poles - \sum Real\ part\ of\ finite\ open\ loop\ zeros}{P - Z}$$

The formula for the angle of **asymptotes θ** is

$$\Theta = \frac{\pm 180(2l+1)}{P-Z}$$

lType equation here.

Where $l = 0, 1, 2, \ldots$

$$G(s)H(s) = \frac{1}{s(s+1)(s+2)}$$

Determine the *n* - *m* asymptotes:

- Locate s = α on the real axis:

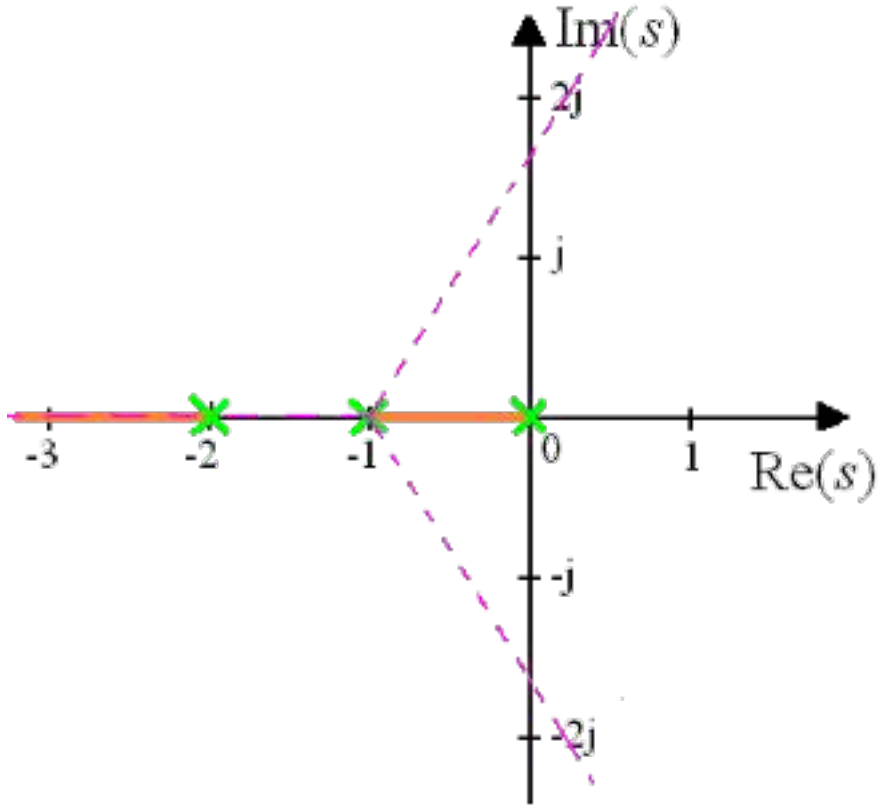
$$s = \alpha = \frac{p_1 + p_2 + p_3}{3 - 0} = \frac{0 - 1 - 2}{3} = -1$$

- Compute and draw angles:

$$\varphi_l = \frac{\pm 180(2l+1)}{n-m} \qquad l = 0,\ 1,\ 2,\ \ldots$$

$$\Rightarrow \begin{cases} \varphi_0 = \dfrac{\pm 180^0(2\times 0+1)}{3-0} = \pm 60^0 \\ \varphi_1 = \dfrac{\pm 180^0(2\times 1+1)}{3-0} = \pm 180^0 \end{cases}$$

- Draw the asymptotes using dash lines.



Rule 5 – Find the intersection points of root locus branches with an imaginary axis.

We can calculate the point at which the root locus branch intersects the imaginary axis and the value of **K** at that point by using the Routh array method and special **case (ii)**.

If all elements of any row of the Routh array are zero, then the root locus branch intersects the imaginary axis and vice-versa.

Identify the row in such a way that if we make the first element as zero, then the elements of the entire row are zero. Find the value of **K** for this combination.

Substitute this **K** value in the auxiliary equation. You will get the intersection point of the root locus branch with an imaginary axis.

$$G(s)H(s) = \frac{1}{s(s+1)(s+2)}$$

$$1 + KG(s)H(s) = s^3 + 3s^2 + 2s + K$$

Routh table

1	2
3	K
(6-K)/3	0
K	0

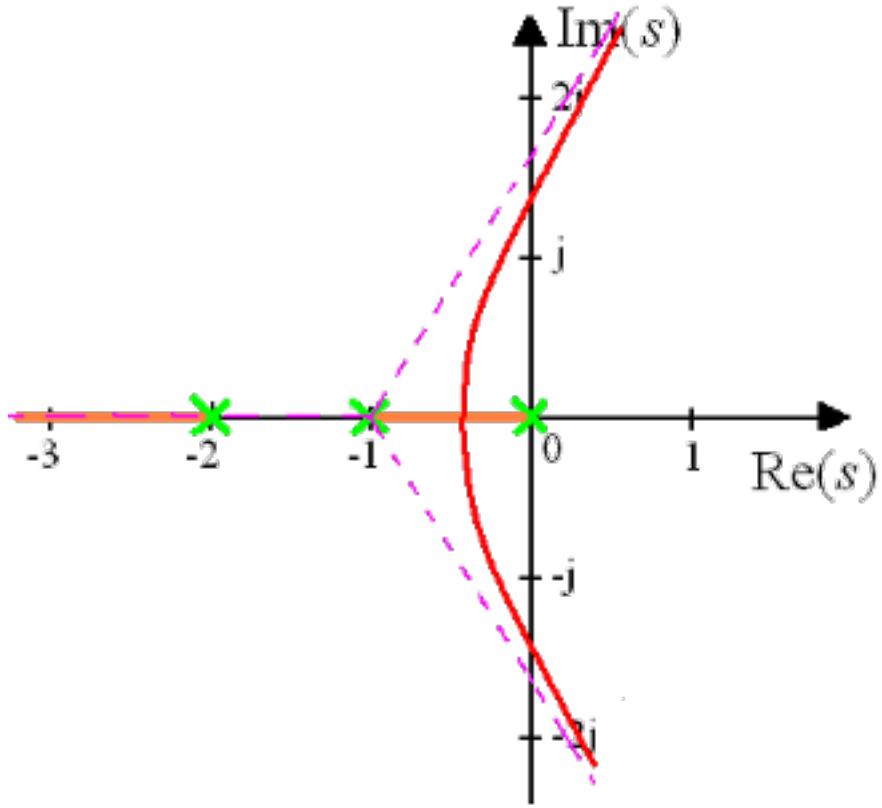
Critical gain K cr=6

Auxiliary equation

$3s^2 + 6 = 0$

$s^2 + 2 = 0$

$s_{1,2} = \pm\sqrt{2}$



- **Rule 6** – Find Break-away and Break-in points.
- If there exists a real axis root locus branch between two open loop poles, then there will be a **break-away point** in between these two open loop poles.
- If there exists a real axis root locus branch between two open loop zeros, then there will be a **break-in point** in between these two open loop zeros.

$$G(s)H(s) = \frac{1}{s(s+1)(s+2)}$$

Find the breakpoints.

- Express *K* such as:

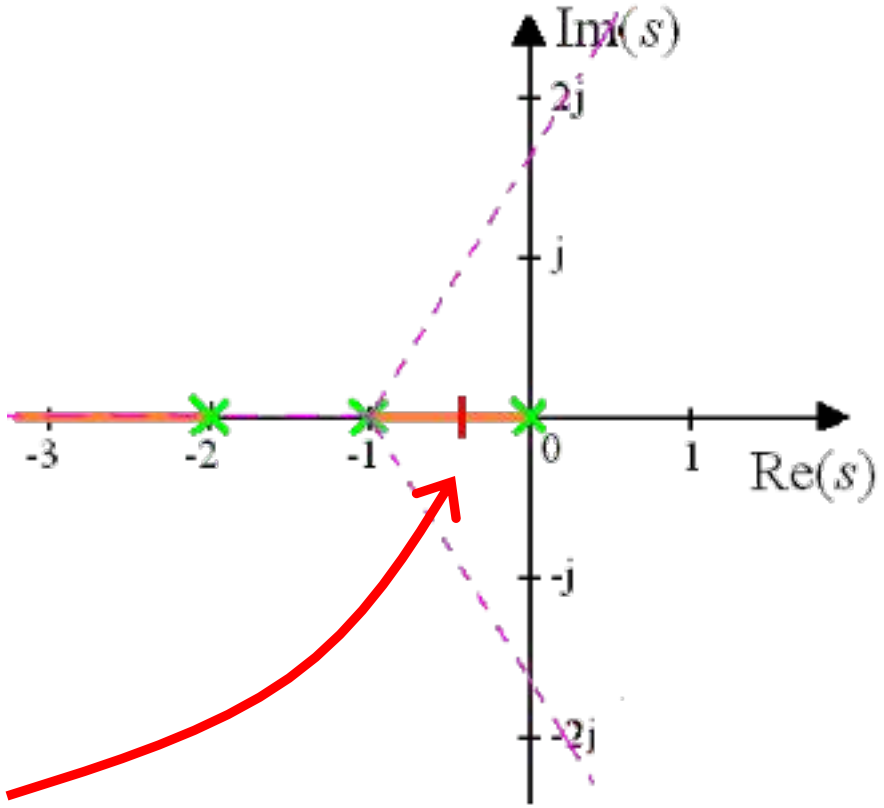
$$K = \frac{-1}{G(s)H(s)} = -s(s+1)(s+2)$$

$$K = -s^3 - 3s^2 - 2s$$

- Set *dK/ds* = 0 and solve for the roots.

$$-3s^2 - 6s - 2 = 0$$

$$s_1 = -1.5774, \quad s_2 = -0.4226$$



Rule 7 :Angles of departure and arrival

*If there are starting points at complex conjugate poles there will be departure angels

•If there are terminating points at complex conjugate zeros there will be arrival angels

$$G(s) = \frac{K(s+2)}{s^2-4s+13}, H(s) = 1.$$

1 zero at -2, 2 poles at $\frac{4\pm\sqrt{16-52}}{2} = 2 \pm 3j$

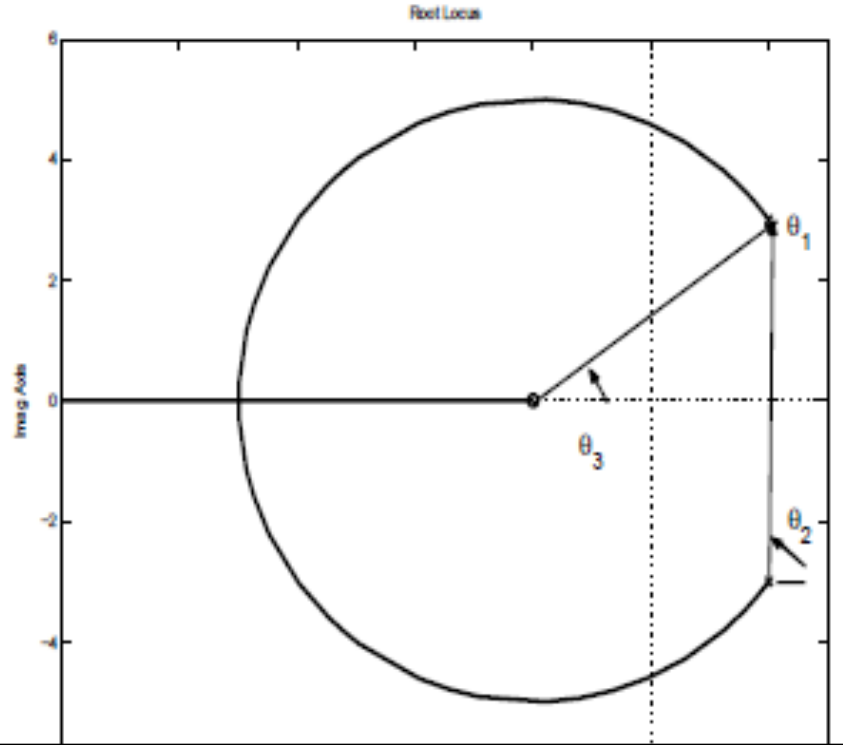
Angles of departure:

$$-\theta_1 - \theta_2 + \theta_3 = (2k+1)180^o$$

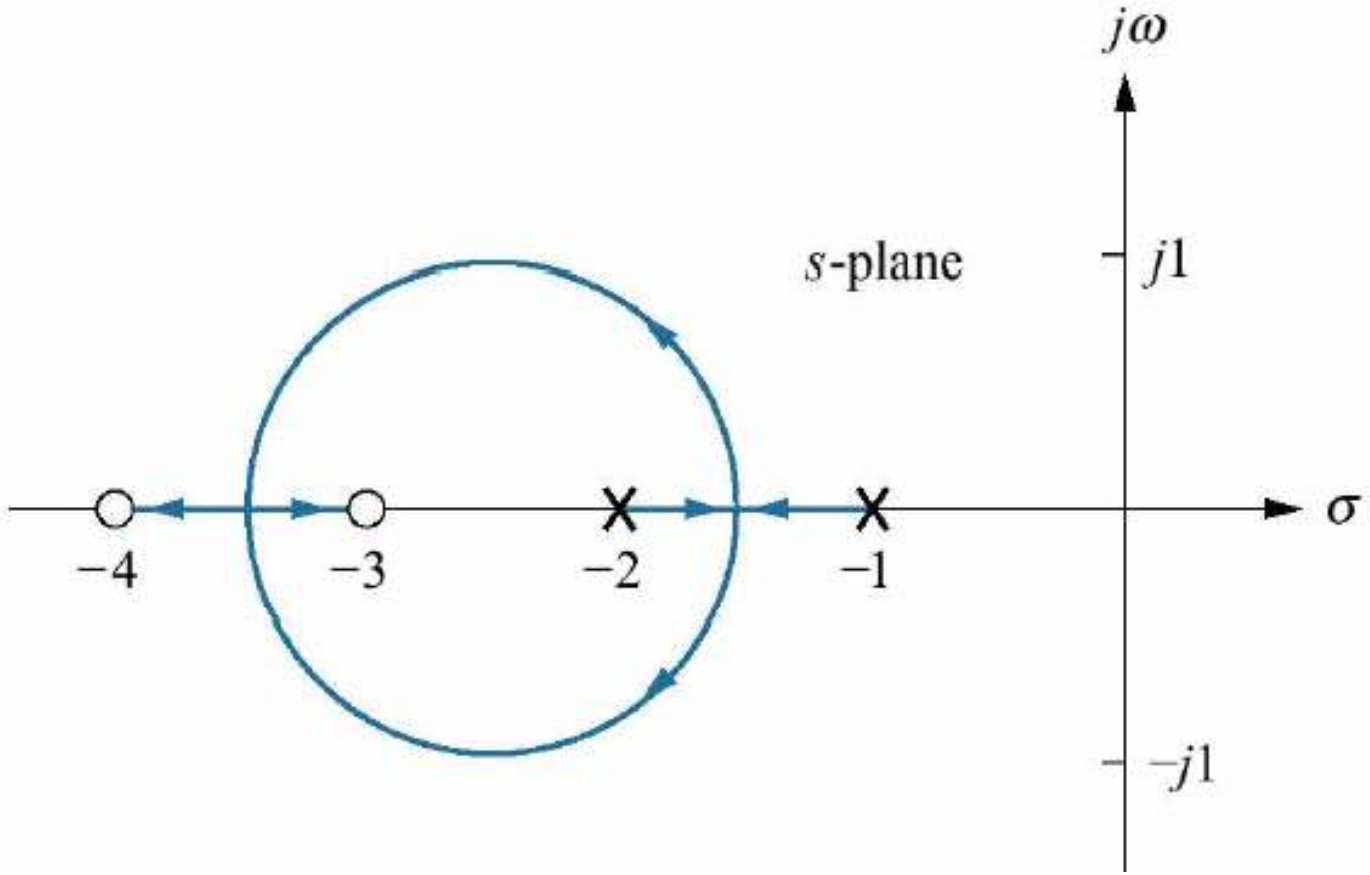
$$-\theta_1 - 90 + \arctan(3/4) = -180^o$$

$$-\theta_1 - 90 + 36.87 = 180^o$$

$$\theta_1 = 126.87^o$$



$$GH(s) = \frac{(s+3)(s+4)}{(s+1)(s+2)}$$



Frequency Response Analysis

What is frequency response

So far, we have described the response and performance of a system in terms of complex frequency variable $s=\sigma+j\omega$ and the location of poles and zeros in the s-plane. An important alternative approach to system analysis and design is the frequency response method.

The **frequency response** of a system is defined as the steady-state response of the system to a sinusoidal input signal with constant amplitude and varying frequency . We will investigate the steady-state response of the system to the sinusoidal input as the frequency varies.

When the input signal is a sinusoid, the resulting output signal for LTI systems is sinusoidal in the steady state, it differs from the input only in amplitude and phase.

Consider the system $Y(s) = T(s)R(s)$ with $r(t) = A\sin\omega\, t$. ⟸ $R(s) = \mathrm{L}\{r(t)\} = \frac{A\omega}{s^2+\omega^2}$

If $T(s) = \frac{m(s)}{(s+p_1)(s+p_2)\cdots(s+p_n)}$, where $p_1, p_2, \ldots, p_n$ are distinctive poles,

then in partial fraction expansion form, we have

$$Y(s) = \frac{k_1}{s+p_1} + \frac{k_2}{s+p_2} + \cdots \frac{k_n}{s+p_n} + \frac{\alpha s+\beta}{s^2+\omega^2}$$

Taking the inverse Laplace transform yields

$$y(t) = k_1 e^{-p_1 t} + k_2 e^{-p_2 t} + \cdots k_n e^{-p_n t} + \mathrm{L}^{-1}\left\{\frac{\alpha s+\beta}{s^2+\omega^2}\right\}$$

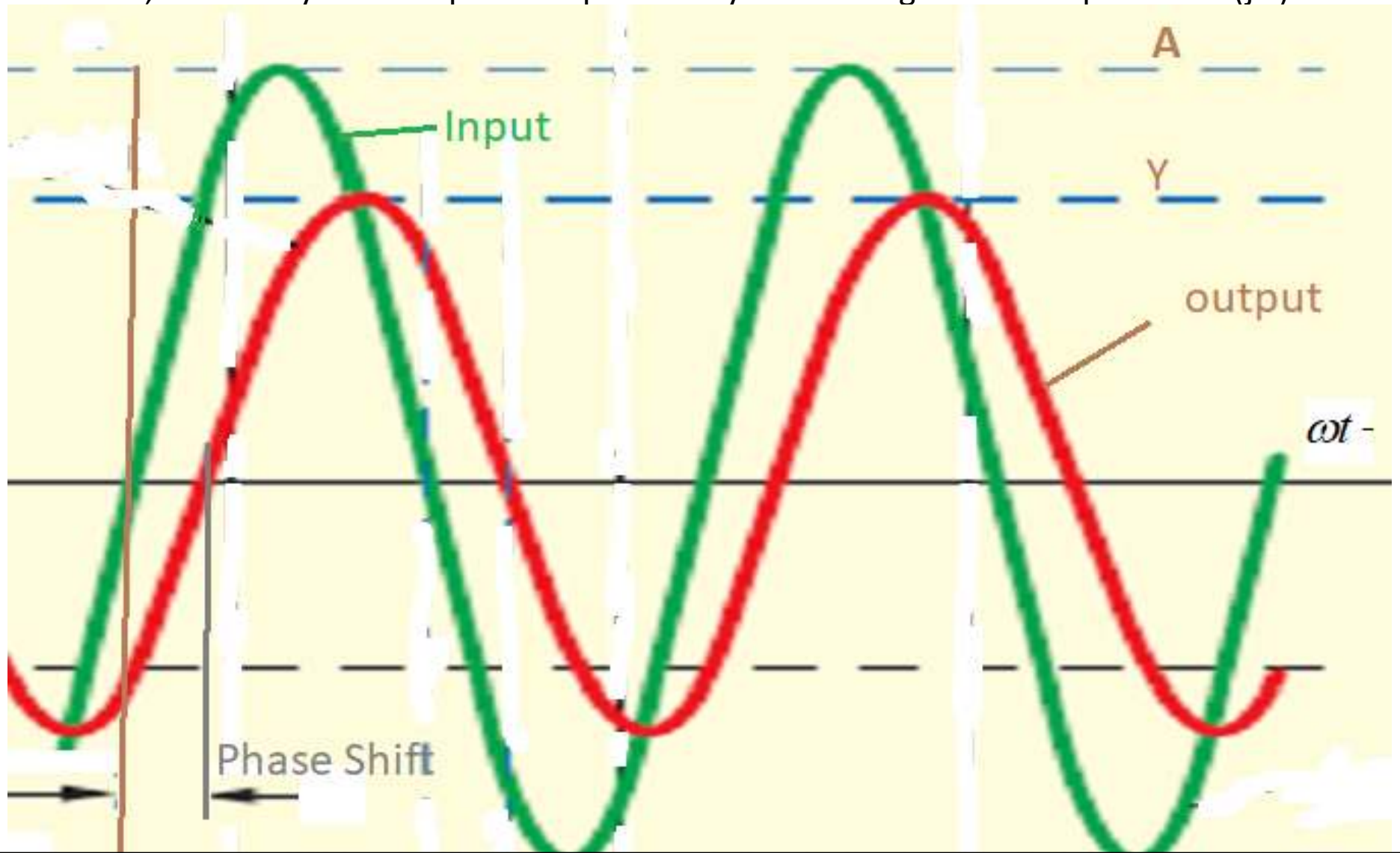
Suppose the system is stable, then all the poles are located in the left half plane and thus the exponential terms decay to zero as t→∞. Hence, the steady-state response of the system is

$$\lim_{t\to\infty} y\,(t) = \mathrm{L}^{-1}\left\{\frac{\alpha s+\beta}{s^2+\omega^2}\right\} = A|T(j\omega)|\sin(\,\omega t+\varphi),\ \text{where } \varphi = \angle T(j\omega).$$

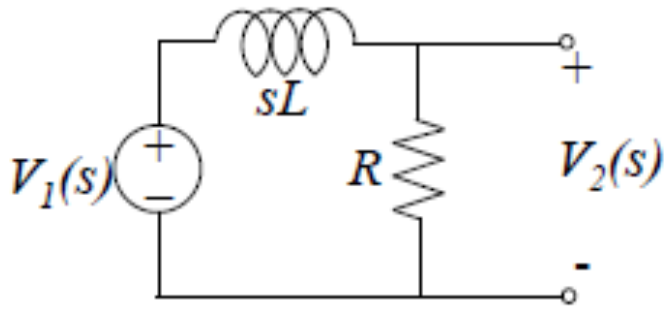
For the system $Y(s) = T(s)R(s)$ with $r(t) = A\sin\omega\, t$,

the steady-state output is $\lim_{t\to\infty} y\,(t) = A|T(j\omega)|\sin(\,\omega t + \varphi),$ where $\varphi = \angle T(j\omega).$

That is, the steady-state response depends only on the magnitude and phase of T(jω).



Plot the frequency response of the following system



Compute the s-domain transfer function T(s)

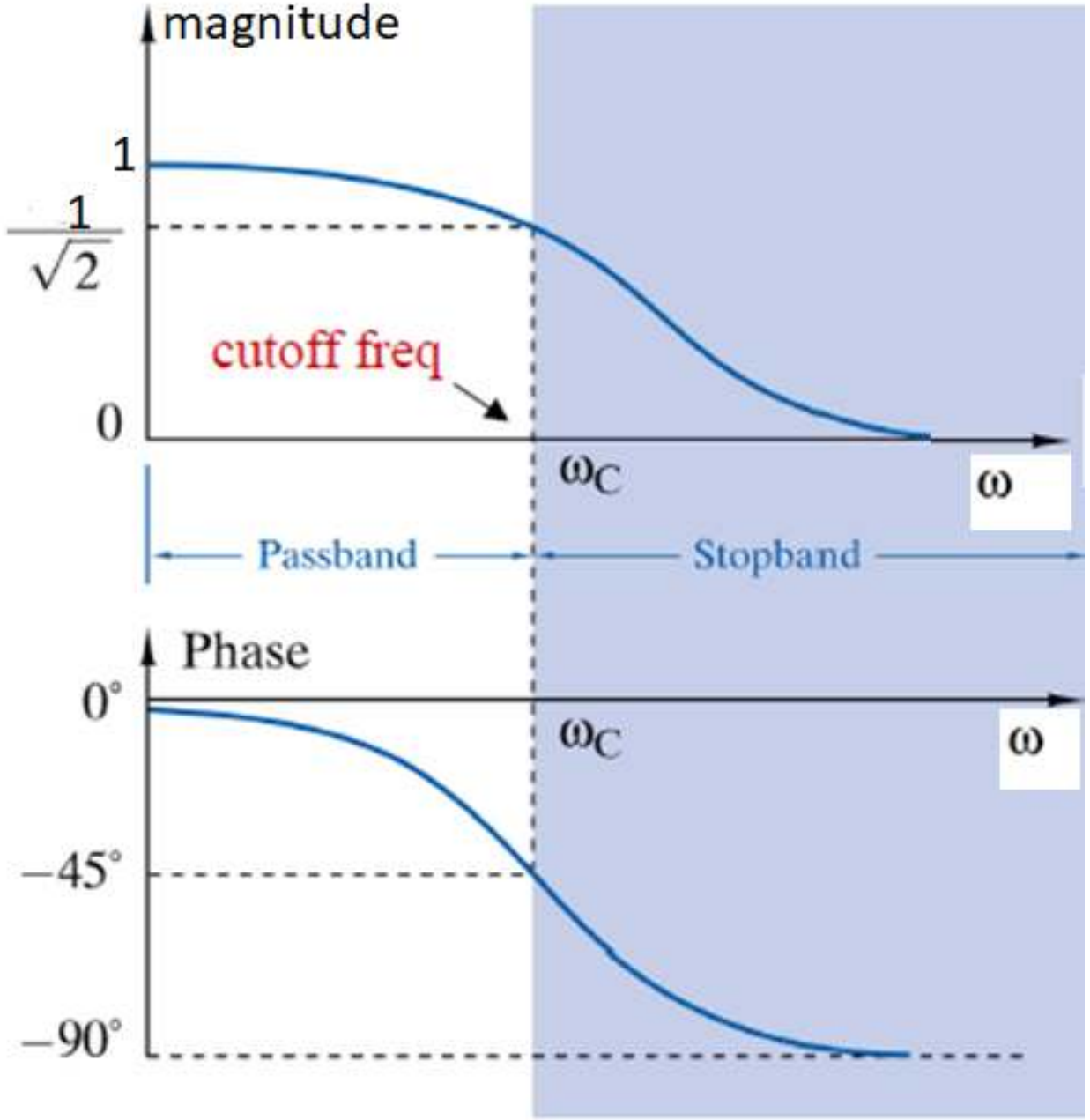
Voltage divider $T(s) = \dfrac{R}{sL + R}$

Compute the frequency response

$$|T(j\omega)| = \frac{R}{\sqrt{R^2 + (\omega L)^2}}, \qquad \angle T(j\omega) = -\tan^{-1}\left(\frac{\omega L}{R}\right)$$

ω		0	R/L	inf	
Magnitude	$\lvert T(j\omega)\rvert$	1	$1/\sqrt{2}$	0	
Phase	$\angle T(j\omega)$	0	-45	-90	

We called $\omega=\omega_c$=R/L the cut off frequency where the magnitude is reduced by $1/\sqrt{2}$


magnitude
1
1/√2
cutoff freq
0
ωC
ω
Passband
Stopband
Phase
0°
ωC
ω
−45°
−90°

• Bode Plot

- The Bode plot is a most useful technique for hand plotting was developed by H.W. Bode at Bell Laboratories between 1932 and 1942.
- This technique allows plotting that is quick and yet sufficiently accurate for control systems design.
- The idea in Bode's method is to plot magnitude curves using a logarithmic scale and phase curves using a linear scale.

The Bode plot consists of two graphs:

i. A logarithmic plot of the magnitude of a transfer function.
ii. A plot of the phase angle.

- Both are plotted against the frequency on a logarithmic scale.
- The standard representation of the logarithmic magnitude of G(jw) is 20log|G(jw)| where the base of the logarithm is 10, and the unit is in decibel (dB).

- The log-magnitude and phase frequency response curves as functions of log ω are called Bode plots or Bode diagrams.

- Sketching Bode plots can be simplified because they can be approximated as a sequence of straight lines.

- Straight-line approximations simplify the evaluation of the magnitude and phase frequency response.

- We call the straight-line approximations as asymptotes.

- The low-frequency approximation is called the low-frequency asymptote, and the high-frequency approximation is called the high-frequency asymptote.

Classes of Factors of Transfer Functions

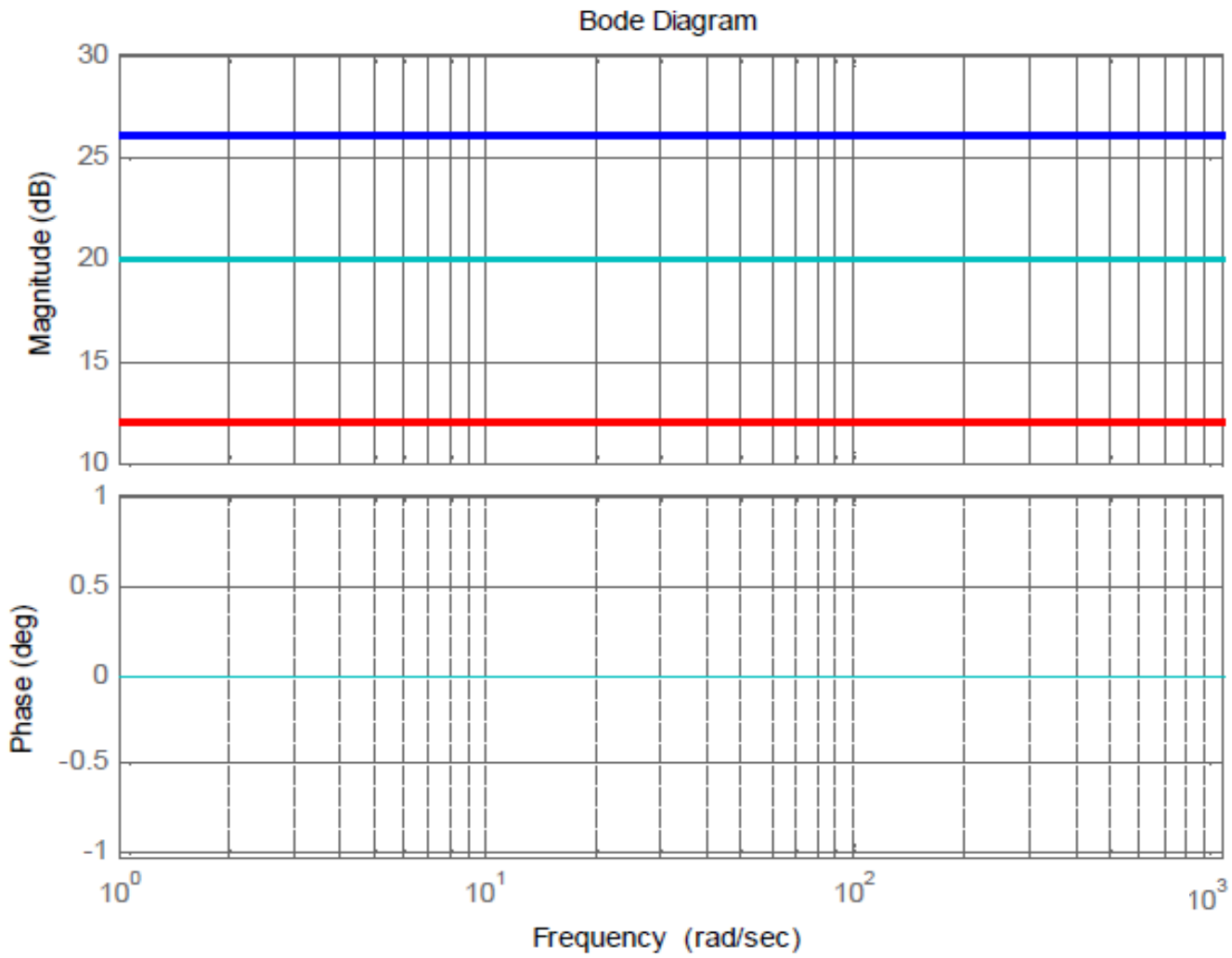
- Basic factors of G(jw)H(jw) that frequently occur in an arbitrarily transfer function are

1. Class-I: Constant Gain factor, $\boldsymbol{K}$
2. Class-II: Integral and derivative factors, $(\boldsymbol{j\omega})^{\mp 1}$
3. Class-III: First order factors, $(\boldsymbol{1 + j\omega})^{\mp 1}$
4. Class-IV: Second order factors, $[\boldsymbol{1 + 2\zeta\left(\frac{j\omega}{\omega_n}\right) + \left(\frac{j\omega}{\omega_n}\right)^2}]^{\mp 1}$

$$G(s)H(s) = \frac{16}{s(s+1)(s+2)}$$

Class-I: The Constant Gain Factor (K)

- If the open loop gain $G(j\omega)H(j\omega) = K$
- Then its **Magnitude** (dB) $|G(j\omega)H(j\omega)|_{dB} = 20log|K|$ = *constant*
- And its **Phase** $\angle G(j\omega)H(j\omega) = 0^o, \quad K > 0$
- The log-magnitude plot for a constant gain K is a horizontal straight line at the magnitude of $20\log K$ decibels.
- The effect of varying the gain K in the transfer function is that it raises or lowers the log-magnitude curve of the transfer function by the corresponding amount.
- The constant gain K has no effect on the phase curve.



$K = 20$

$K = 10$

$K = 4$

$K = 4$, 10, and 20

Class-II: The Integral Factor $(j\omega)^{-1}$

- If the open loop gain $G(j\omega)H(j\omega) = (j\omega)^{-1} = \frac{1}{j\omega}$,
  it means a pole is exist at the orign.

- **Magnitude (dB)** $|G(j\omega)H(j\omega)|_{dB} = 20log\left|\frac{1}{j\omega}\right| = 20\,log(1) - 20\,log(\omega) = -20log(\omega)$

- When the above equation is plotted against the frequency logarithmic, the *magnitude* plot produced is a straight line with a negative slope of 20 dB/ decade.

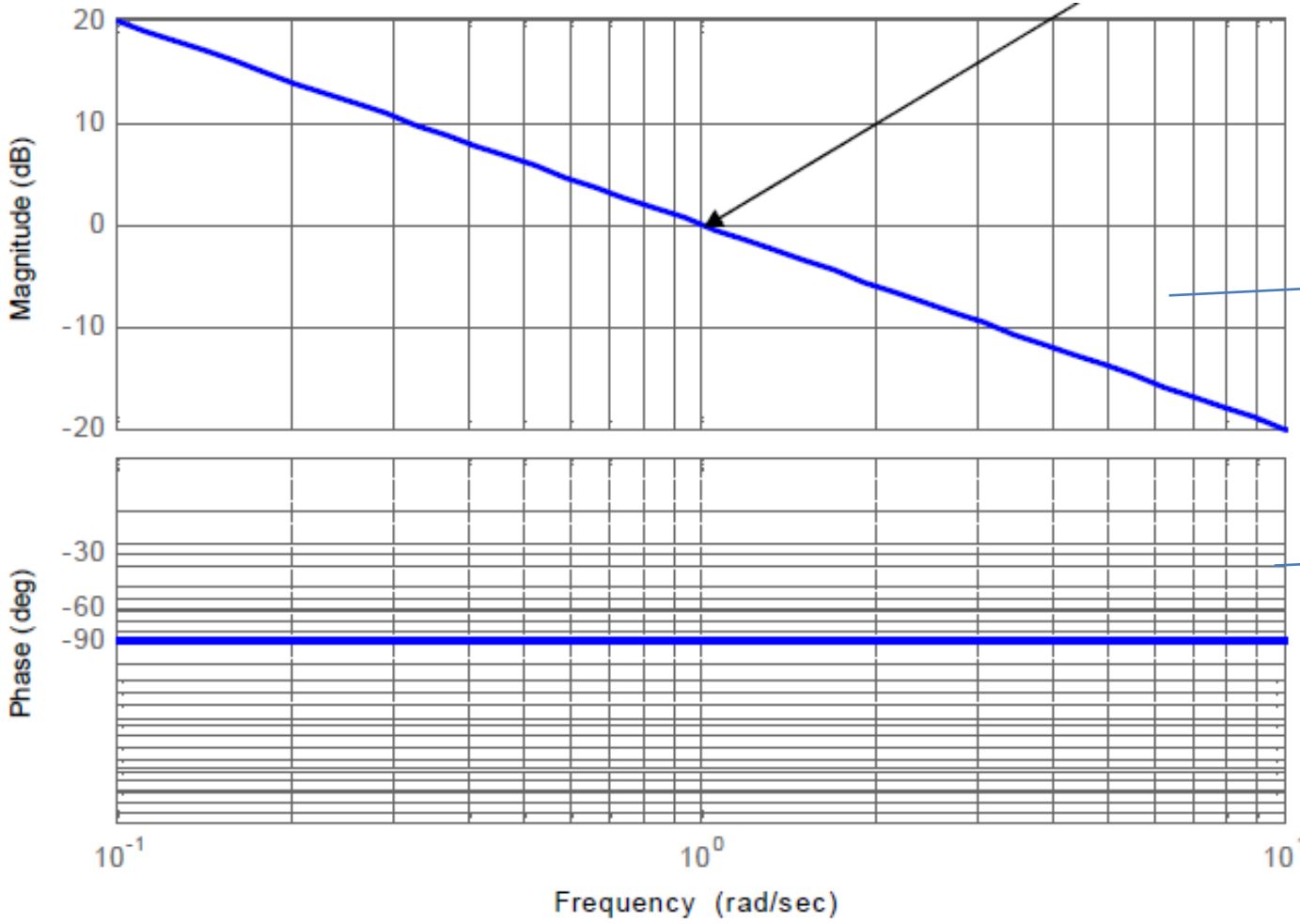
- **Phase** $\angle G(j\omega)H(j\omega) = \angle\frac{1}{j\omega} = -90^o$

- When the above equation is plotted against the frequency logarithmic, the *phase* plot produced is a straight line at -90°.

- ***Corner frequency or break point ω = 1 at the magnitude of 0 dB.***

ω{rad/sec]	.01	0,1	1	10	100	
Log M[dB]	40	20	0	-20	-40	

The slope intersects with 0 dB line at frequency ω =1



A slope of -20 dB/dec for magnitude plot of factor $\frac{1}{j\omega}$

A straight horizontal line at - 90° for phase plot of factor $\frac{1}{j\omega}$

Class-II: The Derivative Factor $j\omega$

- If the open loop gain $G(j\omega)H(j\omega) = j\omega$, $it\ means\ a\ zero\ is\ exist\ at\ the\ orign$.

- $Magnitude\ (dB)\ |G(j\omega)H(j\omega)|_{dB} = 20log|j\omega| = 20\,log(\omega)$

- When the above equation is plotted against the frequency logarithmic, the *magnitude* plot produced is a straight line with a positive slope of 20 dB/ decade.

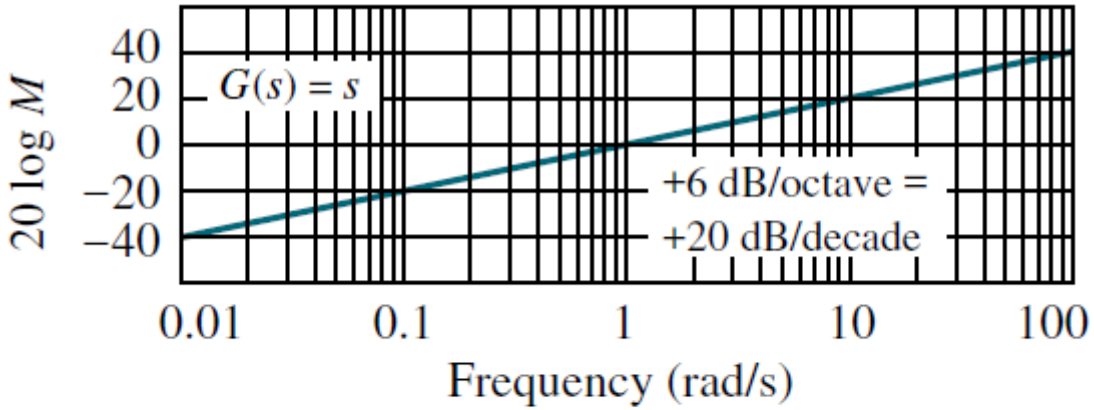
- ***Phase*** $\angle G(j\omega)H(j\omega) = \angle j\omega = 90^o$

- When the above equation is plotted against the frequency logarithmic, the *phase* plot produced is a straight line at 90°.

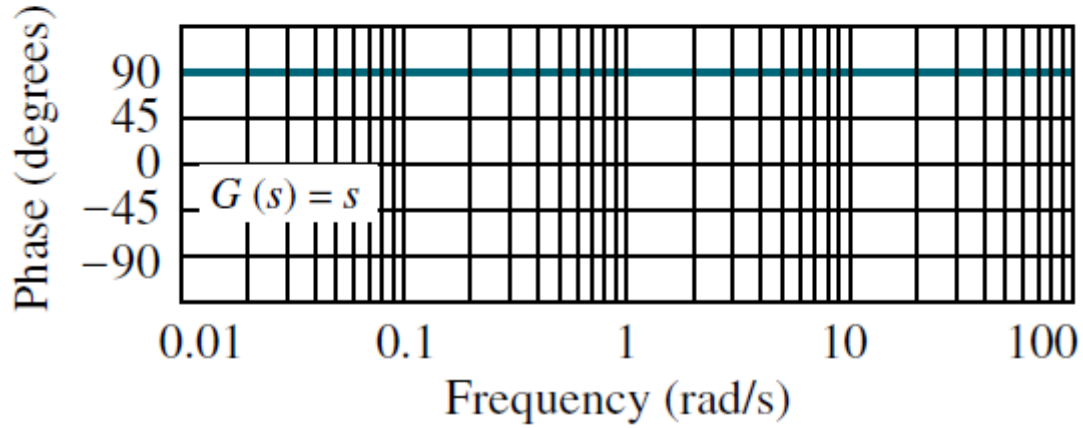
- ***Corner frequency or break point ω = 1 at the magnitude of 0 dB.***

The frequency response of the function ***G(s) = s***, is shown in the Figure.
G(s) = s has only a high-frequency asymptote, where s = jω.

The Bode magnitude plot is a straight line with a +20 dB/decade slope passing through 0 dB at ω = 1.



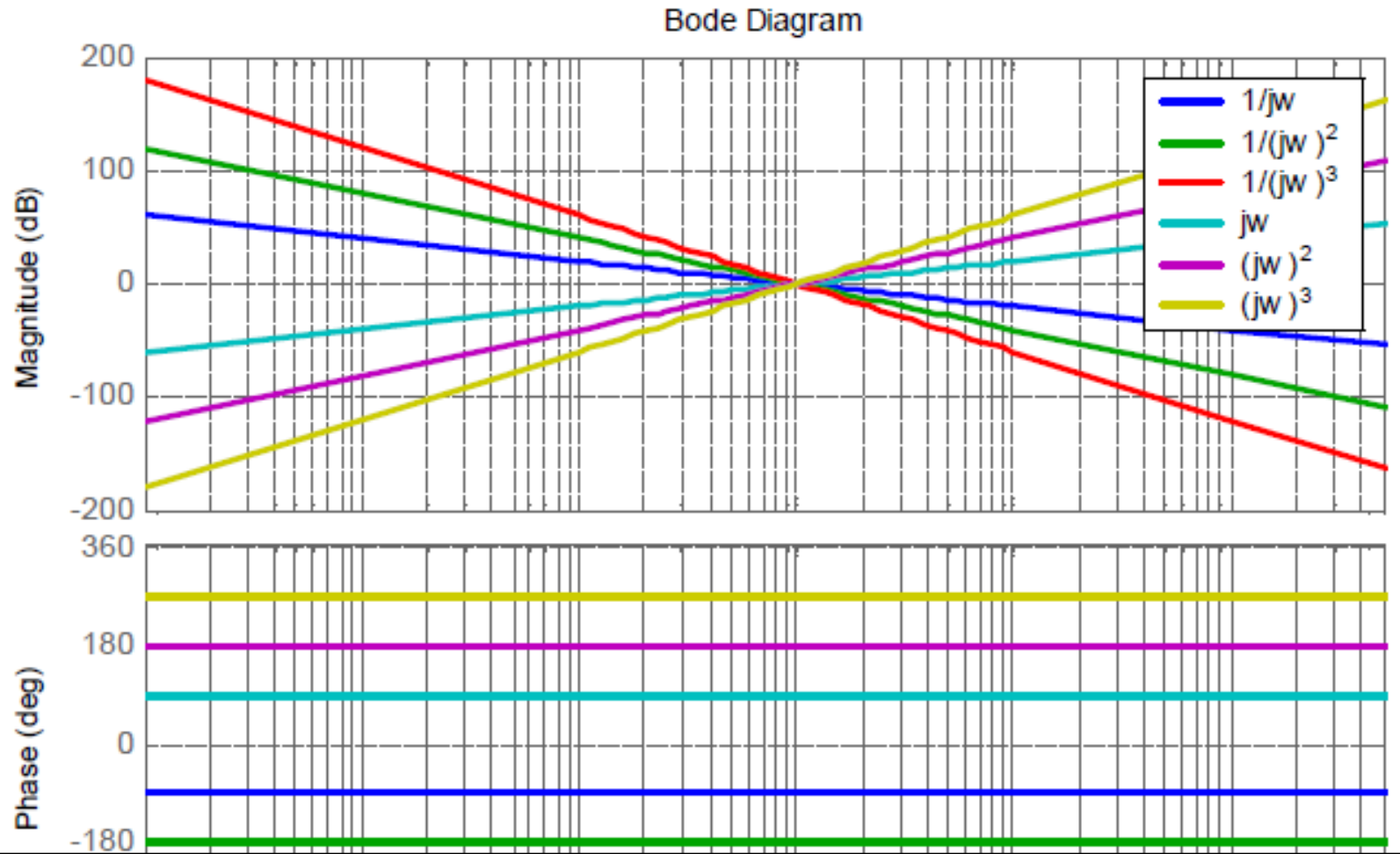
The Bode phase plot is equal to a constant +90°.



Example1 of Class-II (Generalize): The Factor $(j\omega)^{\pm P}$

$\mathrm{GH(s)} = 1/s^2 \longrightarrow \mathrm{GH(j\omega)} = 1/j\omega^2 \longrightarrow \mathrm{M}=1/\omega^2 \longrightarrow LogM = -20\log(1/\omega^2 = -40\mathrm{llog}\omega$

$\Theta = \Theta_1 + \Theta_2 = -90-90 = -180$



Class-III: First Order Factors A lag factor G(s)H(s)=1/(Ts+1)

- If the open loop gain $\boldsymbol{G(j\omega)H(j\omega) = (1 + j\omega T)^{-1} = \frac{1}{1+j\omega T}}$, *where T* is a real constant.

- **Magnitude (dB)** $\boldsymbol{|G(j\omega)H(j\omega)|_{dB} = 20log\left|\frac{1}{1+j\omega T}\right| = 20\,log|1| - 20\,log|1 + j\omega T| = -20log\sqrt{1^2 + \omega^2 T^2}}$

ω_c

Low-Frequency Asymptote (letting frequency S → 0

- When $\omega \ll \frac{1}{T}$, then magnitude $|G(j\omega)H(j\omega)|_{dB} = 20\log(1) = 0$ dB,

The magnitude plot is a horizontal straight line at 0 dB at low frequency (ωT << 1).

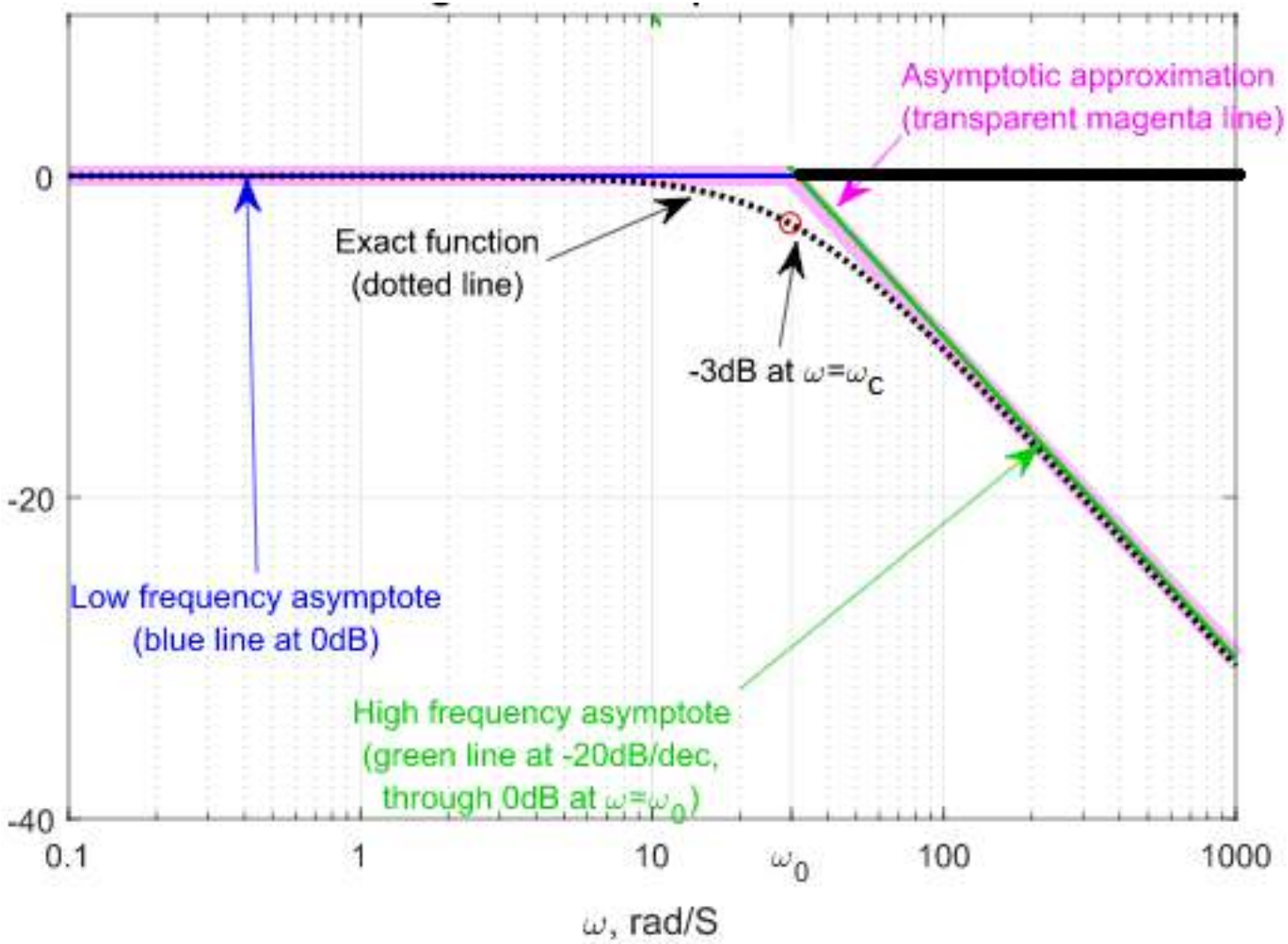
High-Frequency Asymptote (letting frequency s → INF

- When $\omega \gg \frac{1}{T}$, then magnitude $|G(j\omega)H(j\omega)|_{dB} = -20\log(\omega T)$

The magnitude plot is a straight line with a slope of -20 dB/decade at high frequency (ωT >> 1).

At corner frequency the asymptotes are equal $0=-20\log T\omega_c$

$\omega_c=1/T$



$$\angle H(j\omega) = -\arctan\left(\frac{\omega}{\omega_c}\right)$$

Low-Frequency

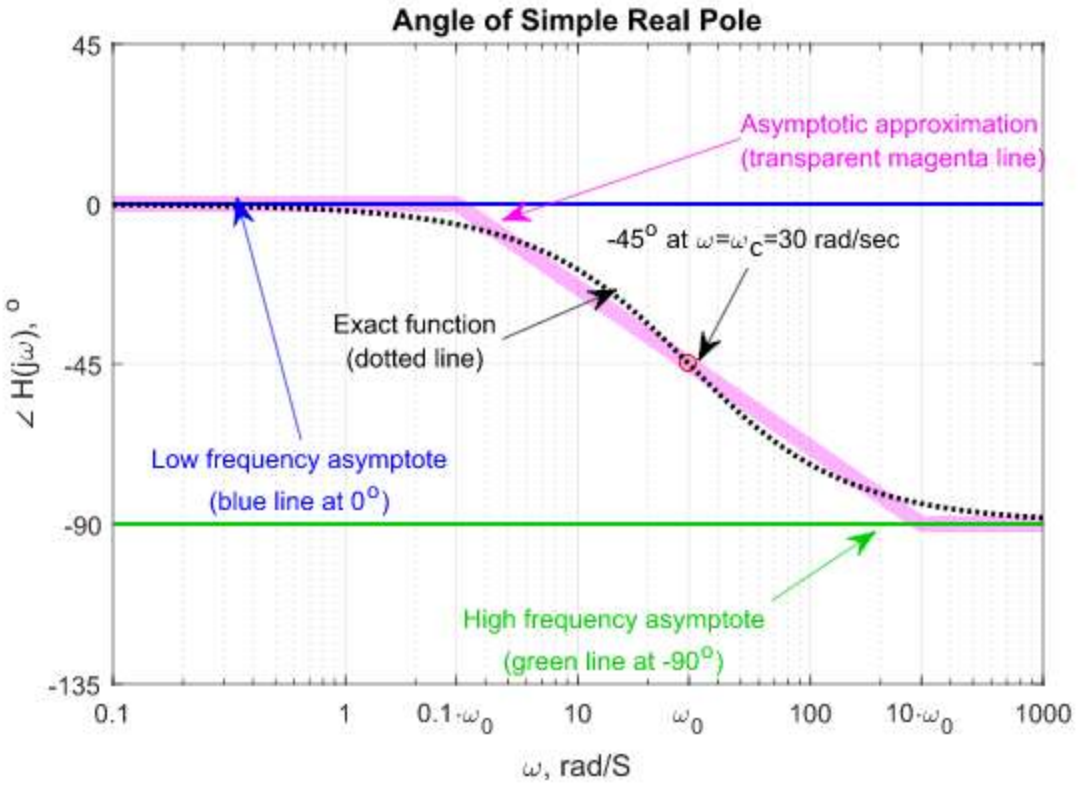
$$\angle H(j\omega) \approx -\arctan(0) = 0^\circ = 0\ rad$$

High-Frequency Asy

$$\angle H(j\omega) \approx -\arctan(\infty) = -90^\circ = -\frac{\pi}{2}\ rad$$

Corner-Frequency

$$\angle H(j\omega) = -\arctan(1) = -45^\circ = -\frac{\pi}{4}\ rad$$



Magnitude of a real pole: The piecewise linear asymptotic Bode plot for magnitude is at 0 dB until the break frequency and then drops at 20 dB per decade as frequency increases (i.e., the slope is -20 dB/decade).

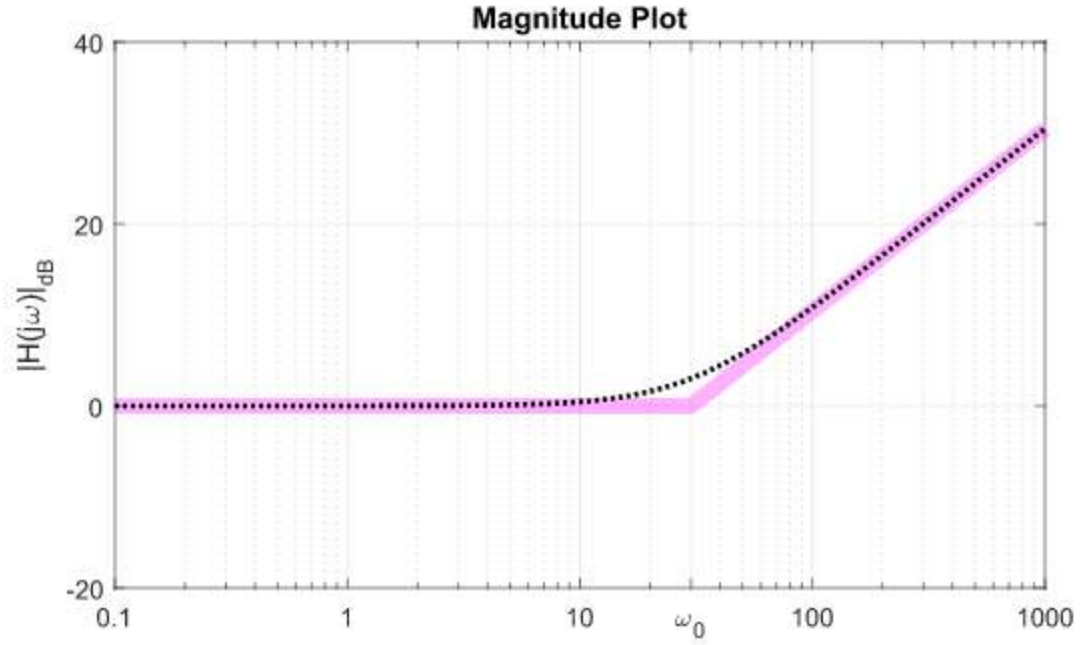
Phase of a real pole: The piecewise linear asymptotic Bode plot for phase follows the low frequency asymptote at 0° until one tenth the break frequency ($0.1 \cdot \omega_c$) then decrease linearly to meet the high frequency asymptote at ten times the break frequency ($10 \cdot \omega_c$). This line is shown above. Note that there is no error at the break frequency and about 5.7° of error at $0.1 \cdot \omega_c$ and $10 \cdot \omega_c$ the break frequency.

Class-III: First Order Factors B: Leed factor G(s)H(s)= Ts+1

Magnitude (dB) $|G(j\omega)H(j\omega)|_{dB} = 20log\sqrt{1^2 + \omega^2T^2}$

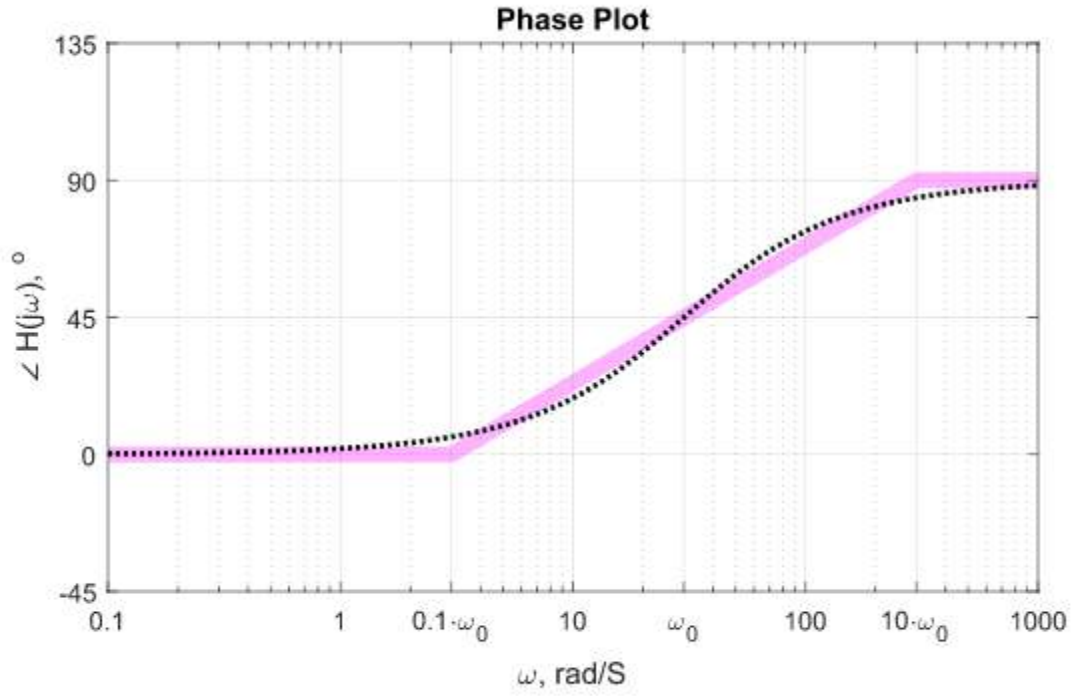
Phase $\angle G(j\omega)H(j\omega) = \tan^{-1}\omega T$

Again, as with the case of the real pole, there are three cases:
1.At low frequencies, $\omega << \omega_0$, the gain is approximately 1 (or 0 dB).
2.At high frequencies, $\omega >> \omega_0$, the gain increases at 20 dB/decade and goes through the break frequency at 0 dB.
3.At the break frequency, $\omega = \omega_c$, the gain is about 3 dB.



The phase of a single real zero also has three cases (which can be derived similarly to those for the real pole, given above):

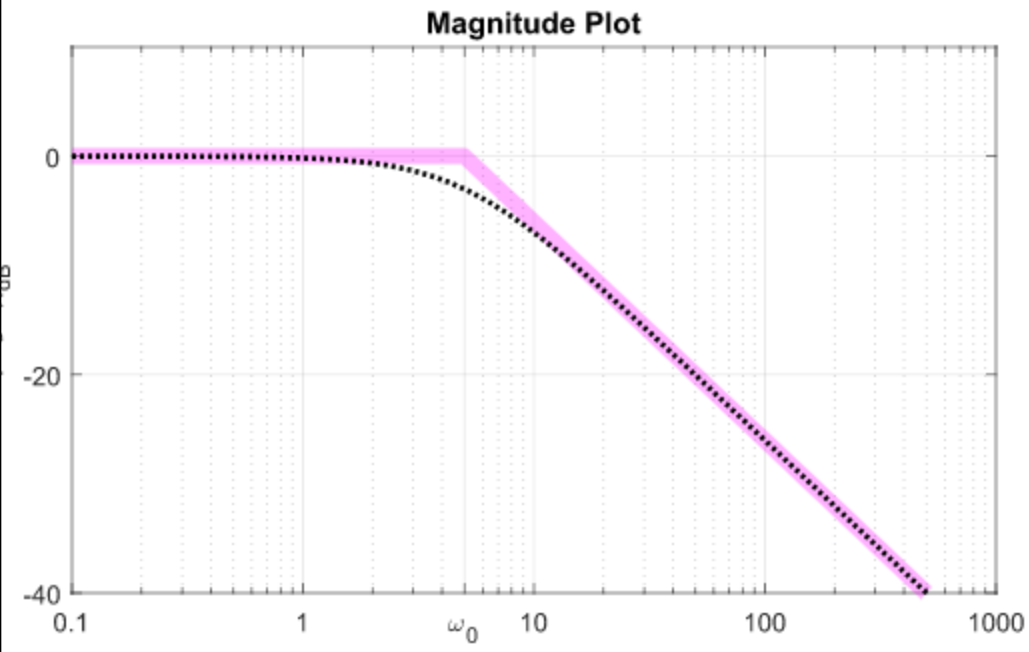
1. At low frequencies, $\omega << \omega_c$, the phase is approximately zero.
2. At high frequencies, $\omega >> \omega_c$, the phase is +90°.
3. At the break frequency, $\omega = \omega_c$, the phase is +45°.

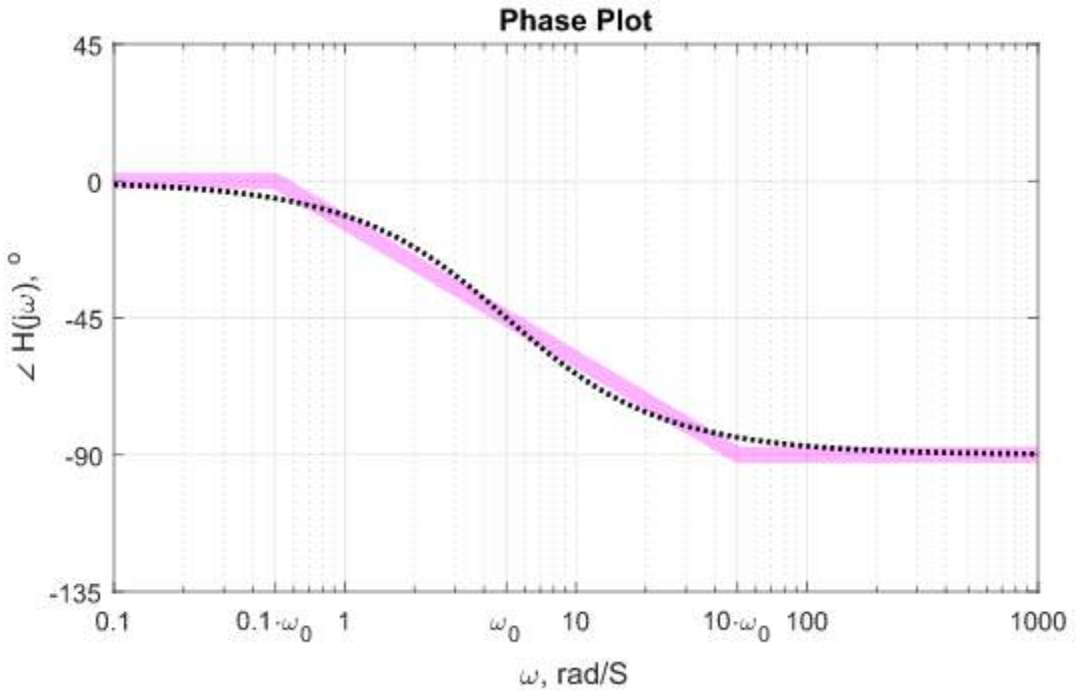


Key Concept: Bode Plot of Real Zero:

- The plots for a real zero are like those for the real pole but mirrored about 0dB or 0°.
- For a simple real *zero* the piecewise linear asymptotic Bode plot for magnitude is at 0 dB until the break frequency and then *rises* at +20 dB per decade (i.e., the slope is +20 dB/decade). An nth order *zero* has a slope of +20·n dB/decade.
- The phase plot is at 0° until one tenth the break frequency and then *rises* linearly to +90° at ten times the break frequency. An nth order *zero rises* to +90°·n.

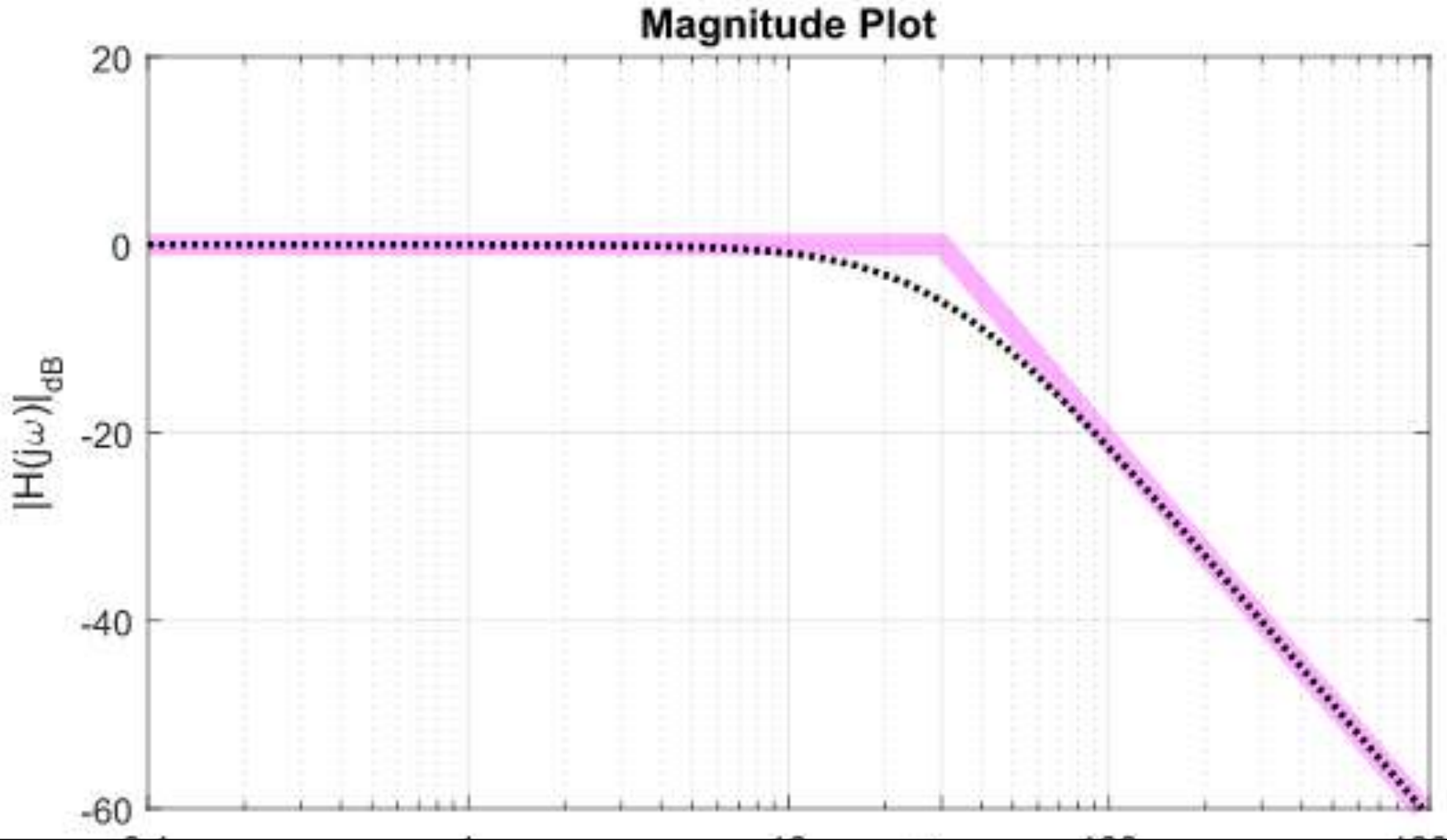
Example :The first example is a simple pole at 5 radians per second. .G(s)=1/(1+s/5)

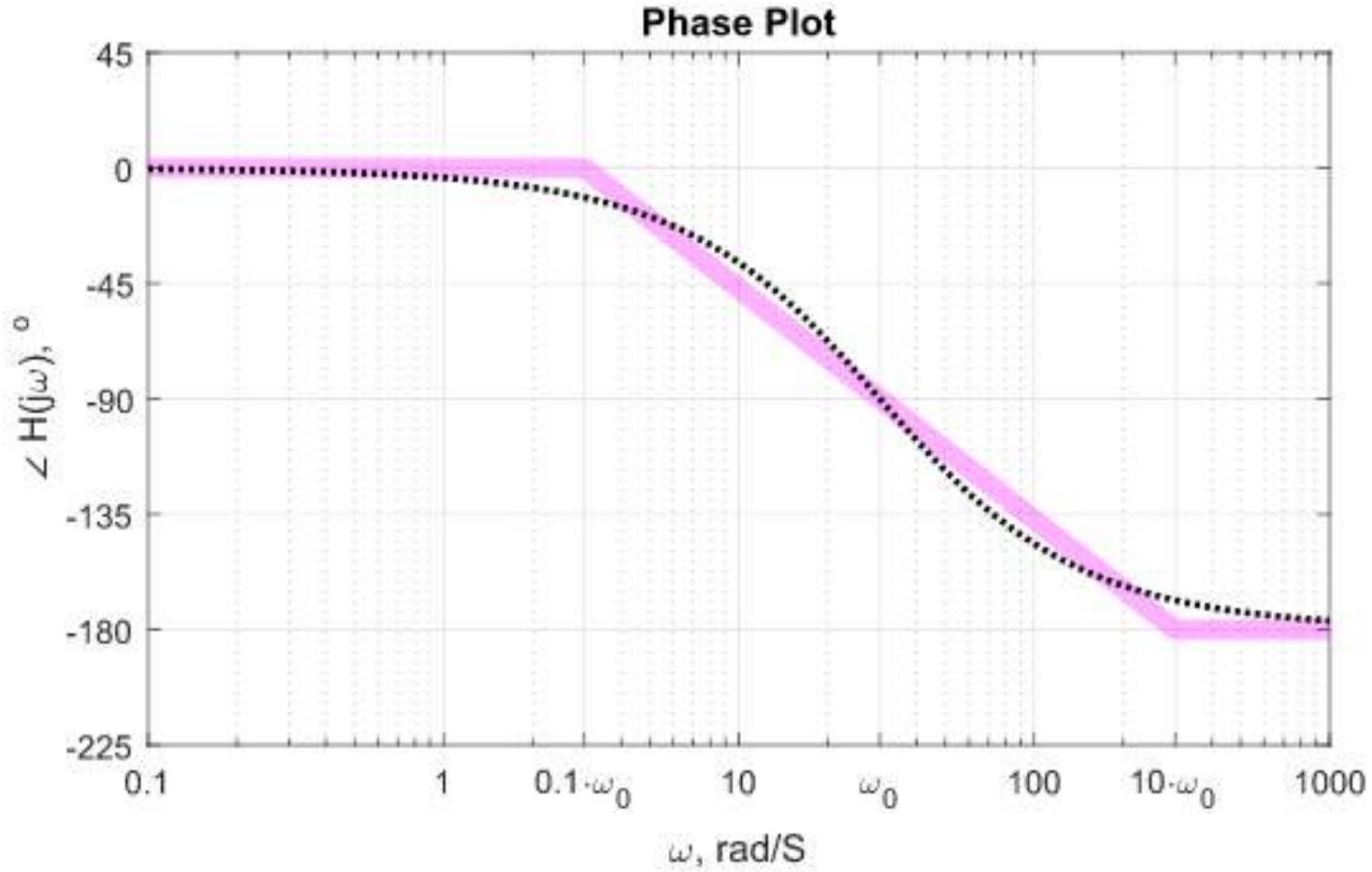




Example

$$H(s) = \frac{1}{\left(1 + \frac{s}{30}\right)^2}$$





Key Concept: Bode Plot for Real Pole

•For a simple real pole the piecewise linear asymptotic Bode plot for magnitude is at 0 dB until the break frequency and then drops at 20 dB per decade (i.e., the slope is -20 dB/decade). An nth order pole has a slope of -20·n dB/decade.

•The phase plot is at 0° until one tenth the break frequency and then drops linearly to -90° at ten times the break frequency. An nth order pole drops to -90°·n.

Example-5: Obtain the Bode plot of the system given by the transfer function;

$$G(s) = \frac{1}{2s+1}$$

- We convert the transfer function in the following format by substituting $s = j\omega$

- We call $\omega = 1/2$, the break point or corner frequency. So for
- So when $\omega << 1$, (i.e., for small values of ω), then $G(j\omega) \approx 1$
- Therefore taking the log magnitude of the transfer function for very small values of ω, we get
- Hence below the break point, the magnitude curve is approximately a constant.
- So when $\omega >> 1$, (i.e., for very large values of ω), then

Quadratic Factors $[1 + 2\zeta(j\omega/\omega_n) + (j\omega/\omega_n)^2]^{\mp 1}$. Control systems often possess quadratic factors of the form

$$G(j\omega) = \frac{1}{1 + 2\zeta\left(j\frac{\omega}{\omega_n}\right) + \left(j\frac{\omega}{\omega_n}\right)^2}$$

The asymptotic frequency-response curve may be obtained as follows: Since

$$20\log\left|\frac{1}{1 + 2\zeta\left(j\frac{\omega}{\omega_n}\right) + \left(j\frac{\omega}{\omega_n}\right)^2}\right| = -20\log\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + \left(2\zeta\frac{\omega}{\omega_n}\right)^2}$$

or low frequencies such that $\omega \ll \omega_n$, the log magnitude becomes

$$-20\log 1 = 0 \text{ dB}$$

The low-frequency asymptote is thus a horizontal line at 0 dB. For high frequencies such that $\omega \gg \omega_n$, the log magnitude becomes

$$-20 \log \frac{\omega^2}{\omega_n^2} = -40 \log \frac{\omega}{\omega_n} \text{ dB}$$

This frequency, ω_n, is the corner frequency for the quadratic factor considered.

The magnitude asymptotes intersect at the 0dB line when $u=\omega/\omega_n=1$.

$$20 \log|G(\omega)| = -20 \log(2\zeta)$$

If u=1,

The difference between the actual magnitude curve and the asymptotic approximation is a function of ζ. The max value of the frequency response occurs at the resonant frequency, ω_r.

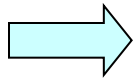
When $\zeta \to 0$, $\omega_r \to \omega_n$.

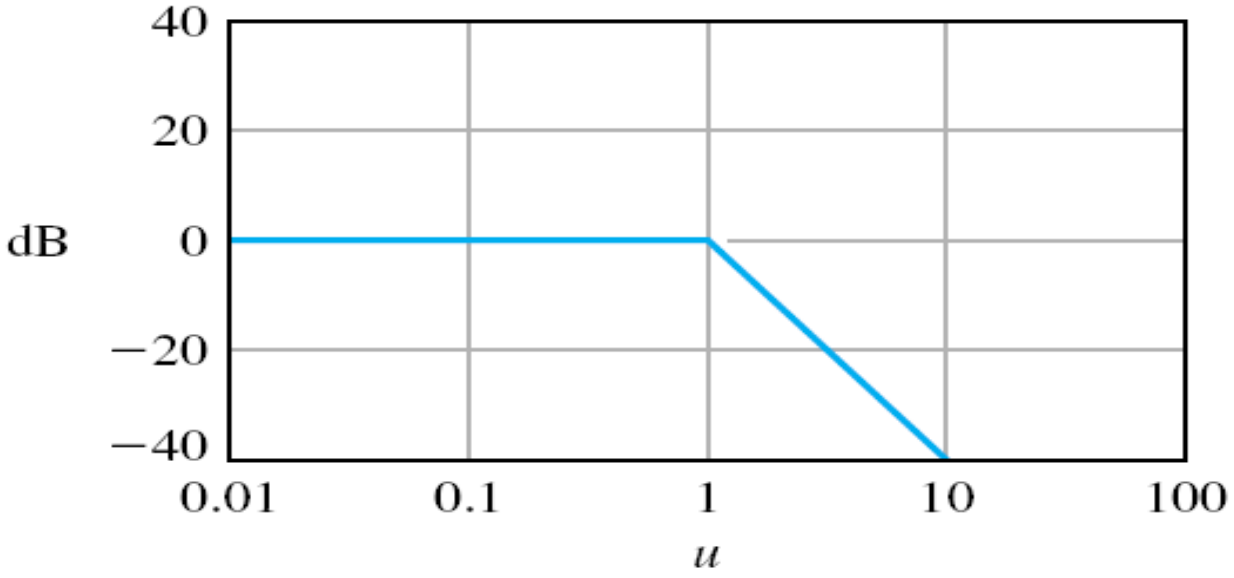
When u<<1, the log magnitude is -10log1=0 dB, and the phase angle approaches 0^0. When u>>1, the log magnitude approaches -10log($u^{4)}$=-40logu, which results in a curve with a slope of -40dB/decade. The phase angle, when u>>1, approaches -180^0. The magnitude asymptotes meet at the 0dB line when

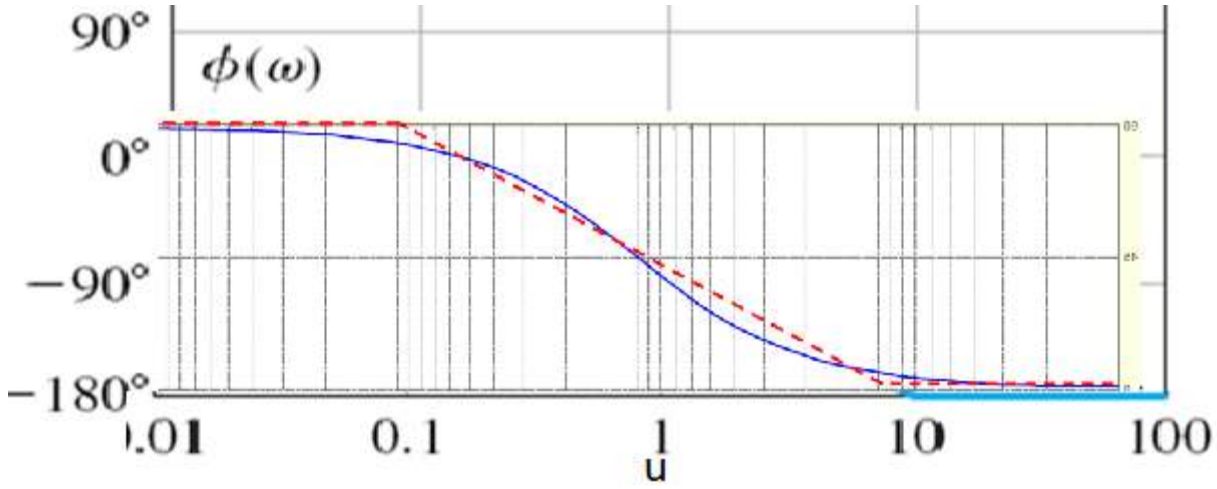
$$u = \frac{\omega}{\omega_n} = 1.$$

asymptotes for

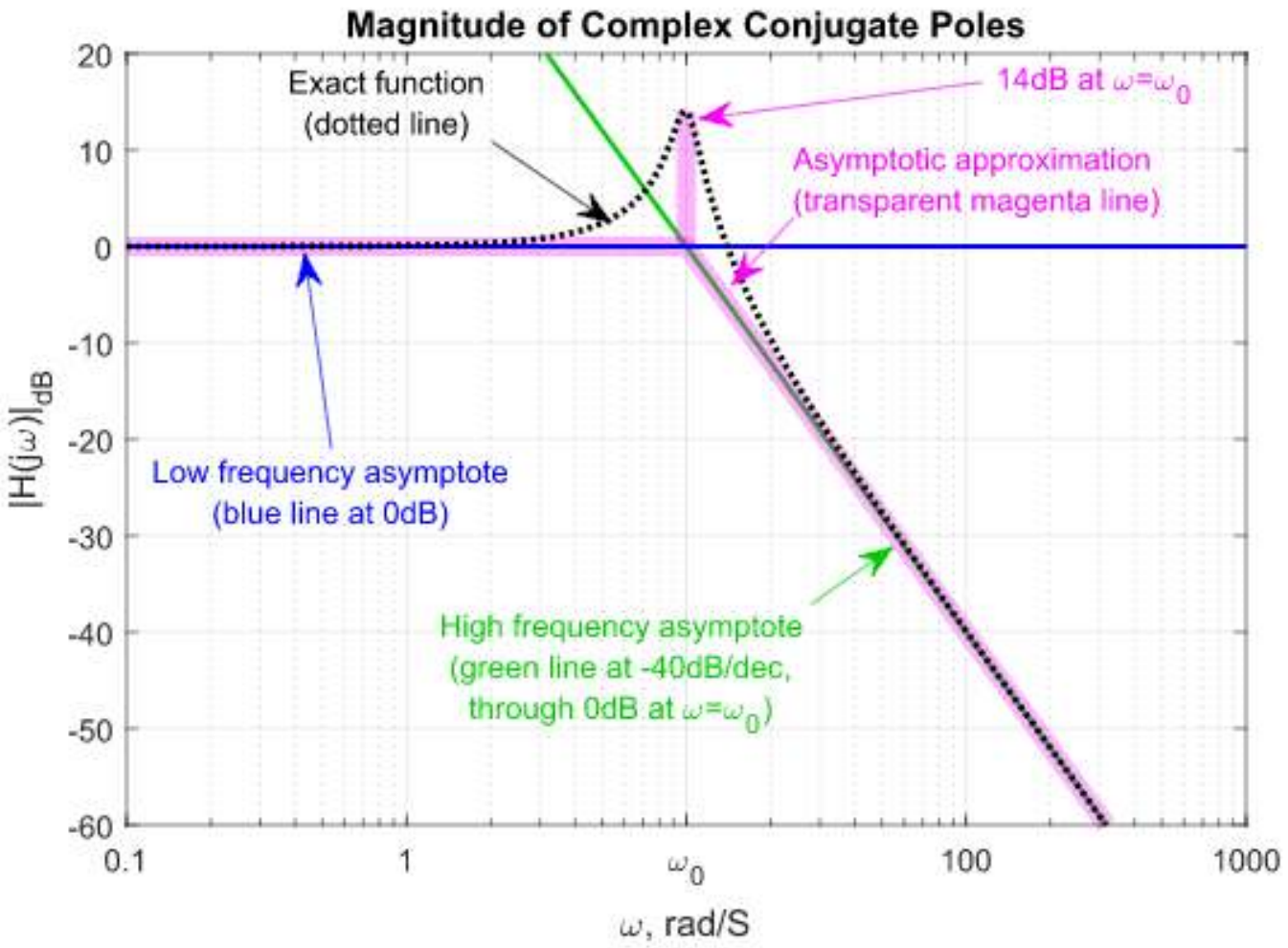
$[1 + j2\zeta u - u^2]^{-1}$,
where $u = \omega/\omega_n$

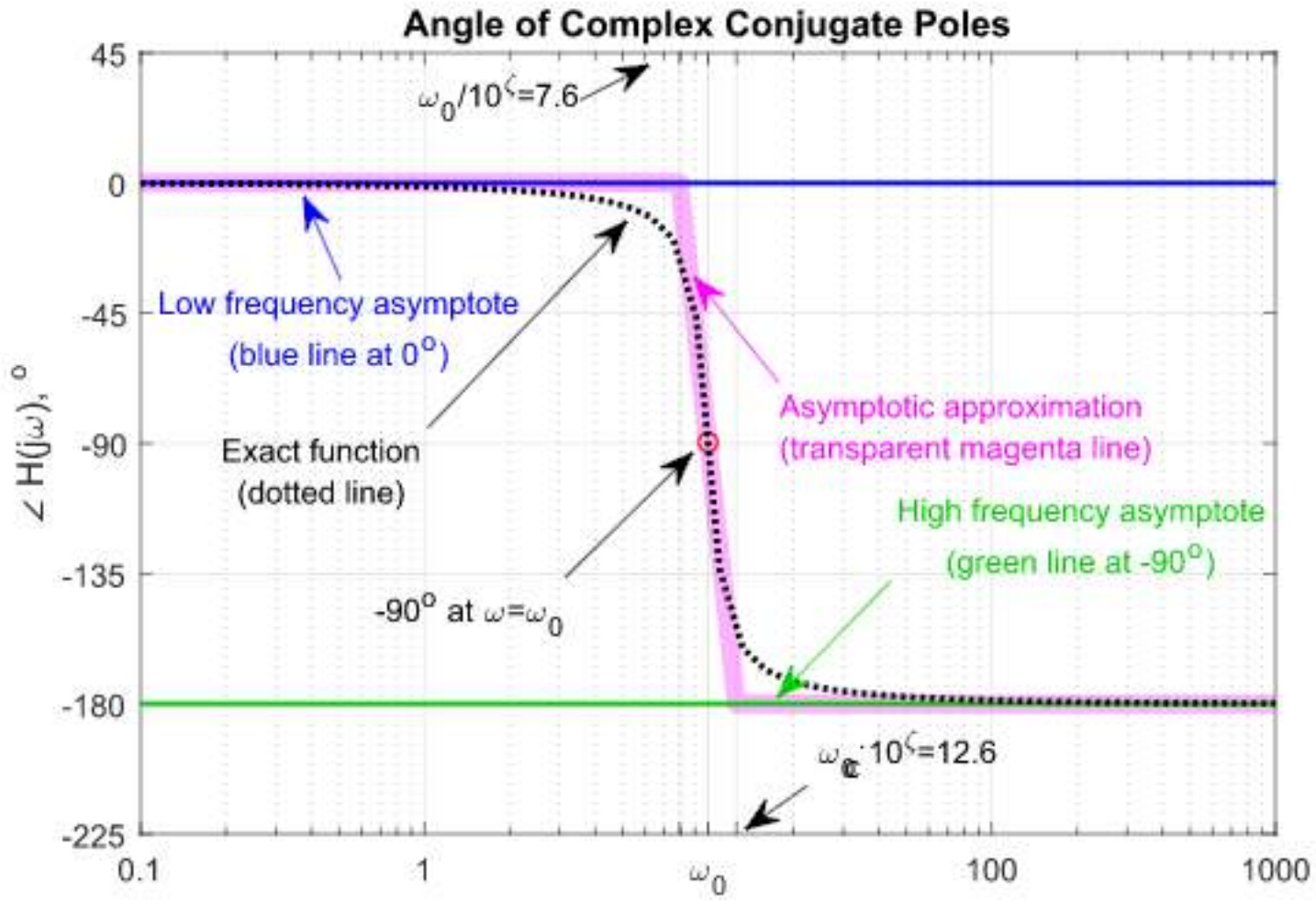






The peak will have an amplitude of 1/(2·ζ)=5.00 or 14 dB.



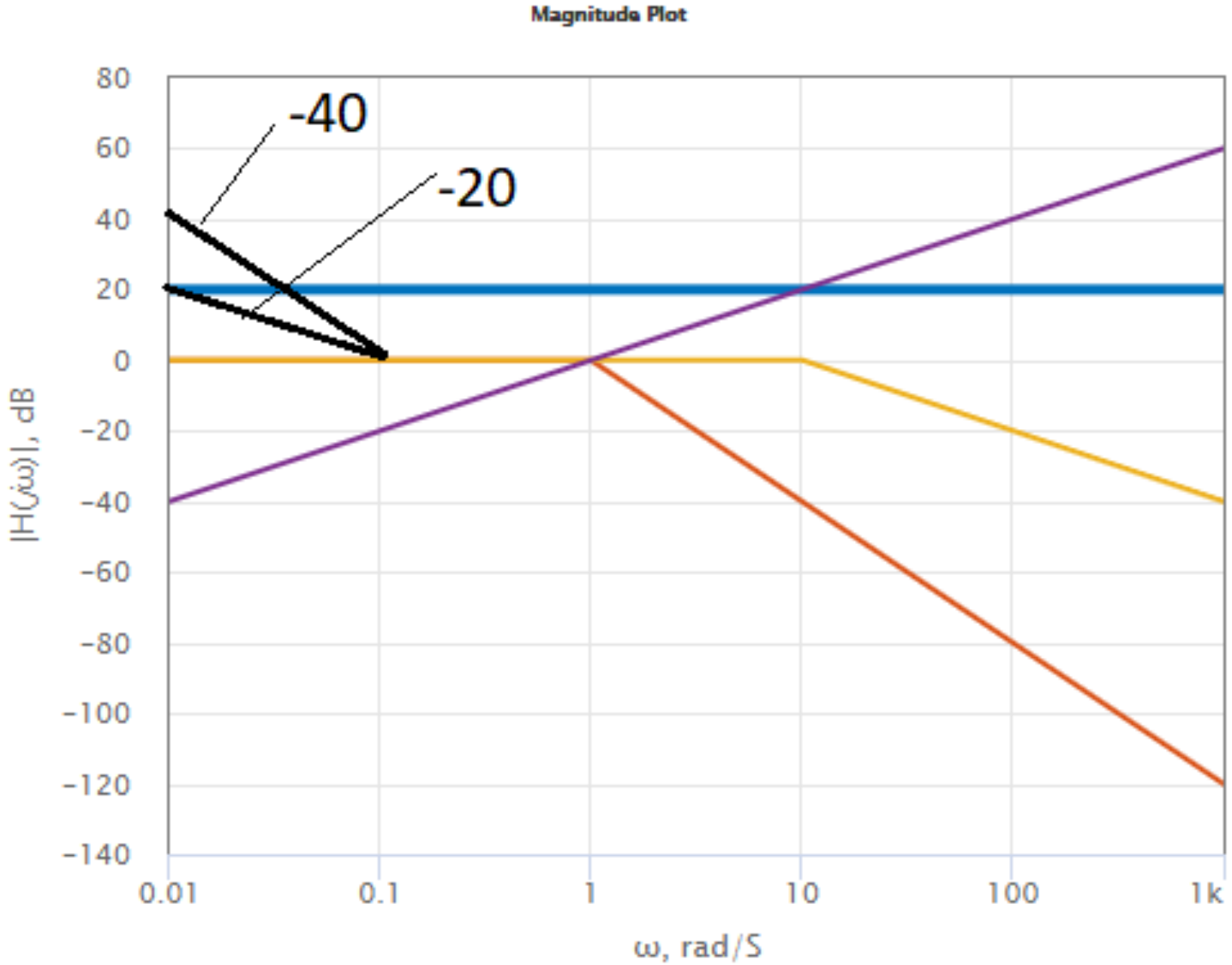

Angle of Complex Conjugate Poles
45
0
-45
-90
-135
-180
-225
0.1
1
100
1000
ω_0
$\angle H(j\omega)$, °
$\omega_0/10^{\zeta}$=7.6
Low frequency asymptote
(blue line at 0°)
Exact function
(dotted line)
-90° at $\omega=\omega_0$
Asymptotic approximation
(transparent magenta line)
High frequency asymptote
(green line at -90°)
$\omega_0 \cdot 10^{\zeta}$=12.6

Draw a Bode diagram for

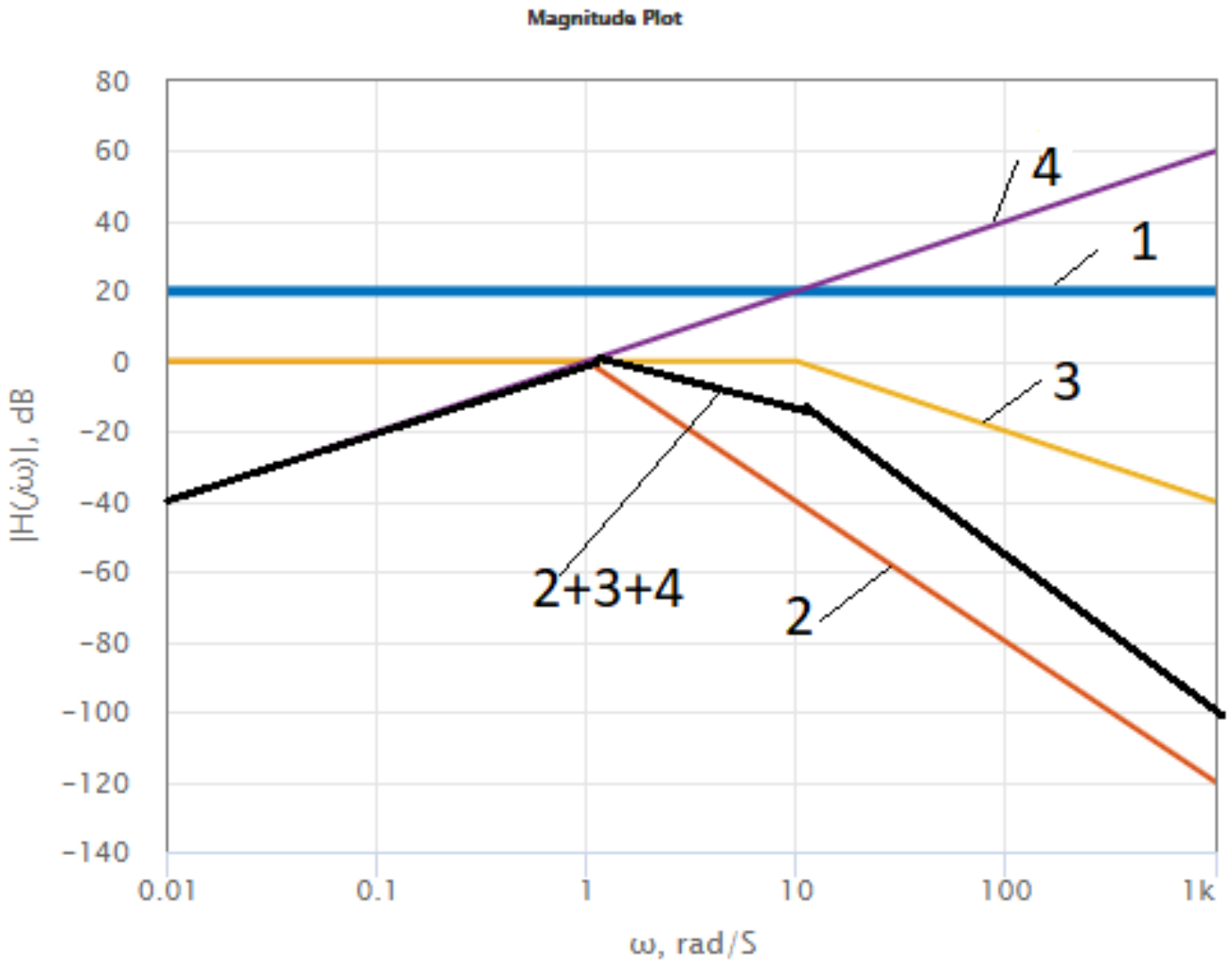
$$G(s)H(s) = \frac{10s}{(1 + s/\omega_{p1})^2(1 + s/\omega_{p2})}$$

Individual Transfer Function Terms:

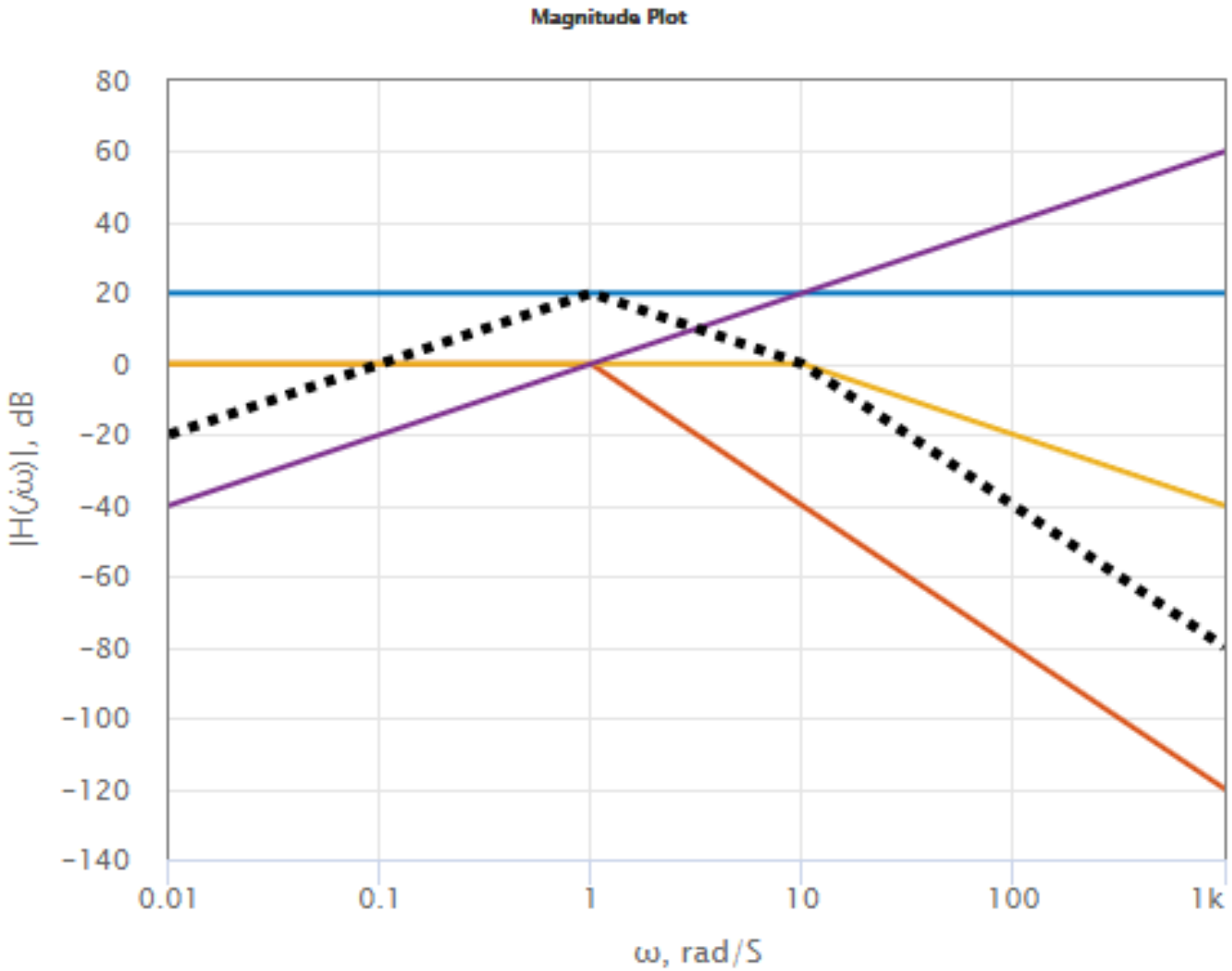
- (●) Constant K=20.0 dB
- ( ) Real Pole, ω_{p1}= 1.00, of muliplicity 2
- ( ) Real Pole, ω_{p2}= 10.0
- ( ) Zero at Origin


Magnitude Plot
-40
-20
80
60
40
20
0
-20
-40
-60
-80
-100
-120
-140
|H(jω)|, dB
0.01
0.1
1
10
100
1k
ω, rad/S

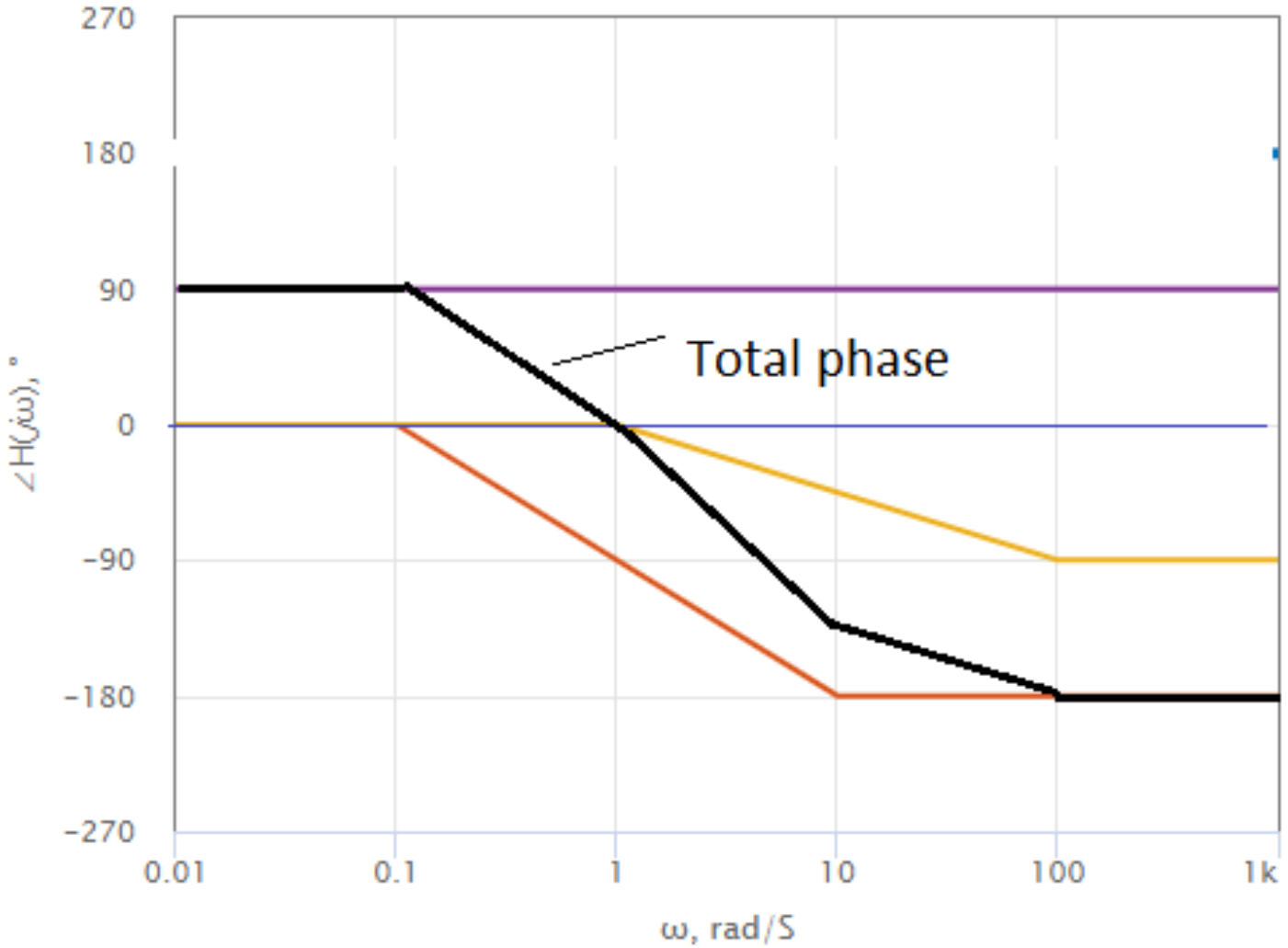
Sum of all plots except the gain



Add the gain to obtained the total plot



Phase plot



Nyquist Plot or Polar Plot

- Nyquist Plots were invented by Nyquist - who worked at Bell Laboratories, the premiere technical organization in the U.S. at the time.
- Nyquist Plots are a way of showing frequency responses of linear systems.
- There are several ways of displaying frequency response data, including Bode' plots and Nyquist plots.
- Bode' plots use frequency as the horizontal axis and use two separate plots to display amplitude and phase of the frequency response.
- Nyquist plots display both amplitude and phase angle on a single plot, using frequency as a parameter in the plot.
- Nyquist plots have properties that allow you to see whether a system is stable or unstable.

- A Nyquist plot is a polar plot of the frequency response function of a linear system.
- That means a Nyquist plot is a plot of the transfer function, G(s) with s = j**ω**. That means you want to plot G(j **ω**).
- G(j **ω**) is a complex number for any angular frequency, **ω**, so the plot is a plot of complex numbers.
- The complex number, G(j **ω**), depends upon frequency, so frequency will be a parameter if you plot the imaginary part of G(j **ω**) against the real part of G(j **ω).**

Sketch the Polar plot of Frequency Response

To sketch the polar plot of $G(j\omega)$ for the entire range of frequency ω, i.e., from 0 to infinity, there are four key points that usually need to be known:

1) The start of plot where $\omega = 0$,
2) The end of plot where $\omega = \infty$,
3) Where the plot crosses the real axis, i.e., $\mathrm{Im}(G(j\omega)) = 0$, and
4) Where the plot crosses the imaginary axis, i.e., $\mathrm{Re}(G(j\omega)) = 0$.

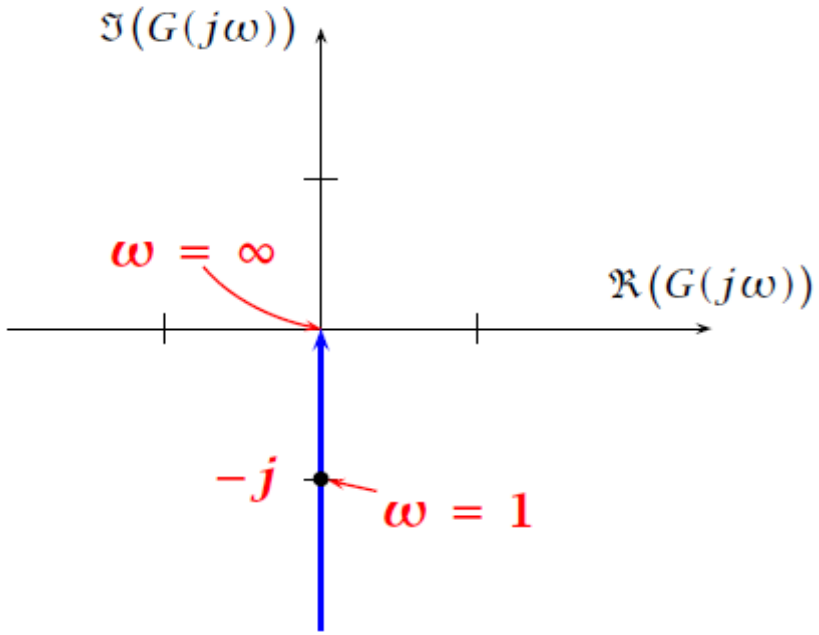
Problem-1: Polar Plot of Integrator

$$G(s) = \frac{1}{s}$$ Representing $G(s)$ in the frequency response form $G(j\omega)$ by replacing $s = j\omega$:

$$G(j\omega) = \frac{1}{j\omega} = \frac{-j}{\omega}$$

The **magnitude** of $G(j\omega)$, i.e., $|G(j\omega)|$, is obtained as; $|G(j\omega)| = 1/\omega$

The **phase** of $G(j\omega)$, denoted by, ϕ , is obtained as; $\angle G(j\omega) = -90^\circ$



Problem-2: Polar Plot of First Order System

Consider a first order system $G(s) = \dfrac{1}{1+sT}$

Representing $G(s)$ in the frequency response form $G(j\omega)$ by replacing **s = jω**:

$$G(j\omega) = \frac{1}{1+j\omega T}$$

The **magnitude** of $G(j\omega)$, i.e., $|G(j\omega)|$, is obtained as; $|G(j\omega)| = \dfrac{1}{\sqrt{1+\omega^2 T^2}}$

The **phase** of $G(j\omega)$, denoted by, ϕ , is obtained as; $\angle G(j\omega) = -\arctan(\omega T)$

The start of plot where ω = 0

$$|G(j\omega)| = \frac{1}{\sqrt{1+0}} = 1, \qquad \phi = \tan^{-1}\left(\frac{0}{1}\right) = 0$$

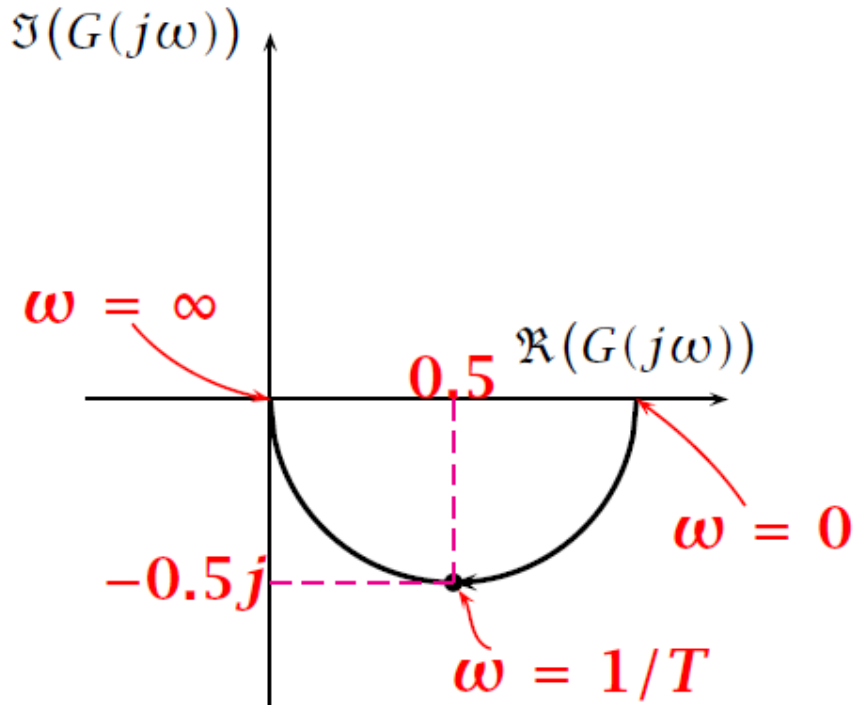
The end of plot where ω = ∞

$$|G(j\omega)| = \frac{1}{\sqrt{1+\infty}} = 0, \qquad \phi = \tan^{-1}\left(\frac{-\infty}{1}\right) = -90^\circ$$

The mid part of plot where ω = 1/T

$$|G(j\omega)| = \frac{1}{\sqrt{1+1}} = \frac{1}{\sqrt{2}}, \qquad \phi = \tan^{-1}\left(\frac{-1}{1}\right) = -45^\circ$$

	$\lvert G(j\omega)\rvert$	$\angle G(j\omega)$
$\omega = 0$	1	0
$\omega = \frac{1}{T}$	$\frac{1}{\sqrt{2}}$	$-45°$
$\omega = \rightarrow \infty$	0	$-90°$



Problem-3: Polar Plot of Second Order System

Consider a second order system

$$G(s) = \frac{1}{(1 + sT_1)(1 + sT_2)}$$

Representing $G(s)$ in the frequency response form $G(j\omega)$ by replacing **$s = j\omega$**:

$$G(j\omega) = \frac{1}{(1 + j\omega T_1)(1 + j\omega T_2)}$$

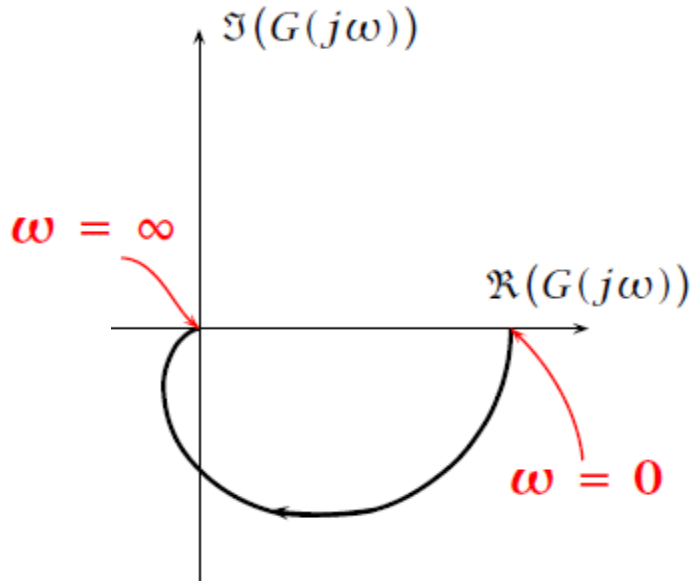
The **magnitude** of $G(j\omega)$, i.e., $|G(j\omega)|$, is obtained as;

$$|G(j\omega)| = \frac{1}{\sqrt{1 + \omega^2 T_1^2}\sqrt{1 + \omega^2 T_2^2}}$$

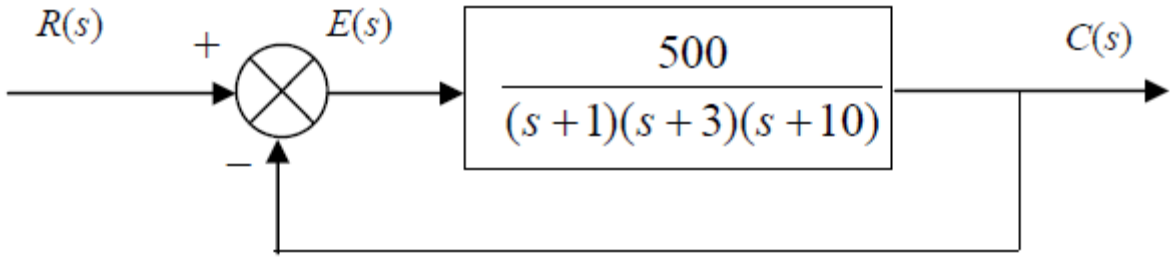
The **phase** of $G(j\omega)$, denoted by, φ , is obtained as;

$$\angle G(j\omega) = -\arctan(\omega T_1) - \arctan(\omega T_2)$$

	$\lvert G(j\omega)\rvert$	$\angle G(j\omega)$
$\omega = 0$	1	0
$\omega = \rightarrow \infty$	0	-180°



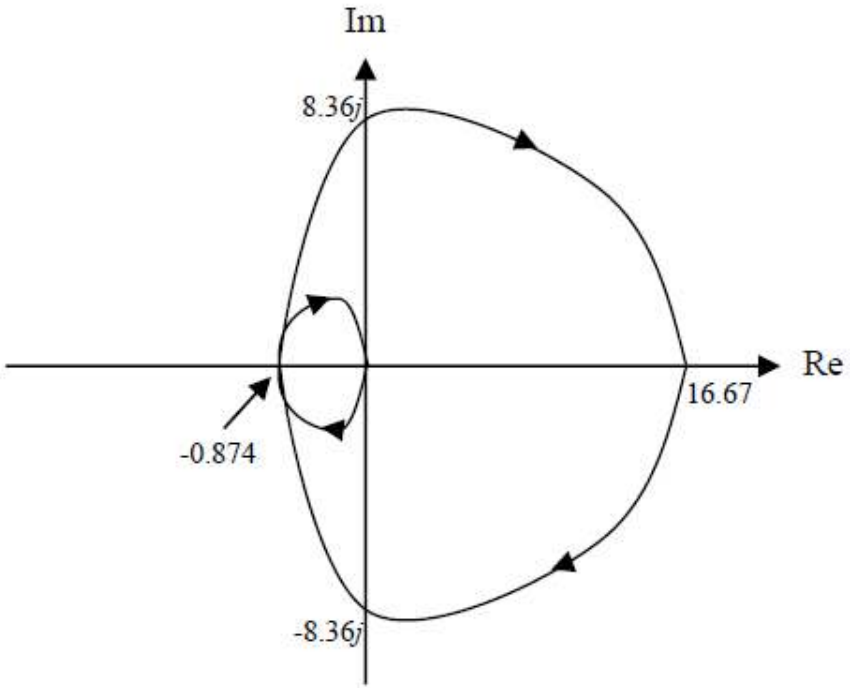
Example :Plot the Nyquist plot for the open loop system



Key Points of the polar plot:

	$\lvert GH \rvert$	$\angle GH$
$\omega = 0$	16.67	0
$\omega = \infty$	0	-270°
Cross Re: $\omega = 0$ $\omega = \infty$ $\omega = 6.56$ rad/s	 See above 0.874	 See above -180°
Cross Re: $\omega = 0$ $\omega = \infty$ $\omega = 1.46$ rad/s	 See above 8.36	 See above -90°

Nyquist diagram



Four Important Points for Derivation the Nyquist Stability Criterion

1. The concept of mapping points;

2. The concept of mapping contours.

3. The relationship between the poles of $1 + G(s)H(s)$ and the poles of $G(s)H(s)$;

4. The relationship between the zeros of $1 + G(s)H(s)$ and the poles of the closed-loop transfer function, $T(s)$;

The Concept of Mapping Points

- If we take a complex number s = x+*j*y on the s-plane and substitute it into a function, F(s), another complex number results. This process is called mapping.

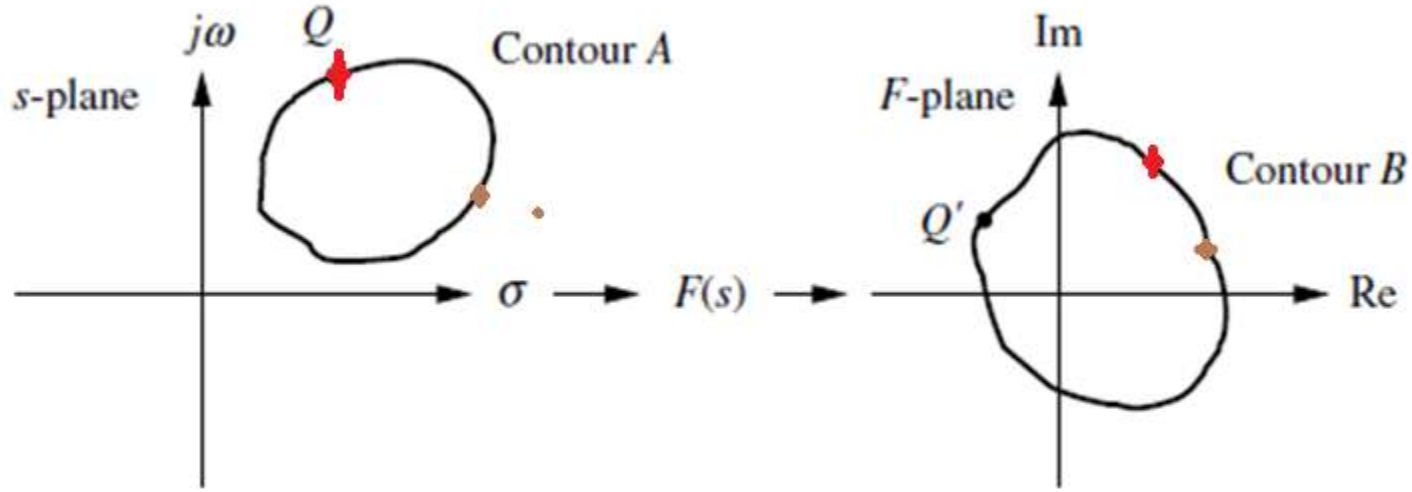
OR

- The term mapping is defined as the substitution of a complex number into a function, F(s), to get another complex number.

- **For example**, substituting **s = 4 + j3** into the function **F(s) = (s^2 +2s + 1)** yields **16+j30**. We say that 4 + j3 maps into 16+j30 through the function s^2 +2s + 1.

The Concept of Mapping Contours

The collection of points, called a **contour**.

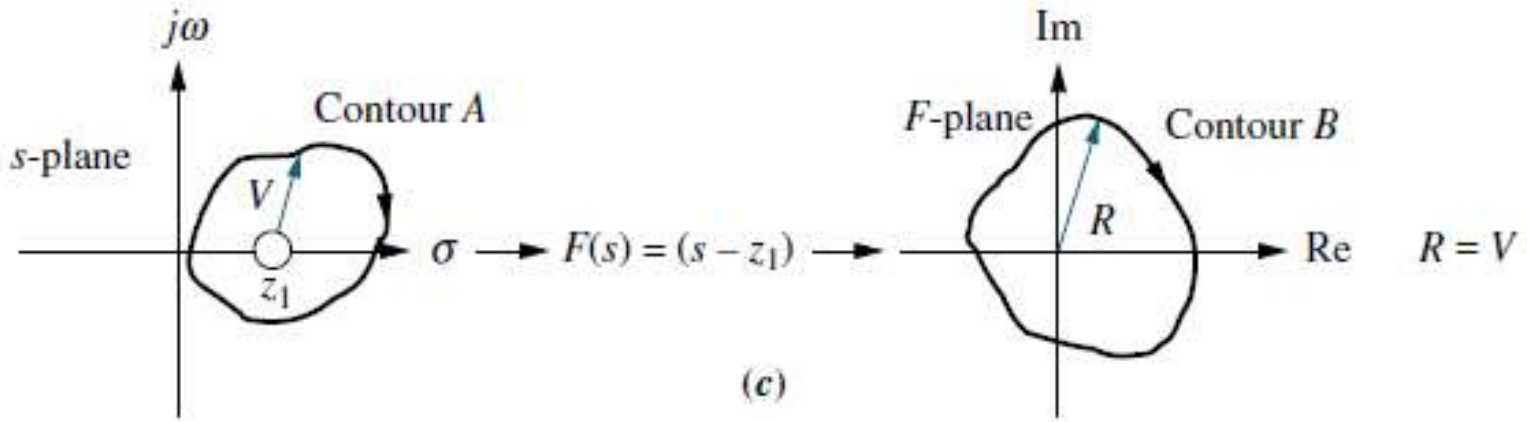


Mapping contour A through function $F(s)$ to contour B

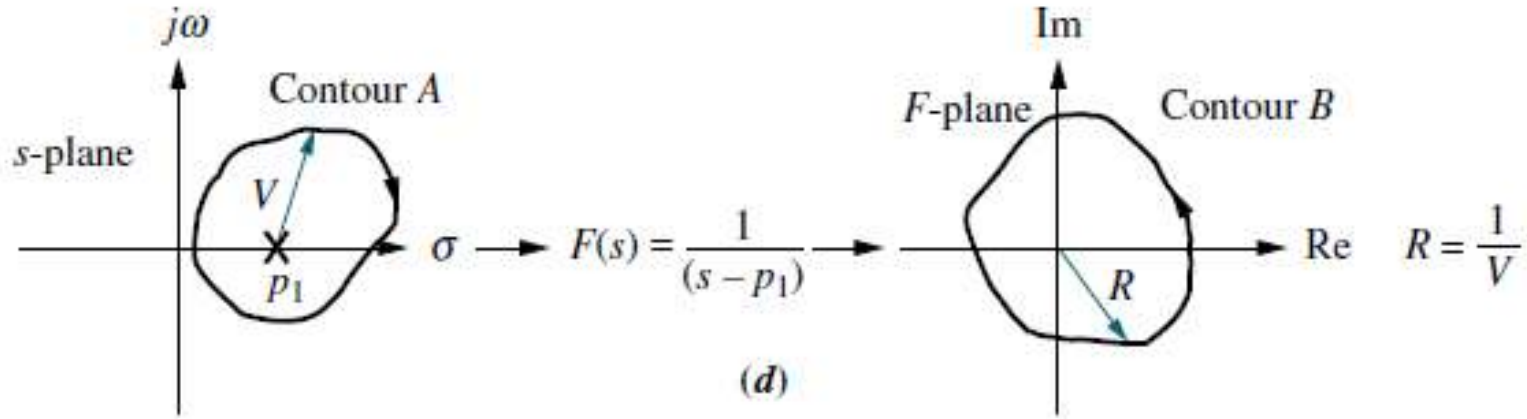
Contour A can be mapped through $F(s)$ into contour B by substituting each point of contour A into the function $F(s)$ and plotting the resulting complex numbers. For example, point Q in Figure 10.21 maps into point Q' through the function $F(s)$.

Examples of Contour Mapping

The **contour B** maps in a counter clockwise direction if **F(s)** has just poles that are encircled by the contour, Also, you should verify that, if the pole or zero of **F(s)** is enclosed by **contour A**, the mapping encircles the origin .



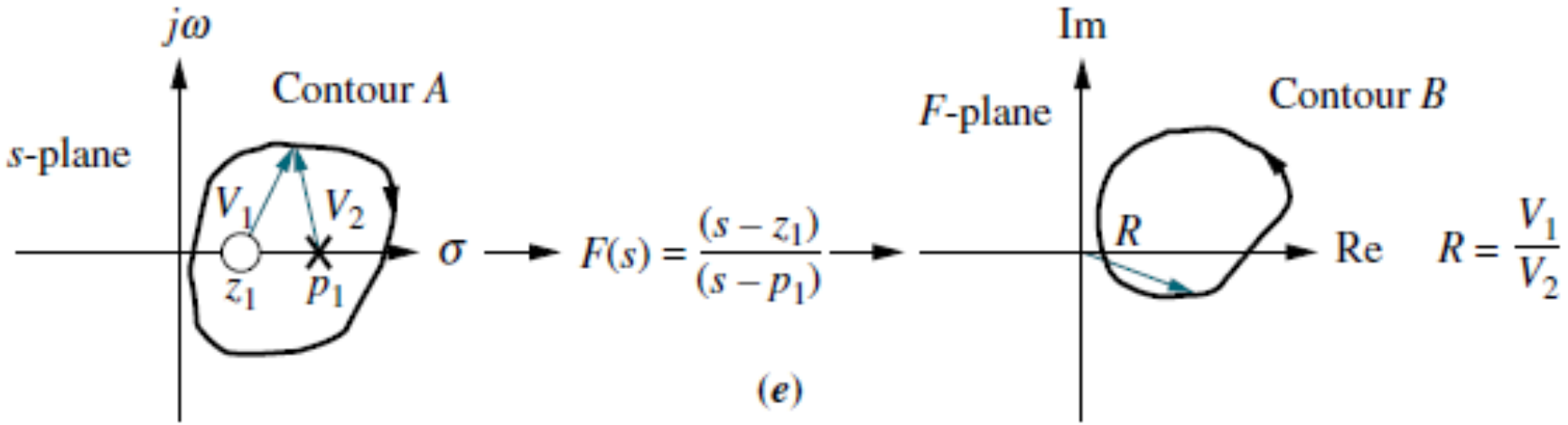
(c)



(d)

Examples of Contour Mapping

In this last case, the pole and zero rotation cancel, and the mapping does not encircle the origin.



The number of encirclements of the origin of F(s) contour is N=Z-P;
Where Z number of zeros and P number of poles of F(s) inside s contour

The Function F(s) for Stability

Letting

$$G(s) = \frac{N_G}{D_G}$$

$$H(s) = \frac{N_H}{D_H}$$

we find

$$G(s)H(s) = \frac{N_G N_H}{D_G D_H}$$

$$F(s) = 1 + G(s)H(s) = 1 + \frac{N_G N_H}{D_G D_H} = \frac{D_G D_H + N_G N_H}{D_G D_H} \quad \text{—} \quad \frac{\text{Z -closd loop poles}}{\text{P-open loop poles}}$$

$$T(s) = \frac{G(s)}{1 + G(s)H(s)} = \frac{N_G D_H}{D_G D_H + N_G N_H}$$

1) The poles of 1 + G(s)H(s) are the same as the poles of G(s)H(s), the open-loop system.

2) The zeros of 1 + G(s)H(s) are the same as the poles of T(s), the closed-loop system.

F(s) combine the characteristic equations of the open loop and closed loop systems

THE NYQUIST STABILITY CRITERION

- **A** linear closed-loop ***continuous control system*** is absolutely stable if the roots of the characteristic equation have negative real parts.

- Equivalently, the poles of the closed-loop transfer function, or the zeros of the denominator, ***1 + GH(s),*** of the closed-loop transfer function, must lie in the left-half plane.

- The Nyquist Stability Criterion establishes the number of zeros of ***1 + GH(s)*** in the right-half plane directly from the Nyquist Stability Plot of ***GH(s)***.

S – Contour for Stability

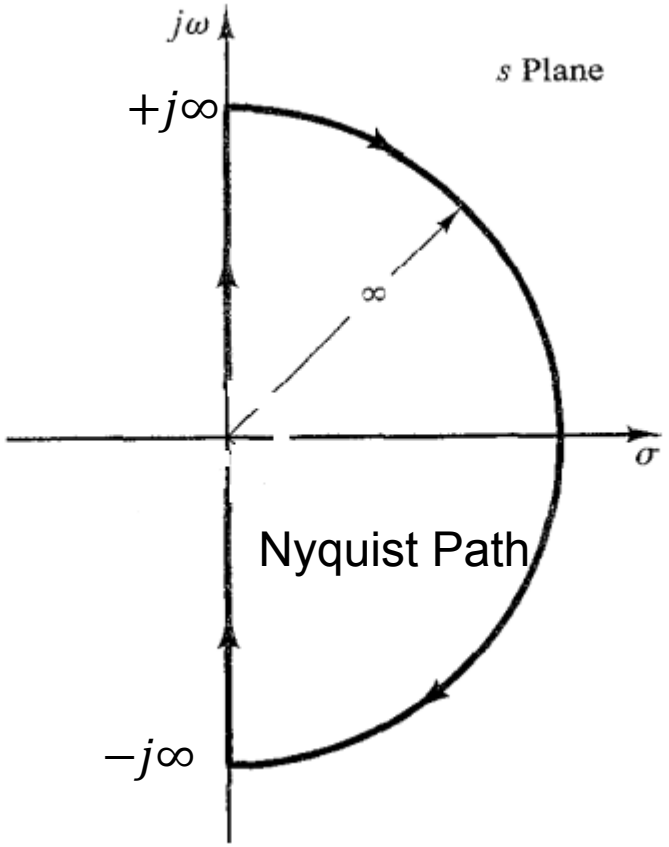
For analyzing the stability of linear control systems, we let the closed contour in the s plane enclose the entire right-half s plane. The contour consists of the entire j ω axis from $-j\infty \leq \omega \leq = +j\infty$ ω
and a semicircular path of infinite radius in the right-half s plane. Such a contour is called the Nyquist path. (The direction of the path is clockwise.) The Nyquist path encloses the entire right-half s plane and encloses all the zeros and poles of 1 + G(s)H(s) that have positive real parts

$$G(s)H(s) = F(S) - 1$$

$$Replacing\ S = j\omega$$

$$G(j\omega)H(j\omega) = F(j\omega) - 1$$

We use the Nyquist plot of the open loop system and the number of encirclement of (-1,j0) point



Nyquist Stability Criterion

Nyquist stability criterion [for a special case when G(s)H(s) has neither poles nor zeros on the jw axis.]: if the open-loop transfer function G(s)H(s) has k poles in the right-half s plane then for stability, the G(jw) H(jw) locus, as w varies from -∞ to +∞, must encircle the (-1 + j0) point k times in the counterclockwise direction.

This criterion can be expressed as $N = Z_0 - P_0$

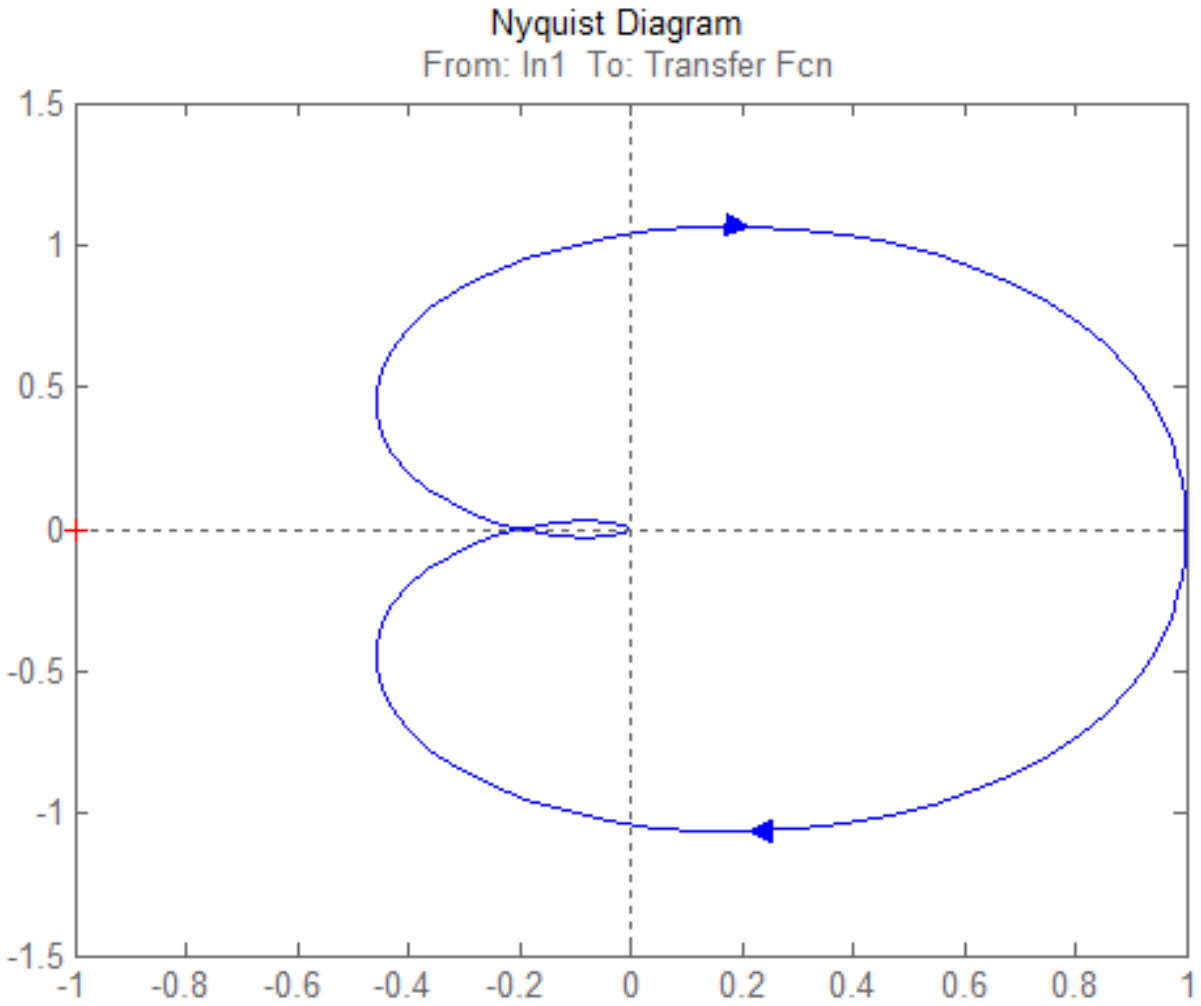
where Z_0 = number of zeros of 1 + G(s)H(s)(number of positive poles of the closed loop system) in the right-half s plane

N = number of clockwise encirclements of the -1 + j0 point

P_0 = number of positive poles of G(s)H (s) (open loop system)in the right-half s plane

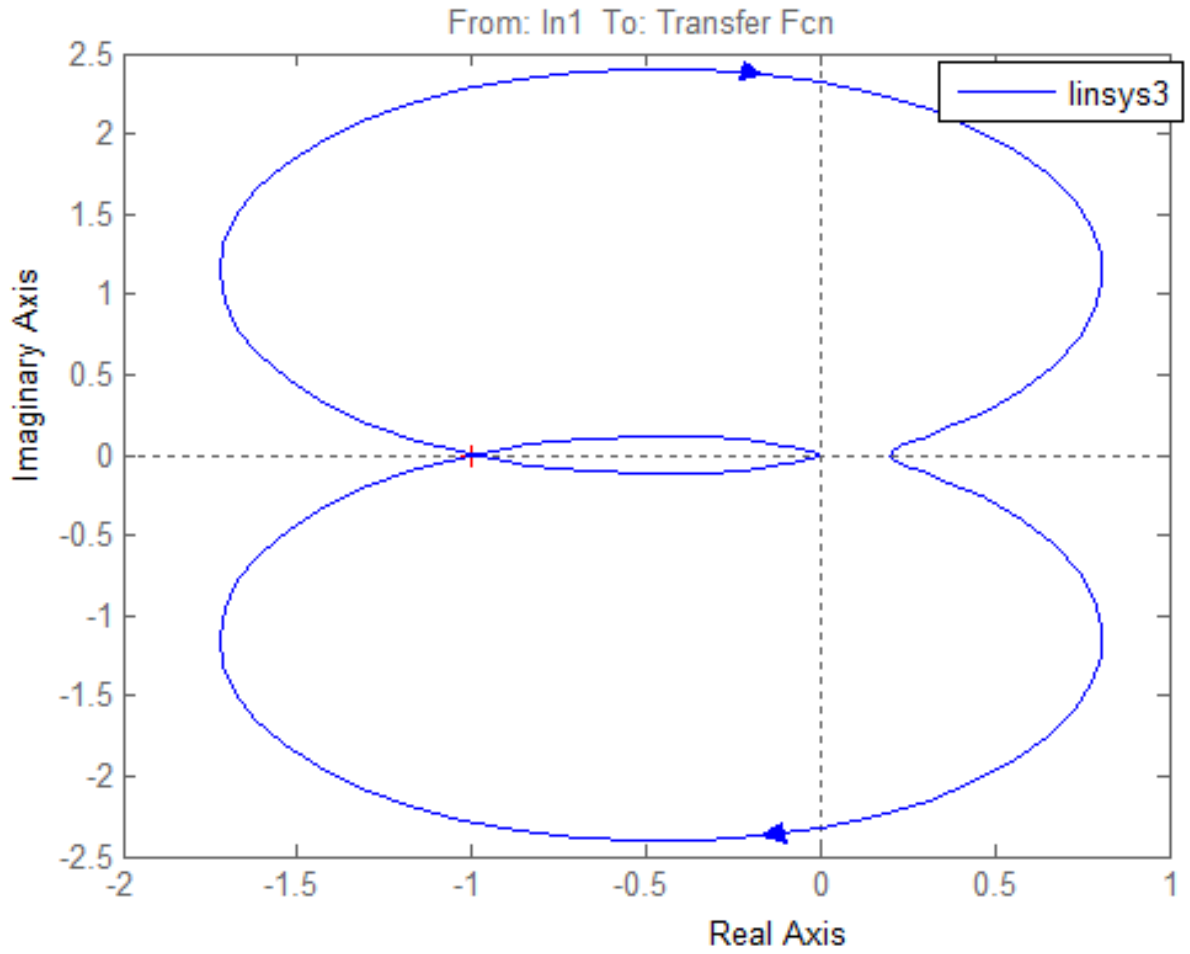
$A-The\ open\ loop\ system\ is\ stable$

G(s)H(s)=$\frac{1}{s^3+\ 2s^2+3s+1}$



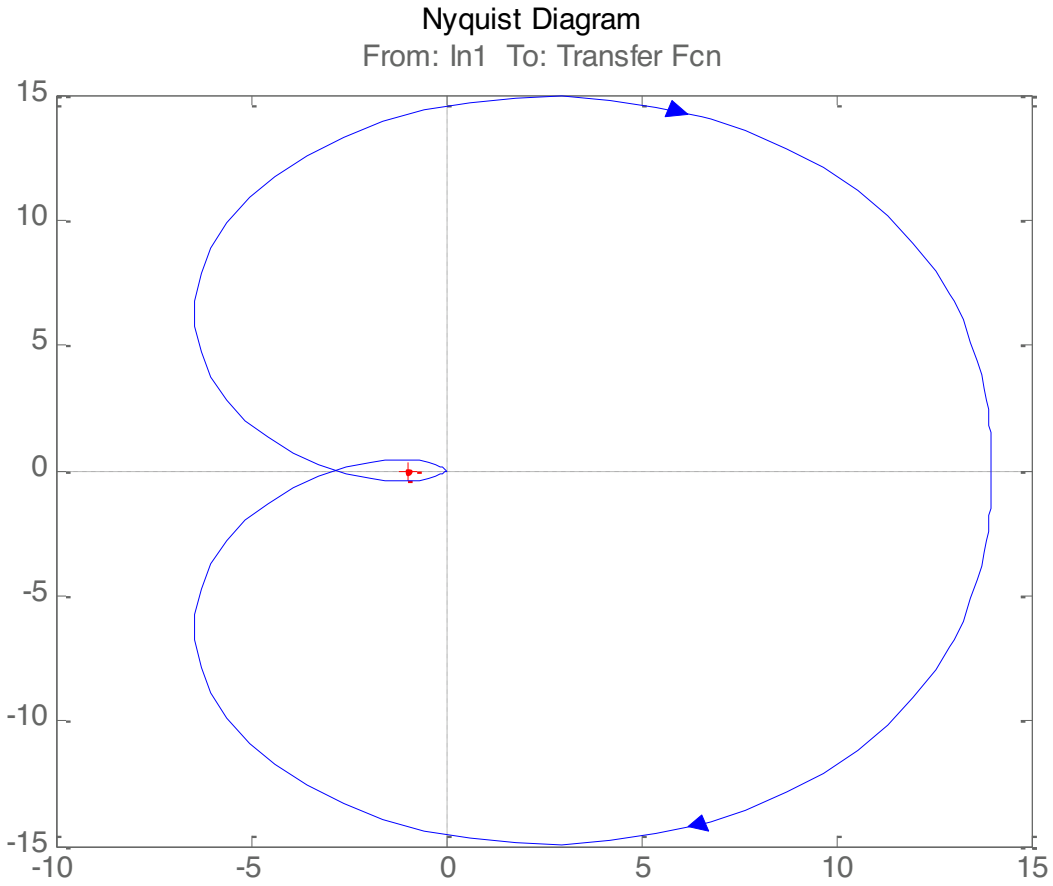
$P_0 = 0,\ then\ Z_0$=N+ P_0 = 0+0 =0 The closed loop system is stable

$$G(s)H(s)=\frac{5}{s^3+\ 2s^2\ +3s\ +1}$$



$P_0 = 0,\ then\ \ N$=?+ $P_0 =$?+0 The closed loop system is critically stable

$$G(s)H(s)=\frac{13}{s^3+\ 2s^2\ +3s\ +1}$$

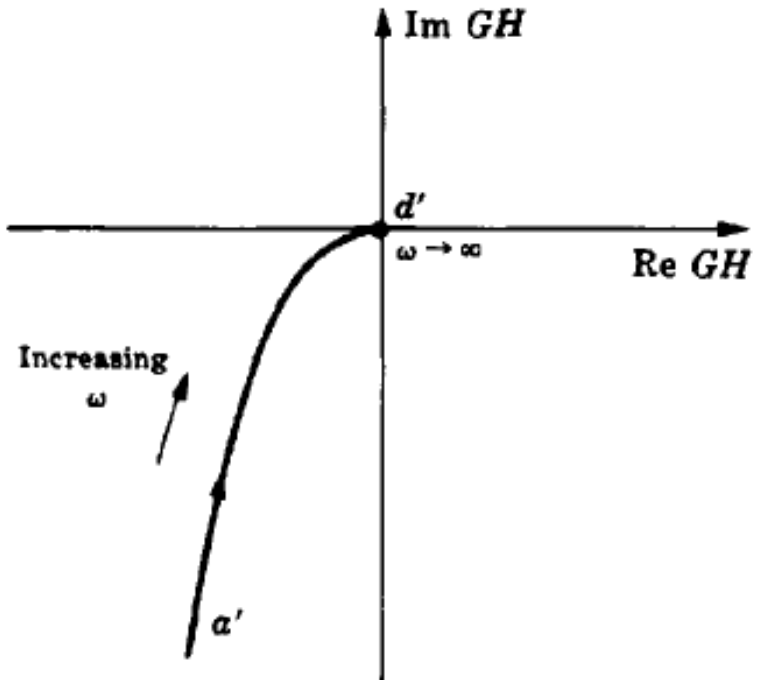


$P_0 = 0,\ then\ Z_0$=N+ $P_0 =$ 2+0 =2 The closed loop system is unstable

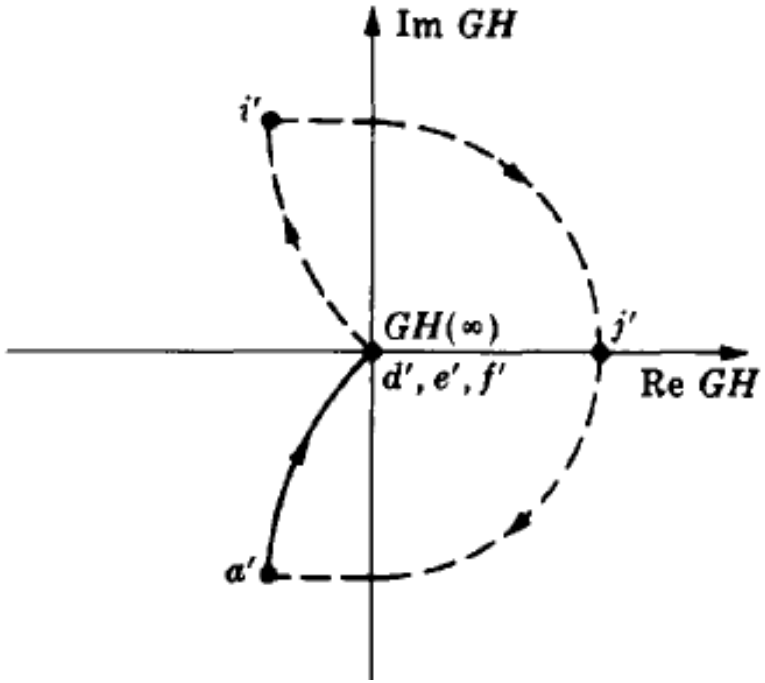
B-The open loop system is critically stable

$$GH(s) = \frac{1}{s(s+1)}$$

Nyquist or Polar Plot



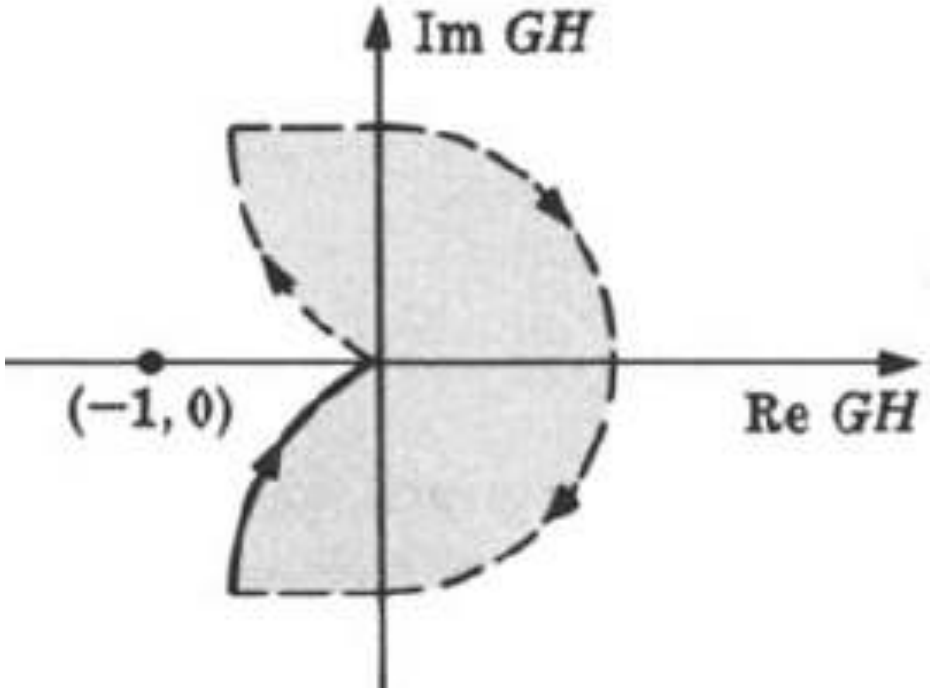
Nyquist Diagram



- The region to the right of the contour has been shaded.

- Clearly, the **(-1,0)** point is not in the shaded region; therefore it is not enclosed by the contour and so $N \leq 0$.

- The poles of ***GH(s)*** are at **s = 0** and **s = -1**, neither of which are in the right-hand-plane RHP; hence $P_o = 0$. Thus $N = -P_o = 0$, and the system is absolutely stable.



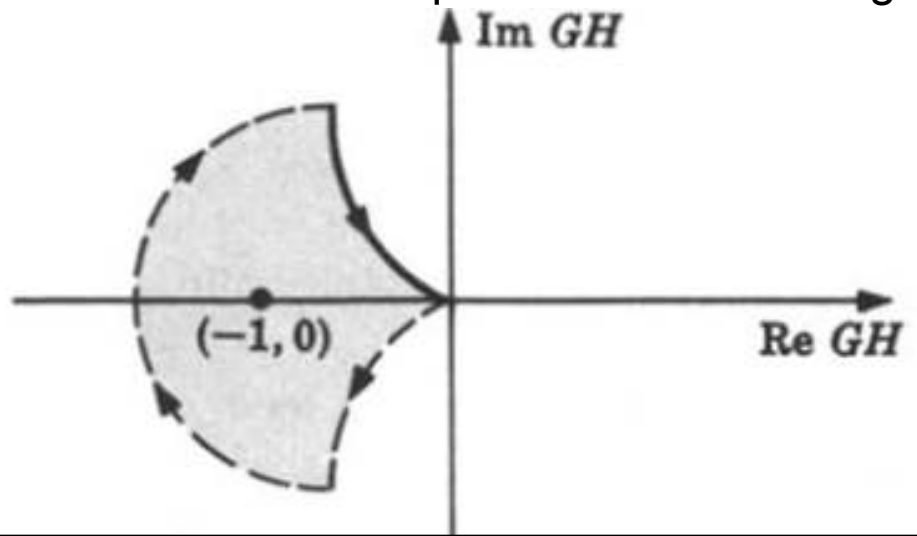
$C - the\ \ open\ loop\ system\ is\ unstable$

The Nyquist Stability Plot for ***GH(s)*** **= 1/s(s-1)** is given in the figure below.

- The region to the right of the contour has been shaded and the **(-1,0)** point is enclosed; then ***N*** **> 0**. (It is clear that ***N*** **= 1**).
-
- The poles of ***GH*** are at **s = 0** and **s = +1**, the latter pole being in the RHP. Hence P_o = **1**.

- $N \neq P_o$ indicates that the **system is unstable**.

- $Z = N + P_o = 1 + 1 = 2,$ therefore the poles of the closed-loop transfer function, or the zeros of the characteristic equation ***1+GH*** in the right-hand plane RHP.



Relative Stability

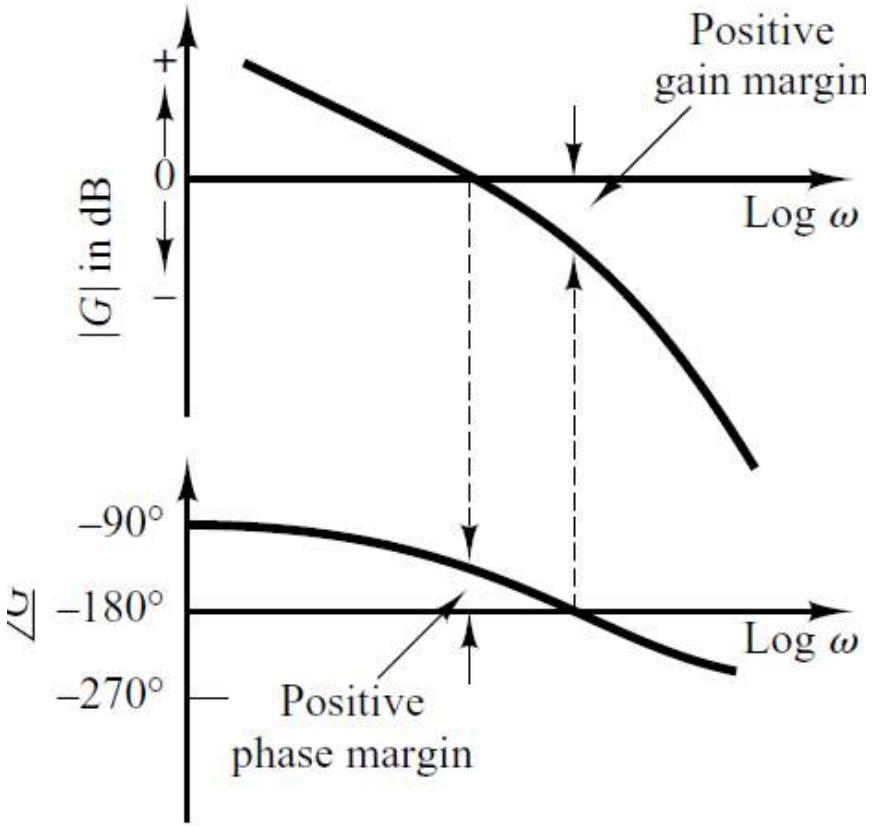
- **Phase Crossover Frequency (ω_p)** is the frequency at which the phase angle of the open-loop transfer function equals –180°.

- **Gain Crossover Frequency (ω_g)** is the frequency at which the magnitude of the open loop transfer function, is unity.

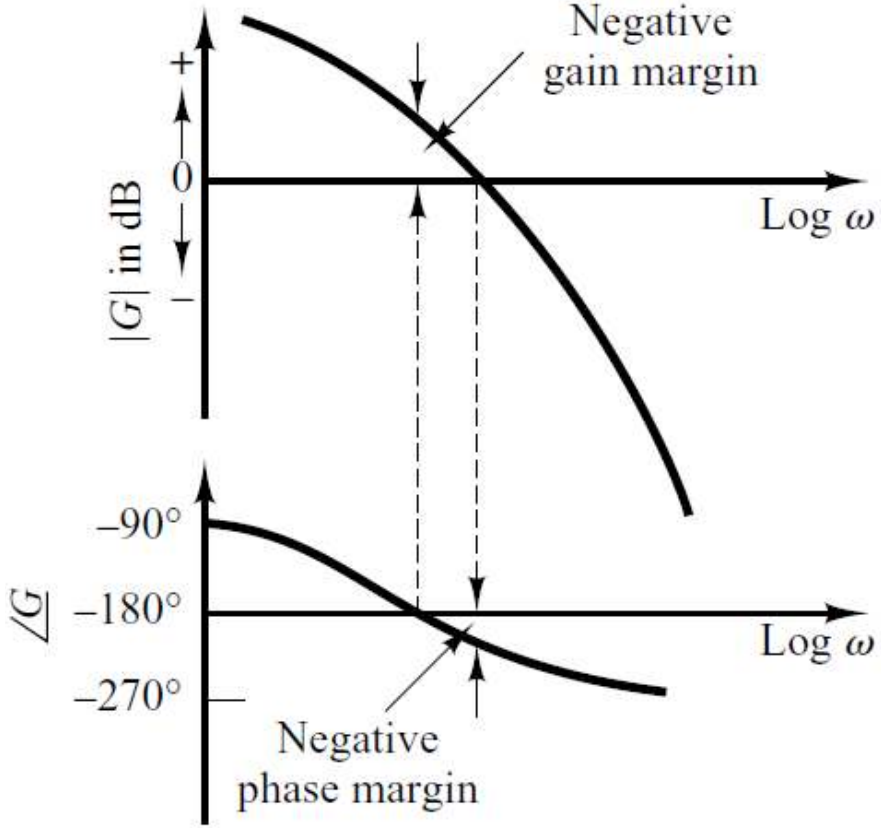
- **Gain margin (K_g)** is the reciprocal of the magnitude of G(jω) at the phase cross over frequency.

$$K_g = \frac{1}{|G(j\omega_1)|} \qquad K_g \text{ dB} = 20 \log K_g = -20 \log |G(j\omega_1)|$$

Phase margin (γ) is that amount of additional phase lag at the gain crossover frequency required to bring the system to the verge of instability.



Stable system



Unstable system

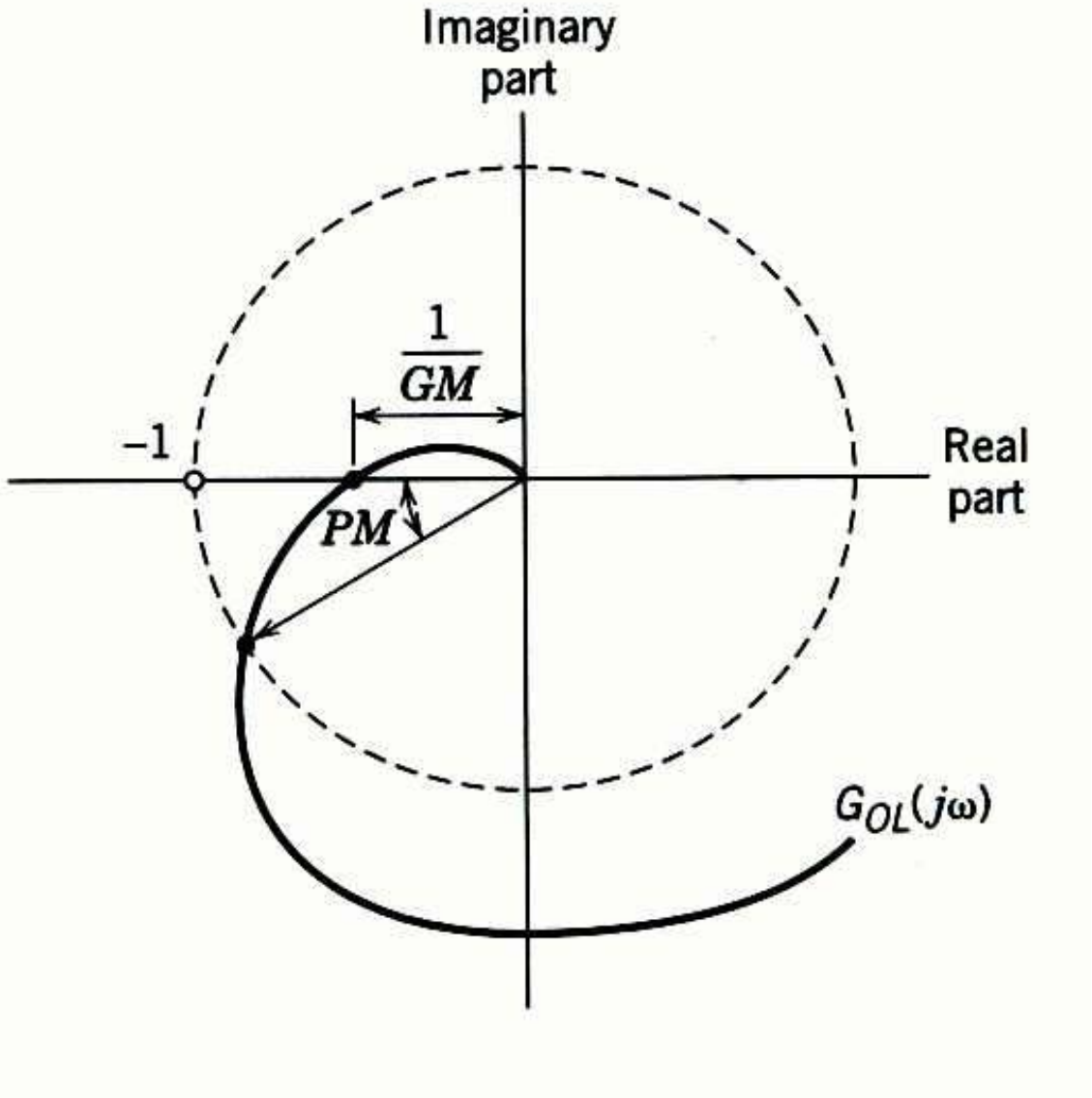


Figure 14.10 Gain and phase margins on a Nyquist plot.

تم تسجل المادة المتبقية وتحميلها على التعليم الالكتروني

انتهى التدريس

اليوم مافيه محاضره

علامة المشاركة

First quiz	10
Second quiz	10
Online test and assignment	10

علامة المشاركة اعلى 20 علامه واعتذر عن قبول اي طلب بهذا الخصوص